

Stochastic Analysis

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December 9, 2025

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Chapter 1

Basic Measure Theory

1.1 Conditional Expectation

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 1.1.1 (Conditional Expectation). Let $X: \Omega \rightarrow \mathbb{R}$ be a L^1 random variable and $\mathcal{G} \subset \mathcal{F}$ be a σ -sub-field. A random variable Y is called the conditional expectation of X given $\mathcal{G}$ if

- (i) Y is $\mathcal{G}$ -measurable,
- (ii) for any $A \in \mathcal{G}$,

$$\int_A Y d\mathbb{P} = \int_A X d\mathbb{P},$$

Theorem 1.1.2. For given X and $\mathcal{G}$, such Y exists and is unique, denoted by $Y = \mathbb{E}[X | \mathcal{G}]$.

Proof. For the uniqueness, let Y' be another conditional expectation. Let

$$A_\varepsilon = \{Y - Y' \geq \varepsilon\} \in \mathcal{G}.$$

for any $\varepsilon > 0$. So

$$\varepsilon \mathbb{P}(A_\varepsilon) \leq \int_{A_\varepsilon} Y - Y' d\mathbb{P} = \int_{A_\varepsilon} X d\mathbb{P} - \int_{A_\varepsilon} X d\mathbb{P} = 0.$$

As $\varepsilon \rightarrow 0$, $Y \leq Y'$ a.e.. Similarly, we have $Y' \leq Y$. So $Y = Y'$.

For existence, WTLG, assume $X \geq 0$. Let

$$\nu(A) = \int_A X d\mathbb{P}, \quad A \in \mathcal{G}.$$

Then ν is a measure on $\mathcal{G}$, which is absolutely continuous with respect to $\mathbb{P}$ on $\mathcal{G}$. So by the Radon-Nikodym theorem, there exists a $\mathcal{G}$ -measurable Y such that

$$\int_A Y d\mathbb{P} = \int_A X d\mathbb{P}, \quad A \in \mathcal{G}. \quad \square$$

Example 1.1.3. Suppose $X \in L^2$. Then

$$\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])^2] = \inf \{ \mathbb{E}[(X - Y)^2] \mid Y \text{ is } \mathcal{G} - \text{measurable.} \}$$

1.2 Change of Measures

Fix $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ satisfying the usual condition. Let $W = (W_t)_{t \geq 0}$ be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$. Denote $\mathbb{E} = \mathbb{E}_{\mathbb{P}}$.

Proposition 1.2.1. *Let $\mathbb{P}, \mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$. Suppose $\mathbb{Q} \ll \mathbb{P}$ and*

$$Z = \frac{d\mathbb{Q}}{d\mathbb{P}}.$$

Then for any σ -algebra $\mathcal{G} \subset \mathcal{F}$, $\mathbb{Q} \ll \mathbb{P}$ on $\mathcal{G}$ and

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{G}} = \mathbb{E}[Z \mid \mathcal{G}].$$

Proof. Absolutely continuity is obvious. For any $A \in \mathcal{G}$, by the property of conditional expectation,

$$\mathbb{Q}(A) = \int_A Z d\mathbb{P} = \int_A \mathbb{E}[Z \mid \mathcal{G}] d\mathbb{P}.$$

So by the uniqueness in Radon-Nikodym Theorem,

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{G}} = \mathbb{E}[Z \mid \mathcal{G}]. \quad \square$$

Proposition 1.2.2. *Let $\mathbb{P}, \mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$. Suppose $\mathbb{P} \sim \mathbb{Q}$ with $Z = \frac{d\mathbb{Q}}{d\mathbb{P}}$. Let $\mathcal{G} \subset \mathcal{F}$ σ -subalgebra. Then for any $\mathcal{F}$ -measurable $Y \geq 0$.*

$$\mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}] = \frac{\mathbb{E}[YZ \mid \mathcal{G}]}{\mathbb{E}[Z \mid \mathcal{G}]}.$$

Remark 1.2.3. For general Y , we need $Y \in L^1$ then $Y = Y^+ - Y^-$.

Proof. For any $A \in \mathcal{G}$,

$$\begin{aligned} \int_A \mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}] d\mathbb{Q} &= \int_A Y d\mathbb{Q} = \int_A Y Z d\mathbb{P} \\ &= \int_A \mathbb{E}[YZ \mid \mathcal{G}] d\mathbb{P} = \int_A \mathbb{E}[YZ \mid \mathcal{G}] \frac{d\mathbb{P}}{d\mathbb{Q}} d\mathbb{Q} \\ &= \int_A \mathbb{E}[YZ \mid \mathcal{G}] \left(\frac{d\mathbb{Q}}{d\mathbb{P}} \right)^{-1} d\mathbb{Q} = \int_A \mathbb{E}[YZ \mid \mathcal{G}] \left(\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{G}} \right)^{-1} d\mathbb{Q} \\ &= \int_A \mathbb{E}[YZ \mid \mathcal{G}] \mathbb{E}[Z \mid \mathcal{G}]^{-1} d\mathbb{Q}. \end{aligned}$$

So

$$\mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{G}] = \frac{\mathbb{E}[YZ \mid \mathcal{G}]}{\mathbb{E}[Z \mid \mathcal{G}]} \quad \square$$

Chapter 2

Discrete Martingale Theory

2.1 Stochastic Process

Definition 2.1.1 (Stochastic Process). A family of $\{X_t : t \in I\}$ of random variables on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called a stochastic process, where

- (1) $I = \mathbb{N} \cup \{0\}$ or
- (2) $I = [0, \infty)$.

Definition 2.1.2 (Finite Dimensional Distribution). A finite distribution of a stochastic process $\{X_t : t \in I\}$ for a give sequence of time $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$ is the probability law of $(X_{t_1}, \dots, X_{t_n})$. If two stochastic processes has same finite distributions, then they are called having the same law.

2.2 Discrete Martingale

Theorem 2.2.1. Let $X_1, \dots, X_n$ be a sequence of random variables. Then Y is $\sigma(X_1, \dots, X_n)$ -measurable if and only if $Y = g(X_1, \dots, X_n)$ for some measurable function g .

Example 2.2.2. (1) Let $\{\mathcal{F}_n\}$ be a filtration and Y be an integrable random variable. Let $Z_n = \mathbb{E}[Y \mid \mathcal{F}_n]$. Clearly, Z_n is $\mathcal{F}_n$ -measurable and by Jensen's Inequality,

$$\mathbb{E}[|Z_n|] \leq \mathbb{E}[Y] < \infty.$$

Furthermore,

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[\mathbb{E}[Y \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] = \mathbb{E}[Y \mid \mathcal{F}_n] = Z_n,$$

i.e. Z_n is a $\{\mathcal{F}_n\}$ -martingale.

- (2) Assume $X_1, X_n, \dots$ are independent, integrable random variables with $a_n = \mathbb{E}[X_n] \neq 0$. Define

$$Z_n = \frac{X_1 X_2 \cdots X_n}{a_1 a_2 \cdots a_n}$$

and $Z_0 = 1$. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. So Z_n is $\mathcal{F}_n$ -measurable and is integrable by the independence. Moreover,

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \frac{X_1 \cdots X_n}{a_1 \cdots a_n a_{n+1}} \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \frac{X_1 \cdots X_n}{a_1 \cdots a_n} = Z_n$$

i.e. Z_n is a $\{\mathcal{F}_n\}$ -martingale.

- (3) Assume $X_1, X_2, \dots$ are independent, integrable random variables valued $1, -1$ with $\mathbb{P}(X_n = 1) = \mathbb{P}(X_n = -1) = \frac{1}{2}$. Define

$$Z_n = S_n^2 - n, \quad S_n = \sum_{i=1}^n X_i.$$

Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Then clearly Z_n is $\mathcal{F}_n$ -measurable and is integrable.

$$\begin{aligned} \mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[(S_n + X_{n+1})^2 \mid \mathcal{F}_n] - n - 1 \\ &= S_n^2 + 2S_n\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + \mathbb{E}[X_{n+1}^2 \mid \mathcal{F}_n] - n - 1 \\ &= S_n^2 + 2S_n\mathbb{E}[X_{n+1}] + \mathbb{E}[X_{n+1}^2] - n - 1 \\ &= S_n^2 - n = Z_n. \end{aligned}$$

So Z_n is a $\{\mathcal{F}_n\}$ -martingale.

Proposition 2.2.3. (1) If $(X_n)_{n \geq 0}$ is a martingale w.r.t. $\{\mathcal{F}_n\}$ and φ is a convex function such that $\varphi(X_n) \in L^1$, then $\{\varphi(X_n)\}_{n \geq 0}$ is a submartingale.

- (2) If $(X_n)_{n \geq 0}$ is a sub(sup)-martingale w.r.t. $\{\mathcal{F}_n\}$ and φ is a increasing(decreasing) convex function such that $\varphi(X_n) \in L^1$, then $\{\varphi(X_n)\}_{n \geq 0}$ is a submartingale. In particular, $\{(X_n - a)_+\}$ is a submartingale.

2.3 Stopping Time

Definition 2.3.1 (Stopping Time). Let $\{\mathcal{F}_n\}_{n \geq 0}$ be a filtration. A random variable $T(\omega) \in \mathbb{N} \cup \{0\} \cup \{\infty\}$ is called a stopping time w.s.t $\{\mathcal{F}_n\}_{n \geq 0}$ if

$$\{T \leq n\} \in \mathcal{F}_n, \quad n \in \mathbb{N} \cup \{0\} \cup \{\infty\}.$$

Remark 2.3.2. By the definition, it is clear that $\{T > n\} \in \mathcal{F}_n$ and so $\{T = n\} = \{T \leq n\} \cap \{T > n-1\} \in \mathcal{F}_n$ and also $\{T < n\} = \bigcup_{k=1}^{n-1} \{T = k\} \in \mathcal{F}_n$.

Definition 2.3.3 (Stopping Process). Let T be a stopping time and $\{Z_n\}_{n \geq 0}$ be a stochastic process. Define

$$Z_{T \wedge n}(\omega) = \begin{cases} Z_n(\omega), & n \leq T(\omega) \\ Z_T(\omega), & n > T(\omega) \end{cases}$$

Then the process $\{Z_{T \wedge n}\}_{n \geq 0}$ is called the stopping process of Z at T .

Theorem 2.3.4. If $\{Z_n\}_{n \geq 0}$ is a (sub or sup-)martingale w.s.t. $\mathbb{F} = \{\mathcal{F}_n\}_{n \geq 0}$ and T is a stopping time w.s.t. $\mathbb{F}$, then the stopping process $\{Z_{T \wedge n}\}_{n \geq 0}$ is also a (sub or sup-)martingale w.s.t. $\mathbb{F}$.

Proof. Let $Y_n = Z_{T \wedge n}$. Then

$$\begin{aligned} Y_n &= Z_{T \wedge n} \mathbb{I}_{\{T \geq n\}} + Z_{T \wedge n} \mathbb{I}_{\{T < n\}} \\ &= Z_n \mathbb{I}_{\{T \geq n\}} + Z_T \mathbb{I}_{\{T < n\}} \\ &= Z_n \mathbb{I}_{\{T \geq n\}} + \sum_{k=0}^{n-1} Z_k \mathbb{I}_{\{T = k\}} \end{aligned}$$

Therefore, Y_n is $\mathcal{F}_n$ -measurable and L^1 . For the martingale property, first note that

$$\begin{aligned} Y_{n+1} &= Z_{T \wedge n+1} \\ &= Z_{T \wedge n} + \mathbb{I}_{\{T \geq n+1\}}(Z_{n+1} - Z_n) \\ &= Y_n + \mathbb{I}_{\{T \geq n+1\}}(Z_{n+1} - Z_n). \end{aligned}$$

Therefore, by $\mathbb{I}_{\{T \geq n+1\}} = \mathbb{I}_{\{T < n\}}^c \in \mathcal{F}_n$,

$$\begin{aligned} \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}[Y_n \mid \mathcal{F}_n] + \mathbb{E}[\mathbb{I}_{\{T \geq n+1\}}(Z_{n+1} - Z_n) \mid \mathcal{F}_n] \\ &= Y_n + \mathbb{I}_{\{T \geq n+1\}}\mathbb{E}[(Z_{n+1} - Z_n) \mid \mathcal{F}_n] \\ &= Y_n. \end{aligned}$$

Similarly, the reasoning is true for sub or sup-martingale. $\square$

Lemma 2.3.5. *Suppose (X_n) is a supermartingale. Let T, S be two bounded stopping times with $S \leq T \leq N$. Then*

$$\int_{S \leq N} X_T \, d\mathbb{P} \leq \int_{S \leq N} X_S \, d\mathbb{P}.$$

Proof. Let $Y_n = X_{T \wedge n} - X_{S \wedge n}$ and note that

$$Y_n - Y_{n-1} = \mathbb{I}_{\{T \geq n > S\}}(X_n - X_{n-1}).$$

It follows that

$$\mathbb{E}[Y_n - Y_{n-1} \mid \mathcal{F}_{n-1}] = \mathbb{E}[\mathbb{I}_{\{T \geq n > S\}}(X_n - X_{n-1}) \mid \mathcal{F}_{n-1}] = \mathbb{I}_{\{T \geq n > S\}}\mathbb{E}[(X_n - X_{n-1}) \mid \mathcal{F}_{n-1}] \leq 0$$

because $\{T \geq n > S\} \in \mathcal{F}_{n-1}$. Therefore, Y_n is a supermartingale, i.e.,

$$\mathbb{E}[Y_N \mid \mathcal{F}_n] \leq Y_n, \quad \forall n \leq N.$$

It implies that

$$\mathbb{I}_{S=n}Y_n \geq \mathbb{E}[\mathbb{I}_{S=n}Y_N \mid \mathcal{F}_n]$$

and by taking expectation we have

$$\mathbb{E}[\mathbb{I}_{S=n}Y_N] \leq \mathbb{E}[\mathbb{I}_{S=n}Y_n] = 0,$$

i.e.,

$$\mathbb{E}[\mathbb{I}_{S=n}Y_N] = \mathbb{E}[\mathbb{I}_{S=n}(X_T - X_S)] \leq 0$$

Taking summation of n from 1 to N , we get

$$\int_{S \leq N} X_T \, d\mathbb{P} \leq \int_{S \leq N} X_S \, d\mathbb{P}. \quad \square$$

2.4 Martingale Convergence Theorem

For a real-valued $\mathbb{F} = (\mathcal{F}_n)$ -adapted process (X_n) and $a < b$, define a sequence of stopping times (τ_n) as follows,

$$\begin{aligned} \tau_1 &:= \min \{n : X_n \leq a\}, \quad \tau_2 := \min \{n \geq \tau_1 : X_n \geq b\}, \dots \\ \tau_{2k+1} &:= \min \{n \geq \tau_{2k} : X_n \leq a\}, \quad \tau_{2k+2} := \min \{n \geq \tau_{2k+1} : X_n \geq b\}, \dots \end{aligned}$$

Set a random variable

$$U_N^X(a, b) := \max \{k : \tau_{2k} \leq N\}$$

Then $U_N^X(a, b)$ is the number of up-crossing of $(X_n)_{n=0}^N$ for the interval $[a, b]$.

Theorem 2.4.1. *Let (X_n) is a supermartingale. We have*

$$\mathbb{P} (U_N^X(a, b) > j) \leq \frac{1}{b-a} \int_{U_N^X(a, b)=j} (X_N - a)^- \, d\mathbb{P}$$

and

$$\mathbb{E} [U_N^X(a, b)] \leq \frac{1}{b-a} \mathbb{E} [(X_N - a)^-].$$

Proof. WTLG, assume $a = 0$ and $(X_n)_{n=0}^N$. Set

$$S = \tau_{2j+1} \wedge (N+1), \quad T = \tau_{2(j+1)} \wedge (N+1).$$

Then

$$\{\tau_{2j+1} \leq N\} = \{S \leq N\},$$

on $\{S \leq N\}$, $X_S = X_{\tau_{2j+1}} \leq a = 0$, and

$$\{\tau_{2(j+1)} \leq N\} = \{U_N^X(a, b) \geq j+1\} = \{U_N^X(a, b) > j\},$$

which follows that

$$\{U_N^X(0, b) > j\} = \{\tau_{2(j+1)} \leq N\} = \{S < N, X_T \geq b\}.$$

On the other hand,

$$\{S < N, X_T < b\} = \{S < N, T = N+1\} \subset \{U_N^X(a, b) = j\}.$$

Then

$$\begin{aligned} b\mathbb{P} (U_N^X(0, b) > j) &= \int_{\{U_N^X(0, b) > j\}} b \, d\mathbb{P} = \int_{\{S < N, X_T \geq b\}} b \, d\mathbb{P} \\ &\leq \int_{\{S < N, X_T \geq b\}} X_T \, d\mathbb{P} \\ &= \int_{\{S < N\}} X_T \, d\mathbb{P} - \int_{\{S < N, X_T < b\}} X_T \, d\mathbb{P} \\ &\leq \int_{\{S < N\}} X_S \, d\mathbb{P} - \int_{\{S < N, T=N+1\}} X_T \, d\mathbb{P} \\ &\leq 0 - \int_{\{S < N, T=N+1\}} X_N \, d\mathbb{P} \leq \int_{\{S < N, T=N+1\}} X_N^- \, d\mathbb{P} \\ &\leq \int_{\{U_N^X(a, b)=j\}} X_N^- \, d\mathbb{P}. \end{aligned}$$

and the second result is by taking the sum of j from 0 to ∞ . □

Theorem 2.4.2 (Martingale Convergence Theorem). *Let (X_n) be a supermartingale with*

$$\sup_n \mathbb{E}[X_n^-] < \infty.$$

Then

$$X(\omega) := \lim_{n \rightarrow \infty} X_n(\omega)$$

exists almost everywhere. In particular, if $\sup_n \mathbb{E}[|X_n|] < \infty$, $X \in L^1$.

Proof. For any $a < b$, let

$$U^X(a, b) = \lim_{N \rightarrow \infty} U_N^X(a, b),$$

which always exists by taking value in $[0, \infty]$ since $U_N^X(a, b)$ is monotone increasing. By MCT,

$$\mathbb{E}[U^X(a, b)] = \lim_{N \rightarrow \infty} \mathbb{E}[U_N^X(a, b)] \leq \frac{1}{b-a} \sup_N \mathbb{E}[(X_N - a)^-] < \infty.$$

Set $W_{a,b} = \{U^X(a, b) = \infty\}$, so $\mathbb{P}(W_{a,b}) = 0$. Define

$$V_{a,b} = \left\{ \liminf_n X_n < a, \limsup_n X_n > b \right\},$$

and so $V_{a,b} \subset W_{a,b}$ and $\mathbb{P}(V_{a,b}) = 0$. Next,

$$\left\{ \liminf_n X_n < \limsup_n X_n \right\} = \bigcup_{a < b \in \mathbb{Q}} V_{a,b}$$

So

$$\mathbb{P}\left\{ \liminf_n X_n < \limsup_n X_n \right\} = 0.$$

When $\sup_n \mathbb{E}[|X_n|] < \infty$, by Fatou's lemma, it is obvious $X \in L^1$. $\square$

Example 2.4.3. (1) Let (X_n) be a martingale with $|X_{n+1} - X_n| \leq M$ for any n . Let

$$C = \left\{ \lim_n X_n \text{ exists and finite.} \right\}$$

and

$$D = \left\{ \liminf_n X_n = -\infty, \limsup_n X_n = \infty \right\}.$$

Then we have

$$\mathbb{P}(C \cup D) = 1.$$

Proof. WTLG, assume $X_0 = 0$. For any $k \in \mathbb{N}$, let

$$N_k := \inf \{n : X_n \leq -k\},$$

which is a stopping time. So $(X_{n \wedge N_k})$ is also a martingale. Note that

$$X_{N_k} = X_{N_k} - X_{N_k-1} + X_{N_k-1} \geq -M - k,$$

so

$$X_{n \wedge N_k} \geq -k - M \Rightarrow X_{n \wedge N_k} + a + M \geq 0$$

Then by the martingale convergence theorem, $\lim_n X_{n \wedge N_k}$ exists. So $\lim_n X_n$ exists on $\{N_k = \infty\}$.

$$\left\{ \liminf_n X_n > -\infty \right\} = \bigcup_k \{N_k = \infty\}$$

It implies that $\lim_n X_n$ exists on $\{\liminf_n X_n > -\infty\}$. Similarly, by considering $-X_n$, $\lim_n X_n$ exists on $\{\limsup_n X_n < \infty\}$. $\square$

(2) Let $(\mathcal{F}_n)$ be a filtration with $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Let $B_n \in \mathcal{F}_n$ be a sequence of events.

$$\bigcap_n \bigcup_{k \geq n} B_k = \left\{ \sum_{n=1}^{\infty} \mathbb{E}[\mathbb{I}_{B_n} \mid \mathcal{F}_{n-1}] = \infty \right\}$$

Proof. Set $X_0 = 0$ and $X_n = \sum_{m=1}^n \mathbb{I}_{B_m}$. Note that

$$\bigcap_n \bigcup_{k \geq n} B_k = \left\{ \sum_{m=1}^{\infty} \mathbb{I}_{B_m} = \infty \right\}$$

Define $M_0 = 0$ and

$$M_n = X_n - \sum_{m=1}^n \mathbb{E}[\mathbb{I}_{B_m} \mid \mathcal{F}_{m-1}] = \sum_{m=1}^n (\mathbb{I}_{B_m} - \mathbb{E}[\mathbb{I}_{B_m} \mid \mathcal{F}_{m-1}])$$

and so (M_n) is a martingale w.s.t. $(\mathcal{F}_n)$. Moreover,

$$|M_{n+1} - M_n| = |\mathbb{I}_{B_{n+1}} - \mathbb{E}[\mathbb{I}_{B_{n+1}} \mid \mathcal{F}_n]| \leq 2$$

By above, it suffices to prove that on C and D . For C , because $\lim_n M_n$ exists,

$$\sum_{m=1}^{\infty} \mathbb{I}_{B_m} = \infty \Leftrightarrow \sum_{m=1}^{\infty} \mathbb{E}[\mathbb{I}_{B_m} \mid \mathcal{F}_{m-1}] = \infty.$$

On D , it is also true. □

2.5 Doob's Decomposition

Definition 2.5.1. Let $(\mathcal{F}_n)$ be a filtration.

- (1) A stochastic process (H_n) is called adapted w.s.t. $(\mathcal{F}_n)$ if H_n is $\mathcal{F}_n$ -measurable.
- (2) A stochastic process (H_n) is called predictable w.s.t. $(\mathcal{F}_n)$ if H_n is $\mathcal{F}_{n-1}$ -measurable.

Theorem 2.5.2 (Doob's Decomposition Theorem). *Any submartingale (X_n) can be uniquely written as*

$$X_n = M_n + A_n$$

where M_n is a martingale and A_n is a predictable increasing process with $A_0 = 0$.

Proof. If $X_n = M_n + A_n$, then

$$\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] = \mathbb{E}[M_n + A_n \mid \mathcal{F}_{n-1}] = M_{n-1} + A_n = X_{n-1} - A_{n-1} + A_n.$$

So

$$A_n - A_{n-1} = \mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}],$$

which implies that by setting A_0

$$A_n = \sum_{k=1}^n \mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}]$$

that is predictable and increasing because (X_n) is a submartingale. Let $M_n := X_n - A_n$.

$$\begin{aligned} \mathbb{E}[M_n \mid \mathcal{F}_{n-1}] &= \mathbb{E}[X_n \mid \mathcal{F}_{n-1}] - A_n \\ &= \mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] - A_n + X_{n-1} \\ &= A_n - A_{n-1} - A_n + X_{n-1} = M_{n-1}. \end{aligned}$$

So M_n is a martingale. □

Remark 2.5.3. Note that if (X_n) is a supermartingale then it can be uniquely written as

$$X_n = M_n - A_n,$$

for a martingale M_n and a predictable increasing process A_t with $A_0 = 0$.

2.6 L^p Convergence

Lemma 2.6.1 (Bounded Optional Stopping Time Theorem). *If (X_n) is a submartingale and N is a finite stopping time with $N \leq K$, then*

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_N] \leq \mathbb{E}[X_K]$$

Proof. We have known $(X_{n \wedge N})$ is a submartingale, i.e.,

$$\mathbb{E}[X_0] = \mathbb{E}[X_{0 \wedge N}] \leq \mathbb{E}[X_{N \wedge K}] = \mathbb{E}[X_N].$$

For the second part, because $N \leq K$, $\Omega = \bigcup_{n=0}^K \{N = n\}$. It follows that

$$\mathbb{E}[X_N] = \sum_{n=0}^K \mathbb{E}[X_N \mathbb{I}_{\{N=n\}}] = \sum_{n=0}^K \mathbb{E}[X_n \mathbb{I}_{\{N=n\}}].$$

Moreover, for any $n \leq K$, because (X_n) is a submartingale

$$X_n \leq \mathbb{E}[X_K \mid \mathcal{F}_n].$$

Because N is a stopping time,

$$X_n \mathbb{I}_{\{N=n\}} \leq \mathbb{E}[X_K \mathbb{I}_{\{N=n\}} \mid \mathcal{F}_n] \Rightarrow \mathbb{E}[X_n \mathbb{I}_{\{N=n\}}] \leq \mathbb{E}[X_K \mathbb{I}_{\{N=n\}}].$$

So

$$\mathbb{E}[X_N] \leq \sum_{n=0}^K \mathbb{E}[X_K \mathbb{I}_{\{N=n\}}] = \mathbb{E}[X_K].$$

□

Theorem 2.6.2 (Doob's Martingale Inequality). *Let (X_n) be a submartingale. Define*

$$A = \left\{ \max_{0 \leq m \leq n} X_m \geq \lambda \right\}, \quad \lambda > 0.$$

Then we have

$$\mathbb{P}(A) \leq \frac{1}{\lambda} \mathbb{E}[X_n \mathbb{I}_A] \leq \frac{1}{\lambda} \mathbb{E}[X_n^+ \mathbb{I}_A] \leq \frac{1}{\lambda} \mathbb{E}[X_n^+]$$

Proof. Define

$$N = \min \{m : X_m \geq \lambda\} \wedge n$$

Clearly, $N \leq n$ is a stopping time. On A , $X_N \geq \lambda$. It follows that

$$\lambda \mathbb{P}(A) = \int_A \lambda \, d\mathbb{P} \leq \int_A X_N \, d\mathbb{P} = \mathbb{E}[X_N \mathbb{I}_A]$$

By above lemma, $\mathbb{E}[X_N] \leq \mathbb{E}[X_n]$. Note that

$$\begin{aligned} \mathbb{E}[X_N] &= \mathbb{E}[X_N \mathbb{I}_A] + \mathbb{E}[X_N \mathbb{I}_{A^c}] \\ &= \mathbb{E}[X_N \mathbb{I}_A] + \mathbb{E}[X_n \mathbb{I}_{A^c}] \end{aligned}$$

and

$$\mathbb{E}[X_n] = \mathbb{E}[X_n \mathbb{I}_A] + \mathbb{E}[X_n \mathbb{I}_{A^c}].$$

So $\mathbb{E}[X_N \mathbb{I}_A] \leq \mathbb{E}[X_n \mathbb{I}_A]$.

□

Theorem 2.6.3. Let (X_n) be a submartingale. Set

$$\bar{X}_n = \max_{0 \leq m \leq n} X_m.$$

Then for any $p > 1$,

$$\mathbb{E} [\bar{X}_n^p] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E} [(X_n^+)^p].$$

In particular, if (Y_n) is a martingale and set $Y_n^* = \max_{0 \leq m \leq n} |Y_m|$, then for $p > 1$

$$\mathbb{E} [(Y_n^*)^p] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E} [|Y_n|^p].$$

Note that this inequality is only useful for $\mathbb{E} [|Y_n|^p] < \infty$.

Proof. For $M > 0$, note that

$$\{\bar{X}_n \wedge M \geq \lambda\} = \{\bar{X}_n \geq \lambda\} \text{ or } \emptyset.$$

First,

$$\begin{aligned} \mathbb{E} [(\bar{X}_n \wedge M)^p] &= \mathbb{E} \left[\int_0^{\bar{X}_n \wedge M} p \lambda^{p-1} d\lambda \right] \\ &= \mathbb{E} \left[\int_0^\infty \mathbb{I}_{\{\bar{X}_n \wedge M \geq \lambda\}} p \lambda^{p-1} d\lambda \right] \\ &= \int_0^\infty p \lambda^{p-1} \mathbb{P}(\{\bar{X}_n \wedge M \geq \lambda\}) d\lambda \\ &= \int_0^\infty p \lambda^{p-1} \mathbb{P}(\{\bar{X}_n \geq \lambda\}) \mathbb{I}_{\{M \geq \lambda\}} d\lambda \\ &\leq \int_0^\infty p \lambda^{p-1} \frac{1}{\lambda} \int X_n^+ \mathbb{I}_{\{\bar{X}_n \wedge M \geq \lambda\}} d\mathbb{P} d\lambda \\ &= \int X_n^+ \int_0^{\bar{X}_n \wedge M} p \lambda^{p-2} d\lambda d\mathbb{P} \\ &= \frac{p}{p-1} \int X_n^+ (\bar{X}_n \wedge M)^{p-1} d\mathbb{P} \\ &\leq \frac{p}{p-1} \mathbb{E} [|X_n^+|^p]^{\frac{1}{p}} \mathbb{E} [(\bar{X}_n \wedge M)^p]^{\frac{p-1}{p}}, \end{aligned}$$

where the final inequality is by the Hölder's Inequality. So

$$\mathbb{E} [(\bar{X}_n \wedge M)^p] \leq \mathbb{E} [|X_n^+|^p].$$

As $M \rightarrow \infty$, by Fatou's lemma,

$$\mathbb{E} [\bar{X}_n^p] \leq \left(\frac{p}{p-1} \right)^p \mathbb{E} [(X_n^+)^p].$$

In particular, when (Y_n) is a martingale, $|Y_n|$ is a submartingale by Jensen's Inequality. $\square$

Theorem 2.6.4. Let (X_n) be a martingale with $\sup_n \mathbb{E} [|X_n|^p] < \infty$ for $p > 1$. Then $X_n \rightarrow X$ a.e. and it is in L^p ($p > 1$), i.e.,

$$\mathbb{E} [|X_n - X|^p] \rightarrow 0.$$

Proof. Define $Y = \sup_n |X_n|$. Then by MCT and by above theorem,

$$\mathbb{E} [Y^p] = \lim_n \mathbb{E} \left[\sup_{0 \leq m \leq n} |X_m|^p \right] \leq \limsup_n \left(\frac{p}{p-1} \right)^p \mathbb{E} [|X_n|^p] \leq \left(\frac{p}{p-1} \right)^p \sup_n \mathbb{E} [|X_n|^p]$$

Because $|X_n - X|^p \rightarrow 0$ a.e. and $|X_n - X|^p \leq c(|X_n|^p + |X|^p) \leq c(Y^p + |X|^p)$, by DCT

$$\mathbb{E} [|X_n - X|^p] \rightarrow 0. \quad \square$$

2.7 UI and L^1 Convergence

Definition 2.7.1 (Uniform Integrability). A family of random variables $(X_i, i \in I)$ is said uniformly integrable (UI) if

$$\lim_{M \rightarrow \infty} \sup_{i \in I} \mathbb{E} [|X_i| \mathbb{I}_{\{|X_i| \geq M\}}] = 0.$$

Example 2.7.2. If $|X_i| \leq Y$ and $Y \in L^1$, then

$$\sup_i \mathbb{E} [|X_i| \mathbb{I}_{\{|X_i| \geq M\}}] \leq \mathbb{E} [|Y| \mathbb{I}_{\{|X_i| \geq M\}}] \leq \mathbb{E} [|Y| \mathbb{I}_{\{|Y| \geq M\}}] \rightarrow 0.$$

Lemma 2.7.3. If $X \in L^1$, then

$$\lim_{\mathbb{P}(A) \rightarrow 0} \int_A |X| \, d\mathbb{P} = 0$$

Proof. Since $X \in L^1$,

$$\int |X| \mathbb{I}_{\{|X| \geq M\}} \, d\mathbb{P} \rightarrow 0, \text{ as } M \rightarrow \infty.$$

For any $M > 0$,

$$\begin{aligned} \int_A |X| \, d\mathbb{P} &= \int_{A \cap \{|X| \leq M\}} |X| \, d\mathbb{P} + \int_{A \cap \{|X| > M\}} |X| \, d\mathbb{P} \\ &\leq M\mathbb{P}(A) + \int_{\{|X| > M\}} |X| \, d\mathbb{P} \end{aligned}$$

For any $\varepsilon > 0$, it can choose M such that $\int_{\{|X| > M\}} |X| \, d\mathbb{P} \leq \varepsilon/2$. For such M , by choosing $\delta \leq \varepsilon/(2M)$, then for any A with $\mathbb{P}(A) \leq \delta$, $\int_A |X| \, d\mathbb{P} \leq \varepsilon$. $\square$

Example 2.7.4. Let $X \in L^1$. Then

$$\{\mathbb{E}[X \mid \mathcal{G}] : \mathcal{G} \subset \mathcal{F}\}$$

is UI.

Proof. For any $\varepsilon > 0$, by above lemma, it can choose $\delta > 0$ so that if $\mathbb{P}(A) < \delta$, then

$$\int_A |X| \, d\mathbb{P} < \delta.$$

Note that

$$\begin{aligned} \int_{\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}} |\mathbb{E}[X \mid \mathcal{G}]| \, d\mathbb{P} &\leq \mathbb{E} [\mathbb{E}[|X| \mathbb{I}_{\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}} \mid \mathcal{G}]] \\ &= \mathbb{E} [\mathbb{I}_{\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}} \mathbb{E}[|X| \mid \mathcal{G}]] = \mathbb{E} [|X| \mathbb{I}_{\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}}] \\ &= \int_{\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}} |X| \, d\mathbb{P} \end{aligned}$$

On the other hand, by Chebyshev's Inequality

$$\mathbb{P}(\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}) \leq \frac{1}{M} \mathbb{E} [\mathbb{E}[|X| \mid \mathcal{G}]] = \frac{\mathbb{E}[|X|]}{M}$$

Therefore, it can choose $M \geq M_0$ such that

$$\mathbb{P}(\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}) \leq \delta$$

So

$$\sup_{\mathcal{G}} \int_{\{|\mathbb{E}[X \mid \mathcal{G}]| \geq M\}} |\mathbb{E}[X \mid \mathcal{G}]| \, d\mathbb{P} \leq \varepsilon. \quad \square$$

Theorem 2.7.5. Let $\varphi(x) \geq 0$ and $\frac{\varphi(x)}{x} \rightarrow \infty$ as $x \rightarrow \infty$. If

$$\sup_{i \in I} \mathbb{E} [\varphi(X_i)] < \infty,$$

then $(X_i, i \in I)$ is UI.

Proof. Let

$$\varepsilon_M := \sup \left\{ \frac{x}{\varphi(x)} : x \geq M \right\},$$

so $\varepsilon_M \rightarrow 0$ as $M \rightarrow \infty$.

$$\begin{aligned} \mathbb{E} [|X_i|, |X_i| \geq M] &= \mathbb{E} \left[\frac{|X_i|}{\varphi(X_i)} \varphi(X_i), |X_i| \geq M \right] \\ &\leq \varepsilon_M \mathbb{E} [\varphi(X_i)] \leq C \varepsilon_M \rightarrow 0. \end{aligned}$$

So it is UI. □

Remark 2.7.6. In particular, for $p > 1$, $\varphi(x) = |x|^p$ is valid. So if

$$\sup_i \|X_i\|_p < \infty$$

for some $p > 1$, $(X_i, i \in I)$ is UI.

Theorem 2.7.7. $(X_i, i \in I)$ is UI if and only if it satisfies the following two conditions:

- (1) $\sup_{i \in I} \mathbb{E} [|X_i|] < \infty$,
- (2) for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that for any $E \in \mathcal{F}$,

$$\mathbb{P}(E) \leq \delta \quad \Rightarrow \quad \int_E |X_i| d\mathbb{P} \leq \varepsilon, \quad \forall i \in I.$$

Proof. First, assume $(X_i, i \in I)$ is UI. Then for (i), there exists $M > 0$ such that $\sup_{i \in I} \mathbb{E} [|X_i| \mathbb{I}_{\{|X_i| \geq M\}}] < 1$

$$\sup_{i \in I} \mathbb{E} [|X_i|] = \sup_{i \in I} \mathbb{E} [|X_i| \mathbb{I}_{\{|X_i| < M\}}] + \sup_{i \in I} \mathbb{E} [|X_i| \mathbb{I}_{\{|X_i| \geq M\}}] \leq M + 1 < \infty.$$

For (ii), choose $M > 0$ such that $\sup_{i \in I} \mathbb{E} [|X_i| \mathbb{I}_{\{|X_i| \geq M\}}] \leq \varepsilon/2$. Then for $\delta = \varepsilon/(2M)$ and $A \in \mathcal{F}$ with $\mathbb{P}(A) \leq \delta$,

$$\begin{aligned} \mathbb{E} [|X_i| \mathbb{I}_A] &= \mathbb{E} [|X_i| \mathbb{I}_{A \cap \{|X_i| < M\}}] + \mathbb{E} [|X_i| \mathbb{I}_{A \cap \{|X_i| \geq M\}}] \\ &\leq M \mathbb{P}(A) + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Conversely, for $M = C/\delta$, by Markov's inequality

$$\mathbb{P}(|X_i| \geq M) \leq \frac{1}{M} \mathbb{E} [|X_i|] \leq \delta.$$

So

$$\sup_{i \in I} \mathbb{E} [|X_i| \mathbb{I}_{\{|X_i| \geq M\}}] \leq \varepsilon. \quad \square$$

Theorem 2.7.8. Let (X_n) be a sequence of L^1 random variables. If $X_n \rightarrow X$ in probability, then TFAE

(1) (X_n) is UI,

(2) $X_n \rightarrow X$ in L^1 ,

(3) $\mathbb{E}[|X_n|] \rightarrow \mathbb{E}[|X|]$.

Proof. (3) $\Rightarrow$ (2): Let $Y_n = |X_n|$, $Y = |X|$, and

$$Z_n = Y_n + Y - |X_n - X| \geq 0.$$

By (3), $Z_n \rightarrow 2Y$ in measure. Then by Fatou's lemma,

$$2\mathbb{E}[Y] \leq \liminf_n \mathbb{E}[Z_n] \leq 2Y - \limsup_n \mathbb{E}[|X_n - X|]$$

Therefore,

$$\limsup_n \mathbb{E}[|X_n - X|] \rightarrow 0$$

(2) $\Rightarrow$ (1):

$$\begin{aligned} \mathbb{E}[|X_n| \mathbb{I}_{\{|X_n| \geq M\}}] &\leq \mathbb{E}[|X_n - X| \mathbb{I}_{\{|X_n| \geq M\}}] + \mathbb{E}[|X| \mathbb{I}_{\{|X_n| \geq M\}}] \\ &\leq \mathbb{E}[|X_n - X|] + \mathbb{E}[|X| \mathbb{I}_{\{|X_n| \geq M\}}] \end{aligned}$$

$X_n \rightarrow X$ in L^1 , so $\mathbb{E}[|X_n - X|] \leq \varepsilon/2$ for sufficiently large $n > N_0$. For the other term, by Chebyshev's Inequality

$$\mathbb{P}(\{|X_n| \geq M\}) \leq \frac{\mathbb{E}[|X_n|]}{M} \leq \frac{C}{M}$$

because X_n is convergent in L^1 . So for $M > \bar{M}$, $\mathbb{P}(\{|X_n| \geq M\}) < \delta$ and $\mathbb{E}[|X| \mathbb{I}_{\{|X_n| \geq M\}}] \leq \varepsilon/2$. For $n = 0, 1, \dots, N_0$, choose M_n such that $\mathbb{E}[|X_n| \mathbb{I}_{\{|X_n| \geq M\}}] \leq \varepsilon$. Then let $M^* = \max\{M_0, \dots, M_{N_0}, \bar{M}\}$. Then for all n ,

$$\mathbb{E}[|X_n| \mathbb{I}_{\{|X_n| \geq M\}}] \leq \varepsilon, \quad \forall M > M^*.$$

(1) $\Rightarrow$ (3): By Fatou's Lemma, $\mathbb{E}[|X|] \leq \liminf_n \mathbb{E}[|X_n|]$. So it suffices to prove that

$$\limsup_n \mathbb{E}[|X_n|] \leq \mathbb{E}[|X|].$$

For any $\varepsilon > 0$, by UI, there exists M such that

$$\mathbb{E}[|X_n| \mathbb{I}_{\{|X_n| \geq M\}}] < \varepsilon.$$

Then

$$\begin{aligned} \mathbb{E}[|X_n|] &= \mathbb{E}[|X_n| \mathbb{I}_{\{|X_n| < M\}}] + \mathbb{E}[|X_n| \mathbb{I}_{\{|X_n| \geq M\}}] \\ &\leq \mathbb{E}[|(X_n - X)| \mathbb{I}_{\{|X_n| < M\}}] + \mathbb{E}[|X|] + \varepsilon, \end{aligned}$$

which follows that

$$\limsup_n \mathbb{E}[|X_n|] \leq \varepsilon + \mathbb{E}[|X|] + \limsup_n \mathbb{E}[|(X_n - X)| \mathbb{I}_{\{|X_n| < M\}}]$$

Because $X_n \rightarrow X$ in probability, by DCT,

$$\limsup_n \mathbb{E}[|(X_n - X)| \mathbb{I}_{\{|X_n| < M\}}] = 0.$$

Therefore,

$$\limsup_n \mathbb{E}[|X_n|] \leq \mathbb{E}[|X|].$$

□

Example 2.7.9 (Random Walk). Let $S_0 = 1$ and $S_n = S_0 + \xi_1 + \cdots + \xi_n$, where ξ_i are i.i.d with $\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = 1/2$. Define

$$N = \inf \{n: S_n = 0\}.$$

that is a stopping time. Because (S_n) is a martingale, $(S_{n \wedge N})$ is also a martingale. Moreover, $S_{n \wedge N} \geq 0$. Then by the Martingale Convergence Theorem, $X_n = S_{n \wedge N} \rightarrow 0$ a.e.. However,

$$\mathbb{E}[X_n] = \mathbb{E}[S_{n \wedge N}] = \mathbb{E}[S_0] = 1,$$

so (X_n) is not UI.

Theorem 2.7.10. *For a submartingale (X_n) , TFAE*

- (1) (X_n) is UI.
- (2) $X_n \rightarrow X$ in L^1 and a.e..
- (3) $X_n \rightarrow X$ in L^1 .

Furthermore, if (X_n) is a martingale, then $X_n = \mathbb{E}[X | \mathcal{F}_n]$.

Proof. It suffices to prove $(1) \Rightarrow (2)$. If (X_n) is UI, then

$$\sup_n \mathbb{E}[|X_n|] < \infty.$$

So by the Martingale Convergence Theorem, $X_n \rightarrow X$ a.e.. Then by above theorem, $X_n \rightarrow X$ in L^1 .

For martingale, for any $n \leq k$ and $A \in \mathcal{F}_n$

$$\mathbb{E}[X_k \mathbb{I}_A] = \mathbb{E}[\mathbb{I}_A \mathbb{E}[X_k | \mathcal{F}_n]] = \mathbb{E}[X_n \mathbb{I}_A]$$

Therefore, for any $A \in \mathcal{F}_n$, because $X_n \rightarrow X$ in L^1 , by DCT,

$$\mathbb{E}[X \mathbb{I}_A] = \mathbb{E}[X_n \mathbb{I}_A].$$

Then by the uniqueness of conditional expectation, $X_n = \mathbb{E}[X | \mathcal{F}_n]$. □

Theorem 2.7.11 (Lévy's upward theorem). *Suppose a sequence of σ -fields $\mathcal{F}_n \uparrow \mathcal{F}_\infty = \sigma(\cup \mathcal{F}_n)$. Assume $\mathbb{E}[|X|] < \infty$. Then*

$$\mathbb{E}[X | \mathcal{F}_n] \rightarrow \mathbb{E}[X | \mathcal{F}_\infty]$$

in L^1 and a.e..

Proof. Let $Y_n = \mathbb{E}[X | \mathcal{F}_n]$. Then Y_n is a martingale w.s.t. $(\mathcal{F}_n)$ and (Y_n) is UI. So

$$Y_n \rightarrow Y_\infty$$

in L^1 and a.e.. It suffices to prove $Y_\infty = \mathbb{E}[X | \mathcal{F}_\infty]$. First, it is obvious that Y_∞ is $\mathcal{F}_\infty$ -measurable. For any n and any $A \in \mathcal{F}_n \subset \mathcal{F}_m$ ($m > n$),

$$\mathbb{E}[Y_m \mathbb{I}_A] = \mathbb{E}[X \mathbb{I}_A].$$

As $m \rightarrow \infty$, because $Y_n \rightarrow Y_\infty$ in L^1 , $\mathbb{E}[Y_\infty \mathbb{I}_A] = \mathbb{E}[X \mathbb{I}_A]$. By the uniqueness of conditional expectation, $Y_\infty = \mathbb{E}[X | \mathcal{F}_\infty]$. □

Theorem 2.7.12. Suppose a sequence of σ -fields $\mathcal{F}_n \uparrow \mathcal{F}_\infty$. Assume that $Y_n \rightarrow Y$ a.e.. If $|Y_n| \leq Z$ for some $Z \in L^1$, then

$$\mathbb{E}[Y_n \mid \mathcal{F}_n] \rightarrow \mathbb{E}[Y \mid \mathcal{F}_\infty]$$

a.e..

Proof. Let $W_N = \sup \{|Y_n - Y_m| : n, m \geq N\}$. So (W_N) is decreasing to 0 and $|W_N| \leq 2Z$. Then

$$\mathbb{E}[W_N \mid \mathcal{F}_\infty] \rightarrow 0.$$

For any N , by Fatou's Lemma and above theorem

$$\begin{aligned} \limsup_n \mathbb{E}[|Y_n - Y| \mid \mathcal{F}_n] &\leq \limsup_n \liminf_m \mathbb{E}[|Y_n - Y_m| \mid \mathcal{F}_n] \\ &\leq \limsup_n \mathbb{E}[W_N \mid \mathcal{F}_n] = \mathbb{E}[W_N \mid \mathcal{F}_\infty] \rightarrow 0, \text{ as } N \rightarrow \infty \end{aligned}$$

Therefore,

$$\lim_n \mathbb{E}[Y_n \mid \mathcal{F}_n] = \lim_n \mathbb{E}[Y \mid \mathcal{F}_n] = \mathbb{E}[Y \mid \mathcal{F}_\infty].$$

□

Example 2.7.13. Let (Y_n) and (Z_n) be independent random variables with the same distribution

$$\begin{aligned} \mathbb{P}(Y_n = 1) &= \frac{1}{n}, \quad \mathbb{P}(Y_n = 0) = 1 - \frac{1}{n}, \\ \mathbb{P}(Z_n = n) &= \frac{1}{n}, \quad \mathbb{P}(Z_n = 0) = 1 - \frac{1}{n}. \end{aligned}$$

Let $X_n = Z_n Y_n$. Then $\mathbb{P}(X_n \geq 0) = 1/n^2$. By the Borel–Cantelli lemma, $X_n \rightarrow 0$ a.e.. Moreover,

$$\mathbb{E}[X_n \mathbb{I}_{\{X_n \geq 1\}}] = \frac{1}{n} \rightarrow 0,$$

which means (X_n) is UI. Let $\mathcal{F}_n = \sigma(Y_0, \dots, Y_n)$.

$$\mathbb{E}[X_n \mid \mathcal{F}_n] = Y_n \mathbb{E}[Z_n] = Y_n$$

But by the Borel–Cantelli lemma, Y_n does not converges to 0.

2.8 Backward Martingale

Fix $n \leq 0$, let $(\mathcal{F}_n)_{n \leq 0}$ be a family of decreasing σ -field as $n \rightarrow -\infty$.

Definition 2.8.1 (Backward Martingale). We say a stochastic process $(X_n)_{n \leq 0}$ is a backward martingale w.s.t. $(\mathcal{F}_n)_{n \leq 0}$ if

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n, \quad n \leq -1.$$

Moreover, “ $\geq$ ” is a backward submartingale and “ $\leq$ ” is a backward supermartingale.

Theorem 2.8.2. If $(X_n)_{n \leq 0}$ is a backward martingale, then

$$X_{-\infty} = \lim_{n \rightarrow -\infty} X_n$$

a.e. and in L^1 .

Proof. Let $U_n(a, b)$ be the number of up-crossing of $X_n, \dots, X_{-1}, X_0$ on $[a, b]$. As before,

$$\mathbb{E}[U_n(a, b)] \leq \frac{1}{b-a} \mathbb{E}[(X_0 - a)^-]$$

Therefore, similarly, we always have

$$X_{-\infty} = \lim_{n \rightarrow -\infty} X_n$$

a.e. by the Martingale Convergence Theorem. By the backward martingale property,

$$X_n = \mathbb{E}[X_0 \mid \mathcal{F}_n].$$

So $(X_n)_{n \leq 0}$ is UI and it implies that $X_{-\infty} = \lim_{n \rightarrow -\infty} X_n$ in L^1 . $\square$

Theorem 2.8.3. *Let $(X_n)_{n \leq 0}$ be a backward martingale. Let $\mathcal{F}_{-\infty} = \cap \mathcal{F}_n$. Then*

$$X_{-\infty} = \lim_{n \rightarrow -\infty} X_n = \mathbb{E}[X_0 \mid \mathcal{F}_{-\infty}].$$

Proof. First, because X_n is $\mathcal{F}_{-\infty}$ -measurable, $X_{-\infty}$ is $\mathcal{F}_{-\infty}$ -measurable. For any $A \in \mathcal{F}_{-\infty}$,

$$\mathbb{E}[X_{-\infty} \mathbb{I}_A] = \lim_{n \rightarrow -\infty} \mathbb{E}[X_n \mathbb{I}_A] = \lim_{n \rightarrow -\infty} \mathbb{E}[\mathbb{I}_A \mathbb{E}[X_0 \mid \mathcal{F}_n]] = \mathbb{E}[X_0 \mathbb{I}_A]. \quad \square$$

Remark 2.8.4. If $(X_n)_{n \leq 0}$ is a backward submartingale with $\sup_n \mathbb{E}[|X_n|] < \infty$, then

$$\lim_{n \rightarrow -\infty} \mathbb{E}[X_n] = X_{-\infty},$$

a.e. and in L^1 and

$$X_{-\infty} \leq \mathbb{E}[X_m \mid \mathcal{F}_{-\infty}], \quad \forall m \in -\mathbb{N}_0.$$

Theorem 2.8.5 (Lévy's downward theorem). *If $\mathcal{F}_n \downarrow \mathcal{F}_{-\infty}$ as $n \rightarrow -\infty$, then for any $Y \in L^1$,*

$$\mathbb{E}[Y \mid \mathcal{F}_n] \rightarrow \mathbb{E}[Y \mid \mathcal{F}_{-\infty}]$$

a.e. and in L^1 .

Proof. Let $X_n = \mathbb{E}[Y \mid \mathcal{F}_n]$ that is a backward martingale. Then by above

$$X_n \rightarrow \mathbb{E}[X_0 \mid \mathcal{F}_{-\infty}] = \mathbb{E}[Y \mid \mathcal{F}_{-\infty}]$$

a.e. and in L^1 . $\square$

Example 2.8.6 (Strong Law of Large Number). Let $\xi_1, \xi_2, \dots$ be a sequence of i.i.d. L^1 random variables. Define

$$X_{-n} = \frac{S_n}{n}, \quad S_n = \xi_1 + \dots + \xi_n.$$

Let $\mathcal{F}_{-n} = \sigma(S_n, S_{n+1}, \dots)$. By symmetry, for any $j, k \leq n+1$,

$$\mathbb{E}[\xi_k \mid \mathcal{F}_{-n-1}] = \mathbb{E}[\xi_j \mid \mathcal{F}_{-n-1}]$$

It follows that

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_{-n-1}] = \frac{1}{n+1} \sum_{k=1}^{n+1} \mathbb{E}[\xi_k \mid \mathcal{F}_{-n-1}] = \frac{1}{n+1} \mathbb{E}[S_{n+1} \mid \mathcal{F}_{-n-1}] = \frac{S_{n+1}}{n+1}$$

So X_{-n} is a backward martingale. So

$$\frac{S_n}{n} \rightarrow \mathbb{E}[X_0 \mid \mathcal{F}_{-\infty}] = \mathbb{E}[X_0].$$

2.9 Doob's Optional Stopping Theorem

Theorem 2.9.1. *If $X = (X_n)$ is a UI submartingale, then for any stopping time N , $(X_{n \wedge N})$ is also UI.*

Proof. Note that (X_n^+) is also a submartingale. So $(X_{n \wedge N}^+)$ is a submartingale. Moreover, because $n \wedge N$ is a bounded stopping time

$$\mathbb{E}[X_{n \wedge N}^+] \leq \mathbb{E}[X_n^+],$$

and because (X_n) is UI,

$$\sup_n \mathbb{E}[X_{n \wedge N}^+] \leq \sup_n \mathbb{E}[X_n^+] = \sup_n \mathbb{E}[|X_n|] < \infty$$

Then by the Martingale Convergence Theorem,

$$X_{n \wedge N} \rightarrow X_N, \text{ a.e.}$$

On the other hand,

$$\mathbb{E}[X_{N \wedge n}^-] = \mathbb{E}[X_{N \wedge n}^+] - \mathbb{E}[X_{N \wedge n}] \leq \mathbb{E}[X_{N \wedge n}^+] - \mathbb{E}[X_0].$$

So

$$\sup_n \mathbb{E}[X_{N \wedge n}^-] \leq \sup_n \mathbb{E}[X_{N \wedge n}^+] - \mathbb{E}[X_0] < +\infty.$$

It follows that $\sup_n \mathbb{E}[|X_{n \wedge N}|] < \infty$. So by Fatou's Lemma,

$$\mathbb{E}[|X_N|] \leq \liminf_n \mathbb{E}[|X_{n \wedge N}|] < \infty$$

To verify the uniform integrability,

$$\begin{aligned} \mathbb{E}[|X_{n \wedge N}|, |X_{n \wedge N}| \geq K] &= \mathbb{E}[|X_N|, |X_N| \geq K, N \leq n] + \mathbb{E}[|X_n|, |X_n| \geq K, N > n] \\ &\leq \mathbb{E}[|X_N|, |X_N| \geq K] + \mathbb{E}[|X_n|, |X_n| \geq K] \end{aligned}$$

Because $X_N \in L^1$ and (X_n) is UI, for any $\varepsilon > 0$, it can find K_1, K_2 such that

$$\mathbb{E}[|X_N|, |X_N| \geq K_1], \mathbb{E}[|X_n|, |X_n| \geq K_2] \leq \frac{1}{2}\varepsilon.$$

Therefore, $(X_{n \wedge N})$ is also UI. □

Remark 2.9.2. For the positive and negative part, because $f^+(x) = \max\{f(x), 0\}$ and $f^-(x) = \max\{-f(x), 0\}$, we know $|f| = f^+ + f^-$ and

$$(-f)^- = f^+, \quad (-f)^+ = f^-$$

Moreover, if $f \leq g$, then

$$f^+ \leq g^+, \quad f^- \geq g^-.$$

Theorem 2.9.3 (Doob's Optional Theorem). *Suppose (X_n) is a submartingale and N is a stopping time. If $X_N \in L^1$ and $(X_n \mathbb{I}_{N > n})$ is UI, then $(X_{n \wedge N})$ is UI and*

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_N].$$

Proof. The uniform integrability is directly obtained by the proof of above theorem. Because of it,

$$X_{n \wedge N} \rightarrow X_N$$

a.e. and in L^1 . Moreover, because

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_{n \wedge N}],$$

as $n \rightarrow \infty$, $\mathbb{E}[X_0] \leq \mathbb{E}[X_N]$. □

Theorem 2.9.4. *If (X_n) is a UI submartingale, then for any stopping time N , we have*

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_N] \leq \mathbb{E}[X_\infty],$$

where $X_\infty = \lim_n X_n$.

Proof. First, we have $X_\infty = \lim_n X_n$ a.e. and in L^1 . Fix any n ,

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_{n \wedge N}] \leq \mathbb{E}[X_n]$$

By above theorem, as $n \rightarrow \infty$, we have

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_N] \leq \mathbb{E}[X_\infty].$$
 □

Corollary 2.9.5. *If (X_n) is a UI submartingale and $M \leq N$ are two stopping times, then*

$$\mathbb{E}[X_0] \leq \mathbb{E}[X_M] \leq \mathbb{E}[X_N] \leq \mathbb{E}[X_\infty]$$

Proof. Consider the submartingale $Y_n = X_{N \wedge n}$ that is a UI submartingale. By applying above theorem to (Y_n) ,

$$\mathbb{E}[Y_0] = \mathbb{E}[X_0] \leq \mathbb{E}[Y_M] = \mathbb{E}[X_M] \leq \mathbb{E}[Y_\infty] = \mathbb{E}[X_N].$$
 □

Theorem 2.9.6. *Suppose (X_n) is a submartingale with $\sup_n \mathbb{E}[|X_{n+1} - X_n| \mid \mathcal{F}_n] \leq B$ for a constant B . If N is a stopping time with $\mathbb{E}[N] < \infty$, then $(X_{N \wedge n})$ is UI and so we have $\mathbb{E}[X_0] \leq \mathbb{E}[X_N]$.*

Proof. Observe that

$$\begin{aligned} |X_{N \wedge n}| &= \left| X_0 + \sum_{m=0}^{N \wedge n-1} (X_{m+1} - X_m) \right| \\ &\leq |X_0| + \sum_{m=0}^{N-1} |X_{m+1} - X_m| = |X_0| + \sum_{m=0}^{\infty} |X_{m+1} - X_m| \mathbb{I}_{\{m \leq N-1\}} =: Y \end{aligned}$$

It suffices to prove $Y \in L^1$. Note that $\{N \geq m+1\} = \{N < m\}^c \in \mathcal{F}_m$

$$\begin{aligned} \mathbb{E}[Y] &= \mathbb{E}[|X_0|] + \sum_{m=0}^{\infty} \mathbb{E}[|X_{m+1} - X_m| \mathbb{I}_{\{N \geq m+1\}}] \\ &= \mathbb{E}[|X_0|] + \sum_{m=0}^{\infty} \mathbb{E}[\mathbb{I}_{\{N \leq m+1\}} \mathbb{E}[|X_{m+1} - X_m| \mid \mathcal{F}_m]] \\ &\leq \mathbb{E}[|X_0|] + B \sum_{m=0}^{\infty} \mathbb{P}(\{N \leq m+1\}) = \mathbb{E}[|X_0|] + B\mathbb{E}[N] < \infty \end{aligned}$$

So $Y \in L^1$ and $(X_{N \wedge n})$ is UI. □

Example 2.9.7 (Gambler's Ruin Problem). Consider A, B play a series of games against each other in which a fair coin is tossed respectively. In each game, gambler A wins or loses 1 dollar with probability $1/2$ and $1/2$. The initial capital of gambler A is a dollars, and that of gambler B is b dollars. They continue play until one of them is ruined. Determine the probability of that A will be ruined and the expected number of games.

Solution: Let $\hat{S}_n$ be the fortune of A after n -th games, so

$$\hat{S}_n = a + X_1 + \cdots + X_n = a + S_n,$$

where X_i are i.i.d. $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = 1/2$. The game will stop of

$$T = \min \{n : S_n = -a \text{ or } S_n = b\}$$

that is a stopping time. Then

$$\{ \text{Gambler } A \text{ is ruined} \} = \{S_T = -a\}.$$

So it needs to find $\mathbb{P}(\{S_T = -a\})$ and $\mathbb{E}[T]$.

We already know (S_n) is a martingale with $S_0 = 0$, so is the stopping process $(S_{T \wedge n})$. Moreover, because

$$|S_{T \wedge n}| \leq a + b,$$

$(S_{T \wedge n})$ is UI. So

$$\mathbb{E}[S_T] = \mathbb{E}[S_0] = 0$$

Note that

$$\mathbb{E}[S_T] = -a\mathbb{P}(S_T = -a) + b\mathbb{P}(S_T = b) = 0$$

Moreover,

$$\mathbb{P}(S_T = -a) + \mathbb{P}(S_T = b) = 1.$$

So

$$\mathbb{P}(S_T = -a) = \frac{b}{a+b}, \quad \mathbb{P}(S_T = b) = \frac{a}{a+b}.$$

First, we need to check $\mathbb{E}[T] < \infty$. By induction it can have

$$\mathbb{P}(T > m(a+b)) \leq \left(1 - \left(\frac{1}{2}\right)^{a+b}\right)^m, m \geq 1$$

So

$$\begin{aligned} \mathbb{E}[T] &= \mathbb{E}[T\mathbb{I}_{\{T \leq a+b\}}] + \mathbb{E}[T\mathbb{I}_{\{T > a+b\}}] \\ &= \mathbb{E}[T\mathbb{I}_{\{T \leq a+b\}}] + \sum_{m=1}^{\infty} \mathbb{E}[T\mathbb{I}_{\{m(a+b) < T \leq (m+1)(a+b)\}}] \\ &\leq a+b + \sum_{m=1}^{\infty} (m+1)(a+b)\mathbb{P}(m(a+b) < T \leq (m+1)(a+b)) \\ &\leq a+b + \sum_{m=1}^{\infty} (m+1)(a+b)\mathbb{P}(T > m(a+b)) \\ &\leq a+b + \sum_{m=1}^{\infty} (m+1)(a+b) \left(1 - \left(\frac{1}{2}\right)^{a+b}\right)^m < +\infty \end{aligned}$$

Note that $(Y_n = S_n^2 - n)$ is also a martingale. So by the bounded optional stopping time theorem

$$\mathbb{E}[S_{T \wedge n}^2 - T \wedge n] = \mathbb{E}[S_0^2] = 0 \Rightarrow \mathbb{E}[S_{T \wedge n}^2] = \mathbb{E}[T \wedge n]$$

Because $|S_{T \wedge n}^2| \leq (a+b)^2$, by DCT, as $n \rightarrow \infty$,

$$\mathbb{E}[T] = \mathbb{E}[S_T^2] = ab.$$

Example 2.9.8 (Random Walk). $\xi_1, \dots, \xi_n, \dots$ is i.i.d with $\mathbb{P}(\xi = 1) = p$ and $\mathbb{P}(\xi = -1) = q$. $S_0 = k < N$ and $S_n = S_0 + \xi_1 + \dots + \xi_n$. Find the probability that the random walk hits 0 before N .

Solution: Let

$$T = \inf \{n: S_n = 0 \text{ or } S_n = N\}.$$

Define $Z_n = \left(\frac{q}{p}\right)^{S_n}$ that can be proved a martingale w.s.t. $\mathcal{F}_n = \sigma(S_0, \dots, S_n)$. Therefore, $(Z_{T \wedge n})$ is also a martingale and it is UI because $|S_T| < \infty$. Then

$$\mathbb{E}[Z_T] = \mathbb{E}[Z_0] = \left(\frac{q}{p}\right)^k$$

Note that

$$\mathbb{E}[Z_T] = \mathbb{E}\left[\left(\frac{q}{p}\right)^{S_T}\right] = \mathbb{P}(S_T = 0) + \left(\frac{q}{p}\right)^N \mathbb{P}(S_T = N)$$

Combining it with $\mathbb{P}(S_T = N) + \mathbb{P}(S_T = 0) = 1$, we have

$$\mathbb{P}(S_T = 0) = \frac{\left(\frac{q}{p}\right)^k - \left(\frac{q}{p}\right)^N}{1 - \left(\frac{q}{p}\right)^N}.$$

Example 2.9.9. Let (X_n) be i.i.d. with $\mathbb{E}[X_n] = \mu$ and $N \in L^1$ be a stopping time w.s.t. $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. How to calculate $\mathbb{E}\left[\sum_{i=1}^N X_i\right]$.

Solution: Let $Y_0 = 0$ and

$$Y_n = \sum_{i=1}^n X_i - n\mu$$

that is clear a martingale. Then $(Y_{N \wedge n})$ is also a martingale. Because

$$\mathbb{E}[|Y_{n+1} - Y_n| \mid \mathcal{F}_n] \leq \mu + \mathbb{E}[|X_{n+1}| \mid \mathcal{F}_n] = \mu + \mathbb{E}[|X_{n+1}|] = 2\mu < \infty,$$

$(Y_{N \wedge n})$ is UI and so

$$\mathbb{E}[Y_N] = \mathbb{E}\left[\sum_{i=1}^N X_i - N\mu\right] = \mathbb{E}[Y_0] = 0.$$

It follows that

$$\mathbb{E}\left[\sum_{i=1}^N X_i\right] = \mathbb{E}[N]\mu.$$

Chapter 3

Discrete Time Markov Theory

3.1 Markov Chain

Definition 3.1.1 (Markov Chain). If the state space is at most countable, a stochastic process $(X_n)_{n \geq 0}$ is said to have Markov property if for any n and any $i_0, \dots, i_{n-1}, i, j \in S$

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

and $(X_n)_{n \geq 0}$ is called a Markov chain. Furthermore,

$$p_{ij}(n) = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

is called transition probability. In particular, if $p_{ij}(n) \equiv p_{ij}$, such Markov chain is called time-homogeneous, otherwise, it is called time-inhomogeneous.

In following, we mainly consider the time-homogeneous Markov chain.

Example 3.1.2 (Ehrenfest Chain). Let A, B be two bottles such that A contains k balls and B contains $r - k$ balls. Each operation is to randomly choose a ball from the r balls and then transfer it from its original bottle into another one. Let X_n be the number of balls in A after n -th operation. Note that the state space $S = \{0, 1, \dots, r\}$. Then

$$p_{kj} = \mathbb{P}(X_{n+1} = j \mid X_n = k) = \begin{cases} 0 & , |k - j| \neq 1 \\ \frac{k}{r} & , j = k + 1 \\ \frac{r-k}{r} & , j = k - 1 \end{cases}.$$

3.2 Constructing Markov Chain

Definition 3.2.1 (Transition Probability). Let $(S, \mathcal{S})$ be the state space (measurable space). A function

$$p: S \times \mathcal{S} \rightarrow \mathbb{R}$$

is called a transition probability if

- (i) For any $x \in S$, $A \rightarrow p(x, A)$ is a probability measure defined on $\mathcal{S}$,
- (ii) For any $A \in \mathcal{S}$, $x \rightarrow p(x, A)$ is a measurable function.

(X_n) is a Markov chain with the transition probability p if

$$\mathbb{P}(X_{n+1} \in B \mid \mathcal{F}_n) = p(X_n, B)$$

When given a transition probability p , let

$$\mathbb{P}_n(B_0 \times \cdots \times B_n) = \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \int_{B_2} p(x_1, dx_2) \cdots \int_{B_n} p(x_{n-1}, dx_n)$$

for $B_0, \dots, B_n \in \mathcal{S}$. Then $\mathbb{P}_n$ on $(S^{n+1}, \mathcal{S}^{n+1})$ are consistent, i.e.,

$$\mathbb{P}_{n+1}(B_0 \times \cdots \times B_n \times S) = \mathbb{P}_n(B_0 \times \cdots \times B_n).$$

So by Kolmogorov Extension Theorem, there exists a measure $\mathbb{P}_\mu$ on $(S^\infty, \mathcal{S}^\infty)$ such that it is the finite dimensional distribution of the coordinate process $(X_n)_{n \geq 0}$ on $(S^\infty, \mathcal{S}^\infty)$, i.e. $X_n(\omega) = \omega_n$ for $\omega \in S^\infty$,

$$\mathbb{P}_\mu(X_0 \in B_0, \dots, X_n \in B_n) = \mathbb{P}_n(B_0 \times \cdots \times B_n)$$

In fact, such $(X_n)_{n \geq 0}$ is a Markov chain with transition probability p , where $\mathcal{F}_n = \mathcal{S}^n$, i.e.

$$\mathbb{P}_\mu(X_{n+1} \in B \mid \mathcal{F}_n) = \mathbb{E} [\mathbb{I}_{\{X_{n+1} \in B\}} \mid \mathcal{F}_n] = p(X_n, B)$$

Proof. First, clearly $p(X_n, B)$ is $\mathcal{F}_n$ -measurable. So it suffices to check for any $A \in \mathcal{F}_n = \mathcal{S}^n$,

$$\mathbb{E} [\mathbb{I}_{\{X_{n+1} \in B\}} \mathbb{I}_A] = \int_A p(X_n, B) d\mathbb{P}_\mu.$$

Because $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$, we can assume

$$A = \{X_0 \in B_0, \dots, X_n \in B_n\}$$

for $B_i \in \mathcal{S}$. So

$$\begin{aligned} \mathbb{E} [\mathbb{I}_{\{X_{n+1} \in B\}} \mathbb{I}_A] &= \mathbb{P}_\mu(X_0 \in B_0, \dots, X_n \in B_n, \{X_{n+1} \in B\}) \\ &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) \int_B p(x_n, dx_{n+1}) \\ &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_n, B) p(x_{n-1}, dx_n) \end{aligned}$$

In fact, we can prove that for any measurable function f

$$\int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} f(x_n) p(x_{n-1}, dx_n) = \int_A f(X_n) d\mathbb{P}_\mu.$$

By approximation, it can assume $f = \mathbb{I}_C$ for some $C \in \mathcal{S}$. Then

$$\begin{aligned} \text{LHS} &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_{n-1}} p(x_{n-1}, B_n \cap C) p(x_{n-2}, dx_{n-1}) \\ &= \mathbb{P}_\mu(X_0 \in B_0, \dots, X_{n-1} \in B_{n-1}, X_n \in B_n \cap C). \end{aligned}$$

and

$$\begin{aligned} \int_A f(X_n) d\mathbb{P}_\mu &= \int_A \mathbb{I}_{\{X_n \in C\}} d\mathbb{P}_\mu \\ &= \int_S \mathbb{I}_{\{X_n \in C\} \cap A} d\mathbb{P}_\mu \\ &= \mathbb{P}_\mu(X_0 \in B_0, \dots, X_{n-1} \in B_{n-1}, X_n \in B_n \cap C). \end{aligned}$$

So LHS = RHS. □

Let $\Omega_0 = S^\infty$ with $\mathcal{F}_\infty = \mathcal{S}^\infty$.

Definition 3.2.2 (Shift Operator). For any $n \in \mathbb{N}$, define

$$\theta_n: \Omega_0 \rightarrow \Omega_0$$

by

$$\theta_n(\omega) = (\omega_n, \omega_{n+1}, \dots)$$

for $\omega = (\omega_0, \omega_1, \dots) \in \Omega_0$.

Proposition 3.2.3. *If $(X_n)_{n \geq 0}$ is a Markov chain with transition probability p , then for any bounded measurable function f on $(S, \mathcal{S})$, we have*

$$\mathbb{E}[f(X_{n+1}) \mid \mathcal{F}_n] = \int_S f(y)p(X_n, dy).$$

Proof. It is clear true for $f = \mathbb{I}_C$ and so for any simple function f . Therefore, by the following Monotone Class Theorem, it is true for any bounded measurable functions. $\square$

Theorem 3.2.4 (Monotone Class Theorem). *Let $\mathcal{A} \subset \mathcal{P}(S)$ be a π -system (i.e. closed under intersection) that contains S . Let $\mathcal{H}$ be a collection of real-valued functions satisfying*

- (1) *if $A \in \mathcal{A}$, then $\mathbb{I}_A \in \mathcal{H}$,*
- (2) *if $f, g \in \mathcal{H}$, then $f + g, cf \in \mathcal{H}$ for any real c ,*
- (3) *if $f_n \in \mathcal{H}$ are nonnegative and $f_n \uparrow f$ for a bounded measurable f , then $f \in \mathcal{H}$.*

Then $\mathcal{H}$ contains all bounded $\sigma(\mathcal{A})$ -measurable functions.

Theorem 3.2.5 (Markov Property). *Given the μ on S , the corresponding $\mathbb{P}_\mu$ on Ω_0 , and the Markov chain $(X_n)_{n \geq 0}$. Let $Y: \Omega_0 \rightarrow \mathbb{R}$ be a bounded and $\sigma(X_0, \dots, X_n, \dots)$ -measurable random variable. Then*

$$\mathbb{E}_\mu[Y \circ \theta_m \mid \mathcal{F}_m] = \mathbb{E}_{X_m}[Y],$$

where $\mathbb{E}_\mu$ is the expectation w.s.t. $\mathbb{P}_\mu$ and $\mathbb{E}_{X_m} = \mathbb{E}_x|_{x=X_m}$. $\mathbb{E}_x$ is the expectation w.s.t. $\mathbb{P}_x$, where $\mathbb{P}_x = \mathbb{P}_{\delta_x}$.

Proof. By the Monotone Class Theorem, we can assume

$$Y = \prod_{k=0}^n g_k(X_k),$$

where g_i is bounded measurable function on $(S, \mathcal{S})$. Because $\mathbb{E}_{X_m}[Y]$ is a function of X_m , it is clear $\mathcal{F}_m$ -measurable. For $A \in \mathcal{F}_m$, it suffices to check

$$\mathbb{E}_\mu[Y \circ \theta_m \cdot \mathbb{I}_A] = \mathbb{E}_\mu[\mathbb{E}_{X_m}[Y] \cdot \mathbb{I}_A].$$

It can assume

$$A = \{X_0 \in A_1, \dots, X_m \in A_m\},$$

Note that $X_k(\theta(\omega)) = X_{k+m}(\omega)$. So

$$\begin{aligned} \mathbb{E}_\mu[Y \circ \theta_m \cdot \mathbb{I}_A] &= \mathbb{E}_\mu[g_0(X_m) \cdots g_n(X_{m+n}) \mathbb{I}_{A_0}(X_0) \cdots \mathbb{I}_{A_m}(X_m)] \\ &= \int_{A_0} \mu(dx_0) \int_{A_1} p(x_0, dx_1) \cdots \int_{A_m} g_0(x_m) p(x_{m-1}, dx_m) \\ &\quad \int_S g_1(x_{m+1}) p(x_m, dx_{m+1}) \cdots \int_S g_n(x_{m+n}) p(x_{m+n-1}, dx_{m+n}) \end{aligned}$$

On the other hand,

$$\begin{aligned}\mathbb{E}_{X_m}[Y] &= \int_S g_0(x_0) \delta_{X_m}(dx_0) \int_S g(x_1) p(x_0, dx_1) \cdots \int_S g(x_n) p(x_{n-1}, dx_n) \\ &= g_0(X_m) \int_S g(x_{m+1}) p(X_m, dx_{m+1}) \cdots \int_S g(x_{m+n}) p(x_{m+n-1}, dx_{m+n})\end{aligned}$$

by replacing x_i by x_{m+i} . So

$$\begin{aligned}\mathbb{E}_\mu[Y \circ \theta_m \cdot \mathbb{I}_A] &= \int_{A_0} \mu(dx_0) \int_{A_1} p(x_0, dx_1) \cdots \int_{A_m} \mathbb{E}_{X_m}[Y] p(x_{m-1}, dx_m) \\ &= \mathbb{E}_\mu[\mathbb{E}_{X_m}[Y] \cdot \mathbb{I}_A].\end{aligned}$$

□

Remark 3.2.6. By the definition of $\mathbb{P}_\mu$, for $f_i(x) = \mathbb{I}_{B_i}(x)$,

$$\mathbb{E}_\mu[f_0(X_0)f_1(X_1)\cdots f_n(X_n)] = \int_S f_0(x_0)\mu(dx_0) \int_S f(x_1)p(x_0, dx_1) \cdots \int_S f_n(x_n)p(x_{n-1}, dx_n).$$

Corollary 3.2.7. *We have*

$$\mathbb{E}_\mu[Y \circ \theta_m \mid \mathcal{F}_m] = \mathbb{E}_\mu[Y \circ \theta_m \mid \sigma(X_m)]$$

Proof. Because $\sigma(X_m) \subset \mathcal{F}_m$,

$$\begin{aligned}\mathbb{E}_\mu[Y \circ \theta_m \mid \sigma(X_m)] &= \mathbb{E}_\mu[\mathbb{E}_\mu[Y \circ \theta_m \mid \mathcal{F}_m] \mid \sigma(X_m)] \\ &= \mathbb{E}_\mu[\mathbb{E}_{X_m}[Y] \mid \sigma(X_m)] \\ &= \mathbb{E}_{X_m}[Y] = \mathbb{E}_\mu[Y \circ \theta_m \mid \mathcal{F}_m].\end{aligned}$$

□

Remark 3.2.8. For any Markov chain $X = (X_n)_{n \geq 0}$ on a space $(\Omega, \mathcal{F})$ and taking values on $(S, \mathcal{S})$, we can still obtain a $\mathbb{P}_\mu$ on $(S^\infty, \mathcal{S}^\infty)$, which is $\mathbb{P}_\mu = X_\# \mathbb{P}$. Or on the other hand, $\mu = (X_0)_\# \mathbb{P}$ and $\mathbb{P}_\mu$ is deduced from such μ . And $(X_n)_{n \geq 0}$ on $(\Omega, \mathbb{P})$ is equivalent to the coordinate process $(\pi_n)_{n \geq 0}$ on $(\mathcal{S}^\infty, \mathbb{P}_\mu)$. Then the Markov property is described as the above theorem.

Definition 3.2.9. If N is a stopping time w.s.t. $(\mathcal{F}_n)_{n \geq 0}$, then define

$$\mathcal{F}_N := \{A: A \cap \{N \leq n\} \in \mathcal{F}_n, \forall n\},$$

which is a σ -field.

Remark 3.2.10. (1) If $A \in \mathcal{F}_N$, then for any n ,

$$A \cap \{N = n\} = (A \cap \{N \leq n\}) \setminus (A \cap \{N \leq n-1\}) \in \mathcal{F}_n.$$

(2) Note that X_N is $\mathcal{F}_N$ -measurable, because for any n ,

$$\{X_N \in B\} \cap \{N = n\} = \{X_n \in B\} \cap \{N = n\} \in \mathcal{F}_n.$$

For a stopping time N , define $\theta_N: \Omega_0 \rightarrow \Omega_0$ by

$$\theta_N(\omega) = \begin{cases} \theta_n(\omega), & \omega \in \{N = n\}, n < \infty \\ *, & \omega \in \{N = \infty\}, \end{cases}$$

where $*$ is an extra point adding to Ω_0 .

Theorem 3.2.11 (Strong Markov Property). *On $\{N < \infty\}$,*

$$\mathbb{E}_\mu [Y \circ \theta_N \mid \mathcal{F}_N] = \mathbb{E}_{X_N} [Y]$$

Proof. First, because $\mathbb{E}_{X_N} [Y]$ is a function of X_N , it is $\mathcal{F}_N$ -measurable. It suffices to check for any $A \in \mathcal{F}_N$,

$$\mathbb{E}_\mu [Y \circ \theta_N, A \cap \{N < \infty\}] = \mathbb{E}_\mu [\mathbb{E}_{X_N} [Y], A \cap \{N < \infty\}].$$

Note that

$$\begin{aligned} \text{LHS} &= \sum_{n=0}^{\infty} \mathbb{E}_\mu [Y \circ \theta_n, A \cap \{N = n\}] \\ &= \sum_{n=0}^{\infty} \mathbb{E}_\mu [\mathbb{E}_{X_n} [Y, A \cap \{N = n\}]] \\ &= \mathbb{E}_\mu [\mathbb{E}_{X_N} [Y, A \cap \{N < \infty\}]]. \end{aligned}$$

□

Define

$$p^k(x, y) = \mathbb{P}(X_k = y \mid X_0 = x) = \mathbb{P}_x(X_k = y)$$

where the second equality is by the Markov property.

Theorem 3.2.12 (Chapman-Kolmogorov Equation). *For any $x, y, z \in S$,*

$$p^{m+n}(x, z) = \mathbb{P}_x(X_{m+n} = z) = \sum_{y \in S} \mathbb{P}_x(X_m = y) \mathbb{P}_y(X_n = z) = \sum_{y \in S} p^m(x, y) p^n(y, z).$$

Proof. By the Markov property

$$\begin{aligned} \mathbb{P}_x(X_{m+n} = z) &= \mathbb{E}_x [\mathbb{I}_{\{X_{m+n}=z\}}] \\ &= \mathbb{E}_x [\mathbb{E}_x [\mathbb{I}_{\{X_{m+n}=z\}} \mid \mathcal{F}_m]] \\ &= \mathbb{E}_x [\mathbb{E}_x [\mathbb{I}_{\{z\}}(X_n \circ \theta_m) \mid \mathcal{F}_m]] \\ &= \mathbb{E}_x [\mathbb{E}_{X_m} [X_n = z]] \\ &= \sum_{y \in S} \mathbb{E}_y [X_n = z] \mathbb{P}_x(X_m = y) \\ &= \sum_{y \in S} \mathbb{P}_y(X_n = z) \mathbb{P}_x(X_m = y) \end{aligned}$$

□

Remark 3.2.13. For any $x \in S$, by the definition of $\mathbb{P}_x$, we have

$$\mathbb{P}_x(X_1 = y_1, X_2 = y_2, \dots, X_n = y_n) = p(x, y_1) p(y_1, y_2) \cdots p(y_{n-1}, y_n).$$

Note that it can be also obtained by the property of conditional probability.

3.3 Classification of States

Let $(X_n)_{n \geq 0}$ be a Markov chain with discrete state space S . Let $y \in S$. Define $T_y^0 = 0$ and for any $k \in \mathbb{N}$

$$T_y^k = \inf \{n > T_y^{k-1} : X_n = y\},$$

i.e., the time of the k -th returning to y . Note that T_y^k is a stopping time. For simplicity, let $T_y = T_y^1$. Define

$$\rho_{xy} = \mathbb{P}_x(T_y < \infty),$$

i.e. the probability of the chain that can reach y with starting from x .

Theorem 3.3.1. *The probability*

$$\mathbb{P}_x(T_y^k < \infty) = \rho_{xy}\rho_{yy}^{k-1}.$$

Proof. By induction, for $k = 1$, it is obvious. Assume $k \geq 2$ and it is true for $k - 1$. Let $Y(\omega) = 1$ if $X_n(\omega) = y$ for some n , otherwise $Y(\omega) = 0$. So

$$\{Y = 1\} = \{T_y < \infty\}.$$

Let $N = T_y^{k-1}$. Then

$$\{Y \circ \theta_N = 1\} = \{T_y^k < \infty\}.$$

By the strong Markov property, on $\{N < \infty\}$,

$$\mathbb{E}_x[Y \circ \theta_N \mid \mathcal{F}_N] = \mathbb{E}_{X_N}[Y] = \mathbb{E}_y[Y] = \rho_{yy}$$

because $X_N = y$ and $\mathbb{E}_y[Y] = \mathbb{P}_y(Y = 1)$. Therefore,

$$\begin{aligned} \mathbb{P}_x(T_y^k < \infty) &= \mathbb{P}_x(N < \infty, Y \circ \theta_N = 1) \\ &= \mathbb{E}_x[Y \circ \theta_N = 1, N < \infty] \\ &= \mathbb{E}_x[\mathbb{E}_{X_N}[Y], N < \infty] \\ &= \rho_{yy}\mathbb{E}_x[\mathbb{I}_{\{N < \infty\}}] = \rho_{yy}\mathbb{P}_x(T_y^{k-1} < \infty). \end{aligned}$$

The by assumption of induction, it is true for k . □

Definition 3.3.2 (Classification of States). Given Markov chain (X_n) valued on discrete $(S, \mathcal{S})$, let $y \in S$.

- (1) y is called recurrent if $\rho_{yy} = 1$.
- (2) y is called transient if $\rho_{yy} < 1$. In this case, there is a positive probability $1 - \rho_{yy}$ that the Markov chain starting from y never return y .

For $y \in S$, let

$$N(y) = \sum_{n=1}^{\infty} \mathbb{I}_{\{X_n=y\}}$$

that is the number of visits to y .

Lemma 3.3.3. *If y is transient,*

$$\mathbb{E}_x[N(y)] = \frac{\rho_{xy}}{1 - \rho_{yy}}.$$

Proof. By definition,

$$\begin{aligned} \mathbb{E}_x[N(y)] &= \sum_{k=1}^{\infty} \mathbb{P}_x(N(y) \geq k) \\ &= \sum_{k=1}^{\infty} \mathbb{P}_x(T_y^k < \infty) \\ &= \sum_{k=1}^{\infty} \rho_{xy}\rho_{yy}^{k-1} = \frac{\rho_{xy}}{1 - \rho_{yy}}. \end{aligned}$$

□

Theorem 3.3.4. $y \in S$ is recurrent if and only if $\mathbb{E}_x[N(y)] = \infty$.

Proof. If y is recurrent, because

$$\mathbb{E}_x[N(y)] = \sum_{k=1}^{\infty} \rho_{xy} \rho_{yy}^{k-1} = \infty.$$

Conversely, assume $y \in S$ is not recurrent, then

$$\mathbb{E}_x[N(y)] = \frac{\rho_{xy}}{1 - \rho_{yy}} = \infty$$

implies that $\rho_{yy} = 1$, inducing a contradiction. $\square$

Theorem 3.3.5. *If x is recurrent and $\rho_{xy} > 0$, then y is recurrent and $\rho_{yx} = 1$.*

Proof. Assume $\rho_{yx} < 1$. Let

$$K = \inf \{k : p^k(x, y) > 0\}.$$

$\rho_{xy} > 0$ implies that $K < \infty$ and there is a $y_1, \dots, y_{K-y}$ such that

$$p(x, y_0)p(y_1, y_2) \cdots p(y_{K-1}, y) > 0,$$

because

$$p^K(x, y) = \sum_{y_1, \dots, y_{K-1} \in S} \mathbb{P}_x(X_1 = y_1, \dots, X_{K-1} = y_{K-1}, X_K = y) > 0.$$

Note that $y_i \neq x$ for $i = 1, \dots, K-1$. Because $\rho_{yx} < 1$,

$$\begin{aligned} \mathbb{P}_x(T_x = \infty) &\geq p(x, y_1) \cdots p(y_{K-1}, y)(1 - \rho_{yx}) \\ &= \mathbb{P}_x(X_1 = y_1, \dots, X_K = y, T_x \circ \theta_K = \infty) > 0, \end{aligned}$$

contradicting to the recurrence of x . Therefore, $\rho_{yx} = 1$.

To check that y is recurrent, it suffices to prove $\mathbb{E}_y[N(y)] = \infty$. Since $\rho_{yx} = 1 > 0$, there exist an $\ell \in \mathbb{N}$ such that

$$p^\ell(y, x) > 0.$$

Note that for $n \geq 1$, by the Chapman-Kolmogorov Equation,

$$p^{\ell+n+K}(y, y) \geq p^\ell(y, x)p^n(x, x)p^K(x, y).$$

So

$$\sum_{n=1}^{\infty} p^{\ell+n+K}(y, y) \geq p^\ell(y, x)p^K(x, y) \sum_{n=1}^{\infty} p^n(x, x).$$

Moreover,

$$\mathbb{E}_x[N(x)] = \sum_{n=1}^{\infty} \mathbb{E}_x[\mathbb{I}_{X_n=x}] = \sum_{n=1}^{\infty} p^n(x, x) = \infty,$$

because of the recurrence of x . It follows that

$$\mathbb{E}_y[N(y)] = \sum_{n=1}^{\infty} p^n(y, y) \geq \sum_{n=1}^{\infty} p^{\ell+n+K}(y, y) = \infty. \quad \square$$

Definition 3.3.6 (Closedness). Let $C \subset S$. C is called closed if for any $x \in C$, $\rho_{xy} > 0$ implies $y \in C$.

Remark 3.3.7. If C is closed and $x \in C$, then $\mathbb{P}_x(X_n \in C) = 1$ for all n . Otherwise, there is a $y \notin C$ such that

$$\mathbb{P}_x(X_n = y) > 0$$

which implies that $\rho_{xy} \geq \mathbb{P}_x(X_n = y) > 0$ and so $y \in C$, contradicting to the assumption.

Definition 3.3.8 (Irreducibility). $D \subset S$ is called irreducible if for any $x, y \in D$, $\rho_{xy} > 0$.

Theorem 3.3.9. Assume $C \subset S$ is finite and closed. Then C contains a recurrent state. In particular, if C is also irreducible, then every state in C is recurrent.

Proof. Assume C contains no recurrent state. Then for all $y \in C$, $\rho_{yy} < 1$ and so

$$\mathbb{E}_x[N(y)] = \frac{\rho_{xy}}{1 - \rho_{yy}} < \infty.$$

It implies that

$$\begin{aligned} \infty &> \sum_{y \in C} \mathbb{E}_x[N(y)] = \sum_{y \in C} \sum_{n=1}^{\infty} p^n(x, y) \\ &= \sum_{n=1}^{\infty} \sum_{y \in C} p^n(x, y) \\ &= \sum_{n=1}^{\infty} 1 = \infty \end{aligned}$$

because C is finite and closed, which induces a contradiction. $\square$

Example 3.3.10. Consider a Markov chain with $|S| = 7$ and the transition matrix $P = (p_{ij} = \mathbb{P}(X_1 = j \mid X_0 = i))$

$$P = \begin{pmatrix} 0.3 & 0 & 0 & 0 & 0.7 & 0 & 0 \\ 0.1 & 0.2 & 0.3 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0.6 & 0 & 0 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0.1 & 0 & 0.1 & 0.8 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

Find all recurrent and transient states.

Solution: First, because $\rho_{21} > 0$ but $\rho_{12} = 0$, 2 is transient. Similarly, $\rho_{34} > 0$ with $\rho_{43} = 0$ implies that 3 is transient. Note that $\{1, 5\}$ is closed and irreducible, so $\{1, 5\}$ are recurrent. $\{4, 6, 7\}$ is also closed and irreducible, so they are transient.

Theorem 3.3.11 (Decomposition Theorem). Let $R = \{x \in S : \rho_{xx} = 1\}$ be the set of all recurrent states. Then

$$R = \bigcup_i R_i,$$

where R_i is closed and irreducible.

Proof. For any $x \in R$, let

$$C_x = \{y : \rho_{xy} > 0\}.$$

By above theorem, $C_x \subset R$.

Claim: Either $C_x \cap C_y = \emptyset$ or $C_x = C_y$.

Suppose $C_x \cap C_y \neq \emptyset$. If $z \in C_x \cap C_y$, then

$$\rho_{xy} \geq \rho_{xz}\rho_{zy} = \rho_{xz} > 0,$$

because $\rho_{xz}, \rho_{yz} > 0$ and $y \in R$. For any $w \in C_y$, we have

$$\rho_{xw} \geq \rho_{xy}\rho_{yw} > 0,$$

which implies that $w \in C_x$. So $C_y \subset C_x$. By symmetry, $C_y = C_x$. Moreover, C_x is closed and irreducible. So

$$R = \bigcup_x C_x. \quad \square$$

Example 3.3.12 (Birth and Death Chain). Let $S = \{0, 1, 2, \dots\}$ and X_n be the size of certain population at time n with

$$\mathbb{P}(X_1 = i + 1 \mid X_0 = i) = p_i, \quad \mathbb{P}(X_1 = i - 1 \mid X_0 = i) = q_i, \quad \mathbb{P}(X_1 = i \mid X_0 = i) = r_i = 1 - p_i - q_i.$$

Note that $q_0 = 0$. Determine under which condition that the state 0 is recurrent.

Solution: Step 1. Construction a function $\varphi: S \rightarrow \mathbb{R}$ such that $(\varphi(X_n))_{n \geq 0}$ is a martingale. Let $\varphi(0) = 0$ and $\varphi(1) = 1$. In order that $\varphi(X_n)$ is a martingale, we have

$$\begin{aligned} \varphi(X_n) &= \mathbb{E}[\varphi(X_{n+1}) \mid \mathcal{F}_n] \\ &= \mathbb{E}[\varphi(X_1 \circ \theta_n) \mid \mathcal{F}_n] \\ &= \mathbb{E}_{X_n}[\varphi(X_1)] \end{aligned}$$

If $X_n = k$, then

$$\varphi(k) = \mathbb{E}_k[\varphi(X_1)] = p_k \varphi(k+1) + q_k \varphi(k-1) + r_k \varphi(k)$$

which implies that

$$p_k(\varphi(k+1) - \varphi(k)) = q_k(\varphi(k) - \varphi(k-1)) \Rightarrow \varphi(k+1) - \varphi(k) = \prod_{j=1}^k \frac{q_j}{p_j}$$

and so

$$\varphi(n) = \sum_{m=0}^{n-1} \prod_{j=1}^m \frac{q_j}{p_j}$$

with $\prod_{j=1}^0 \frac{q_j}{p_j} = 1$, that is a increasing function.

Step 2. Let $T_c = \inf \{n \geq 1: X_n = c\}$. Then we will prove that if $a < x < b$

$$\mathbb{P}_x(T_a < T_b) = \frac{\varphi(b) - \varphi(x)}{\varphi(b) - \varphi(a)}$$

and so

$$\mathbb{P}_x(T_b < T_a) = \frac{\varphi(x) - \varphi(a)}{\varphi(b) - \varphi(a)}.$$

Let $T = T_a \wedge T_b$ that is a stopping time. Note that $(\varphi(X_{T \wedge n}))_{n \geq 0}$ is a martingale. Moreover,

$$|\varphi(X_{T \wedge n})| \leq \varphi(a) + \varphi(b).$$

So it is UI. Then

$$\mathbb{E}_x[\varphi(X_0)] = \mathbb{E}_x[\varphi(X_T)].$$

It follows that

$$\begin{aligned} \varphi(x) &= \varphi(a)\mathbb{P}_x(X_T = a) + \varphi(b)\mathbb{P}_x(X_T = b) \\ &= \varphi(a)\mathbb{P}_x(T_a < T_b) + \varphi(b)(1 - \varphi(a)\mathbb{P}_x(T_a < T_b)) \end{aligned}$$

Step 3. Assume $a = 0$ and $b = M$. Then

$$\mathbb{P}_x(T_M < T_0) = \frac{\varphi(x) - \varphi(0)}{\varphi(M) - \varphi(0)}$$

Note that $T_M \geq M \rightarrow \infty$ as $M \rightarrow \infty$. So

$$\mathbb{P}_x(T_0 = \infty) = \frac{\varphi(x)}{\varphi(\infty)}$$

Claim: 0 is recurrent if and only if $\varphi(\infty) = \infty$.

If 0 is recurrent, then because $\rho_{0x} > 0$, $\rho_{x0} = 1$ that implies that $\mathbb{P}_x(T_0 = \infty) = 0$. Conversely, if $\mathbb{P}_x(T_0 = \infty) = 0$, then $\mathbb{P}_x(T_0 < \infty) = 1$. It is also true for $x = 1$.

$$\{T_0 < \infty\} = \{X_1 = 0, T_0 < \infty\} \cup \{X_1 = 1, T_0 < \infty\},$$

So

$$\begin{aligned} \mathbb{P}_0(T_0 < \infty) &= \mathbb{P}_0(X_1 = 0, T_0 < \infty) + \mathbb{P}_0(X_1 = 1, T_0 < \infty) \\ &\leq \mathbb{P}_0(X_1 = 0) + \mathbb{P}_0(X_1 = 1)\mathbb{P}_1(T_0 < \infty) < \infty. \end{aligned}$$

Example 3.3.13 (Symmetric Random Walk). $(X_n)_{n \geq 0}$ is called a random walk if $X_n = x_0 + \sum_{i=1}^n \xi_i$, where ξ_i are i.i.d.. In general, X_n represents the position of a particle at n . A symmetric (simple) random walk on $\mathbb{Z}^d$ is that each transition probability is equal. Note that a symmetric random walk is Markov and irreducible. So one state is recurrent and all states are recurrent. For a symmetric random walk with $x_0 = 0$, let $\tau_0 = 0$ and

$$\tau_n = \inf \{k > \tau_{n-1} : X_k = 0\},$$

i.e., the n -th returning time of 0. By the strong Markov property,

$$\mathbb{P}_0(\tau_n < \infty) = \mathbb{P}_0(\tau_1 < \infty)^n$$

Proof. It is true for $n = 1$. Assume it is true for n . Note that

$$\tau_{n+1} = \tau_n + \tau_1 \circ \theta_{\tau_n}$$

Because $\{\tau_{n+1} < \infty\} \subset \{\tau_n < \infty\}$,

$$\begin{aligned} \mathbb{P}_0(\tau_{n+1} < \infty) &= \mathbb{P}_0(\tau_n < \infty, \tau_{n+1} < \infty) \\ &= \mathbb{P}_0(\tau_n < \infty, \tau_n + \tau_1 \circ \theta_{\tau_n} < \infty) \\ &= \mathbb{P}_0(\tau_n < \infty, \tau_1 \circ \theta_{\tau_n} < \infty) \\ &= \mathbb{E}_0 [\mathbb{I}_{\{\tau_n < \infty\}} \mathbb{I}_{\{\tau_1 < \infty\}} \circ \theta_{\tau_n}] \\ &= \mathbb{E}_0 [\mathbb{E}_0 [\mathbb{I}_{\{\tau_n < \infty\}} \mathbb{I}_{\{\tau_1 < \infty\}} \circ \theta_{\tau_n} \mid \mathcal{F}_{\tau_n}]] \\ &= \mathbb{E}_0 [\mathbb{I}_{\{\tau_n < \infty\}} \mathbb{E}_0 [\mathbb{I}_{\{\tau_1 < \infty\}} \circ \theta_{\tau_n} \mid \mathcal{F}_{\tau_n}]] \\ &= \mathbb{E}_0 [\mathbb{I}_{\{\tau_n < \infty\}} \mathbb{E}_{X_{\tau_n}} [\mathbb{I}_{\{\tau_1 < \infty\}}]] \\ &= \mathbb{E}_0 [\mathbb{I}_{\{\tau_n < \infty\}} \mathbb{E}_0 [\mathbb{I}_{\{\tau_1 < \infty\}}]] \\ &= \mathbb{E}_0 [\mathbb{I}_{\{\tau_n < \infty\}} \mathbb{P}_0 \{\tau_1 < \infty\}] \\ &= \mathbb{P}_0 \{\tau_1 < \infty\} \mathbb{P}_0(\tau_n < \infty). \end{aligned}$$

Then by induction, it is true. □

Theorem 3.3.14. For any random walk, TFAE.

$$(1) \mathbb{P}_0(\tau_1 < \infty) = 1.$$

$$(2) \mathbb{P}_0(X_n = 0, i.o.) = 1.$$

$$(3) \sum_{n=0}^{\infty} \mathbb{P}_0(X_n = 0) = \infty.$$

Proof. (1) $\Rightarrow$ (2): By above

$$\mathbb{P}_0(\tau_n < \infty) = \mathbb{P}_0(\tau_1 < \infty)^n = 1,$$

which implies (2).

(2) $\Rightarrow$ (3): Let $N(0) = \sum_{m=0}^{\infty} \mathbb{I}_{\{X_m=0\}}$. Then (2) means

$$\mathbb{P}_0(N(0) = \infty) = 1.$$

Then

$$\mathbb{E}_0[N(0)] = \sum_{m=0}^{\infty} \mathbb{P}_0(X_m = 0) = \infty.$$

(3) $\Rightarrow$ (1): Note that

$$N(0) = \sum_{n=0}^{\infty} \mathbb{I}_{\tau_n < \infty},$$

which implies that

$$\mathbb{E}_0[N(0)] = \sum_{n=0}^{\infty} \mathbb{P}_0(\tau_n < \infty) = \sum_{n=0}^{\infty} \mathbb{P}_0(\tau_1 < \infty)^n = \infty$$

So $\mathbb{P}_0(\tau_1 < \infty) = 1$. □

Theorem 3.3.15. *Let $(X_n)_{n \geq 0}$ be a simple random walk in $\mathbb{Z}^d$. $(X_n)_{n \geq 0}$ is recurrent if $d \leq 2$. $(X_n)_{n \geq 0}$ is transient if $d \geq 3$.*

Proof. By above theorem, $(X_n)_{n \geq 0}$ is recurrent if and only if

$$\sum_{n=0}^{\infty} \mathbb{P}_0(X_n = 0) = \infty.$$

(i) $d = 1$: $\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = 1/2$. First, it's obvious

$$\mathbb{P}_0(X_{2m+1} = 0) = 0.$$

For $n = 2m$,

$$\mathbb{P}_0(X_m = 0) = \binom{2m}{m} \left(\frac{1}{2}\right)^{2m}$$

By Stirling's formula,

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \text{ as } n \rightarrow \infty,$$

we have

$$\binom{2m}{m} \left(\frac{1}{2}\right)^{2m} \sim m^{-\frac{1}{2}}.$$

So

$$\sum_{n=0}^{\infty} \mathbb{P}_0(X_n = 0) = \infty.$$

(ii) $d = 2$: First, it's obvious

$$\mathbb{P}_0(X_{2n+1} = 0) = 0.$$

To make $X_{2n} = 0$, there exists $0 \leq m \leq n$ such that m steps up with m steps down, and $n - m$ steps left with $n - m$ steps right. So

$$\mathbb{P}_0(X_{2n} = 0) = \sum_{m=0}^n \frac{(2n)!}{m!m!(n-m)!(n-m)!} \left(\frac{1}{4}\right)^{2n} = \left(\frac{1}{4}\right)^{2n} \frac{(2n)!}{n!n!} \sum_{m=0}^n \frac{n!n!}{m!m!(n-m)!(n-m)!}$$

Let

$$C_n = \frac{(2n)!}{n!n!} = \binom{2n}{n}.$$

Then

$$\sum_{m=0}^n \frac{n!n!}{m!m!(n-m)!(n-m)!} = \sum_{m=0}^n \binom{n}{m} \binom{n}{n-m} = \binom{2n}{n} = C_n.$$

So

$$\mathbb{P}_0(X_{2n} = 0) = \left(\frac{1}{4}\right)^{2n} C_n^2 \sim \frac{1}{n} 4^{2n}$$

(iii) $d = 3$: First,

$$\mathbb{P}_0(X_{2n+1} = 0) = 0.$$

Similarly, we have

$$\begin{aligned} \mathbb{P}_0(X_{2n} = 0) &= \sum_{j,k=0}^n \frac{(2n)!}{j!j!k!k!(n-j-k)!(n-j-k)!} \left(\frac{1}{6}\right)^{2n} \\ &= 2^{-2n} \binom{2n}{n} \sum_{j,k} \left(3^{-n} \frac{n!}{j!k!(n-j-k)!}\right)^2 \\ &\leq 2^{-2n} \binom{2n}{n} \max_{j,k} \left(3^{-n} \frac{n!}{j!k!(n-j-k)!}\right) \sum_{j,k} 3^{-n} \frac{n!}{j!k!(n-j-k)!} \\ &= 2^{-2n} \binom{2n}{n} \max_{j,k} \left(3^{-n} \frac{n!}{j!k!(n-j-k)!}\right) \end{aligned}$$

because

$$(a+b+c)^n = \sum_{j,k} a^j b^k c^{n-j-k} \frac{n!}{j!k!(n-j-k)!}.$$

Moreover, because the maximum should be taken at $i = j \approx n/3$, by Stirling's formula,

$$\max_{j,k} \frac{n!}{j!k!(n-j-k)!} \leq C 3^n$$

Then

$$\mathbb{P}_0(X_{2n} = 0) \leq C' n^{-\frac{3}{2}}.$$

So it is transient. □

3.4 Stationary Measure

Definition 3.4.1 (Stationary/Invariant Measure). A measure μ on $(S, \mathcal{S})$ is said to be a stationary or invariant measure if

$$\sum_x \mu(x)p(x, y) = \mu(y), \quad \forall y \in S.$$

In matrix form, $\mu P = \mu$ for $P = (p(x, y))$ and $\mu = (\mu(x))$. Furthermore, if μ is a probability measure, it is called a stationary distribution.

Remark 3.4.2. Note that

$$\sum_x \mu(x)p(x, y) = \mathbb{P}_\mu(X_1 = y) = \mathbb{E}_\mu[\mathbb{P}_{X_0}(X_1 = y)] = \mu(y),$$

i.e., starting from μ , $X_1 \sim \mu$. Then by Markov property, $X_n \sim \mu$.

Example 3.4.3 (Random Walk). $X_n = x_0 + \xi_1 + \cdots + \xi_n$ on $\mathbb{Z}^d$ with ξ_i i.i.d $\mathbb{P}(\xi = z) = f(z)$. In such case,

$$p(x, y) = \mathbb{P}_x(X_1 = y) = \mathbb{P}(\xi_1 = y - x) = f(y - x).$$

Let $\mu(x) \equiv 1$ for any $x \in S$. Then μ is a stationary measure because

$$\sum_x \mu(x)p(x, y) = \sum_x f(y - x) = \sum_x f(x) = 1.$$

Example 3.4.4 (1-dim Random Walk). $X_n = \xi_1 + \cdots + \xi_n$ on $\mathbb{Z}$ with ξ_i i.i.d $\mathbb{P}(\xi = 1) = p$ and $\mathbb{P}(\xi = -1) = q$. Assume $p \neq q$. Let

$$\mu(x) = \left(\frac{p}{q}\right)^x, \quad \forall x \in \mathbb{Z}.$$

Then μ is a stationary measure.

First, the transition probability

$$p(x, y) = \begin{cases} p, & y = x + 1, \\ q, & y = x - 1, \\ 0, & \text{otherwise.} \end{cases}$$

So

$$\begin{aligned} \sum_x \mu(x)p(x, y) &= \mu(y-1)p(y-1, y) + \mu(y+1)p(y+1, y) \\ &= \left(\frac{p}{q}\right)^y = \mu(y). \end{aligned}$$

Example 3.4.5 (Birth and Death Process). Let $S = \{0, 1, 2, \cdots\}$ and X_n be the size of certain population at time n with

$$p(x, x+1) = p_x, \quad p(x, x-1) = q_x, \quad p(x, x) = r_x = 1 - p_x - q_x.$$

with $q_0 = 0$. Let

$$\mu(x) = \prod_{k=1}^x \frac{p_{k-1}}{q_k}.$$

Then μ is an invariant measure.

Assume $y > 0$.

$$\begin{aligned}
\sum_x \mu(x)p(x, y) &= \mu(y-1)p(y-1, y) + \mu(y+1)p(y+1, y) + \mu(y)p(y, y) \\
&= p_{y-1} \prod_{k=1}^{y-1} \frac{p_{k-1}}{q_k} + q_{y+1} \prod_{k=1}^{y+1} \frac{p_{k-1}}{q_k} + r_y \prod_{k=1}^y \frac{p_{k-1}}{q_k} \\
&= \mu(y).
\end{aligned}$$

It is also true for $y = 0$.

Definition 3.4.6 (Reversible Markov Chain). A measure μ on $(S, \mathcal{S})$ is said to be a reversible or symmetric measure if

$$\mu(x)p(x, y) = \mu(y)p(y, x), \quad \forall x, y \in S.$$

Remark 3.4.7. Note that if μ is reversible, then it is obvious invariant.

Theorem 3.4.8. Assume μ is invariant and the Markov chain $(X_n)_{n \geq 0}$ with $X_0 \sim \mu$ and transition probability p . Then for any fixed n , let

$$Y_m = X_{n-m}, \quad m = 0, 1, \dots, n$$

Then (Y_m) is also a Markov chain with $Y_0 \sim \mu$. Moreover, its transition probability is

$$q(x, y) = \frac{\mu(y)p(y, x)}{\mu(x)}.$$

In particular, if μ is reversible, $p = q$.

Proof. The Markov property can be easily obtained by using the Bayesian rule. For the transition probability, because $X_n \sim \mu$ for all n ,

$$\begin{aligned}
q(x, y) &= \mathbb{P}(Y_{m+1} = y \mid Y_m = x) \\
&= \mathbb{P}(X_{n-m-1} = y \mid X_{n-m} = x) \\
&= \frac{\mathbb{P}(X_{n-m} = x \mid X_{n-m-1} = y) \mathbb{P}(X_{n-m-1} = y)}{\mathbb{P}(X_{n-m} = x)} \\
&= \frac{p(y, x) \mu(y)}{\mu(x)}.
\end{aligned}$$

The followings are obvious. □

Theorem 3.4.9 (Existence). Assume x is a recurrent state. Let $T = \inf \{m \geq 1 : X_m = x\}$. Then

$$\mu_x(y) := \mathbb{E}_x \left[\sum_{n=0}^{T-1} \mathbb{I}_{\{X_n=y\}} \right] = \mathbb{E}_x \left[\sum_{n=0}^{\infty} x \mathbb{I}_{\{X_n=y, n < T\}} \right] = \sum_{n=0}^{\infty} \mathbb{P}_x(X_n = y, T > n)$$

is an invariant measure.

Proof. It can see $\mu_x(x) = 1$. Let

$$\bar{p}_n(x, y) = \mathbb{P}_x(X_n = y, T > n).$$

Then $\mu_x(y) = \sum_{n=0}^{\infty} \bar{p}_n(x, y)$. It should to check

$$\sum_y \mu_x(y) p(y, z) = \mu_x(z).$$

By Markov property,

$$\begin{aligned} \mathbb{P}_x(X_n = y, X_{n+1} = z, T > n) &= \mathbb{E}_x [\mathbb{I}_{\{X_n=y, T>n\}} \mathbb{I}_{\{X_{n+1}=z\}}] \\ &= \mathbb{E}_x [\mathbb{E}_x [\mathbb{I}_{\{X_n=y, T>n\}} \mathbb{I}_{\{X_{n+1}=z\}} \mid \mathcal{F}_n]] \\ &= \mathbb{E}_x [\mathbb{I}_{\{X_n=y, T>n\}} \mathbb{E}_x [\mathbb{I}_{\{X_{n+1}=z\}} \mid \mathcal{F}_n]] \\ &= \mathbb{E}_x [\mathbb{I}_{\{X_n=y, T>n\}} \mathbb{E}_{X_n} [\mathbb{I}_{\{X_1=z\}}]] \\ &= \mathbb{E}_x [\mathbb{I}_{\{X_n=y, T>n\}} p(X_n, z)] \\ &= \mathbb{P}_x(X_n = y, T > n) p(y, z) \end{aligned}$$

Consider the following two cases:

Case 1: $z \neq x$. So $\mathbb{P}_x(X_n = y, X_{n+1} = z, T > n+1) = \mathbb{P}_x(X_n = y, X_{n+1} = z, T > n)$, we have

$$\begin{aligned} \sum_y \mu_x(y) p(y, z) &= \sum_y \sum_{n=0}^{\infty} \bar{p}_n(x, y) p(y, z) \\ &= \sum_{n=0}^{\infty} \left(\sum_y \mathbb{P}_x(X_n = y, T > n) p(y, z) \right) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(X_{n+1} = z, T > n+1) \\ &= \mu_x(z). \end{aligned}$$

Case 2: $z = x$. The right-hand side is 1.

$$\begin{aligned} \sum_y \mu_x(y) p(y, z) &= \sum_{n=0}^{\infty} \sum_y \mathbb{P}_x(X_n = y, X_{n+1} = x, T > n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(X_{n+1} = x, T > n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(T = n+1) \\ &= \mathbb{P}_x(T < \infty) = 1, \end{aligned}$$

because x is recurrent. □

Remark 3.4.10. If S is finite, by above theorem, it always has a recurrent state. So for finite case, Markov chain always has an invariant measure.

Theorem 3.4.11 (Uniqueness). *If the Markov chain is irreducible and recurrent, then the invariant measure is unique up to constant multiples.*

Proof. Fix $a \in S$ that is obvious recurrent. So we have an invariant measure μ_a . Given any invariant measure ν . First, we have for any z ,

$$\nu(z) = \sum_y \nu(y) p(y, z) = \nu(a) p(a, z) + \sum_{y \neq a} \nu(y) p(y, z).$$

Using that for multiple times, we have

$$\begin{aligned}
\nu(z) &= \nu(a)p(a, z) + \sum_{y \neq a} \left(\nu(a)p(a, y) + \sum_{x \neq a} \nu(x)p(x, y) \right) p(y, z) \\
&= \nu(a)p(a, z) + \sum_{y \neq a} \nu(a)p(a, y)p(y, z) + \sum_{y \neq a} \sum_{x \neq a} \nu(x)p(x, y)p(y, z) \\
&= \nu(a)\mathbb{P}_z(X_1 = z) + \nu(a) \sum_{y \neq a} \mathbb{P}_a(X_1 = y, X_2 = z) \\
&\quad + \sum_{y \neq a} \sum_{x \neq a} \left(\nu(a)p(a, x) + \sum_{w \neq x} \nu(w)p(w, x) \right) p(x, y)p(y, z) \\
&= \nu(a)\mathbb{P}_a(X_1 = z) + \nu(a)\mathbb{P}_a(X_1 \neq a, X_2 = z) + \nu(a)\mathbb{P}_a(X_1 \neq a, X_2 \neq a, X_3 = z) \\
&\quad + \sum_{y \neq a} \sum_{x \neq a} \sum_{w \neq a} \nu(w)p(w, x)p(x, y)p(y, z) \\
&= \dots \\
&\geq \nu(a) \sum_{n=1}^{\infty} \mathbb{P}_a(X_k \neq a, 1 \leq k < n, X_n = z) \\
&= \nu(a) \sum_{n=1}^{\infty} \mathbb{P}_a(T_a > n, X_n = z) \\
&= \nu(a)\mu_a(z).
\end{aligned}$$

Conversely, because ν is invariant

$$\begin{aligned}
\nu(a) &= \sum_x \nu(x)p^n(x, a) \\
&\geq \sum_x \nu(a)\mu_a(x)p^n(x, a) \\
&= \nu(a)\mu_a(a) = \nu(a).
\end{aligned}$$

Therefore,

$$\sum_x (\nu(x) - \nu(a)\mu_a(x)) p^n(x, a) = 0.$$

For any $y \in S$,

$$(\nu(y) - \nu(a)\mu_a(y)) p^n(y, a) + \sum_{x \neq y} (\nu(x) - \nu(a)\mu_a(x)) p^n(x, a) = 0.$$

Because of the irreducibility, there exists a n such that $p^n(y, a) > 0$. So

$$\nu(y) = \nu(a)\mu_a(y).$$

□

Theorem 3.4.12. *If π is a stationary distribution, then all states y that $\pi(y) > 0$ is recurrent.*

Proof. For any $n \in \mathbb{N}$, because π is stationary,

$$\sum_{n=1}^{\infty} \sum_{x \in S} \pi(x)p^n(x, y) = \sum_{n=1}^{\infty} \pi(y) = \infty.$$

On the other hand,

$$\begin{aligned}\sum_{n=1}^{\infty} \sum_{x \in S} \pi(x) p^n(x, y) &= \sum_{x \in S} \pi(x) \sum_{n=1}^{\infty} p^n(x, y) \\ &= \sum_{x \in S} \pi(x) \frac{\rho_{xy}}{1 - \rho_{yy}} \\ &\leq \frac{1}{1 - \rho_{yy}}.\end{aligned}$$

So $\rho_{yy} = 1$. □

Theorem 3.4.13. *If the Markov chain is irreducible and it has a stationary distribution π , then $\pi(x) > 0$ for all x and*

$$\pi(x) = \frac{1}{\mathbb{E}_x[T_x]},$$

where $T_x = \inf \{n > 0 : X_n = x\}$.

Proof. Claim: For all x , $\pi(x) > 0$.

Because π is invariant,

$$\pi(x) = \sum_{y \in S} \pi(y) p^n(y, x).$$

Because there is a z such that $\pi(z) > 0$ and the irreducibility implies that $p^n(z, x) > 0$ for some n ,

$$\pi(x) > \pi(z) p^n(z, x) > 0.$$

So all x are recurrent, which means the Markov chain is irreducible and recurrent. By the uniqueness of the stationary distribution, for any x ,

$$\mu_x(y) = c\pi(y).$$

So

$$\sum_{y \in S} \mu_x(y) = \sum_{n=0}^{\infty} \sum_{y \in S} \mathbb{P}_x(X_n = y, T_x > n) = \sum_{n=0}^{\infty} \mathbb{P}_x(T_x > n) = \mathbb{E}_x[T_x]$$

On the other hand,

$$c = \mathbb{E}_x[T_x] = \frac{\mu_x(y)}{\pi(y)}.$$

In particular, let $y = x$.

$$\pi(x) = \frac{1}{\mathbb{E}_x[T_x]}.$$

□

Definition 3.4.14 (Positive Recurrence). Let x be a recurrent state. If $\mathbb{E}_x[T_x] < \infty$, then x is called positively recurrent, otherwise, it is called null recurrent.

Theorem 3.4.15. *If the Markov chain is irreducible, then TFAE.*

- (1) *There exists a positively recurrent state.*
- (2) *There exists a stationary distribution.*
- (3) *All states are positively recurrent state.*

Proof. (1) $\Rightarrow$ (2): If x is positively recurrent, then define

$$\pi(y) := \frac{\mu_x(y)}{\mathbb{E}_x[T_x]},$$

which is a stationary distribution.

(2) $\Rightarrow$ (3): By above theorem,

$$\pi(y) = \frac{1}{\mathbb{E}_y[T_y]} > 0,$$

so $\mathbb{E}_y[T_y] < \infty$.

□

Chapter 4

Poisson Process

4.1 Construction

Definition 4.1.1 (Poisson Process). A stochastic process $(N_t)_{t \geq 0}$ with $N_0 = 0$ is called a Poisson process of rate λ if

- (i) (Independent increasing) for any $t_1 < s_1 < t_2 < s_2 < \cdots < t_n < s_n$,

$$N_{s_1} - N_{t_1}, N_{s_2} - N_{t_2}, \dots, N_{s_n} - N_{t_n}$$

are independent.

- (ii) for any $s < t$, $N_t - N_s \sim \text{Pois}(\lambda(t - s))$.

Remark 4.1.2. In general, N_t = the number of times an event occurs in $[0, t]$.

The next problem is how to construct a Poisson process: Given a $\lambda > 0$, let $\xi_1, \dots, \xi_n, \dots$ be i.i.d. with exponential distribution $\exp(\lambda)$, i.e.

$$\mathbb{P}(\xi_i > t) = e^{-\lambda t},$$

(In fact, ξ_i is the time between incidents). Let $T_0 = 0$ and $T_n = \xi_1 + \cdots + \xi_n$ that is the time at which the n -th incident occurs. Define

$$N_t = \sup \{n > 0: T_n \leq t\}.$$

Then $(N_t)_{t \geq 0}$ is a Poisson process.

Proof. Step 1: $N_t \sim \text{Pois}(\lambda t)$.

Note that $T_n \sim \Gamma(n, \lambda)$, i.e., its density function is

$$f_{T_n}(s) = \frac{\lambda^n s^{n-1}}{(n-1)!} e^{-\lambda s}.$$

As we know,

$$\{N_t = 0\} = \{T_1 > t\} = \{\xi_1 > t\}$$

which implies

$$\mathbb{P}(N_t = 0) = e^{-\lambda t}.$$

For $N_t = n$ with $n \geq 1$,

$$\begin{aligned}
\mathbb{P}(N_t = n) &= \mathbb{P}(T_n \leq t < T_{n+1}) \\
&= \mathbb{P}(T_n \leq t < T_n + \xi_{n+1}) \\
&= \iint_{s \leq t < s+u} f_{T_n}(s) f_{\xi_{n+1}}(u) ds du \\
&= \frac{(\lambda t)^n}{n!} e^{-\lambda t}.
\end{aligned}$$

Step 2: Fix t , let

$$T'_1 = T_{N_t+1} - t, \quad T'_2 = T_{N_t+2} - T_{N_t+1}, \quad \dots \quad T'_k = T_{N_t+k} - T_{N_t+k-1}, \quad \dots$$

Claim: $T'_1, T'_2, \dots$ are i.i.d. $\exp(\lambda)$ and they are independent with N_t .

First,

$$\begin{aligned}
\mathbb{P}(T_{n+1} - t \geq s \mid N_t = n) &= \frac{\mathbb{P}(T_{n+1} - t \geq s, N_t = n)}{\mathbb{P}(N_t = n)} \\
&= \frac{\mathbb{P}(T_{n+1} - t \geq s, T_n \leq t)}{\mathbb{P}(N_t = n)} \\
&= \frac{\mathbb{P}(T_n + \xi_{n+1} - t \geq s, T_n \leq t)}{\mathbb{P}(N_t = n)} \\
&= \frac{e^{-\lambda(t+s)} \frac{(\lambda t)^2}{n!}}{\mathbb{P}(N_t = n)} \\
&= e^{-\lambda s}.
\end{aligned}$$

Then consider

$$\begin{aligned}
&\mathbb{P}(T_n \leq t, T_{n+1} - t \geq s, T_{n+k} - T_{n+k-1} \geq v_k, k = 2, 3, \dots, m) \\
&= \mathbb{P}(T_n \leq t, T_{n+1} - t \geq s, \xi_{n+k} \geq v_k, k = 2, 3, \dots, m) \\
&= \mathbb{P}(T_n \leq t, T_{n+1} - t \geq s) \prod_{k=2}^m \mathbb{P}(\xi_{n+k} \geq v_k),
\end{aligned}$$

which implies that

$$\begin{aligned}
&\mathbb{P}(T_{n+1} - t \geq s, T_{n+k} - T_{n+k-1} \geq v_k, k = 2, 3, \dots, m \mid N_t = n) \\
&= \frac{\mathbb{P}(T_n \leq t, T_{n+1} - t \geq s, \xi_{n+k} \geq v_k, k = 2, 3, \dots, m)}{\mathbb{P}(N_t = n)} \\
&= \frac{\mathbb{P}(T_n \leq t, T_{n+1} - t \geq s)}{\mathbb{P}(N_t = n)} \prod_{k=2}^m \mathbb{P}(\xi_{n+k} \geq v_k) \\
&= e^{-\lambda s} \prod_{k=2}^m e^{-\lambda v_k}
\end{aligned}$$

For the independence,

$$\begin{aligned}
&\mathbb{P}(T'_1 \geq s, T'_k \geq v_k, k = 2, 3, \dots, m, N_t \leq \ell) \\
&= \sum_{n=0}^{\ell} \mathbb{P}(T'_1 \geq s, T'_k \geq v_k, k = 2, 3, \dots, m, N_t = \ell) \\
&= \sum_{n=0}^{\ell} \mathbb{P}(T_{N_t+1} - t \geq s, T_{N_t+k} - T_{N_t+k-1} \geq v_k, k = 2, 3, \dots, m \mid N_t = n) \mathbb{P}(N_t = n)
\end{aligned}$$

$$\begin{aligned}
&= \sum_{n=0}^{\ell} \mathbb{P}(T_{n+1} - t \geq s, T_{n+k} - T_{n+k-1} \geq v_k, k = 2, 3, \dots, m \mid N_t = n) \mathbb{P}(N_t = n) \\
&= e^{-\lambda s} \prod_{k=2}^m e^{-\lambda v_k} \mathbb{P}(N_t \leq \ell).
\end{aligned}$$

Step 3: For any $t_0 < t_1 < t_2 < \dots < t_n$, it suffices to check

$$\mathbb{P}(N_{t_i} - N_{t_{i-1}} \geq k_i, i = 1, \dots, n) = \prod_{i=1}^n e^{-\lambda(t_i - t_{i-1})} \frac{(\lambda(t_i - t_{i-1}))^{k_i}}{k_i!}.$$

It only needs to prove for $N_{t_2} - N_{t_1}$ and N_{t_1} . Let $T'_1 = T_{N_{t_1}+1} - t_1$ and $T'_k = T_{N_{t_1}+k} - T_{N_{t_1}+k-1}$. Then by Step 2, $T'_1, \dots, T'_k$ are independent with N_{t_1} . Note that

$$\begin{aligned}
\{N_{t_2} - N_{t_1} = m\} &= \{T_{N_{t_1}+m} \leq t_2, T_{N_{t_1}+m+1} > t_2\} \\
&= \{T_{N_{t_1}+m} - t_1 \leq t_2 - t_1, T_{N_{t_1}+m+1} - t_1 > t_2 - t_1\} \\
&= \left\{ T'_1 + \sum_{k=2}^m T'_k \leq t_2 - t_1, T'_1 + \sum_{k=2}^{m+1} T'_k > t_2 - t_1 \right\},
\end{aligned}$$

which follows that $N_{t_2} - N_{t_1}$ is independent with N_{t_1} and moreover

$$\mathbb{P}(N_{t_2} - N_{t_1} = m) = e^{-\lambda(t_2 - t_1)} \frac{(\lambda(t_2 - t_1))^m}{m!}$$

by Step 2. □

4.2 Compound Poisson Process

Definition 4.2.1 (Compound Poisson Process). Let $(N_t)_{t \geq 0}$ be a Poisson process with λ and $Y_1, \dots, Y_n, \dots$ be i.i.d. and independent with N_t . Then

$$S(t) = \sum_{k=1}^{N_t} Y_k$$

is called a compound Poisson process.

Theorem 4.2.2. Let $Y_1, \dots, Y_n, \dots$ be i.i.d. and $N \geq 0$ be an integer-valued and independent random variable. Let

$$S = Y_1 + Y_2 + \dots + Y_N,$$

and $S = 0$ if $N = 0$. Then

$$(i) \quad \mathbb{E}[S] = \mathbb{E}[N] \mathbb{E}[Y_i].$$

$$(ii) \quad \text{Var}(S) = \mathbb{E}[N] \text{Var}(Y_i) + \text{Var}(N) (\mathbb{E}[Y_i])^2. \text{ In particular, if } N \sim \text{Pois}(\lambda), \text{ then } \text{Var}(S) = \lambda \mathbb{E}[Y_i^2].$$

Proof. First, by independence,

$$\begin{aligned}
\mathbb{E}[S] &= \sum_{n=0}^{\infty} \mathbb{E}[S \mathbb{I}_{\{N=n\}}] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[Y_i] \mathbb{E}[\mathbb{I}_{\{N=n\}}] \\
&= \mathbb{E}[Y_i] \mathbb{E}[N].
\end{aligned}$$

For $\text{Var}(S) = \mathbb{E}[S^2] - (\mathbb{E}[S])^2$,

$$\begin{aligned}
\mathbb{E}[S^2] &= \sum_{n=1}^{\infty} \mathbb{E}[S^2 \mathbb{I}_{\{N=n\}}] \\
&= \sum_{n=1}^{\infty} \mathbb{E}[(Y_1 + \dots + Y_n)^2 \mathbb{I}_{\{N=n\}}] \\
&= \sum_{n=1}^{\infty} \mathbb{E}[S_n^2] \mathbb{E}[\mathbb{I}_{\{N=n\}}] \\
&= \sum_{n=1}^{\infty} (n \text{Var}(Y_i) + (n\mathbb{E}[Y_i])^2) \mathbb{P}(N = n) \\
&= \text{Var}(Y_i) \mathbb{E}[N] + \mathbb{E}[Y_i]^2 \mathbb{E}[N^2]
\end{aligned}$$

for $S_n = Y_1 + \dots + Y_n$. Furthermore, $(\mathbb{E}[S])^2 = \mathbb{E}[Y_i]^2 \mathbb{E}[N]^2$. So it is obtained. $\square$

Theorem 4.2.3. Suppose $(N_t)_{t \geq 0}$ is a Poisson process with rate λ , which describes the number of points come by time t . We keep a point that lands at s with probability p_s . Let $\bar{N}_t$ be the number of points landing at s by time t . Then $(\bar{N}_t)$ is also a Poisson process with rate λp_s .

Proof. **Independent Increasing:** Because $\bar{N}_{t_1} - \bar{N}_{t_2}$ is determined by $N_{t_1} - N_{t_2}$.

Poisson Distribution: First,

$$\begin{aligned}
\mathbb{P}(\bar{N}_t = m) &= \mathbb{P}(\bar{N}_t = m, N_t \geq m) \\
&= \sum_{k=m}^{\infty} \mathbb{P}(\bar{N}_t = m, N_t = k) \\
&= \sum_{k=m}^{\infty} \mathbb{P}(\bar{N}_t = m \mid N_t = k) \mathbb{P}(N_t = k) \\
&= \sum_{k=m}^{\infty} \binom{k}{m} (p_s)^m (1 - p_s)^{k-m} e^{-\lambda t} \frac{(\lambda t)^k}{k!} \\
&= \frac{e^{-\lambda t} (\lambda t)^m}{m!} (p_s)^m \sum_{k=m}^{\infty} \frac{(\lambda t (1 - p_s)^{k-m})}{(k - m)!} \\
&= \frac{e^{-\lambda t} (\lambda t)^m}{m!} (p_s)^m e^{\lambda t (1 - p_s)} = e^{-\lambda p_s t} \frac{(\lambda p_s t)^m}{m!}.
\end{aligned}$$

It is similar for others. $\square$

Chapter 5

Brownian Motion

5.1 Definition and Properties

Definition 5.1.1. Let $(B_t)_{t \geq 0}$ be a stochastic process. It is called a (standard when $B_0 = 0$) Brownian motion if

- (1) $t \rightarrow B_t(\omega)$ is continuous a.e.
- (2) it is independent increments.
- (3) for any $s < t$, $B_t - B_s \sim \mathcal{N}(0, t - s)$.

Remark 5.1.2. Note that for $s \leq t$

$$\mathbb{E}[B_s B_t] = \mathbb{E}[B_s] \mathbb{E}[B_t - B_s] + \mathbb{E}[B_s^2] = s.$$

So for any $s, t \geq 0$, $\mathbb{E}[B_s B_t] = s \wedge t$.

Theorem 5.1.3 (Finite-dimensional Distribution). *Let $(B_t)_{t \geq 0}$ be a standard Brownian motion. For any $0 = t_0 < t_1 < \dots < t_n$,*

$$(B_{t_1}, \dots, B_{t_n}) \sim \mathcal{N}(\mu, \Sigma)$$

with

$$p(x_1, x_2, \dots, x_n) = \frac{(2\pi)^{-\frac{n}{2}}}{\sqrt{t_1(t_2 - t_1) \dots (t_n - t_{n-1})}} \exp \left(- \sum_{i=1}^n \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})} \right),$$

where $x_0 = 0$.

Proof. For any measurable f , let $X_i = B_{t_i} - B_{t_{i-1}} \sim \mathcal{N}(0, t_i - t_{i-1})$,

$$\begin{aligned} \mathbb{E}[f(B_{t_1}, \dots, f(B_{t_n}))] &= \mathbb{E} \left[f \left(B_{t_1}, B_{t_2} - B_{t_1} + B_{t_1}, \dots, \sum_{i=1}^n B_{t_i} - B_{t_{i-1}} \right) \right] \\ &= \mathbb{E} \left[f \left(X_1, X_1 + X_2, \dots, \sum_i X_i \right) \right] \\ &= \int f(y_1, y_1 + y_2, \dots) \prod_i f_{X_i}(y_i) dy_1 \dots dy_n. \end{aligned}$$

Let $x_1 = y_1$, $x_2 = y_1 + y_2$, $\dots$, and $x_n = \sum_i y_i$. Then

$$f_{X_i}(y_i) = f_{X_i}(x_i - x_{i-1}) = \frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} \exp \left(- \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})} \right),$$

which implies the desired property. □

Theorem 5.1.4. Suppose $(B_t)_{t \geq 0}$ is a Brownian motion.

(1) $(-B_t)_{t \geq 0}$ is also a Brownian motion.

(2) For any $\lambda > 0$, the process

$$B_t^\lambda = \frac{1}{\lambda} B_{\lambda^2 t}$$

is also a Brownian motion.

(3) For any $s > 0$,

$$B_t^s = B_{t+s} - B_s$$

is also a Brownian motion and independent of $\mathcal{F}_s = \sigma(B_t : t \leq s)$.

Proof. (1) is obvious. For (2), because

$$B_{\lambda^2 t} - B_{\lambda^2 s} \sim \mathcal{N}(0, \lambda^2(t - s)),$$

$B_t^\lambda - B_s^\lambda \sim \mathcal{N}(0, t - s)$. For (3), it is obvious a Brownian motion. The independence is directly obtained by

$$(B_{s+t_1} - B_s, B_{s+t_2} - B_s) = (B_{s+t_1} - B_s, B_{s+t_2} - B_{s+t_1} + B_{s+t_1} - B_s)$$

independent of B_s . □

Remark 5.1.5. A direct corollary for (3) is, for any $t_0 < t_1 < \dots < t_n$, the joint $(B_{t_1}, \dots, B_{t_n})$ is independent of B_{t_0} .

Given a Brownian motion $B = (B_t)_{t \geq 0}$, a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ is called a Brownian filtration if it is a filtration and B_t is $\mathcal{F}_t$ -adapted and $B_t - B_s$ is $\mathcal{F}_s$ -independent. It is not hard to see B is a $\mathbb{F}$ -martingale.

5.2 Properties of Path

Theorem 5.2.1 (0-1 Law). Let $(B_t)_{t \geq 0}$ be a standard Brownian motion with the nature filtration $\mathcal{F}_t = \sigma(B_s : s \leq t)$. Define

$$\mathcal{F}_{0+} = \bigcap_{t>0} \mathcal{F}_t.$$

Then for any $A \in \mathcal{F}_{0+}$, either $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$.

Proof. First, for any $0 < t_1 < \dots < t_n$ and any bounded continuous function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, we will show that $\mathbb{I}_A$ is independent with $f(B_{t_1}, \dots, B_{t_n})$. By the continuity of path and the continuity of f ,

$$\mathbb{E}[\mathbb{I}_A f(B_{t_1}, \dots, B_{t_n})] = \lim_{\epsilon \rightarrow 0} \mathbb{E}[\mathbb{I}_A f(B_{t_1} - B_\epsilon, \dots, B_{t_n} - B_\epsilon)].$$

For $0 < \epsilon < t_1$, $B_{t_1} - B_\epsilon, \dots, B_{t_n} - B_\epsilon$ are independent with B_s for all $s \leq \epsilon$, which means they are independent with $\mathcal{F}_\epsilon$. Because $A \in \mathcal{F}_\epsilon$, $f(B_{t_1} - B_\epsilon, \dots, B_{t_n} - B_\epsilon)$ is independent with $\mathbb{I}_A$. So

$$\begin{aligned} \mathbb{E}[\mathbb{I}_A f(B_{t_1}, \dots, B_{t_n})] &= \lim_{\epsilon \rightarrow 0} \mathbb{E}[\mathbb{I}_A] \mathbb{E}[f(B_{t_1} - B_\epsilon, \dots, B_{t_n} - B_\epsilon)] \\ &= \mathbb{E}[\mathbb{I}_A] \mathbb{E}[f(B_{t_1}, \dots, B_{t_n})] \end{aligned}$$

So $\mathbb{I}_A$ is independent of $\sigma(B_s : s > 0) = \sigma(B_s : s \geq 0)$ because $B_0 = \lim_{t \rightarrow 0} B_t$, which implies that A is independent of itself. □

Remark 5.2.2. In general, if

$$Y = \limsup_n X_n = \inf_{n \geq 1} \sup_{k \geq n} X_k$$

then Y is $\sigma(X_n : n \in \mathbb{N})$ -measurable, because

$$\{Y > a\} = \left\{ \inf_n \sup_{k \geq n} X_k > a \right\} = \bigcap_{n=1}^{\infty} \left\{ \sup_{k \geq n} X_k > a \right\} = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \{X_k > a\}.$$

Theorem 5.2.3. *Let $(B_t)_{t \geq 0}$ be a standard Brownian motion.*

(1) *We have almost surely for every $\varepsilon > 0$*

$$\sup_{0 \leq s \leq \varepsilon} B_s > 0, \quad \inf_{0 \leq s \leq \varepsilon} B_s < 0.$$

(2) *For every $a \in \mathbb{R}$, let*

$$T_a = \inf \{t \geq 0 : B_t = a\}.$$

Then $\mathbb{P}(T_a < \infty) = 1$.

Proof. (1) Let ε_p be a sequence of positive numbers decreasing to 0. Let

$$A = \bigcap_{p > 0} \left\{ \sup_{0 \leq \varepsilon \leq \varepsilon_p} B_\varepsilon > 0 \right\}.$$

Note for any $p_0 > 0$,

$$A = \bigcap_{p \geq p_0} \left\{ \sup_{0 \leq \varepsilon \leq \varepsilon_p} B_\varepsilon > 0 \right\}.$$

So $A \in \mathcal{F}_{\varepsilon_{p_0}}$ because ε_p is decreasing in p , which follows that

$$A \in \bigcap_{p_0} \mathcal{F}_{\varepsilon_{p_0}} = \mathcal{F}_{0+}.$$

On the other hand,

$$\mathbb{P}(A) = \lim_{p \rightarrow \infty} \mathbb{P} \left(\sup_{0 \leq \varepsilon \leq \varepsilon_p} B_\varepsilon > 0 \right).$$

Moreover, for any p ,

$$\mathbb{P} \left(\sup_{0 \leq \varepsilon \leq \varepsilon_p} B_\varepsilon > 0 \right) \geq \mathbb{P}(B_{\varepsilon_p} > 0) = \frac{1}{2}$$

So $\mathbb{P}(A) \geq \frac{1}{2}$. By 0-1 Law,

$$\mathbb{P}(A) = 1 \Rightarrow \mathbb{P} \left(\sup_{0 \leq \varepsilon \leq \varepsilon_p} B_\varepsilon > 0 \right) = 1.$$

For the other one, it is because of $(-B_t)$ also a Brownian motion.

(2) First,

$$\{T_a < \infty\} = \bigcup_{t=0}^{\infty} \{B_t = a\}.$$

It follows that we only need

$$\mathbb{P} \left(\bigcup_{t=0}^{\infty} \{B_t = a\} \right) = 1.$$

Claim: For any $M > 0$,

$$\mathbb{P}(\sup_s B_s > M) = 1, \quad \mathbb{P}(\inf_s B_s < -M) = 1.$$

By (1), we have

$$1 = \mathbb{P}(\sup_{0 \leq s \leq 1} B_s > 0) = \lim_{\delta \rightarrow 0} \mathbb{P}(\sup_{0 \leq s \leq 1} B_s > \delta).$$

For the right hand side

$$\mathbb{P}(\sup_{0 \leq s \leq 1} B_s > \delta) = \mathbb{P} \left(\sup_{0 \leq s \leq 1} \frac{1}{\delta} B_s > 1 \right).$$

Because $\frac{1}{\delta} B_s \stackrel{d}{=} B_{\frac{1}{\delta^2} s}$,

$$\mathbb{P}(\sup_{0 \leq s \leq 1} B_s > \delta) = \mathbb{P} \left(\sup_{0 \leq s \leq 1} B_{\frac{1}{\delta^2} s} > 1 \right) = \mathbb{P} \left(\sup_{0 \leq u \leq 1/\delta^2} B_u > 1 \right).$$

Therefore,

$$\lim_{\delta \rightarrow 0} \mathbb{P} \left(\sup_{0 \leq u \leq 1/\delta^2} B_u > 1 \right) = \mathbb{P} \left(\sup_s B_s > 1 \right) = 1.$$

Then for any $M > 0$,

$$\mathbb{P} \left(\sup_s B_s > M \right) = \mathbb{P} \left(\sup_s \frac{1}{M} B_s > 1 \right) = \mathbb{P} \left(\sup_s B_{\frac{1}{M^2} s} > 1 \right) = \mathbb{P} \left(\sup_s B_s > 1 \right) = 1.$$

For the infimum, it is because $(-B_t)$ is also a Brownian motion.

Then if $a > 0$, there is an M such that $a < M$. By the continuity of path and $B_0 = 0$,

$$\left\{ \sup_s B_s > M \right\} \subset \bigcup_{t=0}^{\infty} \{B_t = a\}$$

So $\mathbb{P}(\cup_t \{B_t = a\}) = 1$. Similarly, for $a \leq 0$, it can get by $\inf B_s$. □

Corollary 5.2.4. For a standard Brownian motion $(B_t)_{t \geq 0}$,

$$\limsup_{t \rightarrow \infty} B_t = \infty, \quad \liminf_{t \rightarrow \infty} B_t = -\infty.$$

Proposition 5.2.5. Let $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = t$ be a sequence of partition of $[0, t]$ such that $\max_i(t_i^n - t_{i-1}^n) \rightarrow 0$ as $n \rightarrow \infty$. Then

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} \left(B_{t_i^n} - B_{t_{i-1}^n} \right)^2 = t$$

in $L^2(\Omega)$.

Proof. To show L^2 convergence, we need

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\left(\sum_{i=1}^{p_n} (B_{t_i^n} - B_{t_{i-1}^n})^2 - t \right)^2 \right] = 0.$$

Let $t = \sum_i t_i^n - t_{i-1}^n$. Then

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{i=1}^{p_n} (B_{t_i^n} - B_{t_{i-1}^n})^2 - t \right)^2 \right] = \mathbb{E} \left[\left(\sum_{i=1}^{p_n} \left((B_{t_i^n} - B_{t_{i-1}^n})^2 - (t_i^n - t_{i-1}^n) \right) \right)^2 \right] \\ &= \mathbb{E} \left[\sum_{i,j} \left((B_{t_i^n} - B_{t_{i-1}^n})^2 - (t_i^n - t_{i-1}^n) \right) \left((B_{t_j^n} - B_{t_{j-1}^n})^2 - (t_j^n - t_{j-1}^n) \right) \right] \\ &= \sum_{i \neq j} \mathbb{E} \left[(B_{t_i^n} - B_{t_{i-1}^n})^2 - (t_i^n - t_{i-1}^n) \right] \mathbb{E} \left[(B_{t_j^n} - B_{t_{j-1}^n})^2 - (t_j^n - t_{j-1}^n) \right] \\ &\quad + \sum_i \mathbb{E} \left[\left((B_{t_i^n} - B_{t_{i-1}^n})^2 - (t_i^n - t_{i-1}^n) \right)^2 \right] \\ &= \sum_i \mathbb{E} \left[\left((B_{t_i^n} - B_{t_{i-1}^n})^2 - (t_i^n - t_{i-1}^n) \right)^2 \right] \\ &\leq 2 \sum_i \left(\mathbb{E} \left[(B_{t_i^n} - B_{t_{i-1}^n})^4 \right] + (t_i^n - t_{i-1}^n)^2 \right) \\ &= 2(c+1) \sum_i (t_i^n - t_{i-1}^n)^2 \leq 2(c+1)t \max_i (t_i^n - t_{i-1}^n) \rightarrow 0 \end{aligned}$$

Note the $X \sim \mathcal{N}(0, \sigma^2)$, $\mathbb{E}[X^4] = c\sigma^4$. □

Corollary 5.2.6. *For a.e. $t \mapsto B_t$ has infinite variation on any finite interval.*

Proof. Let $0 = t_0^n < t_1^n < \dots < t_{p_n}^n = t$ be a sequence of partition of $[0, t]$.

$$\sum_i (B_{t_i^n} - B_{t_{i-1}^n})^2 \leq \max_i |B_{t_i^n} - B_{t_{i-1}^n}| \sum_i |B_{t_i^n} - B_{t_{i-1}^n}|$$

By the continuity, $\max_i |B_{t_i^n} - B_{t_{i-1}^n}| \rightarrow 0$. If $\sum_i |B_{t_i^n} - B_{t_{i-1}^n}| < \infty$,

$$\sum_i (B_{t_i^n} - B_{t_{i-1}^n})^2 \rightarrow 0,$$

which induces a contradiction. □

Theorem 5.2.7. *Given a Brownian motion $B = (B_t)_{t \geq 0}$, for a.e. $\omega \in \Omega$,*

$$\limsup_{t \downarrow 0} \frac{W_t(\omega)}{\sqrt{2t \log \log(1/t)}} = 1, \quad \liminf_{t \downarrow 0} \frac{W_t(\omega)}{\sqrt{2t \log \log(1/t)}} = -1$$

and

$$\limsup_{t \rightarrow \infty} \frac{W_t(\omega)}{\sqrt{2t \log \log(1/t)}} = 1, \quad \liminf_{t \rightarrow \infty} \frac{W_t(\omega)}{\sqrt{2t \log \log(1/t)}} = -1.$$

5.3 Strong Markov Property

Given a standard Brownian motion $(B_t)_{t \geq 0}$, let $\mathcal{F}_t = \sigma(B_s : s \leq t)$ and $\mathcal{F}_\infty = \sigma(B_t : t \geq 0)$, i.e. $(\mathcal{F}_t)_{t \geq 0}$ is the natural filtration of B_t .

First, for Markov property, we already know $B_{t+s} - B_s$ is independent with B_s , which directly implies the Markov property by the following lemma.

Lemma 5.3.1. *Let X and Y be two random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G}$ is a σ -subalgebra of $\mathcal{F}$. If X is $\mathcal{G}$ -measurable and Y is independent with $\mathcal{G}$, then for any Borel measurable function $g: \mathbb{R}^2 \rightarrow \mathbb{R}$,*

$$\mathbb{E}[g(X, Y) \mid \mathcal{G}] = \mathbb{E}[g(X, Y) \mid \sigma(X)]$$

Remark 5.3.2. First, assume $g(x, y) = \mathbb{I}_A(x)\mathbb{I}_B(y)$ and it can clearly true so that it is also true for all simple function g . Then by applying the Monotone Class Theorem, it can prove that.

So

$$\begin{aligned} \mathbb{E}[f(B_{t+s}) \mid \mathcal{F}_t] &= \mathbb{E}[f(B_t + (B_{t+s} - B_t)) \mid \mathcal{F}_t] \\ &= \mathbb{E}[f(B_t + (B_{t+s} - B_t)) \mid \sigma(B_t)] \\ &= \mathbb{E}[f(B_{t+s}) \mid \sigma(B_t)] \end{aligned}$$

For the strong Markov property, first, we need the stopping time.

Definition 5.3.3 (Stopping Time). A random time $T: \Omega \rightarrow [0, \infty]$ is a stopping time with respect to $(\mathcal{F}_t)_{t \geq 0}$ if for any t ,

$$\{T \leq t\} \in \mathcal{F}_t.$$

Remark 5.3.4. Note that

$$\{T < t\} = \bigcup_{q \in \mathbb{Q}, q < t} \{T \leq q\} \in \mathcal{F}_t.$$

so that $\{T \geq t\} \in \mathcal{F}_t$.

Example 5.3.5. (1) For any $a \in \mathbb{R}$,

$$T_a = \inf \{s \geq 0 : B_s = a\}$$

is a stopping time because

$$\{T \leq t\} = \left\{ \inf_{0 \leq s \leq t} |B_s - a| = 0 \right\} \in \mathcal{F}_t.$$

(2) Let

$$T = \sup \{s \leq 1 : B_s = 0\}.$$

Then it is not a stopping time because it needs information in $[0, 1]$.

Definition 5.3.6. Given a stopping time T ,

$$\mathcal{F}_T = \{A \in \mathcal{F}_\infty : A \cap \{T \leq t\} \in \mathcal{F}_t, \forall t \geq 0\},$$

which is a σ -field.

Remark 5.3.7. (1) T is $\mathcal{F}_T$ -measurable, where the reasoning is as same as that of the discrete case.

(2) For any $s \geq 0$, $B_s \mathbb{I}_{s \leq T}$ is $\mathcal{F}_T$ -measurable.

For any $A \in \mathcal{R}$ and WTLG assuming $0 \notin A$ (otherwise considering A^c),

$$\{B_s \mathbb{I}_{s \leq T} \in A\} \cap \{T \leq t\} = \begin{cases} \emptyset, & t < s \\ \{B_s \in A\} \cap \{s \leq T \leq t\}, & s \leq T \leq t. \end{cases}$$

Because $s \leq t$, $\{B_s \in A\} \in \mathcal{F}_t$. Furthermore, $\{T \geq s\} \in \mathcal{F}_s \subset \mathcal{F}_t$, so $\{B_s \mathbb{I}_{s \leq T} \in A\} \cap \{T \leq t\} \in \mathcal{F}_t$.

For a stopping time T , consider $\mathbb{I}_{T < \infty} B_T$, which is $\mathcal{F}_T$ -measurable. Let $n \in \mathbb{N}$. If

$$\frac{k}{2^n} \leq T \leq \frac{k+1}{2^n},$$

then define $B_T^n = B_{k/2^n}$. By the continuity of path, $\lim_n B_T^n = B^T$. So

$$\begin{aligned} \mathbb{I}_{\{T < \infty\}} B_T &= \lim_{n \rightarrow \infty} \sum_{i=0}^{\infty} \mathbb{I}_{\{\frac{i}{2^n} \leq T \leq \frac{i+1}{2^n}\}} B_{\frac{i}{2^n}} \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^{\infty} \mathbb{I}_{\{T \geq \frac{i}{2^n}\}} \mathbb{I}_{\{T \leq \frac{i+1}{2^n}\}} B_{\frac{i}{2^n}}. \end{aligned}$$

Both $\mathbb{I}_{\{T \leq \frac{i+1}{2^n}\}} B_{\frac{i}{2^n}}$ and $\mathbb{I}_{\{T \geq \frac{i}{2^n}\}}$ are $\mathcal{F}_T$ -measurable, which implies that $\mathbb{I}_{T < \infty} B_T$ is $\mathcal{F}_T$ -measurable.

Theorem 5.3.8 (Strong Markov Property). *Give a stopping time T . Assume $\mathbb{P}(T < \infty) > 0$. Set*

$$B_t^{(T)} = \mathbb{I}_{\{T < \infty\}} (B_{T+t} - B_T), \quad t \geq 0.$$

Then under the probability $\mathbb{P}(\cdot \mid T < \infty)$, $(B_t^{(T)})_{t \geq 0}$ is a Brownian motion and independent of $\mathcal{F}_T$.

Proof. WTLG assume $\mathbb{P}(T < \infty) = 1$. For any $A \in \mathcal{F}_T$ and $0 \leq t_1 < t_2 < \dots < t_p$ and any bounded continuous function $F: \mathbb{R}^p \rightarrow \mathbb{R}$, it suffices to show that

$$\mathbb{E} \left[\mathbb{I}_A F(B_{t_1}^{(T)}, \dots, B_{t_p}^{(T)}) \right] = \mathbb{P}(A) \mathbb{E} \left[F(B_{t_1}, \dots, B_{t_p}) \right].$$

Define $[t]_n = k/2^n$ if $(k-1)/2^n < t \leq k/2^n$. Observe that

$$F(B_{t_1}^{(T)}, \dots, B_{t_p}^{(T)}) = \lim_{n \rightarrow \infty} F(B_{t_1}^{([T]_n)}, \dots, B_{t_p}^{([T]_n)})$$

by the continuity of F and B_t .

$$\begin{aligned} \mathbb{E} \left[\mathbb{I}_A F(B_{t_1}^{(T)}, \dots, B_{t_p}^{(T)}) \right] &= \lim_{n \rightarrow \infty} \mathbb{E} \left[\mathbb{I}_A F(B_{t_1}^{([T]_n)}, \dots, B_{t_p}^{([T]_n)}) \right] \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^{\infty} \mathbb{E} \left[\mathbb{I}_A \mathbb{I}_{\{\frac{k-1}{2^n} \leq T \leq \frac{k}{2^n}\}} F(B_{t_1}^{(k/2^n)}, \dots, B_{t_p}^{(k/2^n)}) \right] \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^{\infty} \mathbb{E} \left[\mathbb{I}_A \mathbb{I}_{\{\frac{k-1}{2^n} \leq T \leq \frac{k}{2^n}\}} F(B_{\frac{k}{2^n} + t_1} - B_{\frac{k}{2^n}}, \dots, B_{\frac{k}{2^n} + t_p} - B_{\frac{k}{2^n}}) \right] \end{aligned}$$

Because T is a stopping time and $A \in \mathcal{F}_T$,

$$A \cap \left\{ \frac{k-1}{2^n} \leq T \leq \frac{k}{2^n} \right\} \in \mathcal{F}_{\frac{k}{2^n}}.$$

Because $B_{\frac{k}{2^n}+t_1} - B_{\frac{k}{2^n}}, \dots, B_{\frac{k}{2^n}+t_p} - B_{\frac{k}{2^n}}$ are independent of $\mathcal{F}_{\frac{k}{2^n}}$,

$$\begin{aligned}\mathbb{E} \left[\mathbb{I}_A F(B_{t_1}^{(T)}, \dots, B_{t_p}^{(T)}) \right] &= \lim_{n \rightarrow \infty} \sum_{k=0}^{\infty} \mathbb{E} \left[\mathbb{I}_A \mathbb{I}_{\{\frac{k-1}{2^n} \leq T \leq \frac{k}{2^n}\}} \right] \mathbb{E} \left[F \left(B_{\frac{k}{2^n}+t_1} - B_{\frac{k}{2^n}}, \dots, B_{\frac{k}{2^n}+t_p} - B_{\frac{k}{2^n}} \right) \right] \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^{\infty} \mathbb{E} \left[\mathbb{I}_A \mathbb{I}_{\{\frac{k-1}{2^n} \leq T \leq \frac{k}{2^n}\}} \right] \mathbb{E} \left[F \left(\tilde{B}_{t_1}, \dots, \tilde{B}_{t_p} \right) \right] \\ &= \mathbb{P}(A) \mathbb{E} \left[F(B_{t_1}, \dots, B_{t_p}) \right],\end{aligned}$$

where the final equality is because $\tilde{B}_t = B_{\frac{k}{2^n}+t} - B_{\frac{k}{2^n}}$ is also a Brownian motion. $\square$

Remark 5.3.9. A direct corollary of this is

$$\mathbb{E}[f(B_{T+s}) \mid \mathcal{F}_T] = \mathbb{E}[f(B_{T+s}) \mid X_T],$$

which is the strong Markov property.

Theorem 5.3.10 (Reflexive Principle). *For any $t > 0$, let*

$$S_t = \sup_{0 \leq s \leq t} B_s \geq 0.$$

If $a \geq 0$ and $b \leq a$, then

$$\mathbb{P}(S_t \geq a, B_t \leq b) = \mathbb{P}(B_t \geq 2a - b).$$

In particular, S_t has the same distribution as $|B_t|$.

Proof. Let $T_a = \inf \{t \geq 0: B_t = a\}$. By the continuity of B_t , $\{S_t \geq a\} = \{T_a \leq t\}$. So

$$\begin{aligned}\mathbb{P}(S_t \geq a, B_t \leq b) &= \mathbb{P}(T_a \leq t, B_{t-T_a+T_a} \leq b) \\ &= \mathbb{P}(T_a \leq t, B_{t-T_a+T_a} - B_{T_a} \leq b - a) \\ &= \mathbb{P}\left(T_a \leq t, B_{t-T_a}^{(T_a)} \leq b - a\right).\end{aligned}$$

Let $B'_t = B_{t-T_a}^{(T_a)}$ that is a Brownian motion independent of T_a because T_a is $\mathcal{F}_{T_a}$ -measurable. So

$$\begin{aligned}\mathbb{P}\left(T_a \leq t, B_t^{(T_a)} \leq b - a\right) &= \mathbb{P}(T_a \leq t) \mathbb{P}(-B'_t \geq a - b) \\ &= \mathbb{P}(T_a \leq t) \mathbb{P}(B'_t \geq a - b) \\ &= \mathbb{P}(T_a \leq t, B'_t \geq a - b) \\ &= \mathbb{P}(T_a \leq t, B_t - B_{T_a} \geq a - b) \\ &= \mathbb{P}(T_a \leq t, B_t \geq 2a - b).\end{aligned}$$

But $\{T_a \leq t\} \subset \{B_t \geq 2a - b\}$ because of $B_t \geq 2a - b \geq a$ and the continuity of B_t . So

$$\mathbb{P}(S_t \geq a, B_t \leq b) = \mathbb{P}(B_t \geq 2a - b).$$

For the other one,

$$\begin{aligned}\mathbb{P}(S_t \geq a) &= \mathbb{P}(S_t \geq a, B_t \geq a) + \mathbb{P}(S_t \geq a, B_t \leq a) \\ &= \mathbb{P}(B_t \geq a) + \mathbb{P}(B_t \geq 2a - a) \\ &= 2\mathbb{P}(B_t \geq a) = \mathbb{P}(|B_t| \geq a).\end{aligned}$$

$\square$

Corollary 5.3.11. T_a has the same distribution as $\frac{a^2}{B_1^2}$ with the density function

$$f(t) = \frac{a}{\sqrt{2\pi t}} e^{-\frac{a^2}{2t}} \mathbb{I}_{\{t>0\}}$$

Proof. Because $\{S_t \geq a\} = \{T_a \leq t\}$,

$$\mathbb{P}(T_a \leq t) = \mathbb{P}(S_t \geq a) = \mathbb{P}(|B_t| \geq a).$$

$\square$

5.4 High-dimensional Brownian Motion

Definition 5.4.1. A d -dimensional stochastic process $(\mathbf{B}_t = (B_t^1, \dots, B_t^d))_{t \geq 0}$ is called a d -dimensional Brownian motion if for each i , $(B_t^i)_{t \geq 0}$ is a Brownian motion and $(B_t^i)_{t \geq 0}$ ($i = 1, \dots, d$) are independent of each other.

Remark 5.4.2. A d -dimensional Brownian motion is a martingale with

$$\langle B^i, B^j \rangle_t = \delta_{ij}t.$$

Theorem 5.4.3 (Lévy Theorem). *Let $\mathbf{M} = (M^1, \dots, M^d)$ be d -dimensional continuous local martingale with respect to $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ and $M_0 = 0$. If*

$$\langle M^i, M^j \rangle = \delta_{ij}t,$$

then $\mathbf{M}$ is a d -dimensional Brownian motion.

Theorem 5.4.4. *Let M be a continuous local martingale w.s.t. $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ with $M_0 = 0$ and*

$$\lim_{t \rightarrow \infty} \langle M \rangle_t = \infty.$$

For each $t \geq 0$, define the stopping time

$$\tau_t := \inf \{s: \langle M \rangle_s > t\}.$$

Then $(M_{\tau_t})_{t \geq 0}$ is a Brownian motion.

Remark 5.4.5. Let $\mathbf{B}$ be a d -dimensional Brownian motion.

- (1) If $d = 1$, we have seen $B_t = 0$ for infinitely many t .
- (2) If $d = 2$, $\mathbf{B}_t \neq 0$ for $t \neq 0$ but it hits every ball centered at 0.
- (3) If $d \geq 3$, $\|\mathbf{B}_t(\omega)\| \rightarrow \infty$ as $t \rightarrow \infty$.

Chapter 6

Continuous Time Martingale

6.1 Filtration

Definition 6.1.1 (Filtration). A filtration is a family of increasing σ -fields $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ and let $\mathcal{F}_\infty = \bigcup_t \mathcal{F}_t$.

Example 6.1.2. For a process $X = (X_t)_{t \geq 0}$, let $\mathcal{F}_t^X = \sigma(X_s : s \leq t)$. Then $(\mathcal{F}_t^X)_{t \geq 0}$ is a filtration, called the natural filtration of X .

Definition 6.1.3 (Right Continuity). Given a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$. For any $t \geq 0$, define

$$\mathcal{F}_{t+} := \bigcap_{s > t} \mathcal{F}_s$$

and $\mathcal{F}_{\infty+} = \mathcal{F}_\infty$. In general, $\mathcal{F}_t \subset \mathcal{F}_{t+}$. If for any $t \geq 0$,

$$\mathcal{F}_t = \mathcal{F}_{t+},$$

then the filtration is called right-continuous.

Definition 6.1.4 (Completeness). Given a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$. Let

$$N = \{A : \exists A' \supset A, A' \in \mathcal{F}_\infty, \mathbb{P}(A') = 0\}.$$

If $N \subset \mathcal{F}_0$, then the filtration is called complete.

Remark 6.1.5. If a filtration is not complete, then define

$$\mathcal{F}'_t = \mathcal{F}_t \vee \sigma(N),$$

the smallest σ -field containing $\mathcal{F}_t$ and $\sigma(N)$. Then the filtration $(\mathcal{F}'_t)_{t \geq 0}$ is complete. So in the following, we always consider complete filtration.

Remark 6.1.6. A filtration $\mathbb{F}$ is said to satisfy the usual condition if $\mathbb{F}$ is right-continuous and complete.

Given a stochastic process $X = (X_t)_{t \geq 0}$, note that

$$X : \Omega \times [0, \infty) \rightarrow \mathbb{R}$$

viewed by $X(\omega, t) = X_t(\omega)$. On $\Omega \times [0, \infty)$, we can consider the product σ -field $\mathcal{F} \times \mathcal{B}([0, \infty))$.

Definition 6.1.7 (Measurability). Given a stochastic process $X = (X_t)_{t \geq 0}$.

- (1) X is said to be measurable if $X(\omega, t)$ is $\mathcal{R} \times \mathcal{F}$ -measurable.
- (2) X is said to be $(\mathcal{F}_t)_{t \geq 0}$ -progressively measurable if for every t , $X: \Omega \times [0, t] \rightarrow \mathbb{R}$ is $\mathcal{F}_t \times \mathcal{B}([0, t])$ -measurable.
- (3) X is called $(\mathcal{F}_t)_{t \geq 0}$ -adapted if X_t is $\mathcal{F}_t$ -measurable for all t .

Remark 6.1.8. Note that if $f: E_1 \times E_2 \rightarrow \mathbb{R}$ is measurable, then for any $x \in E_1$, $f(x, \cdot): E_2 \rightarrow \mathbb{R}$ is also measurable. So if X is progressively measurable, then it is adapted. Moreover, it is also measurable. Conversely, if X is measurable and adapted, then it has a progressively measurable modification.

Proposition 6.1.9. *Suppose a stochastic process $X = (X_t)_{t \geq 0}$ is $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ -adapted and $t \mapsto X_t(\omega)$ is right-continuous (or left-continuous) a.e.. Then X is $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ -progressively measurable.*

Proof. Fix $t > 0$. Consider the process X on $\Omega \times [0, t]$. Define for $s < t$

$$X_s^n := X_{\frac{kt}{n}}, \quad \text{if } s \in \left[\frac{(k-1)t}{n}, \frac{kt}{n} \right)$$

and $X_t^n = X_t$. Then by the right-continuity, $X_s^n \rightarrow X_s$ a.e. (Similar for the left-continuity by taking the left-end point). Now for $A \in \mathcal{R}$,

$$\{(\omega, s): X_s^n(\omega) \in A\} = (\{X_t \in A\} \times \{t\}) \cup \bigcup_{k=1}^n \left(\left\{ X_{\frac{kt}{n}} \in A \right\} \times \left[\frac{(k-1)t}{n}, \frac{kt}{n} \right) \right) \in \mathcal{F}_t \times \mathcal{B}([0, t]),$$

because X is adapted. So $X^n = (X_t^n)_{t \geq 0}$ is progressively measurable. By taking limits, X is progressively measurable. $\square$

Remark 6.1.10. A σ -field on $\Omega \times [0, \infty)$ is generated by all $(\mathcal{F}_t)_{t \geq 0}$ -progressively measurable process, called $(\mathcal{F}_t)_{t \geq 0}$ -progressively measurable σ -field.

6.2 Stopping Time

Fix a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$.

Definition 6.2.1 (Stopping Time). A random variable $T: \Omega \rightarrow [0, \infty]$ is called a stopping time w.s.t. $(\mathcal{F}_t)_{t \geq 0}$ if $\{T \leq t\} \in \mathcal{F}_t$ for all t .

Remark 6.2.2. As mentioned before, $\{T < t\} \in \mathcal{F}_t$ and so $\{T \geq t\} \in \mathcal{F}_t$. Note that the converse is not true.

Definition 6.2.3. Given a stopping time T , let

$$\mathcal{F}_T := \{A \in \mathcal{F}_\infty: \forall t \geq 0, A \cap \{T \leq t\} \in \mathcal{F}_t\}.$$

Let $\mathcal{G}_t = \mathcal{F}_{t+}$ that is also a filtration and $\mathcal{G}_{t+} = \mathcal{G}_t$.

Proposition 6.2.4. (1) T is a stopping time w.s.t. $(\mathcal{G}_t)_{t \geq 0}$ if and only if for any $t > 0$,

$$\{T < t\} \in \mathcal{F}_t,$$

which is also equivalent to $T \wedge t$ is $\mathcal{F}_t$ -measurable.

(2) Let T be a $(G_t)_{t \geq 0}$ -stopping time. Then

$$\mathcal{G}_T = \{A \in \mathcal{F}_\infty : \forall t > 0, A \cap \{T < t\} \in \mathcal{F}_t\}$$

In some way, $\mathcal{F}_{T+} := \mathcal{G}_T$.

Proof. (1) Assume T is a stopping time w.s.t. $(G_t)_{t \geq 0}$.

$$\{T < t\} = \bigcup_{q \in \mathbb{Q}, q < t} \{T \leq q\}.$$

Because

$$\{T \leq q\} \in \mathcal{G}_q = \mathcal{F}_{q+} \subset \mathcal{F}_t,$$

$$\{T < t\} \in \mathcal{F}_t.$$

Conversely, for any $t \geq 0$ and any $s > t$,

$$\{T \leq t\} = \bigcap_{q \in \mathbb{Q}, t < q < s} \{T < q\} \in \mathcal{F}_s,$$

because

$$\{T < q\} \in \mathcal{F}_q \subset \mathcal{F}_s.$$

So

$$\{T \leq t\} \in \bigcap_{s > t} \mathcal{F}_s = \mathcal{G}_t.$$

(2) Let $A \in \mathcal{G}_T$. Then for all $t > 0$,

$$A \cap \{T \leq t\} \in \mathcal{G}_t.$$

Hence

$$A \cap \{T < t\} = \bigcup_{q \in \mathbb{Q}, q < t} A \cap \{T \leq q\} \in \mathcal{F}_t.$$

Conversely, for any $A \in \mathcal{F}_\infty$ with $A \cap \{T < t\} \in \mathcal{F}_t$ for all $t > 0$,

$$A \cap \{T \leq t\} = \bigcap_{q \in \mathbb{Q}, t < q < s} A \cap \{T < q\} \in \mathcal{F}_s.$$

So

$$A \cap \{T \leq t\} \in \bigcap_{s > t} \mathcal{F}_s = \mathcal{G}_t.$$

□

Proposition 6.2.5. (1) For any stopping time T ,

$$\mathcal{F}_T \subset \mathcal{G}_T = \mathcal{F}_{T+}$$

(2) If $T = t$, $\mathcal{F}_T = \mathcal{F}_t$.

(3) T is $\mathcal{F}_T$ -measurable.

Proof. (1) It is because

$$A \cap \{T < t\} = \bigcup_{q \in \mathbb{Q}, q < t} A \cap \{T \leq q\}.$$

(2) It is because

$$A \cap \{T \leq s\} = A \text{ or } \emptyset.$$

(3) For any $a > 0$, let $A = \{T > a\}$.

$$\begin{aligned} A \cap \{T \leq t\} &= \{T > a\} \cap \{T \leq t\} \\ &= \begin{cases} \emptyset, & a \geq t \\ \{a < T \leq t\}, & a < t. \end{cases} \end{aligned}$$

Because

$$\{a < T \leq t\} = \{T \leq a\}^c \cap \{T \leq t\} \in \mathcal{F}_a \cap \mathcal{F}_t \subset \mathcal{F}_t,$$

$$A \cap \{T \leq t\} \in \mathcal{F}_t.$$

□

Proposition 6.2.6. (1) Let T be a stopping time and $A \in \mathcal{F}_\infty$. Define

$$T^A(\omega) := \begin{cases} T(\omega), & \omega \in A, \\ \infty, & \omega \notin A. \end{cases}$$

Then T^A is a stopping time if and only if $A \in \mathcal{F}_T$.

(2) If T is a stopping time, then $T + s$ is a stopping time for any constant $s \geq 0$.

(3) Let S, T be stopping times with $S \leq T$. Then

$$\mathcal{F}_S \subset \mathcal{F}_T, \quad \mathcal{F}_{S+} \subset \mathcal{F}_{T+}.$$

(4) Let S, T be stopping times. Then $S \vee T$ and $S \wedge T$ are stopping times. Moreover, $\mathcal{F}_{S \wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$ and

$$\{S \leq T\} \in \mathcal{F}_{S \wedge T}.$$

So $\{T \leq S\} \in \mathcal{F}_{S \wedge T}$ and $\{S = T\} \in \mathcal{F}_{S \wedge T}$.

Proof. (1) Note that for any t ,

$$\{T^A \leq t\} = A \cap \{T \leq t\} \in \mathcal{F}_t.$$

(2) It is because

$$\{T + s \leq t\} = \{T \leq t - s\} \in \mathcal{F}_{t-s} \subset \mathcal{F}_t.$$

(3) Note that $\{T \leq t\} \subset \{S \leq t\}$. For any $A \in \mathcal{F}_S$,

$$A \cap \{T \leq t\} = (A \cap \{S \leq t\}) \cap \{T \leq t\} \in \mathcal{F}_t,$$

and similarly for $\mathcal{F}_{S+} \subset \mathcal{F}_{T+}$.

(4) Note that

$$\{S \vee T \leq t\} = \{S \leq t\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

Similarly,

$$\{S \wedge T \leq t\} = \{S \leq t\} \cup \{T \leq t\} \in \mathcal{F}_t.$$

By (3),

$$\mathcal{F}_{S \wedge T} \subset \mathcal{F}_S \cap \mathcal{F}_T.$$

Conversely, let $A \in \mathcal{F}_S \cap \mathcal{F}_T$. Then

$$A \cap \{S \wedge T \leq t\} = (A \cap \{S \leq t\}) \cup (A \cap \{T \leq t\}) \in \mathcal{F}_t.$$

So $\mathcal{F}_S \cap \mathcal{F}_T \subset \mathcal{F}_{S \wedge T}$.

For any $t > 0$,

$$\{S \leq T\} \cap \{S \wedge T \leq t\} = (\{S \leq T\} \cap \{S \leq t\}) \cup (\{S \leq T\} \cap \{T \leq t\}).$$

Because

$$\{S \leq T\} \cap \{T \leq t\} = \{S \leq t\} \cap \{T \leq t\} \cap \{S \wedge t \leq T \wedge t\}$$

and note that $S \wedge t, T \wedge t$ are $\mathcal{F}_t$ -measurable,

$$\{S \leq T\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

For the other term, similarly

$$\{S \leq T\} \cap \{S \leq t\} = \{S \wedge t \leq T \wedge t\} \cap \{S \leq t\} \in \mathcal{F}_t.$$

So

$$\{S \leq T\} \cap \{S \wedge T \leq t\} \in \mathcal{F}_t. \quad \square$$

Remark 6.2.7. For stopping times $S \leq T$, it can define the stochastic interval

$$\llbracket S, T \rrbracket := \{(t, \omega) \in [0, \infty] \times \Omega : S(\omega) < t \leq T(\omega)\},$$

so is similarly $\llbracket S, T \rrbracket, \llbracket S, T \rrbracket$.

Proposition 6.2.8. (1) If $\{S_n\}_{n \geq 0}$ is an increasing stopping time and $S_n \rightarrow S$, then S is a stopping time.

(2) If $\{S_n\}_{n \geq 0}$ is a decreasing stopping time and $S_n \rightarrow S$, then S is a $(\mathcal{F}_{t+})$ -stopping time and

$$\mathcal{F}_{S+} = \bigcap_n \mathcal{F}_{S_n+}.$$

Proof. (1) Because $S_n \uparrow S$,

$$\{S \leq t\} = \bigcap_n \{S_n \leq t\} \in \mathcal{F}_t.$$

So S is a stopping time.

(2) Because $S_n \downarrow S$,

$$\{S < t\} = \bigcup_n \{S_n < t\} \in \mathcal{F}_t, \text{ and } A \cap \{S < t\} = \bigcup_n (A \cap \{S_n < t\}).$$

So S is a $(\mathcal{F}_{t+})$ -stopping time and $\mathcal{F}_{S+} \supset \bigcap_n \mathcal{F}_{S_n+}$. For the other side,

$$A \cap \{S_n < t\} = A \cap \{S < t\} \cap \{S_n < t\} \in \mathcal{F}_t. \quad \square$$

Proposition 6.2.9. Let T be a stopping time. A random variable Y defined on $\{T < \infty\}$ is $\mathcal{F}_T$ -measurable if and only if for any $t \geq 0$, $Y|_{\{T \leq t\}}$ is $\mathcal{F}_t$ -measurable.

Proof. For any $A \in \mathcal{R}$,

$$\{Y \in A\} \cap \{T \leq t\} = \{Y|_{\{T \leq t\}} \in A\}. \quad \square$$

Remark 6.2.10. If $X = (X_t)_{t \geq 0}$ is progressively measurable and T is a stopping time, then $X_T \mathbb{I}_{\{T < \infty\}}$ is $\mathcal{F}_T$ -measurable.

Proposition 6.2.11. *Let T be a stopping time.*

(1) *Let S be a $\mathcal{F}_T$ -measurable random variable with values $[0, \infty]$ such that $S \geq T$. Then S is also a stopping time.*

(2) *Define*

$$T_n(\omega) = \sum_{k=0}^{\infty} \frac{k+1}{2^n} \mathbb{I}_{\{\frac{k}{2^n} < T(\omega) \leq \frac{k+1}{2^n}\}} + \infty \mathbb{I}_{\{T(\omega) = \infty\}}.$$

Then T_n is a sequence of stopping times that decreases to T .

Proof. (1) For any $t \geq 0$, because $T \leq S$,

$$\{S \leq t\} = \{S \leq t\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

(2) Note that $T_n \geq T$ is $\mathcal{F}_T$ -measurable so by (1) it is a stopping time. □

Example 6.2.12. Let $(X_t)_{t \geq 0}$ be an adapted stochastic process.

(1) Assume $t \mapsto X_t(\omega)$ is right-continuous. Let O be an open set.

$$T_O := \inf \{t \geq 0: X_t \in O\}$$

is a stopping time with respect to $(\mathcal{F}_{t+})_{t \geq 0}$.

(2) Assume $t \mapsto X_t(\omega)$ is continuous. Let F be a closed set.

$$T_F = \inf \{t \geq 0: X_t \in F\}$$

is a stopping time (w.s.t. $(\mathcal{F}_t)$).

Proof. (1) For any $t \geq 0$,

$$\{T_O < t\} = \bigcup_{q \in [0, t) \cap \mathbb{Q}} \{X_q \in O\}.$$

To prove that, first, for any ω such that $T_O(\omega) = \alpha = \inf \{t: X_t(\omega) \in O\} < t$, we can choose a rational sequence $t > t_n \downarrow \alpha$. By the right continuity of X_t , $X_{t_n}(\omega) \rightarrow X_\alpha(\omega) \in O$. Because O is open, there is a large n such that $X_{t_n}(\omega) \in O$. So $\{T_O < t\} \subset \bigcup_{q \in [0, t) \cap \mathbb{Q}} \{X_q \in O\}$. The converse is obvious.

So it directly has

$$\{T_O < t\} = \bigcup_{q \in [0, t) \cap \mathbb{Q}} \{X_q \in O\} \in \mathcal{F}_t.$$

(2) For any $t \geq 0$,

$$\{T_F \leq t\} = \left\{ \inf_{0 \leq s \leq t} d(X_s, F) = 0 \right\}.$$

First, “ $\subset$ ” is obvious. Conversely, there is a $\{t_n\} \subset [0, t]$ such that $t_n \rightarrow t_0 \leq t$ and

$$\lim_n d(X_{t_n}(\omega), F) = d(\lim_n X_{t_n}(\omega), F) = d(X_{t_0}(\omega), F) = 0$$

because of the continuity. Since F is closed, $X_{t_0}(\omega) \in F$ so $T_F(\omega) \leq t$.

Then

$$\{T_F \leq t\} = \left\{ \inf_{0 \leq s \leq t} d(X_s, F) = 0 \right\} = \left\{ \inf_{s \in [0, t] \cap \mathbb{Q}} d(X_s, F) = 0 \right\} \in \mathcal{F}_t. \quad \square$$

6.3 Martingale

Fix a filtration $(\mathcal{F}_t)_{t \geq 0}$.

Definition 6.3.1. A stochastic process $(X_t)_{t \geq 0}$ is called a submartingale if

- (1) $X_t \in L^1$ and $(X_t)_{t \geq 0}$ is adapted,
- (2) for any $s < t$,

$$\mathbb{E}[X_t \mid \mathcal{F}_s] \geq X_s.$$

If $\leq$ in (2), it is called a supermartingale. If $(X_t)_{t \geq 0}$ is a sub and super martingale, it is called a martingale.

Example 6.3.2. Let $(B_t)_{t \geq 0}$ be a Brownian motion and $(\mathcal{F}_t)_{t \geq 0}$ be the natural filtration.

- (1) $(B_t)_{t \geq 0}$ is a martingale.
- (2) Let

$$Y_t = B_t^2 - t.$$

Then $(Y_t)_{t \geq 0}$ is a martingale.

- (3) Let.

$$Z_t = \exp\left(\theta B_t - \frac{1}{2}\theta^2 t\right)$$

Then $(Z_t)_{t \geq 0}$ is also a martingale.

Proof. (1) For any $s < t$,

$$\begin{aligned} \mathbb{E}[B_t \mid \mathcal{F}_s] &= \mathbb{E}[B_t - B_s \mid \mathcal{F}_s] + \mathbb{E}[B_s \mid \mathcal{F}_s] \\ &= \mathbb{E}[B_t - B_s] + B_s \\ &= B_s. \end{aligned}$$

- (2) For $s < t$, by (1),

$$\begin{aligned} \mathbb{E}[Y_t \mid \mathcal{F}_s] &= \mathbb{E}[B_t^2 \mid \mathcal{F}_s] - t \\ &= \mathbb{E}[(B_t - B_s)^2 \mid \mathcal{F}_s] + 2\mathbb{E}[B_t B_s \mid \mathcal{F}_s] - \mathbb{E}[B_s^2 \mid \mathcal{F}_s] - t \\ &= \mathbb{E}[(B_t - B_s)^2] + 2B_s \mathbb{E}[B_t \mid \mathcal{F}_s] - B_s^2 - t \\ &= t - s + 2B_s^2 - B_s^2 - t \\ &= B_s^2 - s = Y_s. \end{aligned}$$

- (3) For any $s < t$,

$$\begin{aligned} \mathbb{E}[Z_t \mid \mathcal{F}_s] &= \exp\left(-\frac{1}{2}\theta^2 t\right) \mathbb{E}[\exp(\theta B_t - \theta B_s) \exp(\theta B_s) \mid \mathcal{F}_s] \\ &= \exp\left(\theta B_s - \frac{1}{2}\theta^2 t\right) \mathbb{E}[\exp(\theta B_t - \theta B_s) \mid \mathcal{F}_s] \\ &= \exp\left(\theta B_s - \frac{1}{2}\theta^2 t\right) \mathbb{E}[\exp(\theta(B_t - B_s))] . \end{aligned}$$

Because $B_t - B_s \sim \mathcal{N}(0, t - s)$,

$$\mathbb{E}[\exp(\theta(B_t - B_s))] = \frac{1}{\sqrt{2\pi(t-s)}} \int \exp(\theta x) \exp(-\frac{1}{2(t-s)}x^2) dx = \exp\left(\frac{1}{2}\theta^2(t-s)\right).$$

So

$$\mathbb{E}[Z_t | \mathcal{F}_s] = \exp\left(\theta B_s - \frac{1}{2}\theta^2 t\right) \exp\left(\frac{1}{2}\theta^2(t-s)\right) = Z_s. \quad \square$$

Example 6.3.3. Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ and $(\mathcal{F}_t)_{t \geq 0}$ be the nature filtration.

(1) $(N_t - \lambda t)$ is a martingale.

(2) Let

$$Z_t = (N_t - \lambda t)^2 - \lambda t.$$

Then $(Z_t)_{t \geq 0}$ is also a martingale.

(3) Given $\alpha > 0$, set β such that

$$L_t = \exp(\alpha N_t - \beta t)$$

is a martingale.

Proof. (1) For $s < t$,

$$\begin{aligned} \mathbb{E}[N_t - \lambda t | \mathcal{F}_s] &= \mathbb{E}[N_t - N_s | \mathcal{F}_s] + \mathbb{E}[N_s | \mathcal{F}_s] - \lambda t \\ &= \mathbb{E}[N_t - N_s] + N_s - \lambda t \\ &= \lambda(t-s) + N_s - \lambda t = N_s - \lambda s. \end{aligned}$$

(2) For $s < t$,

$$\begin{aligned} \mathbb{E}[Z_t | \mathcal{F}_s] &= \mathbb{E}[(N_t - \lambda t)^2 | \mathcal{F}_s] - \lambda t \\ &= \mathbb{E}[N_t^2 | \mathcal{F}_s] - 2\lambda t \mathbb{E}[N_t | \mathcal{F}_s] + (\lambda t)^2 - \lambda t \\ &= \mathbb{E}[(N_t - N_s)^2 | \mathcal{F}_s] + 2N_s \mathbb{E}[N_t | \mathcal{F}_s] - \mathbb{E}[N_s^2 | \mathcal{F}_s] \\ &\quad - 2\lambda t \mathbb{E}[N_t - \lambda t | \mathcal{F}_s] - (\lambda t)^2 - \lambda t \\ &= \mathbb{E}[(N_t - N_s)^2] + 2N_s \mathbb{E}[N_t - N_s | \mathcal{F}_s] + N_s^2 \\ &\quad - 2\lambda t(N_s - \lambda s) - (\lambda t)^2 - \lambda t \\ &= \lambda(t-s) + \lambda^2(t-s)^2 + 2\lambda(t-s)N_s + N_s^2 \\ &\quad - 2\lambda t(N_s - \lambda s) - (\lambda t)^2 - \lambda t \\ &= (N_s - \lambda s)^2 - \lambda s. \end{aligned}$$

(3) For $s < t$,

$$\begin{aligned} \mathbb{E}[L_t | \mathcal{F}_s] &= \exp(-\beta t) \mathbb{E}[\exp(\alpha N_t) | \mathcal{F}_s] \\ &= \exp(-\beta t) \mathbb{E}[\exp(\alpha(N_t - N_s)) \exp(\alpha N_s) | \mathcal{F}_s] \\ &= \exp(\alpha N_s - \beta t) \mathbb{E}[\exp(\alpha(N_t - N_s))] \\ &= \exp(\alpha N_s - \beta t) \exp(\lambda(t-s)(e^\alpha - 1)). \end{aligned}$$

because $N_t - N_s \sim \text{Pois}(\lambda(t-s))$. So when $\beta = \lambda(e^\alpha - 1)$,

$$\mathbb{E}[L_t | \mathcal{F}_s] = \exp(\alpha N_s - \beta s) = L_s. \quad \square$$

Proposition 6.3.4. *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be convex function.*

- (1) *If $X = (X_t)_{t \geq 0}$ is a martingale and $f(X_t) \in L^1$, then $\{f(X)_t\}_{t \geq 0}$ is a submartingale.*
- (2) *If $X = (X_t)_{t \geq 0}$ is a submartingale and f is increasing and $f(X_t) \in L^1$, then $\{f(X)_t\}_{t \geq 0}$ is a submartingale.*

Remark 6.3.5. In general, we take $f(x) = |x|^p$ with $p \geq 1$ and $f(x) = x^+$.

Theorem 6.3.6. *Let $X = (X_t)_{t \geq 0}$ be a sub(super)martingale. Then*

$$\sup_{s \in [0, t]} \mathbb{E}[|X_s|] < \infty.$$

Remark 6.3.7. Note that if X is a martingale, by above $|X_t|$ is a submartingale so

$$\mathbb{E}[|X_s|] \leq \mathbb{E}[\mathbb{E}[|X_t| \mid \mathcal{F}_s]] \leq \mathbb{E}[|X_t|] < \infty$$

Proof. Assume X is a submartingale. So $(X_t^+)_{t \geq 0}$ is also a submartingale. So

$$\mathbb{E}[X_s^+] \leq \mathbb{E}[X_t^+] < \infty.$$

On the other hand,

$$\mathbb{E}[X_s^-] = \mathbb{E}[X_s^+] - \mathbb{E}[X_s] \leq \mathbb{E}[X_t^+] - \mathbb{E}[X_0] < \infty.$$

So $\sup_{s \in [0, t]} \mathbb{E}[|X_s|] = \sup_{s \in [0, t]} \mathbb{E}[X_s^+] + \mathbb{E}[X_s^-] \leq 2\mathbb{E}[X_t^+] - \mathbb{E}[X_0] < \infty.$ □

Theorem 6.3.8. *Let $X = (X_t)_{t \geq 0}$ be a positive submartingale (or martingale) with right-continuous paths.*

- (1) *Maximum inequality: For any $\lambda > 0$,*

$$\mathbb{P}\left(\sup_{s \in [0, t]} |X_s| > \lambda\right) \leq \frac{1}{\lambda} \mathbb{E}[|X_t|].$$

- (2) *Doob's inequality: Further assume X is a martingale. For any $p > 1$ and $t > 0$,*

$$\mathbb{E}\left[\sup_{s \in [0, t]} |X_s|^p\right] \leq \left(\frac{p}{1-p}\right)^p \mathbb{E}[|X_t|^p].$$

Note that this inequality is only useful for $\mathbb{E}[|X_t|^p] < \infty$.

Proof. Fix $t > 0$. Consider a countable dense subset $D \subset [0, t]$ containing 0 and t . Let $D_m = \{0 = t_0^m < t_1^m < \dots < t_m^m = t\}$ such that $D_m \uparrow D$. By the continuity of path

$$\mathbb{P}\left(\sup_{s \in [0, t]} |X_s| > \lambda\right) = \mathbb{P}\left(\sup_{s \in D} |X_s| > \lambda\right), \quad \mathbb{E}\left[\sup_{s \in [0, t]} |X_s|^p\right] = \mathbb{E}\left[\sup_{s \in D} |X_s|^p\right].$$

And by the convergence,

$$\mathbb{P}\left(\sup_{s \in D} |X_s| > \lambda\right) = \lim_{m \rightarrow \infty} \mathbb{P}\left(\sup_{s \in D_m} |X_s| > \lambda\right), \quad \mathbb{E}\left[\sup_{s \in D} |X_s|^p\right] = \lim_{m \rightarrow \infty} \mathbb{E}\left[\sup_{s \in D_m} |X_s|^p\right].$$

When $X = (X_t)_{t \geq 0}$ be a positive submartingale (or martingale), $(|X_{t_n^m}|)$ is also a submartingale. Then by the discrete case, we have the inequalities. □

Remark 6.3.9. If X is a nonnegative supermartingale with right-continuous paths, by the proof of discrete case, we clearly have

$$\mathbb{P} \left(\sup_{s \in [0, t]} X_s > \lambda \right) \leq \frac{1}{\lambda} \mathbb{E}[X_0].$$

Moreover, if X is a supermartingale with right-continuous paths,

$$\mathbb{P} \left(\sup_{s \in [0, t]} |X_s| > \lambda \right) \leq \frac{1}{\lambda} (\mathbb{E}[|X_0|] + 2\mathbb{E}[|X_t|]).$$

Proof. By above, we only need to prove the discrete case for a supermartingale $(X_n)_{n \geq 0}$. First, let $N = \min \{n : X_n > \lambda\} \wedge k$. Then by the optional stopping time theorem,

$$\begin{aligned} \mathbb{E}[X_0] &\geq \mathbb{E}[X_N] = \mathbb{E} \left[X_N \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n > \lambda\}} \right] + \mathbb{E} \left[X_N \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n \leq \lambda\}} \right] \\ &\geq \lambda \mathbb{P} \left(\max_{0 \leq n \leq k} X_n > \lambda \right) + \mathbb{E} \left[X_k \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n \leq \lambda\}} \right] \\ &\geq \lambda \mathbb{P} \left(\max_{0 \leq n \leq k} X_n > \lambda \right) - \mathbb{E}[|X_k|]. \end{aligned}$$

So we have

$$\lambda \mathbb{P} \left(\max_{0 \leq n \leq k} X_n > \lambda \right) \leq \mathbb{E}[X_0] + \mathbb{E}[|X_k|].$$

On the other hand, let $T = \min \{n : X_n \leq -\lambda\} \wedge k$. Then

$$\begin{aligned} \mathbb{E}[X_k] &\leq \mathbb{E}[X_T] = \mathbb{E} \left[X_T \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n < -\lambda\}} \right] + \mathbb{E} \left[X_T \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n \geq -\lambda\}} \right] \\ &\leq -\lambda \mathbb{P} \left(\max_{0 \leq n \leq k} X_n < -\lambda \right) + \mathbb{E} \left[X_k \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n \geq -\lambda\}} \right] \end{aligned}$$

So

$$\begin{aligned} \lambda \mathbb{P} \left(\max_{0 \leq n \leq k} X_n < -\lambda \right) &\leq - \left(\mathbb{E}[X_k] - \mathbb{E} \left[X_k \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n \geq -\lambda\}} \right] \right) \\ &= \mathbb{E} \left[-X_k \mathbb{I}_{\{\max_{0 \leq n \leq k} X_n < -\lambda\}} \right] \\ &\leq \mathbb{E}[|X_k|]. \end{aligned}$$

Therefore, we have

$$\lambda \mathbb{P} \left(\max_{0 \leq n \leq k} |X_n| > \lambda \right) \leq 2\mathbb{E}[|X_k|] + \mathbb{E}[|X_0|]. \quad \square$$

6.4 Path Regularity

Definition 6.4.1. Let $f : I \rightarrow \mathbb{R}$ with $I \subset \mathbb{R}_+$. For $a < b$, define

$$M_{a,b}^f(I) \stackrel{\text{def}}{=} \sup \{k \in \mathbb{N}_+ : \exists \{s_1 < t_1 < s_2 < t_2 < \dots < s_k < t_k\} \subset I \text{ s.t. } f(s_i) \leq a, f(t_i) \geq b, \forall i \in [k]\}$$

be the number of up-crossing of f on (a, b) .

Lemma 6.4.2. Let $D \subset \mathbb{R}_+$ be a countable dense subset of $\mathbb{R}_+$ and $f : D \rightarrow \mathbb{R}$. Assume that for any $T \in D$,

(i) f is bounded on $D \cap [0, T]$,

(ii) $M_{a,b}^f(D) < \infty$ for any $a, b \in \mathbb{Q}$ with $a < b$.

Then

(1) for any $t \geq 0$,

$$\lim_{s \downarrow t, s \in D} f(s) =: f(t+)$$

exists.

(2) for any $t > 0$,

$$\lim_{s \uparrow t, s \in D} f(s) =: f(t-)$$

exists.

Furthermore, define $g(t) = f(t+)$ then g is càdlàg (or RLCC), i.e. right-continuous with left-limit.

Proof. Assume for $t > 0$,

$$\lim_{s \downarrow t, s \in D} f(s)$$

does not exist. Then by the boundedness of f ,

$$\liminf_{s \downarrow t, s \in D} f(s) < \limsup_{s \downarrow t, s \in D} f(s),$$

which implies that there exist $a, b \in \mathbb{Q}$ such that

$$\liminf_{s \downarrow t, s \in D} f(s) < a < b < \limsup_{s \downarrow t, s \in D} f(s).$$

It follows that

$$M_{a,b}^f(D) = \infty. \quad \square$$

Theorem 6.4.3. Let $X = (X_t)_{t \geq 0}$ be a supermartingale and D be a countable dense subset of $\mathbb{R}_+$. Then

(1) for a.e. $\omega \in \Omega$,

$$X_{t+}(\omega) := \lim_{s \downarrow t, s \in D} X_s(\omega), \quad X_{t-}(\omega) := \lim_{s \uparrow t, s \in D} X_s(\omega)$$

exist.

(2) for every $t \in \mathbb{R}_+$, $X_{t+} \in L^1$ and

$$X_t \geq \mathbb{E}[X_{t+} \mid \mathcal{F}_t],$$

where “=” if and only if $t \mapsto \mathbb{E}[X_t]$ is right-continuous.

(3) $(X_{t+})_{t \geq 0}$ is a supermartingale w.s.t. $(\mathcal{F}_{t+})_{t \geq 0}$. Moreover, it is a martingale if X is a martingale.

Proof. (1) First, give any $T > 0$ and $\lambda > 0$, we have

$$\mathbb{P} \left(\sup_{t \in [0, T]} |X_t| > \lambda \right) \leq \frac{1}{\lambda} (\mathbb{E}[|X_0|] + 2\mathbb{E}[|X_T|]).$$

As $\lambda \rightarrow \infty$,

$$\mathbb{P} \left(\sup_{t \in [0, T]} |X_t| < \infty \right) = 1.$$

Therefore, $\sup_{t \in [0, T]} |X_t| < \infty$ a.e.. Second, choose a sequence (D_m) of finite subsets of D such that $D_m \uparrow D$ and $T \in D_m$. By the upcrossing inequality of the discrete case of $(X_{s_k}, k \in D_m \cap [0, T])$,

$$\mathbb{E} [M_{a,b}^X(D_m \cap [0, T])] \leq \frac{1}{b-a} \mathbb{E} [(X_T - a)^-].$$

As $m \rightarrow \infty$, by DCT,

$$\mathbb{E} [M_{a,b}^X(D \cap [0, T])] \leq \frac{1}{b-a} \mathbb{E} [(X_T - a)^-] < \infty,$$

which implies that $M_{a,b}^X(D \cap [0, T]) < \infty$ a.e.. By choosing the union of such zero measure set, the above lemma implies X_{t+}, X_{t-} exist a.e..

- (2) First, choose a sequence $(t_n)_{n \in \mathbb{N}} \subset D$ such that $t_n \downarrow t$ and $t_n \leq t+1$, so $X_{t+} = \lim_{n \rightarrow \infty} X_{t_n}$. Note that (X_{t_n}) is a backward supermartingale (by let $Y_{-n} = X_{t_n}$) and

$$\sup_n \mathbb{E}[|X_{t_n}|] \leq \sup_{s \in [0, t+1]} \mathbb{E}[|X_t|] < \infty.$$

By Martingale Convergence Theorem for discrete case (backward case),

$$\lim_{n \rightarrow \infty} X_{t_n} = X_{t+}$$

in L^1 and so $X_{t+} \in L^1$. Note that

$$X_t \geq \mathbb{E}[X_{t_n} | \mathcal{F}_t].$$

Because $X_{t_n} \rightarrow X_{t+}$ in L^1 ,

$$X_t \geq \mathbb{E}[X_{t+} | \mathcal{F}_t].$$

Note that if $X_1 \geq X_2$ and $\mathbb{E}[X_1] = \mathbb{E}[X_2]$, then $X_1 = X_2$. Assume $t \mapsto \mathbb{E}[X_t]$ is right-continuous. Then

$$\mathbb{E}[X_t] = \lim_{t_n \rightarrow t} \mathbb{E}[X_{t_n}] = \mathbb{E}[X_{t+}] = \mathbb{E}[\mathbb{E}[X_{t+} | \mathcal{F}_t]].$$

It follows that $X_t = \mathbb{E}[X_{t+} | \mathcal{F}_t]$. Conversely, it is obvious.

- (3) Let $s < t$. Choose $(s_n)_n \subset D$ and $(t_n)_n \subset D$ such that $s_n \downarrow s$ and $t_n \downarrow t$ and $s_n \leq t_n$. Then

$$X_{s_n} \geq \mathbb{E}[X_{t_n} | \mathcal{F}_{s_n}], \quad \forall n.$$

Now for any $A \in \mathcal{F}_{s+} = \bigcap_n \mathcal{F}_{s_n}$, we have

$$\mathbb{E}[X_{s_n} \mathbb{I}_A] \geq \mathbb{E}[X_{t_n} \mathbb{I}_A].$$

As $n \rightarrow \infty$, by (2),

$$\mathbb{E}[X_{s+} \mathbb{I}_A] \geq \mathbb{E}[X_{t+} \mathbb{I}_A] = \mathbb{E}[\mathbb{I}_A \mathbb{E}[X_{t+} | \mathcal{F}_{s+}]].$$

Because X_{s+} and $\mathbb{E}[X_{t+} | \mathcal{F}_{s+}]$ are $\mathcal{F}_{s+}$ -measurable,

$$X_{s+} \geq \mathbb{E}[X_{t+} | \mathcal{F}_{s+}].$$

□

Theorem 6.4.4 (Regularizing Path). *Assume $(\mathcal{F}_t)_{t \geq 0}$ is right-continuous and complete. Let $X = (X_t)_{t \geq 0}$ be a supermartingale such that $t \mapsto \mathbb{E}[X_t]$ is right-continuous. Then there is a $(Y_t)_{t \geq 0}$ such that it has càdlàg path and it is a supermartingale and $Y_t = X_t$ a.e. (called a modification of X).*

Proof. Let $Y_t = X_{t+}$. Then it is a supermartingale w.s.t. $(\mathcal{F}_{t+})_{t \geq 0} = (\mathcal{F}_t)_{t \geq 0}$, which has càdlàg path. Moreover, by the right-continuity of $t \mapsto \mathbb{E}[X_t]$,

$$X_t = \mathbb{E}[X_{t+} \mid \mathcal{F}_t] = \mathbb{E}[Y_t \mid \mathcal{F}_t] = Y_t. \quad \square$$

6.5 Convergence Theorem

Theorem 6.5.1 (Martingale Convergence Theorem). *Let $X = (X_t)_{t \geq 0}$ be a supermartingale with right-continuous paths such that*

$$\sup_t \mathbb{E}[|X_t|] < \infty$$

Then there exists a $X_\infty \in L^1$ such that

$$\lim_{t \rightarrow \infty} X_t = X_\infty$$

a.e..

Proof. Let D be a countable and dense subset of $\mathbb{R}_+$. For any $T \in D$ and $a < b$,

$$\mathbb{E}[M_{a,b}^X(D \cap [0, T])] \leq \frac{1}{b-a} \mathbb{E}[(X_T - a)^-].$$

So

$$\mathbb{E}[M_{a,b}^X(D \cap [0, T])] \leq \frac{1}{b-a} \sup_T \mathbb{E}[(X_T - a)^-] = M < \infty.$$

As $T \rightarrow \infty$, by MCT,

$$\mathbb{E}[M_{a,b}^X(D)] \leq M < \infty \quad \Rightarrow \quad M_{a,b}^X(D) < \infty, \quad \forall a, b$$

Therefore, by above lemma,

$$X_\infty = \lim_{D \ni t \rightarrow \infty} X_t.$$

By Fatou's lemma,

$$\mathbb{E}[|X_\infty|] \leq \liminf_{D \ni t \rightarrow \infty} \mathbb{E}[|X_t|] \leq \sup_t \mathbb{E}[|X_t|] < \infty.$$

Therefore, for any $\varphi > 0$, there exists N such that for all $D \ni t \geq N$,

$$|X_t - X_\infty| < \varepsilon.$$

Then for any $s \geq N$, let $s_n \downarrow s$ in D , so

$$|X_{s_n} - X_\infty| < \varepsilon.$$

As $n \rightarrow \infty$, because (X_t) is right-continuous,

$$|X_s - X_\infty| < \varepsilon,$$

which implies that

$$X_\infty = \lim_{t \rightarrow \infty} X_t. \quad \square$$

Definition 6.5.2 (Closedness). A martingale $X = (X_t)_{t \geq 0}$ is called closed if there exists a $Z \in L^1$ such that

$$X_t = \mathbb{E}[Z \mid \mathcal{F}_t], \quad \forall t \geq 0.$$

Theorem 6.5.3. Let $X = (X_t)_{t \geq 0}$ be a martingale with RLCC path. TFAE.

(1) X is closed.

(2) X is UI.

(3) X_t converges a.e. and in L^1 .

Moreover, in such cases, $X_t = \mathbb{E}[X_\infty \mid \mathcal{F}_t]$ for $X_\infty = \lim_t X_t$.

Proof. (1) $\Rightarrow$ (2): It has been proved in above chapter.

(2) $\Rightarrow$ (3): Because X is UI,

$$\sup_t \mathbb{E}[|X_t|] < \infty.$$

Then by above theorem,

$$\lim_{t \rightarrow \infty} X_t = X_\infty$$

a.e.. Then because of UI, it is in L^1 .

(3) $\Rightarrow$ (1): For $t < T$,

$$X_t = \mathbb{E}[X_T \mid \mathcal{F}_t].$$

As $T \rightarrow \infty$, because $X_\infty = \lim_t X_t$ in L^1 ,

$$X_t = \mathbb{E}[X_\infty \mid \mathcal{F}_t].$$

□

Remark 6.5.4. In such case, for any stopping time T , we can define

$$X_T(\omega) := \mathbb{I}_{T(\omega) < \infty} X_{T(\omega)}(\omega) + \mathbb{I}_{T(\omega) = \infty} X_\infty(\omega).$$

6.6 Optional Stopping Time

Theorem 6.6.1. Let $(Y_n)_{n \geq 0}$ be a discrete UI martingale. Then for any stopping times $M \leq N$,

$$Y_M = \mathbb{E}[Y_N \mid \mathcal{F}_M].$$

In particular, when M, N are bounded, no UI is required.

Proof. First, $Y_M \in L^1$ is by $(Y_{n \wedge M})$ is a UI martingale and $Y_{n \wedge M} \rightarrow Y_M$ (If M, N bounded, $Y_N = \sum_k \mathbb{I}_{\{N=k\}} Y_k \in L^1$). We already know Y_M is $\mathcal{F}_M$ measurable. For any $A \in \mathcal{F}_M \subset \mathcal{F}_N$, consider

$$M^A = \begin{cases} M, & \omega \in A \\ \infty, & \omega \in A^c \end{cases}, \quad N^A = \begin{cases} N, & \omega \in A \\ \infty, & \omega \in A^c \end{cases},$$

and they are stopping times because $A \in \mathcal{F}_M, \mathcal{F}_N$. $M^A \leq N^A$. Moreover, by optional stopping time theorem

$$\mathbb{E}[Y_0] = \mathbb{E}[Y_{M^A}] = \mathbb{E}[Y_{N^A}].$$

Note that

$$\mathbb{E}[Y_{M^A}] = \mathbb{E}[Y_M \mathbb{I}_A] + \mathbb{E}[Y_\infty \mathbb{I}_{A^c}], \quad \mathbb{E}[Y_{N^A}] = \mathbb{E}[Y_N \mathbb{I}_A] + \mathbb{E}[Y_\infty \mathbb{I}_{A^c}].$$

So

$$\mathbb{E}[Y_M \mathbb{I}_A] = \mathbb{E}[Y_N \mathbb{I}_A].$$

It follows that

$$Y_M = \mathbb{E}[Y_N \mid \mathcal{F}_M].$$

□

Remark 6.6.2. For super(sub)martingale, it has similar result. Let $(Z_n)_{n \geq 0}$ be a discrete UI supermartingale. Then for any stopping times $S \leq T$,

$$\mathbb{E}[Z_T \mid \mathcal{F}_S] \leq Z_S$$

In particular, when $S, T \leq m$ are bounded, no UI is required. First, $Z_S, Z_T \in L^1$ as same as above. Let $A \in \mathcal{F}_S$. Note that $\{Z_{T \wedge n}\}_{n \geq 0}$ is still a supermartingale. First, $A \cap \{S = k\} \in \mathcal{F}_k$. So

$$\begin{aligned} \mathbb{E}[Z_T \mathbb{I}_A] &= \mathbb{E}[Z_{T \wedge m} \mathbb{I}_A] \\ &= \sum_{k=0}^m \mathbb{E}[\mathbb{I}_{A \cap \{S=k\}} Z_{T \wedge m}] \\ &= \sum_{k=0}^m \mathbb{E}[\mathbb{I}_{A \cap \{S=k\}} \mathbb{E}[Z_{T \wedge m} \mid \mathcal{F}_k]] \\ &\leq \sum_{k=0}^m \mathbb{E}[\mathbb{I}_{A \cap \{S=k\}} Z_{T \wedge k}] \\ &= \sum_{k=0}^m \mathbb{E}[\mathbb{I}_{A \cap \{S=k\}} Z_{T \wedge S}] \\ &= \sum_{k=0}^m \mathbb{E}[\mathbb{I}_{A \cap \{S=k\}} Z_S] \\ &= \mathbb{E}[\mathbb{I}_A Z_S]. \end{aligned}$$

Theorem 6.6.3 (Doob's Optional Stopping Time). *Let $X = (X_t)_{t \geq 0}$ be a UI martingale with right-continuous paths. Let S and T be two stopping times with $S \leq T$. Then $X_S, X_T \in L^1$ and*

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof. First, it is obvious X_S is L^1 and $\mathcal{F}_S$ measurable. Set for any integer $n > 0$,

$$T_n = \sum_{k=0}^{\infty} \frac{k+1}{2^n} \mathbb{I}_{\frac{k}{2^n} < T \leq \frac{k+1}{2^n}} + \infty \mathbb{I}_{T=\infty},$$

and

$$S_n = \sum_{k=0}^{\infty} \frac{k+1}{2^n} \mathbb{I}_{\frac{k}{2^n} < S \leq \frac{k+1}{2^n}} + \infty \mathbb{I}_{S=\infty}.$$

Note that T_n and S_n are stopping times with $T_n \downarrow T$ and $S_n \downarrow S$ and $S_n \leq T_n$. Since $(X_{\frac{k}{2^n}})_{k \geq 0}$ is a UI martingale, by above theorem,

$$X_{S_n} = \mathbb{E}[X_{T_n} \mid \mathcal{F}_{S_n}].$$

Let $A \in \mathcal{F}_S$. Because $\mathcal{F}_S \subset \mathcal{F}_{S_n}$,

$$\mathbb{E}[\mathbb{I}_A X_{S_n}] = \mathbb{E}[\mathbb{I}_A X_{T_n}].$$

Let $n \rightarrow \infty$. By the right-continuity of path and UI,

$$\mathbb{E}[\mathbb{I}_A X_S] = \mathbb{E}[\mathbb{I}_A X_T].$$

□

Corollary 6.6.4 (Bounded Optional Stopping Time). *Let $X = (X_t)_{t \geq 0}$ be a martingale with right-continuous paths. Let $S \leq T$ be two bounded stopping times. Then $X_S, X_T \in L^1$ and*

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof. Assume $S \leq T \leq a$ for some constant a . Consider the martingale $Y_t = X_{t \wedge a}$. Then because $Y_t = \mathbb{E}[X_a \mid \mathcal{F}_t]$, (Y_t) is UI. So

$$Y_S = X_{S \wedge a} = X_S, \quad Y_T = X_{T \wedge a} = X_T \in L^1,$$

and $X_S = \mathbb{E}[X_T \mid \mathcal{F}_S]$. □

Corollary 6.6.5. *Let $X = (X_t)_{t \geq 0}$ be a martingale with right-continuous paths. Let T be a stopping time. Then*

(1) *the process $(X_{t \wedge T})_{t \geq 0}$ is also a martingale.*

(2) *if X is UI, then $(X_{t \wedge T})_{t \geq 0}$ is UI and*

$$X_{t \wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_t].$$

Proof. For (2), note $t \wedge T$ is also a stopping time with $t \wedge T \leq T$ and $X_{t \wedge T}$ is $\mathcal{F}_{t \wedge T}$ -measurable and sp $\mathcal{F}_t$ -measurable. For any $A \in \mathcal{F}_t$, let

$$A = (A \cap \{T \leq t\}) \cup (A \cap \{T > t\}).$$

Then

$$\begin{aligned} \mathbb{E}[X_{t \wedge T} \mathbb{I}_A] &= \mathbb{E}[X_{t \wedge T} \mathbb{I}_{A \cap \{T \leq t\}}] + \mathbb{E}[X_{t \wedge T} \mathbb{I}_{A \cap \{T > t\}}], \\ &= \mathbb{E}[X_T \mathbb{I}_{A \cap \{T \leq t\}}] + \mathbb{E}[X_{t \wedge T} \mathbb{I}_{A \cap \{T > t\}}], \\ \mathbb{E}[X_T \mathbb{I}_A] &= \mathbb{E}[X_T \mathbb{I}_{A \cap \{T \leq t\}}] + \mathbb{E}[X_T \mathbb{I}_{A \cap \{T > t\}}], \end{aligned}$$

so to prove $\mathbb{E}[X_{t \wedge T} \mathbb{I}_A] = \mathbb{E}[X_T \mathbb{I}_A]$, it suffices to show

$$\mathbb{E}[X_{t \wedge T} \mathbb{I}_{A \cap \{T > t\}}] = \mathbb{E}[X_T \mathbb{I}_{A \cap \{T > t\}}].$$

Because X is UI, by above

$$X_{t \wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_{t \wedge T}].$$

So we only need to show $A \cap \{T > t\} \in \mathcal{F}_{t \wedge T}$.

$$(A \cap \{T > t\}) \cap \{t \wedge T \leq s\} = A \cap \{T > t\} \cap \{t \leq s\} \in \mathcal{F}_t \subset \mathcal{F}_s$$

It follows

$$X_{t \wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_t].$$

And $X_T \in L^1$ by above theorem, so $(X_{t \wedge T})_{t \geq 0}$ is UI.

For (1), let $a > 0$ and consider $(X_{t \wedge a})_{t \geq 0}$ that is obvious a UI martingale. So by (2),

$$X_{t \wedge a \wedge T} = X_{t \wedge T}$$

is a martingale for any $t \geq a$. □

Remark 6.6.6. In fact, if $X = (X_t)_{t \geq 0}$ be a submartingale with right-continuous paths, then $(X_{t \wedge T})_{t \geq 0}$ is also a submartingale for any stopping time T by discretization. Moreover, by the same reasoning as the discrete case, if $X = (X_t)_{t \geq 0}$ is a UI submartingale, so is $(X_{t \wedge T})_{t \geq 0}$.

Example 6.6.7. Let $(B_t)_{t \geq 0}$ be a standard Brownian motion. Let $a \in \mathbb{R}$ and define

$$T_a = \inf \{t \geq 0 : B_t = a\}.$$

Assume $a < 0 < b$. Define $T = T_a \wedge T_b$.

- (1) Determine the following probability: $\mathbb{P}(T_a < T_b)$ and $\mathbb{P}(T_b \leq T_a)$.
- (2) Find $\mathbb{E}[T]$.
- (3) For $\lambda > 0$, find $\mathbb{E}[e^{-\lambda T_a}]$.

Solution:

- (1) Because $N_t = B_{t \wedge T}$ is a martingale and bounded by $|a| + b$, it is UI. So by optional stopping time theorem,

$$\mathbb{E}[N_T] = \mathbb{E}[B_T] = \mathbb{E}[N_0] = 0$$

On the other hand,

$$\mathbb{E}[B_T] = a\mathbb{P}(T_a < T_b) + b\mathbb{P}(T_b \leq T_a) = 0$$

So

$$\mathbb{P}(T_a < T_b) = \frac{b}{b-a}, \quad \mathbb{P}(T_b \leq T_a) = \frac{-a}{b-a}.$$

- (2) We already know $M_t = B_t^2 - t$ is a martingale. So $M_{t \wedge T} = B_{t \wedge T}^2 - t \wedge T$ is also a martingale.

$$\mathbb{E}[M_{t \wedge T}] = 0 \Rightarrow \mathbb{E}[B_{t \wedge T}^2] = \mathbb{E}[t \wedge T].$$

As $t \rightarrow \infty$, by the DCT on the LHS and the MCT on the RHS,

$$\mathbb{E}[T] = \mathbb{E}[B_T^2] = -ab.$$

- (3) Fix a $b \in \mathbb{R}$. Consider the martingale

$$N_t^b = \exp\left(bB_t - \frac{1}{2}b^2t\right).$$

Assume $b > 0$. Note that $(N_{t \wedge T_a}^b)_{t \geq 0}$ is also a martingale. Moreover, because

$$|N_{t \wedge T_a}^b| = \exp\left(bB_{t \wedge T_a} - \frac{1}{2}b^2t \wedge T_a\right) \leq \exp(bB_{t \wedge T_a}) \leq \exp(b|a|),$$

$Y_t = N_{t \wedge T_a}^b$ is UI. So by the optional stopping theorem,

$$\mathbb{E}[Y_\infty] = \mathbb{E}[N_{T_a}^b] = \mathbb{E}[Y_0] = 1.$$

On the other hand,

$$\mathbb{E}[N_{T_a}^b] = \mathbb{E}\left[\exp\left(bB_{T_a} - \frac{1}{2}b^2T_a\right)\right] = \mathbb{E}\left[\exp\left(ba - \frac{1}{2}b^2T_a\right)\right] = \exp(ab)\mathbb{E}\left[e^{-\frac{1}{2}b^2T_a}\right].$$

So let $b = \sqrt{2\lambda}$.

$$\mathbb{E}[e^{-\lambda T_a}] = e^{-\sqrt{2\lambda}a}.$$

Theorem 6.6.8. Assume $Z = (Z_t)_{t \geq 0}$ is a nonnegative supermartingale with right-continuous paths. Let U, V be stopping times with $U \leq V$. Then $Z_U, Z_V \in L^1$ and

$$Z_U \geq \mathbb{E}[Z_V \mid \mathcal{F}_U].$$

Proof. (i) First, assume $U \leq V \leq P$ for some integer P . Let

$$U_n = \sum_{k=0}^{\infty} \frac{k+1}{2^n} \mathbb{I}_{\{\frac{k}{2^n} < U \leq \frac{k+1}{2^n}\}}, \quad V_n = \sum_{k=0}^{\infty} \frac{k+1}{2^n} \mathbb{I}_{\{\frac{k}{2^n} < V \leq \frac{k+1}{2^n}\}}.$$

So they are stopping times with $U_n \downarrow U$, $V_n \downarrow V$, and $U_n \leq V_n$. Because of the right-continuity of paths,

$$Z_{U_n} \rightarrow Z_U, \quad Z_{V_n} \rightarrow Z_V.$$

Moreover, we can consider $(Y_k = Z_{U_{-k}})_{k \leq 0}$. Because $U_{n+1} \leq U_n$, by the optional stopping time theorem (bounded case),

$$\mathbb{E}[Z_{U_n} \mid \mathcal{F}_{U_{n+1}}] \leq Z_{U_{n+1}},$$

so (Y_k) is a backward supermartingale with $\mathbb{E}[Y_k] \leq \mathbb{E}[Z_0] < \infty$. It follows that (Y_k) is UI and so $Z_{U_n} \rightarrow Z_U$ in L^1 , so is $Z_{V_n} \rightarrow Z_V$ in L^1 .

Because $U_n \leq V_n$, by the optional stopping time theorem (bounded case),

$$\mathbb{E}[Z_{V_n} \mid \mathcal{F}_{U_n}] \leq Z_{U_n},$$

which implies that

$$\mathbb{E}[Z_{V_n}] \leq Z_{U_n}.$$

As $n \rightarrow \infty$, by L^1 -convergence

$$\mathbb{E}[Z_V] \leq \mathbb{E}[Z_U].$$

(ii) Next, consider general $U \leq V$. It is obvious Z_U is $\mathcal{F}_U$ -measurable. To prove

$$\mathbb{E}[Z_V \mid \mathcal{F}_U] \leq Z_U,$$

it suffices to prove that for any $A \in \mathcal{F}_U$,

$$\mathbb{E}[Z_V \mathbb{I}_A] \leq \mathbb{E}[Z_U \mathbb{I}_A].$$

Define

$$U^A = \begin{cases} U, & \omega \in A \\ \infty, & \omega \in A^c \end{cases}, \quad V^A = \begin{cases} V, & \omega \in A \\ \infty, & \omega \in A^c \end{cases}.$$

They are stopping times with $U^A \leq V^A$ because $A \in \mathcal{F}_U \subset \mathcal{F}_V$. For any $p \geq 1$, by above step, we have

$$\mathbb{E}[Z_{V^A \wedge p}] \leq \mathbb{E}[Z_{U^A \wedge p}]$$

For the RHS,

$$\begin{aligned} \mathbb{E}[Z_{U^A \wedge p}] &= \mathbb{E}[Z_{U^A \wedge p} \mathbb{I}_A] + \mathbb{E}[Z_{U^A \wedge p} \mathbb{I}_{A^c}] \\ &= \mathbb{E}[Z_{U \wedge p} \mathbb{I}_A] + \mathbb{E}[Z_p \mathbb{I}_{A^c}], \end{aligned}$$

and we have the similar formula for the LHS. So

$$\mathbb{E}[Z_{V \wedge p} \mathbb{I}_A] \leq \mathbb{E}[Z_{U \wedge p} \mathbb{I}_A]$$

Note that

$$\mathbb{I}_A = \mathbb{I}_{A \cap \{U \leq p\}} + \mathbb{I}_{A \cap \{U > p\}}.$$

So

$$\mathbb{E}[Z_{U \wedge p} \mathbb{I}_A] = \mathbb{E}[\mathbb{I}_{A \cap \{U \leq p\}} Z_U] + \mathbb{E}[\mathbb{I}_{A \cap \{U > p\}} Z_p]$$

and

$$\mathbb{E}[Z_{V \wedge p} \mathbb{I}_A] = \mathbb{E}[\mathbb{I}_{A \cap \{U \leq p\}} Z_{V \wedge p}] + \mathbb{E}[\mathbb{I}_{A \cap \{U > p\}} Z_p].$$

It follows that

$$\mathbb{E}[\mathbb{I}_{A \cap \{U \leq p\}} Z_U] \geq \mathbb{E}[\mathbb{I}_{A \cap \{U \leq p\}} Z_{V \wedge p}] \geq \mathbb{E}[\mathbb{I}_{A \cap \{V \leq p\}} Z_{V \wedge p}]$$

because Z is nonnegative. By MCT, as $p \rightarrow \infty$,

$$\mathbb{E}[Z_V \mathbb{I}_A] \leq \mathbb{E}[Z_U \mathbb{I}_A].$$

□

Remark 6.6.9. Note that if $Z = (Z_t)_{t \geq 0}$ is a UI supermartingale, then above is also clearly true by as $p \rightarrow \infty$ in (ii).

Proposition 6.6.10. *Let $X = (X_t)_{t \geq 0}$ be a adapted and right-continuous and integrable stochastic process satisfying $X_T \in L^1$ for all bounded stopping time T . Then X is a martingale if and only if*

$$\mathbb{E}[X_T] = \mathbb{E}[X_0],$$

Proof. For any $0 \leq s < t$ and any $A \in \mathcal{F}_s$, let

$$T = s\mathbb{I}_{A^c} + t\mathbb{I}_A.$$

Then T is a bounded stopping time and

$$\mathbb{E}[X_0] = \mathbb{E}[X_T] = \mathbb{E}[X_s \mathbb{I}_{A^c}] + \mathbb{E}[X_t \mathbb{I}_A] = \mathbb{E}[X_s] + \mathbb{E}[(X_t - X_s) \mathbb{I}_A].$$

Because $\mathbb{E}[X_0] = \mathbb{E}[X_s]$,

$$\mathbb{E}[X_t \mathbb{I}_A] = \mathbb{E}[X_s \mathbb{I}_A].$$

So

$$X_s = \mathbb{E}[X_t | \mathcal{F}_s].$$

□

Remark 6.6.11. Furthermore, if above conditions are satisfied for all stopping times T , then X is UI.

Chapter 7

Continuous Time Markov Theory

7.1 Transition Semigroup

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space. Let $(E, \mathcal{E})$ be a measurable space.

Definition 7.1.1 (Markov Process). A E -valued stochastic process $(X_t)_{t \geq 0}$ is called $(\mathcal{F}_t)_{t \geq 0}$ -Markov process if

- (i) X_t is $\mathcal{F}_t$ -adapted,
- (ii) for any $t > s$ and any $f \in \mathcal{B}_b(E)$ (bounded measurable function),

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = \mathbb{E}[f(X_t) \mid \sigma(X_s)].$$

Remark 7.1.2. If $(X_t)_{t \geq 0}$ is a Markov process, it is obvious a Markov process w.s.t. its natural filtration $\mathcal{F}_t^X = \sigma(X_s : s \leq t)$.

Definition 7.1.3 (Transition Kernel). A Markov transition kernel from E to E is a map

$$Q: E \times \mathcal{E} \rightarrow [0, 1]$$

such that

- (i) for any $x \in E$, $Q(x, \cdot)$ is a probability measure on $(E, \mathcal{E})$.
- (ii) for any $A \in \mathcal{E}$, $Q(\cdot, A)$ is $\mathcal{E}$ -measurable.

Remark 7.1.4. Given a Markov transition kernel Q , it can define

$$Q: \mathcal{B}_b(E) \rightarrow \mathcal{B}_b(E)$$

as

$$Qf(x) := \int_E f(y)Q(x, dy),$$

which is a linear operator.

Definition 7.1.5 (Transition Semigroup). A collection $(Q_t)_{t \geq 0}$ of transition kernels on E is called a transition semigroup if

- (i) for $x \in E$, $Q_0(x, dy) = \delta_x(dy)$,

(ii) for $s, t \geq 0$ and $A \in \mathcal{E}$,

$$Q_{t+s}(x, A) = \int_E Q_t(x, dy) Q_s(y, A).$$

(Chapman-Kolmogorov equation)

(iii) for any $A \in \mathcal{E}$, $(t, x) \mapsto Q_t(x, A)$ is measurable.

Remark 7.1.6. Note that a transition semigroup induces a semigroup of operators $(Q_t)_{t \geq 0}$. Let $\mathcal{B}_b(E)$ be equipped with $\|\cdot\| = \|\cdot\|_\infty$. Then

(i) $Q_0 f(x) = \int_E f(y) \delta_x(dy) = f(x)$, i.e., $Q_0 = \text{Id}$.

(ii) $Q_t \mathbf{1}(x) = \int_E \mathbf{1}(y) Q_t(x, dy) = 1$, i.e., $Q_t \mathbf{1} = \mathbf{1}$.

(iii) for any $f \geq 0$, $Q_t f \geq 0$.

(iv) for any $s, t \geq 0$,

$$\begin{aligned} Q_{t+s} f(x) &= \int_E f(y) Q_{t+s}(x, dy) \\ &= \int_E f(y) \int_E Q_t(x, dz) Q_s(z, dy) \\ &= \int_E \left(\int_E f(y) Q_s(z, dy) \right) Q_t(x, dz) \\ &= \int_E Q_s f(z) Q_t(x, dz) \\ &= Q_t(Q_s f)(x), \end{aligned}$$

i.e., $Q_{t+s} = Q_t \circ Q_s = Q_t Q_s$.

Definition 7.1.7. A Markov process $X = (X_t)_{t \geq 0}$ with transition semigroup $(Q_t)_{t \geq 0}$ is a $(\mathcal{F}_t)_{t \geq 0}$ -adapted process with values in E such that for any $s, t \geq 0$ and any $f \in \mathcal{B}_b(E)$,

$$\mathbb{E}[f(X_{t+s}) \mid \mathcal{F}_s] = Q_t f(X_s).$$

Remark 7.1.8. Note that it is true Markov because

$$\begin{aligned} \mathbb{E}[f(X_{t+s}) \mid \sigma(X_s)] &= \mathbb{E}[\mathbb{E}[f(X_{t+s}) \mid \mathcal{F}_s] \mid \sigma(X_s)] \\ &= \mathbb{E}[Q_t f(X_s) \mid \sigma(X_s)] \\ &= Q_t f(X_s) = \mathbb{E}[f(X_{t+s}) \mid \mathcal{F}_s]. \end{aligned}$$

Moreover, if $f = \mathbb{I}_A$, then

$$Q_t(X_s, A) = \mathbb{P}(X_{t+s} \in A \mid \mathcal{F}_s).$$

Theorem 7.1.9 (Finite-dimensional Distribution). *Given a Markov process $X = (X_t)_{t \geq 0}$ with transition semigroup $(Q_t)_{t \geq 0}$ and $X_0 \sim \gamma(dx)$. For any $0 < t_1 < \dots < t_p$,*

$$\begin{aligned} &\mathbb{P}(X_0 \in A_0, X_{t_1} \in A_1, \dots, X_{t_p} \in A_p) \\ &= \int_{A_0} \gamma(dx) \int_{A_1} Q_{t_1}(x, dx_1) \int_{A_2} Q_{t_2-t_1}(x_1, dx_2) \dots \int_{A_p} Q_{t_p-t_{p-1}}(x_{p-1}, dx_p). \end{aligned}$$

More generally, for any $f_i \in \mathcal{B}_b(E)$ ($i = 0, 1, \dots, p$),

$$\begin{aligned} &\mathbb{E}[f_0(X_0) f_1(X_{t_1}) \dots f_p(X_{t_p})] \\ &= \int_E f_0(X_0) \gamma(dx) \int_E f_1(x_1) Q_{t_1}(x, dx_1) \int_E f_2(x_2) Q_{t_2-t_1}(x_1, dx_2) \dots \int_E f_p(x_p) Q_{t_p-t_{p-1}}(x_{p-1}, dx_p). \end{aligned}$$

Proof. For $p = 1$,

$$\begin{aligned}
\mathbb{E}[f_0(X_0)f_1(X_{t_1})] &= \mathbb{E}[f_0(X_0)[f_1(X_{t_1}) | \mathcal{F}_0]] \\
&= \mathbb{E}[f_0(X_0)Q_{t_1}f_1(X_0)] \\
&= \int_E f_0(x)Q_{t_1}f_1(x)\gamma(dx) \\
&= \int_E f_0(x)\gamma(dx) \int_E f_1(x_1)Q_{t_1}(x, dx_1).
\end{aligned}$$

Assume it is true for $p - 1$. Then

$$\begin{aligned}
&\mathbb{E}[f_0(X_0)f_1(X_{t_1}) \cdots f_p(X_{t_p})] \\
&= \mathbb{E}[\mathbb{E}[f_0(X_0)f_1(X_{t_1}) \cdots f_p(X_{t_p}) | \mathcal{F}_{t_{p-1}}]] = \mathbb{E}[f_0(X_0) \cdots f_{p-1}(X_{t_{p-1}})\mathbb{E}[f_p(X_{t_p}) | \mathcal{F}_{t_{p-1}}]] \\
&= \mathbb{E}[f_0(X_0) \cdots f_{p-1}(X_{t_{p-1}})Q_{t_p-t_{p-1}}f_p(X_{t_{p-1}})] \\
&= \int_E f_0(X_0)\gamma(dx) \cdots \int_E f_{p-1}(x_{p-1})Q_{t_p-t_{p-1}}f_p(x_{p-1})Q_{t_{p-1}-t_{p-2}}(x_{p-2}, dx_{p-1}) \\
&= \int_E f_0(X_0)\gamma(dx) \cdots \int_E f_{p-1}(x_{p-1})Q_{t_{p-1}-t_{p-2}}(x_{p-2}, dx_{p-1}) \int_E f_p(x_p)Q_{t_p-t_{p-1}}(x_{p-1}, dx_p). \quad \square
\end{aligned}$$

Construction of Markov Process: Given a transition semigroup $(Q_t)_{t \geq 0}$ and an initial distribution γ . First, let

$$\Omega^* = E^{[0, \infty)} := \{\omega : \omega(\cdot) : [0, \infty) \rightarrow E\}$$

with the coordinate process $X = (X_t)_{t \geq 0}$ defined as

$$X_t : \Omega^* \rightarrow E, \quad X_t(\omega) = \omega(t).$$

Then σ -field $\mathcal{F}^* := \sigma(X_t : t \geq 0)$. For any finite subset $U = \{0 \leq t_1 < t_2 < \cdots < t_p\}$ of $[0, \infty)$, define a probability measure on $E^U \cong E^p$,

$$\mu^U(A_1 \times \cdots \times A_p) := \int_{A_0} \gamma(dx) \int_{A_1} Q_{t_1}(x, dx_1) \int_{A_2} Q_{t_2-t_1}(x_1, dx_2) \cdots \int_{A_p} Q_{t_p-t_{p-1}}(x_{p-1}, dx_p).$$

Note that for $\{\mu^U : U \text{ finite.}\}$, if $U \subset V$ and let $\pi_U^V : E^V \rightarrow E^U$ be the natural projection, then

$$\mu^U = (\pi_U^V)_\# \mu^V, \text{ i.e. } \mu^U(A_1 \times \cdots \times A_{p_U}) = \mu^V(A_1 \times \cdots \times A_{p_U} \times E \times \cdots \times E).$$

Then by the Kolmogorov Extension Theorem, there exists a unique $\mathbb{P}^*$ on $(\Omega^*, \mathcal{F}^*)$ such that

$$\mathbb{P}^*(X_0 \in A_0, X_{t_1} \in A_1, \cdots, X_{t_p} \in A_p) = \mu^U(A_1 \times \cdots \times A_p).$$

Therefore, the coordinate process $(X_t)_{t \geq 0}$ is a Markov process on $(\Omega^*, \mathcal{F}^*, \mathbb{P}^*)$ with semigroup $(Q_t)_{t \geq 0}$. Because $\mathbb{P}^*$ is determined by γ , $\mathbb{P}^* = \mathbb{P}_\gamma$. In particular, $\gamma(dy) = \delta_x(dy)$, $\mathbb{P}_\gamma = \mathbb{P}_x$.

Remark 7.1.10. For $A \in \mathcal{E}^U$, let

$$\{\omega \in \Omega^* : (\omega(t_1), \cdots, \omega(t_p)) \in A\}$$

be called a finite-dimensional cylinder. Let $\mathcal{C}$ be the set of finite-dimensional cylinders. Then in fact $\mathcal{F}^* = \sigma(\mathcal{C})$.

Remark 7.1.11. For any Markov process $(X_t)_{t \geq 0}$ on $(\Omega, \mathbb{P})$ with semigroup $(Q_t)_{t \geq 0}$ and $X_0 \sim \gamma$, we can construct $\mathbb{P}_\gamma$ on $(\Omega^*, \mathcal{F}^*)$ by $(Q_t)_{t \geq 0}$. Then we have $X_\# \mathbb{P} = \mathbb{P}_\gamma$ and $(X_t)_{t \geq 0}$ has the same finite-dimensional distribution as the coordinate process $(\pi_t)_{t \geq 0}$ on $(\Omega^*, \mathcal{F}^*, \mathbb{P}^*)$.

Example 7.1.12. If

$$Q_t(x, dy) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} dy$$

then the Markov process with $X_0 = 0$, then the corresponding Markov process is a standard Brownian motion.

7.2 Resolvent

Definition 7.2.1 (Resolvent). Let $\lambda > 0$. The λ -resolvent of the transition semigroup $(Q_t)_{t \geq 0}$ is a linear operator $R_\lambda: \mathcal{B}_b(E) \rightarrow \mathcal{B}_b(E)$ defined as

$$R_\lambda f(x) := \int_0^\infty e^{-\lambda t} Q_t f(x) dt,$$

or formally, $R_\lambda = \int_0^\infty e^{-\lambda t} Q_t dt$.

Proposition 7.2.2. *Given a transition semigroup $(Q_t)_{t \geq 0}$ and the corresponding R_λ .*

$$(1) \|R_\lambda f\| \leq \frac{1}{\lambda} \|f\|.$$

$$(2) \text{ If } 0 \leq f \leq 1, 0 \leq \lambda R_\lambda f \leq 1.$$

$$(3) \text{ If } \lambda, \mu > 0, \text{ then}$$

$$R_\lambda - R_\mu + (\lambda - \mu) R_\lambda R_\mu = 0.$$

Proof. (1) For $t \geq 0$,

$$\begin{aligned} \|Q_t f\| &= \sup_x \left| \int_E f(y) Q_t(x, dy) \right| \\ &\leq \sup_x \int_E |f(y)| Q_t(x, dy) \\ &\leq \|f\|. \end{aligned}$$

Therefore,

$$\begin{aligned} \|R_\lambda f\| &= \left\| \int_0^\infty e^{-\lambda t} Q_t f dt \right\| \\ &\leq \int_0^\infty e^{-\lambda t} \|Q_t f\| dt \\ &\leq \|f\| \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda} \|f\|. \end{aligned}$$

(2) It is obvious by (1).

(3) By definition,

$$\begin{aligned} R_\lambda R_\mu f(x) &= \int_0^\infty e^{-\lambda t} Q_t \left(\int_0^\infty e^{-\mu s} Q_s f(x) ds \right) dt \\ &= \int_0^\infty \int_0^\infty e^{-\lambda t} e^{-\mu s} Q_{t+s} f(x) ds dt \\ &= \int_0^\infty \frac{e^{-\mu r} - e^{-\lambda r}}{\lambda - \mu} Q_r f dr \\ &= \frac{R_\mu - R_\lambda}{\lambda - \mu} f(x). \end{aligned}$$

□

Lemma 7.2.3. *Let $(X_t)_{t \geq 0}$ be a Markov process with semigroup $(Q_t)_{t \geq 0}$. Let $h \in \mathcal{B}_b(E)$ and $h \geq 0$. For $\lambda > 0$,*

$$Y_t = e^{-\lambda t} R_\lambda h(X_t)$$

is a supermartingale.

Proof. For $s > 0$,

$$\begin{aligned} Q_s(R_\lambda h) &= Q_s \left(\int_0^\infty e^{-\lambda t} Q_t h dt \right) \\ &= \int_0^\infty e^{-\lambda t} Q_{t+s} h dt \\ &= e^{\lambda s} \int_s^\infty e^{-\lambda u} Q_u h du. \end{aligned}$$

So

$$e^{-\lambda s} Q_s(R_\lambda h) = \int_s^\infty e^{-\lambda u} Q_u h du \leq \int_0^\infty e^{-\lambda u} Q_u h du = R_\lambda h.$$

Then

$$\begin{aligned} \mathbb{E}[Y_{t+s} \mid \mathcal{F}_s] &= \mathbb{E}[e^{-\lambda(t+s)} R_\lambda h(X_{t+s}) \mid \mathcal{F}_s] \\ &= e^{-\lambda(t+s)} \mathbb{E}[R_\lambda h(X_{t+s}) \mid \mathcal{F}_s] \\ &= e^{-\lambda(t+s)} Q_t R_\lambda h(X_s) = e^{-\lambda s} e^{-\lambda t} Q_t R_\lambda h(X_s) \\ &\leq e^{-\lambda s} R_\lambda h(X_s) = Y_s. \end{aligned}$$

So $(Y_t)_{t \geq 0}$ is a supermartingale. □

7.3 Feller Semigroup and Generator

Let E be a metric space that is locally compact. Moreover, assume E is a union of countably many compact sets, which implies that there exists compact $K_n \uparrow E$ and any compact subset of E is contained in some K_n . That is E is a σ -compact metric space. In such case, a function $f: E \rightarrow \mathbb{R}$ is called trending to 0 at infinity if for any $\varepsilon > 0$, there exists a compact K such that $|f(x)| \leq \varepsilon$ for all $x \notin K$, which is equivalent to

$$\lim_{n \rightarrow \infty} \sup_{x \in E \setminus K_n} |f(x)| < \varepsilon.$$

Let

$$C_0(E) := \{f: f \in C(E), f \text{ trends to 0 at infinity.}\}$$

Then $C_0(E)$ is a Banach space with the norm defined as

$$\|f\| = \sup_{x \in E} |f(x)|.$$

Definition 7.3.1. A transition semigroup $(Q_t)_{t \geq 0}$ if

- (i) for any $f \in C_0(E)$, $Q_t f \in C_0(E)$,
- (ii) for any $f \in C_0(E)$, $\|Q_t f - f\| \rightarrow 0$ as $t \rightarrow 0$.

Remark 7.3.2. (i) It follows that for $f \in C_0(E)$,

$$R_\lambda f = \int_0^\infty e^{-\lambda t} Q_t f dt \in C_0(E).$$

- (ii) Note that given $f \in C_0(E)$, $t \mapsto Q_t f$ is uniformly continuous because

$$\|Q_{t+s} f - Q_t f\| \leq \|Q_s f - f\| \rightarrow 0$$

which is independent of t as $s \rightarrow 0$.

Example 7.3.3. Consider a standard Brownian motion $(B_t)_{t \geq 0}$,

$$Q_t f(x) = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} f(y) dy.$$

For any $f \in C_0(\mathbb{R})$, it is obvious $Q_t f \in C(\mathbb{R})$. Moreover, choose a K such that

$$Q_t f(x) = \int_K + \int_{\mathbb{R} \setminus K} \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} f(y) dy \leq \int_K + \varepsilon$$

Then as $|x| \rightarrow \infty$, by DCT, $|Q_t f(x)| \rightarrow 0$. So $Q_t f \in C_0(\mathbb{R})$.

Proposition 7.3.4. Let $(Q_t)_{t \geq 0}$ be a Feller semigroup. For any $\lambda > 0$, let

$$\mathcal{D} = \{R_\lambda f : f \in C_0(E)\}.$$

Then $\mathcal{D}$ is independent of λ and $\mathcal{D} \subset C_0(E)$ is dense.

Proof. For any $\lambda, \mu > 0$, because

$$R_\lambda f = R_\mu f + (\mu - \lambda) R_\mu R_\lambda f = R_\mu(f + (\mu - \lambda) R_\lambda f),$$

$\text{Im } R_\lambda \subset \text{Im } R_\mu$. So $\text{Im } R_\lambda = \text{Im } R_\mu$. For density, for any $f \in C_0(E)$,

$$\begin{aligned} R_\lambda(\lambda f) &= \lambda R_\lambda f = \lambda \int_0^\infty e^{-\lambda t} Q_t f dt \\ &= \int_0^\infty e^{-u} Q_{\frac{u}{\lambda}} f du \rightarrow f \end{aligned}$$

as $\lambda \rightarrow \infty$ by MCT. □

Definition 7.3.5 (Generator). Let $(Q_t)_{t \geq 0}$ be a Feller semigroup. Set

$$\mathcal{D}(L) := \left\{ f \in C_0(E) : \lim_{t \rightarrow 0} \frac{Q_t f - f}{t} \text{ converges in } C_0(E) \right\}.$$

that is a linear subspace. Then for any $f \in \mathcal{D}(L)$,

$$L f := \lim_{t \rightarrow 0} \frac{Q_t f - f}{t}.$$

L is called the generator of $(Q_t)_{t \geq 0}$, a linear operator.

Proposition 7.3.6. Let $f \in \mathcal{D}(L)$. Then for any $s \geq 0$, $Q_s f \in \mathcal{D}(L)$ and

$$L(Q_s f) = Q_s(L f).$$

Proof. Because Q_s is bounded,

$$\lim_{t \rightarrow 0} \frac{Q_t Q_s f - Q_s f}{t} = \lim_{t \rightarrow 0} Q_s \frac{Q_t f - f}{t} = Q_s L f.$$

So $Q_s f \in \mathcal{D}(L)$ and the LHS

$$L(Q_s f) = Q_s L f. \quad \square$$

Corollary 7.3.7. If $f \in \mathcal{D}(L)$, for any $t \geq 0$,

$$Q_t f - f = \int_0^t Q_s(L f) ds = \int_0^t L(Q_s f) ds.$$

Proof. Consider $t \mapsto Q_t f$,

$$\frac{d}{dt} Q_t f = \lim_{s \rightarrow 0} \frac{Q_{t+s} f - Q_t f}{s} = Q_t L f. \quad \square$$

Proposition 7.3.8. *Let $\lambda > 0$.*

(1) *For any $g \in C_0(E)$, $R_\lambda g \in \mathcal{D}(L)$ and*

$$(\lambda - L)R_\lambda g = g.$$

(2) *If $f \in D(L)$,*

$$R_\lambda(\lambda - L)f = f.$$

It follows that $\text{Im } R_\lambda = \mathcal{D}(L)$ and $R_\lambda = (\lambda - L)^{-1}$.

Proof. (1) Note that

$$\begin{aligned} Q_\varepsilon R_\lambda g &= Q_\varepsilon \int_0^\infty e^{-\lambda t} Q_t g dt \\ &= \int_0^\infty e^{-\lambda t} Q_{\varepsilon+t} g dt \\ &= \int_\varepsilon^\infty e^{-\lambda(u-\varepsilon)} Q_u g du. \end{aligned}$$

So

$$\frac{1}{\varepsilon}(Q_\varepsilon R_\lambda g - R_\lambda g) = \frac{e^{\lambda\varepsilon} - 1}{\varepsilon} R_\lambda g - e^{\lambda\varepsilon} \frac{1}{\varepsilon} \int_0^\varepsilon e^{-\lambda t} Q_t g dt.$$

As $\varepsilon \rightarrow 0$,

$$\lim_{\varepsilon \rightarrow 0} (Q_\varepsilon R_\lambda g - R_\lambda g) = \lambda R_\lambda g - g = L R_\lambda g.$$

and so $R_\lambda g \in \mathcal{D}(L)$.

(2) First, for $f \in \mathcal{D}(L)$,

$$\frac{d}{dt} Q_t f = Q_t L f = L Q_t f.$$

and so

$$Q_t f - f = \int_0^t Q_s(Lf) ds.$$

Therefore,

$$\begin{aligned} R_\lambda f &= \int_0^\infty e^{-\lambda t} Q_t f dt \\ &= \int_0^\infty e^{-\lambda t} \left(f + \int_0^t Q_s(Lf) ds \right) dt \\ &= \frac{1}{\lambda} f + \int_0^\infty \frac{e^{-\lambda s}}{\lambda} Q_s L f ds \\ &= \frac{1}{\lambda} f + \frac{1}{\lambda} R_\lambda L f. \end{aligned} \quad \square$$

Remark 7.3.9. If we have $(L, \mathcal{D}(L))$, then define for $\lambda \geq 0$

$$R_\lambda = (\lambda - L)^{-1}.$$

Such R_λ determines $(Q_t)_{t \geq 0}$ because R_λ is the Laplace transform of $(Q_t)_{t \geq 0}$.

Example 7.3.10. Consider a standard Brownian motion $(B_t)_{t \geq 0}$. Then

$$Q_t f(x) = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} f(y) dy.$$

So the resolvent

$$\begin{aligned} R_\lambda f &= \int_0^\infty e^{-\lambda t} \left(\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} f(y) dy \right) dt \\ &= \int_{\mathbb{R}} f(y) \left(\int_0^\infty e^{-\lambda t} \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} dt \right) dy \\ &= \int_{\mathbb{R}} f(y) \frac{1}{\sqrt{2\lambda}} e^{-\sqrt{2\lambda}|y-x|} dy. \end{aligned}$$

Assume $f \in C_0(E)$ and f'' exists.

$$\begin{aligned} & \left. \frac{d}{dt} Q_t f(x) \right|_{t=0} \\ &= \lim_{t \rightarrow 0} \frac{\int_{\mathbb{R}} \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} (f(y) - f(x)) dy}{t} \\ &= \lim_{t \rightarrow 0} \frac{1}{t\sqrt{2\pi t}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{2t}} \left(f'(x)(y-x) + f''(x) \frac{(y-x)^2}{2} + f'''(\theta_{x,y}) \frac{(y-x)^3}{6} \right) dy \\ &= \lim_{t \rightarrow 0} \frac{1}{t} \left(f'(x) \mathbb{E}[B_t^0] + \frac{1}{2} f''(x) \mathbb{E}[(B_t^0)^2] + \int_{\mathbb{R}} \left[f'''(\theta_{x,y}) \frac{(y-x)^3}{6} \right] \frac{1}{\sqrt{2\pi t}} \exp \left\{ -\frac{(y-x)^2}{2t} \right\} dy \right) \\ &= \frac{1}{2} f''(x) + \lim_{t \rightarrow 0} \frac{1}{t} \int_{\mathbb{R}} \left[f'''(\theta_{x,y}) \frac{(y-x)^3}{6} \right] \frac{1}{\sqrt{2\pi t}} \exp \left\{ -\frac{(y-x)^2}{2t} \right\} dy \\ &= \frac{1}{2} f''(x), \end{aligned}$$

(It is because $f \in C_0(E)$ implies that $f^{(n)} \in C_0(E)$). So

$$Lf(x) = f''(x).$$

Theorem 7.3.11. Given a Markov process $(X_t^x)_{t \geq 0}$ with $X_0^x = x$ and it has RLCC paths. Let $h, g \in C_0(E)$. TFAE.

(1) $h \in \mathcal{D}(L)$ and $Lh = g$.

(2) For any $x \in E$,

$$M_t = h(X_t^x) - \int_0^t g(X_s^x) ds$$

is a martingale.

Proof. (1) $\Rightarrow$ (2) : Note that

$$\begin{aligned} \mathbb{E}[M_{t+s} | \mathcal{F}_t] &= \mathbb{E} \left[h(X_{t+s}^x) - \int_0^{t+s} g(X_u^x) du \mid \mathcal{F}_t \right] \\ &= \mathbb{E} [h(X_{t+s}^x) | \mathcal{F}_t] - \mathbb{E} \left[\int_0^t g(X_u^x) du \mid \mathcal{F}_t \right] - \mathbb{E} \left[\int_t^{t+s} g(X_u^x) du \mid \mathcal{F}_t \right] \\ &= Q_s h(X_t^x) - \int_0^t g(X_u^x) du - \int_t^{t+s} \mathbb{E} [g(X_u^x) | \mathcal{F}_t] du \end{aligned}$$

$$\begin{aligned}
&= Q_s h(X_t^x) - \int_0^t g(X_u^x) du - \int_t^{t+s} Q_{u-t} g(X_t^x) du \\
&= Q_s h(X_t^x) - \int_0^t g(X_u^x) du - \int_0^s Q_u g(X_t^x) du.
\end{aligned}$$

By above,

$$Q_t h = h + \int_0^t Q_s g ds.$$

So

$$\mathbb{E}[M_{t+s} | \mathcal{F}_t] = h(X_t^x) - \int_0^t g(X_u^x) du = M_t. \quad \square$$

(2) $\Rightarrow$ (1) : First,

$$\mathbb{E}[M_t] = \mathbb{E}[M_0] = h(x).$$

On the other hand,

$$\begin{aligned}
\mathbb{E}[M_t] &= \mathbb{E}[h(X_t^x)] - \mathbb{E}\left[\int_0^t g(X_s^x) ds\right] \\
&= Q_t h(x) - \int_0^t Q_s g(x) ds
\end{aligned}$$

Therefore,

$$\int_0^t Q_s g(x) ds = \int_0^t Q_s Lh(x) ds \Rightarrow Q_t(g - Lh) = 0$$

because $t \mapsto Q_t f$ is uniform continuous. Because Q_t is invertible, $g = Lh$.

7.4 Markov Property

Definition 7.4.1. For two processes $(X_t)_{t \geq 0}$ and $(X'_t)_{t \geq 0}$,

(1) If for any $t \geq 0$,

$$\mathbb{P}(X_t = X'_t) = 1,$$

then $(X_t)_{t \geq 0}$ is called a modification of $(X'_t)_{t \geq 0}$.

(2) If

$$\mathbb{P}(X_t = X'_t, t \geq 0) = 1,$$

then $(X_t)_{t \geq 0}$ and $(X'_t)_{t \geq 0}$ are called indistinguishable.

Theorem 7.4.2. Assume $(X_t)_{t \geq 0}$ is a Markov process with Feller semigroup $(Q_t)_{t \geq 0}$. Then $(X_t)_{t \geq 0}$ has a Markov modification $(X'_t)_{t \geq 0}$ that is càdlàg.

Sketch of Proof. Consider $\{R_\lambda f\}$ such that $Y_t = e^{-\lambda} R_\lambda f(X_t)$ is a supermartingale that has a càdlàg modification. Because such family is rich enough, $(X_t)_{t \geq 0}$ has a càdlàg modification $\square$

Assume E is a metric space. Given a semigroup $(Q_t)_{t \geq 0}$. For $x \in E$, $(X_t^x)_{t \geq 0}$ is a Markov process with $X_0^x = x$ associated with $(Q_t)_{t \geq 0}$. Assume $(X_t^x)_{t \geq 0}$ is càdlàg. Let

$$D(E) := \{f: [0, \infty) \rightarrow E: f \text{ is càdlàg.}\} (= E^{[0, \infty)}).$$

equipped with the σ -field $\mathcal{D}$ generated by the coordinate process $W_t(\omega) = \omega(t)$ for $\omega \in D(E)$.

Remark 7.4.3. If $X = (X_t)_{t \geq 0}$ is a càdlàg process on $(\Omega, \mathcal{F}, \mathbb{P})$, then $X: \Omega \rightarrow D(E)$ i.e. X can be viewed as $D(E)$ -value random variable. Furthermore, let $\mathbb{P}_X = X_{\#}\mathbb{P}$ be the law of X on $D(E)$.

Definition 7.4.4 (Shift Operator). Fix $t \geq 0$,

$$\theta_t: D(E) \rightarrow D(E)$$

is defined as for any $\omega \in D(E)$,

$$\theta_t(\omega)(s) := \omega(t + s).$$

Theorem 7.4.5 (Markov Property). *Let $X = (X_t)_{t \geq 0}$ be a càdlàg Markov process associated with semigroup $(Q_t)_{t \geq 0}$. Let $s \geq 0$ and $\Phi: D(E) \rightarrow \mathbb{R}$ be a measurable and bounded function. Then*

$$\mathbb{E}[\Phi(\theta_s \circ X) \mid \mathcal{F}_s] = \mathbb{E}_{X_s}[\Phi].$$

Remark 7.4.6. Note that because $\mathbb{E}_{X_s}[\Phi]$ is $\sigma(X_s)$ -measurable,

$$\mathbb{E}[\Phi(\theta_s \circ X) \mid \mathcal{F}_s] = \mathbb{E}[\Phi(\theta_s \circ X) \mid \sigma(X_s)].$$

Proof. By Monotone Class Theorem, assume

$$\Phi(f) = \varphi_1(f(t_1)) \cdots \varphi_p(f(t_p)).$$

So the RHS is

$$\begin{aligned} \mathbb{E}_{X_s}[\Phi] &= \mathbb{E}_{X_s}[\varphi_1(W_{t_1}) \cdots \varphi_p(W_{t_p})] \\ &= \int_E \delta_{X_s}(dx_0) \int_E \varphi_1(x_1) Q_{t_1}(x_0, dx_1) \int_E \varphi_2(x_2) Q_{t_2-t_1}(x_1, dx_2) \cdots \int_E \varphi_p(x_p) Q_{t_p-t_{p-1}}(x_{p-1}, dx_p) \\ &= \int_E \varphi_1(x_1) Q_{t_1}(X_s, dx_1) \int_E \varphi_2(x_2) Q_{t_2-t_1}(x_1, dx_2) \cdots \int_E \varphi_p(x_p) Q_{t_p-t_{p-1}}(x_{p-1}, dx_p) \end{aligned}$$

the LHS is

$$\mathbb{E}[\Phi(\theta_s \circ X) \mid \mathcal{F}_s] = \mathbb{E}[\varphi_1(X_{t_1+s}) \cdots \varphi_p(X_{t_p+s}) \mid \mathcal{F}_s].$$

For $p = 1$,

$$\mathbb{E}[\varphi_1(X_{t_1+s}) \mid \mathcal{F}_s] = Q_{t_1} \varphi_1(X_s) = \int_E \varphi_1(x_1) Q_{t_1}(X_s, dx_1).$$

So it is true. Assume it is true for $p - 1$.

$$\begin{aligned} &\mathbb{E}[\varphi_1(X_{t_1+s}) \cdots \varphi_p(X_{t_p+s}) \mid \mathcal{F}_s] \\ &= \mathbb{E}[\mathbb{E}[\varphi_1(X_{t_1+s}) \cdots \varphi_p(X_{t_p+s}) \mid \mathcal{F}_{t_{p-1}+s}] \mid \mathcal{F}_s] \\ &= \mathbb{E}[\varphi_1(X_{t_1+s}) \cdots \varphi_{p-1}(X_{t_{p-1}+s}) \mathbb{E}[\varphi_p(X_{t_p+s}) \mid \mathcal{F}_{t_{p-1}+s}] \mid \mathcal{F}_s] \\ &= \mathbb{E}[\varphi_1(X_{t_1+s}) \cdots \varphi_{p-1}(X_{t_{p-1}+s}) Q_{t_p-t_{p-1}} \varphi_p(X_{t_{p-1}+s}) \mid \mathcal{F}_s] \\ &= \int_E \varphi_1(x_1) Q_{t_1}(X_s, dx_1) \cdots \int_E \varphi_{p-1}(x_{p-1}) Q_{t_{p-1}-t_{p-2}}(x_{p-2}, dx_{p-1}) \\ &= \int_E \varphi_1(x_1) Q_{t_1}(X_s, dx_1) \cdots \int_E \varphi_{p-1}(x_{p-1}) Q_{t_{p-1}-t_{p-2}}(x_{p-2}, dx_{p-1}) \int_E \varphi_p(x_p) Q_{t_p-t_{p-1}}(x_{p-1}, dx_p). \square \end{aligned}$$

Theorem 7.4.7 (Strong Markov Property). *Let $(Q_t)_{t \geq 0}$ be a Feller semigroup and $(X_t)_{t \geq 0}$ be the corresponding Markov process with RLCC paths. Let T be a stopping time and $\Phi: D(E) \rightarrow \mathbb{R}$ be a measurable and bounded function.*

$$\mathbb{E}[\mathbb{I}_{\{T < \infty\}} \Phi(\theta_T \circ X) \mid \mathcal{F}_T] = \mathbb{I}_{\{T < \infty\}} \mathbb{E}_{Y_T}[\Phi].$$

Proof. By Monotone Class Theorem, assume

$$\Phi(f) = \varphi_1(f(t_1)) \cdots \varphi_p(f(t_p)).$$

The integrability and measurability are obvious. It suffices to show that for any $A \in \mathcal{F}_T$,

$$\mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \Phi(\theta_T \circ X)] = \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \mathbb{E}_{X_T}[\Phi]] .$$

and it is sufficient to consider $p = 1$, i.e.

$$\mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \varphi_1(X_{t_1+T})] = \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \mathbb{E}_{X_T}[\Phi]] .$$

Note that

$$\mathbb{E}_{X_T}[\Phi] = Q_{t_1} \varphi_1(X_T).$$

So the RHS is

$$\mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \mathbb{E}_{X_T}[\Phi]] = \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} Q_{t_1} \varphi_1(X_T)] ,$$

and our goal is to show

$$\mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \varphi_1(X_{t_1+T})] = \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} Q_{t_1} \varphi_1(X_T)] .$$

Let

$$T_n = \sum_{i=0}^{\infty} \frac{i+1}{2^n} \mathbb{I}_{\{\frac{i}{2^n} < T \leq \frac{i+1}{2^n}\}} + \infty \mathbb{I}_{\{T = \infty\}} .$$

Then $T_n \downarrow T$ stopping time. By Monotone Class Theorem, we further assume φ_1 is continuous. So by the continuity of X_t and Feller property of Q_t ,

$$\begin{aligned} \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \varphi_1(X_{t_1+T})] &= \lim_{n \rightarrow \infty} \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} \varphi_1(X_{t_1+T_n})] \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^{\infty} \mathbb{E} [\mathbb{I}_{A \cap \{\frac{i}{2^n} < T \leq \frac{i+1}{2^n}\}} \varphi_1(X_{t_1+\frac{i}{2^n}})] \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^{\infty} \mathbb{E} [\mathbb{I}_{A \cap \{\frac{i}{2^n} < T \leq \frac{i+1}{2^n}\}} \mathbb{E} [\varphi_1(X_{t_1+\frac{i}{2^n}}) \mid \mathcal{F}_{\frac{i+1}{2^n}}]] \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^{\infty} \mathbb{E} [\mathbb{I}_{A \cap \{\frac{i}{2^n} < T \leq \frac{i+1}{2^n}\}} Q_{t_1} \varphi_1(X_{\frac{i+1}{2^n}})] \\ &= \lim_{n \rightarrow \infty} \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} Q_{t_1}(X_{T_n})] \\ &= \mathbb{E} [\mathbb{I}_{A \cap \{T < \infty\}} Q_{t_1} \varphi_1(X_T)] . \end{aligned}$$

For $p > 1$, it can be done by the Markov property and induction. □

7.5 Jump Process and Lévy Process

Jump Markov Process. Assume the state space E is finite equipping with the discrete metric $d(x, y) = \delta_x(y)$ and σ -field $\mathcal{P}(E)$. Let $f \in D(E)$, i.e. $f: [0, \infty) \rightarrow E$ is càdlàg. Note that for $\{y_n\} \in E$,

$$y_n \rightarrow y \quad \Leftrightarrow \quad \exists m, y_n = y, \quad \forall n \geq m.$$

Therefore, there exists $t \in (0, \infty]$ such that $f(s) = f(0)$ for all $s \in (0, t)$. Let

$$t_1 = \max \{t > 0: f(s) = f(0), \quad \forall s \in (0, t)\} .$$

If $t_1 < \infty$, there exists $t_2 > t_1$ such that

$$t_2 = \max \{t > t_1 : f(s) = f(t_1), \forall s \in (t_1, t)\}.$$

Therefore, there exist $0 < t_1 < t_2 < \dots$ such that

$$f(t) = f(t_n), \quad \forall t \in [t_n, t_{n+1}).$$

Let $(Q_t)_{t \geq 0}$ be a semigroup on E . Because $C(E) = B(E)$, $(Q_t)_{t \geq 0}$ is a Feller semigroup. So we can construct a measure space $(\Omega, \mathcal{F})$ on which there is a family $(\mathbb{P}_x : x \in E)$ and a process $(X_t)_{t \geq 0}$ with càdlàg paths such that $(X_t)_{t \geq 0}$ is a Markov process associated with $(Q_t)_{t \geq 0}$ when $X_0 = x$. For every $\omega \in \Omega$, there exists a sequence

$$0 = T_0(\omega) < T_1(\omega) < \dots < T_n(\omega) < \dots,$$

such that

$$X_t(\omega) = X_{T_n}(\omega), \quad \forall t \in [T_n, T_{n+1}).$$

Moreover, T_n is a stopping time, like

$$\{T_1 < t\} = \bigcup_{q \in [0, t) \cap \mathbb{Q}} \{X_q \neq X_0\} \in \mathcal{F}_t^X.$$

Note that for a t , on the set of $\{\omega : t < T_1(\omega)\}$, then $T_1(\omega) = t + T_1 \circ \theta_t$. So we have

$$T_2 = T_1 + T_1 \circ \theta_{T_1}.$$

Lemma 7.5.1. *Let $x \in E$. There exists a $q(x) \geq 0$ such that T_1 is exponentially distributed with parameter $q(x)$ under $\mathbb{P}_x$. Furthermore, if $q(x) > 0$, then T_1 and X_{T_1} are independent.*

Proof. First,

$$\begin{aligned} \mathbb{P}_x(T_1 > s + t) &= \mathbb{P}_x(T_1 > s + t, T_1 > s) \\ &= \mathbb{P}_x(s + T_1 \circ \theta_s > s + t, T_1 > s) \\ &= \mathbb{P}_x(T_1 \circ \theta_s > t, T_1 > s) \\ &= \mathbb{E}_x [\mathbb{I}_{\{T_1 \circ \theta_s > t\}} \mathbb{I}_{\{T_1 > s\}}] \\ &= \mathbb{E}_x [\mathbb{E}_x [\mathbb{I}_{\{T_1 \circ \theta_s > t\}} \mathbb{I}_{\{T_1 > s\}} \mid \mathcal{F}_s]] \\ &= \mathbb{E}_x [\mathbb{I}_{\{T_1 > s\}} \mathbb{E}_x [\mathbb{I}_{\{T_1 \circ \theta_s > t\}} \mid \mathcal{F}_s]] \\ &= \mathbb{E}_x [\mathbb{I}_{\{T_1 > s\}} \mathbb{E}_{X_s} [\mathbb{I}_{\{T_1 > t\}}]] \\ &= \mathbb{E}_x [\mathbb{I}_{\{T_1 > s\}} \mathbb{E}_x [\mathbb{I}_{\{T_1 > t\}}]] \\ &= \mathbb{P}_x(T_1 > s) \mathbb{P}_x(T_1 > t), \end{aligned}$$

which implies that there exists a $q(x) \geq 0$ such that

$$\mathbb{P}_x(T_1 > t) = e^{-q(x)t}.$$

When $q(x) > 0$, $T_1 < \infty$. Let $y \in E$. Consider

$$\begin{aligned} \mathbb{P}_x(T_1 > t, X_{T_1} = y) &= \mathbb{P}_x(T_1 > t, X_{t+T_1 \circ \theta_t} = y) \\ &= \mathbb{P}_x(T_1 > t, X_{T_1} \circ \theta_t = y) \\ &= \mathbb{E}_x [\mathbb{I}_{\{T_1 > t\}} \mathbb{I}_{\{X_{T_1} \circ \theta_t = y\}}] \\ &= \mathbb{E}_x [\mathbb{I}_{\{T_1 > t\}} \mathbb{E}_{X_t} [\mathbb{I}_{\{X_{T_1} = y\}}]] \\ &= \mathbb{P}_x(T_1 > t) \mathbb{P}_x(X_{T_1} = y). \end{aligned}$$

□

Note that if $q(x) = 0$, $X_t \equiv x$. If $q(x) > 0$, for $x, y \in E$, define

$$\pi(x, y) = \mathbb{P}_x(X_{T_1} = y).$$

So $(\pi(x, y))_{x, y \in E}$ is a transition matrix.

Proposition 7.5.2. *Let L be the generator of $(Q_t)_{t \geq 0}$. Then $\mathcal{D}(L) = C(E) = B(E)$. And for any $\varphi \in C(E)$, $x \in E$, if $q(x) = 0$, then $L\varphi(x) = 0$, and if $q(x) > 0$,*

$$L\varphi(x) = q(x) \sum_{y \in E, y \neq x} \pi(x, y) (\varphi(y) - \varphi(x)).$$

Proof. Note that

$$L\varphi(x) = \lim_{t \rightarrow 0} \frac{Q_t\varphi(x) - \varphi(x)}{t}.$$

If $q(x) = 0$, $X_t \equiv x$ and $T_1 = \infty$. So

$$Q_t\varphi(x) = \mathbb{E}_x[\varphi(X_t)] = \mathbb{E}_x[\varphi(x)] = \varphi(x).$$

So $L\varphi(x) = 0$.

Assume $q(x) > 0$. Then $T_1 < \infty$.

Claim: We claim

$$\mathbb{P}_x(T_2 \leq t) = O(t^2), \quad t \rightarrow 0$$

In fact,

$$\begin{aligned} \mathbb{P}_x(T_2 \leq t) &= \mathbb{P}_x(T_1 \leq t, T_1 + T_1 \circ \theta_{T_1} \leq t) \\ &\leq \mathbb{P}_x(T_1 \leq t, T_1 + T_1 \circ \theta_{T_1} \leq t + T_1) \\ &= \mathbb{P}_x(T_1 \leq t, T_1 \circ \theta_{T_1} \leq t) \\ &= \mathbb{E}_x [\mathbb{I}_{\{T_1 \leq t\}} \mathbb{E}_{X_{T_1}} [\mathbb{I}_{\{T_1 \leq t\}}]] \\ &\leq \mathbb{E}_x \left[\mathbb{I}_{\{T_1 \leq t\}} \sup_{y \in E} \mathbb{P}_y(T_1 \leq t) \right] \\ &= \sup_{y \in E} \mathbb{P}_y(T_1 \leq t) \mathbb{P}_x(T_1 \leq t) \\ &= \sup_{y \in E} (1 - e^{-q(y)t}) (1 - e^{-q(x)t}) \leq Ct^2, \end{aligned}$$

when $t \rightarrow 0$.

Then we have

$$\begin{aligned} Q_t\varphi(x) &= \mathbb{E}_x [\varphi(X_t)] \\ &= \mathbb{E}_x [\varphi(X_t) \mathbb{I}_{\{t < T_1\}}] + \mathbb{E}_x [\varphi(X_t) \mathbb{I}_{\{t \geq T_1\}}] \\ &= \varphi(x) \mathbb{P}_x(T_1 > t) + \mathbb{E}_x [\varphi(X_t) \mathbb{I}_{\{t \geq T_1\}} \mathbb{I}_{\{t < T_2\}}] + \mathbb{E}_x [\varphi(X_t) \mathbb{I}_{\{t \geq T_1\}} \mathbb{I}_{\{t \geq T_2\}}] \\ &= \varphi(x) \mathbb{P}_x(T_1 > t) + E_x [\varphi(X_{T_1}) \mathbb{I}_{\{T_1 \leq t < T_2\}}] + O(t^2) \\ &= \varphi(x) \mathbb{P}_x(T_1 > t) + E_x [\varphi(X_{T_1}) \mathbb{I}_{\{T_1 \leq t\}}] - E_x [\varphi(X_{T_1}) \mathbb{I}_{\{T_2 \leq t\}}] + O(t^2) \\ &= \varphi(x) \mathbb{P}_x(T_1 > t) + E_x [\varphi(X_{T_1}) \mathbb{I}_{\{T_1 \leq t\}}] + O(t^2) \\ &= \varphi(x) e^{-q(x)t} + \mathbb{E}_x [\varphi(X_{T_1})] \mathbb{P}_x(T_1 \leq t) + O(t^2) \\ &= \varphi(x) e^{-q(x)t} + (1 - e^{-q(x)t}) \sum_{y \in E, y \neq x} \pi(x, y) \varphi(y) + O(t^2). \end{aligned}$$

Therefore,

$$L\varphi(x) = q(x) \sum_{y \in E, y \neq x} \pi(x, y) (\varphi(y) - \varphi(x)).$$

□

Theorem 7.5.3. *If $q(y) > 0$ for all y , then $T_1 < T_2 < \dots < \infty$ a.e.. Moreover, $(X_{T_n})_{n \geq 0}$ is a Markov chain with the transition matrix $\pi(x, y)$.*

Proof. First,

$$\begin{aligned}
\mathbb{P}_x(X_{T_1} = z_1, X_{T_2} = z_2) &= \mathbb{P}_x(X_{T_1} = z_1, X_{T_1+T_1 \circ \theta_{T_1}} = z_2) \\
&= \mathbb{P}_x(X_{T_1} = z_1, X_{T_1} \circ \theta_{T_1} = z_2) \\
&= \mathbb{P}_x(X_{T_1} = z_1, E_{X_{T_1}}[X_{T_1} = z_2]) \\
&= \mathbb{P}_x(X_{T_1} = z_1, E_{z_1}[X_{T_1} = z_2]) \\
&= \mathbb{P}_x(X_{T_1} = z_1) P_{z_1}(X_{T_1} = z_2) \\
&= \pi(x, z_1) \pi(z_1, z_2)
\end{aligned}$$

Then by induction, we have

$$P_x(X_{T_1} = z_1, X_{T_2} = z_2, \dots, X_{T_n} = z_n) = \pi(x, z_1) \pi(z_1, z_2) \dots \pi(z_{n-1}, z_n).$$

□

Lévy Process. Let $Y = (Y_t)_{t \geq 0}$ be a stochastic process such that

- (i) $Y_0 = 0$ a.e.
- (ii) for any $s \leq t$, $Y_t - Y_s$ is independent of $\sigma(Y_r : r \leq s)$,
- (iii) $Y_t \rightarrow 0$ in probability as $t \rightarrow 0$.

Then Y is called a Lévy process.

Theorem 7.5.4. *For $t \geq 0$, let $Q_t(x, dy)$ be the law of $Y_t + x$, i.e.,*

$$Q_t f(x) = \mathbb{E}[f(Y_t + x)].$$

$(Q_t)_{t \geq 0}$ is a Feller semigroup and Y is a Markov process associated with $(Q_t)_{t \geq 0}$.

Chapter 8

Stochastic Integral

8.1 Local Martingale

Definition 8.1.1 (Local Martingale). An $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ adapted stochastic process $X = (X_t)_{t \geq 0}$ is called a local martingale if there exists a sequence stopping times $\{T_n\}_{n \in \mathbb{N}}$ with $T_n \uparrow \infty$ such that $(X_{t \wedge T_n} \mathbb{I}_{\{T_n > 0\}})_{t \geq 0}$ is a UI martingale w.s.t. $\mathbb{F}$.

Remark 8.1.2. If $X_0 = 0$, we only need to require $(X_{t \wedge T_n})_{t \geq 0}$ is a UI martingale w.s.t. $\mathbb{F}$. Let $T_n = n$, clearly martingales are local martingales but the converse is not true.

Notation: Let $\mathcal{M}^{loc}$ be the set of all local martingales and $\mathcal{M}_0^{loc} \subset \mathcal{M}^{loc}$ be the set of all local martingales with $X_0 = 0$.

Lemma 8.1.3. *A local martingale with*

$$\sup_{s \leq t} |X_s| \in L^1$$

for any t is a martingale.

Proof. Assume $X_0 = 0$. Let $T_n \uparrow \infty$ be stopping times such that $(X_{t \wedge T_n})_{t \geq 0}$ be UI martingales. Then for $s \leq t$,

$$\mathbb{E}[X_{t \wedge T_n} \mid \mathcal{F}_s] = X_{s \wedge T_n}$$

By Martingale Convergence Theorem, as $n \rightarrow \infty$,

$$X_{t \wedge T_n} \rightarrow X_t, \quad X_{s \wedge T_n} \rightarrow X_s$$

a.e.. Because

$$\sup_n |X_{t \wedge T_n}| \leq \sup_{0 \leq r \leq t} |X_r| \in L^1,$$

by DCT,

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s.$$

□

Remark 8.1.4. In fact, from the proof, it is not difficult to see a local martingale $X = (X_t)_{t \geq 0}$ is a martingale if

$$\{X_T : T \in \mathcal{S}_t\},$$

is UI for all t , where $\mathcal{S}_t := \{T \text{ stopping time} : T \leq t\}$. Such condition is called class (DL).

Proposition 8.1.5. *Every nonnegative local martingale is a supermartingale.*

Proof. Assume $X_0 = 0$. Let $T_n \uparrow \infty$ be stopping times such that $(X_{t \wedge T_n})_{t \geq 0}$ be UI martingales. Then for $s \leq t$,

$$\begin{aligned} X_s &= \lim_{n \rightarrow \infty} X_{s \wedge T_n} = \lim_{n \rightarrow \infty} \mathbb{E}[X_{t \wedge T_n} \mid \mathcal{F}_s] \\ &\geq \mathbb{E}\left[\lim_{n \rightarrow \infty} X_{t \wedge T_n} \mid \mathcal{F}_s\right] = \mathbb{E}[X_t \mid \mathcal{F}_s] \end{aligned}$$

by Fatou's lemma. □

Definition 8.1.6. Let $\bar{\Omega} = [0, \infty) \times \Omega$.

- (1) A σ -field $\mathcal{P}$ over $\bar{\Omega}$ is called predictable if it is generated by all left-continuous and adapted process $X: [0, \infty) \times \Omega \rightarrow \mathbb{R}$.
- (2) A stochastic process $X = (X_t)_{t \geq 0}$ is called predictable if X is $\mathcal{P}$ -measurable on $\bar{\Omega}$.

Remark 8.1.7. (i) Every predictable process is adapted.

(ii) Every continuous and increasing process is predictable.

(iii) If filtration $\mathbb{F}$ satisfies the usual condition, every predictable process is adapted to $\mathbb{F}_- = (\mathcal{F}_{t-})_{t \geq 0}$.

Notation: Let $\mathcal{M}^2$ be the set of all càdlàg martingales $X = (X_t)_{t \geq 0}$ such that

$$\sup_{t \geq 0} \mathbb{E}[X_t^2] < \infty.$$

Let $\mathcal{M}_0^2 \subset \mathcal{M}^2$ be all $X \in \mathcal{M}^2$ with $X_0 = 0$. Let $\mathcal{M}_0^{2,c} \subset \mathcal{M}_0^2$ be all $X \in \mathcal{M}^2$ that is continuous.

Remark 8.1.8. Note that if $M \in \mathcal{M}^2$, then M is UI. So by convergence theorem, $M_\infty = \lim_t M_t$ in L^1 and $M_t = \mathbb{E}[M_\infty \mid \mathcal{F}_t]$.

Theorem 8.1.9 (Doob-Meyer Decomposition). *Let $X = (X_t)_{t \geq 0}$ be a right-continuous supermartingale. Assume*

$$\{X_T : T \in \mathcal{S}\},$$

is UI, where $\mathcal{S} := \{T \text{ stopping time} : T < \infty\}$ (X is called of class (D)). Then X admits a unique decomposition

$$X_t = X_0 + M_t - A_t,$$

where M is a right-continuous UI martingale with $M_0 = 0$ and A is a increasing and right-continuous and predictable process with $A_0 = 0$.

Remark 8.1.10. Note that any càdlàg martingale is of class (DL), but a càdlàg martingale is of class (D) if and only if it is UI. These results are also true for a càdlàg positive submartingale.

Corollary 8.1.11. *Let $M \in \mathcal{M}^2$. Then there exists a unique right-continuous predictable process $\langle M \rangle = (\langle M \rangle_t)_{t \geq 0}$ with $\langle M \rangle = 0$ such that $M^2 - \langle M \rangle$ is a martingale.*

Proof. M is a martingale so $-M^2$ is a supermartingale. To show of class (D), it suffices to show the UI of $(M_t^2)_{t \geq 0}$ because it is a positive submartingale. Because $M \in \mathcal{M}^2$, M is UI so $M_\infty = \lim_t M_t$ in L^1 . So $M_\infty^2 = \lim_t M_t^2$ in L^1 , which implies that M^2 is UI. Then by Doob-Meyer decomposition.

$$-M^2 = \text{martingale} - \langle M \rangle. \quad \square$$

Remark 8.1.12. Because $\langle M \rangle$ is uniquely determined by M , it is called the quadratic variation of M . And $\sup_t \mathbb{E}[\langle M \rangle_t] = \mathbb{E}[M_0^2] < \infty$.

Lemma 8.1.13. Let $M \in M^{2,c}$. For partition Π of $[0, t]$, we have

$$\lim_{\|\Pi\| \rightarrow 0} \sum_{t_k \in \Pi} |M_{t_k} - M_{t_{k-1}}|^2 = \langle M \rangle_t$$

in probability.

Remark 8.1.14. In fact, it is the general definition of quadratic variation.

Definition 8.1.15. For $M, N \in \mathcal{M}^2$, the process

$$\langle M, N \rangle = \frac{1}{4} (\langle M + N \rangle - \langle M - N \rangle).$$

is called the cross variation (or quadratic covariation) of M, N .

Remark 8.1.16. (i) Note that $\langle M, M \rangle_t = \langle M \rangle_t$.

(ii) By definition, it is not hard to see $MN - \langle M, N \rangle$ is a martingale.

(iii) If $M, N \in M^{2,loc}$ right-continuous, then there exists a unique increasing right-continuous predictable process $\langle M \rangle$ and there exists a unique right-continuous predictable process $\langle M, N \rangle$ of bounded variation such that

$$M^2 - \langle M \rangle, \quad MN - \langle M, N \rangle,$$

are local martingales.

(iv) Moreover,

$$\lim_{\|\Pi\| \rightarrow 0} \sum_{k=1}^m (M_{t_k} - M_{t_{k-1}}) (N_{t_k} - N_{t_{k-1}}) = \langle M, N \rangle_t.$$

Example 8.1.17 (Variation of Brownian Motion). Given a Brownian motion $B = (B_t)_{t \geq 0}$, there are two ways to calculate its quadratic variation.

(1) By directly calculating the quadratic total variation,

$$\langle B \rangle_t = \lim_{n \rightarrow \infty} \sum_{i=1}^{p_n} (B_{t_i^n} - B_{t_{i-1}^n})^2 = t.$$

(2) We already know $(B_t^2 - t)_{t \geq 0}$ is a martingale. Then, by Doob-Meyer decomposition, we directly have

$$\langle B \rangle_t = t.$$

So we usually denote

$$dB_t dB_t = t, \quad dB_t dt = dt dt = 0.$$

Example 8.1.18. If $\mathbf{B} = (B^1, \dots, B^d)$ is a d -dim Brownian motion, then

$$\langle B^i, B^j \rangle_t = \delta_{ij} t.$$

Given a partition Π of $[0, t]$ and let

$$S_{\Pi} = \sum_{k=1}^n (B_{t_k}^i - B_{t_{k-1}}^i) (B_{t_k}^j - B_{t_{k-1}}^j), \quad i \neq j.$$

Then $S_{\Pi} \rightarrow \langle B^i, B^j \rangle_t$ by definition. But by independence

$$\mathbb{E} [S_{\Pi}^2] = \sum_{k=1}^n \mathbb{E} [(\Delta_k B^i)^2] \mathbb{E} [(\Delta_k B^j)^2] = \sum_{k=1}^n (\Delta t_k)^2 \leq t \|\Pi\| \rightarrow 0.$$

Definition 8.1.19 (Semimartingale). A process $X = (X_t)_{t \geq 0}$ is called a semimartingale if

$$X_t = X_0 + M_t + A_t,$$

where $M \in \mathcal{M}_0^{loc}$ and A is adapted and càdlàg of bounded variation, i.e. there exists increasing adapted process A^+, A^- such that

$$A = A^+ - A^-.$$

Remark 8.1.20. This decomposition may be not unique. But if X is continuous, it is unique.

Lemma 8.1.21. *A continuous local martingale with bounded variation is constant a.e..*

Remark 8.1.22. A continuous non-constant local martingale is of unbounded variation. Therefore, we cannot use the usual Riemannian-Stieltjes to define the stochastic integral w.s.t. martingale.

8.2 Integral w.s.t. Martingale

For $M \in \mathcal{M}^2$, the goal is to define $\int_0^T H_t dM_t$. By employing the idea from Riemannian-Stieltjes

$$\int_0^T H_t dM_t := \lim_n \sum_{i=0}^n H_{\alpha_i} (M_{t_{i+1}} - M_{t_i}).$$

But because M is not of bounded variation, it is not well-defined. So there are three choices:

- (i) $\alpha_i = t_i$: Itô integral.
- (ii) $\alpha_i = (t_{i+1} - t_i)/2$: Stratonovich integral.
- (iii) $\alpha_i = t_{i+1}$: backward Itô integral.

In the following, we mainly consider the Itô integral.

Notation: Let $\mathcal{E}^b$ be the set of all bounded predictable simple process, i.e., if $H_t \in \mathcal{E}^b$, then

$$H_t(\omega) = \sum_{i=0}^{n-1} h^i(\omega) \mathbb{I}_{(t_i, t_{i+1}]}(t),$$

where h^i is bounded and $\mathcal{F}_{t_i}$ -measurable.

Integral of simple process.

Definition 8.2.1. For $H \in \mathcal{E}^b$ with $H_t(\omega) = \sum_{i=0}^{n-1} h^i(\omega) \mathbb{I}_{(t_i, t_{i+1}]}(t)$, the stochastic integral with respect to M is

$$(H \cdot M)_t = \int_0^t H_s dM_s = \sum_{i=0}^{n-1} h^i (M_{t_{i+1} \wedge t} - M_{t_i \wedge t}).$$

Lemma 8.2.2. *Let $H^1, H^2 \in \mathcal{E}^b$ and $c_1, c_2 \in \mathbb{R}$. Then $c_1 H^1 + c_2 H^2 \in \mathcal{E}^b$ and*

$$(c_1 H^1 + c_2 H^2) \cdot M = c_1 (H^1 \cdot M) + c_2 (H^2 \cdot M).$$

Proposition 8.2.3. *For $H \in \mathcal{E}^b$ and $M \in \mathcal{M}^2$, $H \cdot M \in \mathcal{M}_0^{2,c}$. Moreover,*

$$\mathbb{E} [(H \cdot M)_\infty^2] = \mathbb{E} \left[\left(\int_0^\infty H_u dM_u \right)^2 \right] = \mathbb{E} \left[\int_0^\infty H_u^2 d\langle M \rangle_u \right].$$

Proof. Let $s \leq t$. If $s = t_k$ and $t = t_\ell$ with $k < \ell$, then

$$\begin{aligned}\mathbb{E}[(H \cdot M)_t - (H \cdot M)_s \mid \mathcal{F}_s] &= \sum_{i=k}^{\ell-1} \mathbb{E}[h^i(M_{t_{i+1}} - M_{t_i}) \mid \mathcal{F}_{t_k}] \\ &= \sum_{i=k}^{\ell-1} \mathbb{E}[\mathbb{E}[h^i(M_{t_{i+1}} - M_{t_i}) \mid \mathcal{F}_{t_i}] \mid \mathcal{F}_{t_k}] \\ &= \sum_{i=k}^{\ell-1} \mathbb{E}[h^i \mathbb{E}[(M_{t_{i+1}} - M_{t_i}) \mid \mathcal{F}_{t_i}] \mid \mathcal{F}_{t_k}] \\ &= 0\end{aligned}$$

It is similar for any $s \leq t$. So $H \cdot M \in \mathcal{M}_0^2$. Next,

$$\begin{aligned}\mathbb{E}[(H \cdot M)_\infty^2] &= \mathbb{E}\left[\left(\sum_{i=0}^{n-1} h^i(M_{t_{i+1}} - M_{t_i})\right)^2\right] \\ &= \sum_{i=0}^{n-1} \mathbb{E}[(h^i)^2(M_{t_{i+1}} - M_{t_i})^2] + 2 \sum_{i < j} \mathbb{E}[h_i h_j (M_{t_{i+1}} - M_{t_i})(M_{t_{j+1}} - M_{t_j})].\end{aligned}$$

Note that

$$\begin{aligned}\mathbb{E}[h_i h_j (M_{t_{i+1}} - M_{t_i})(M_{t_{j+1}} - M_{t_j})] &= \mathbb{E}[\mathbb{E}[h_i h_j (M_{t_{i+1}} - M_{t_i})(M_{t_{j+1}} - M_{t_j}) \mid \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[h_i h_j (M_{t_{i+1}} - M_{t_i}) \mathbb{E}[(M_{t_{j+1}} - M_{t_j}) \mid \mathcal{F}_{t_j}]] \\ &= 0.\end{aligned}$$

So

$$\begin{aligned}\mathbb{E}[(H \cdot M)_\infty^2] &= \sum_{i=0}^{n-1} \mathbb{E}[(h^i)^2(M_{t_{i+1}} - M_{t_i})^2] \\ &= \sum_{i=0}^{n-1} \mathbb{E}[(h^i)^2 \mathbb{E}[(M_{t_{i+1}} - M_{t_i})^2 \mid \mathcal{F}_{t_i}]].\end{aligned}$$

Note that

$$\begin{aligned}\mathbb{E}[(M_{t_{i+1}} - M_{t_i})^2 \mid \mathcal{F}_{t_i}] &= \mathbb{E}[M_{t_{i+1}}^2 \mid \mathcal{F}_{t_i}] + M_{t_i}^2 - 2M_{t_i} \mathbb{E}[M_{t_i} \mid \mathcal{F}_{t_i}] \\ &= \mathbb{E}[M_{t_{i+1}}^2 - M_{t_i}^2 \mid \mathcal{F}_{t_i}].\end{aligned}$$

Because $M \in \mathcal{M}^2$, $M^2 - \langle M \rangle$ is a martingale, which implies that

$$\mathbb{E}[M_{t_{i+1}}^2 - M_{t_i}^2 \mid \mathcal{F}_{t_i}] = \mathbb{E}[\langle M \rangle_{t_{i+1}} - \langle M \rangle_{t_i} \mid \mathcal{F}_{t_i}].$$

Therefore,

$$\begin{aligned}\mathbb{E}[(H \cdot M)_\infty^2] &= \sum_{i=0}^{n-1} \mathbb{E}[(h^i)^2 \mathbb{E}[\langle M \rangle_{t_{i+1}} - \langle M \rangle_{t_i} \mid \mathcal{F}_{t_i}]] \\ &= \sum_{i=0}^{n-1} \mathbb{E}[(h^i)^2 (\langle M \rangle_{t_{i+1}} - \langle M \rangle_{t_i})]\end{aligned}$$

$$= \mathbb{E} \left[\int_0^\infty H_u^2 d\langle M \rangle_u \right].$$

Also because H is bounded and $\mathbb{E}[\langle M \rangle] < \infty$,

$$\mathbb{E}[(H \cdot M)_t^2] \leq \mathbb{E}[(H \cdot M)_\infty^2] < \infty.$$

□

We omit the proof of continuity.

Corollary 8.2.4. For $H \in \mathcal{E}^b$ and $B = (B_t)_{t \geq 0}$ a Brownian motion,

$$\mathbb{E} \left[\int_a^b H_u dB_u \right] := \mathbb{E}[(H \cdot M)_b] - \mathbb{E}[(H \cdot M)_a] = 0,$$

and

$$\mathbb{E} \left[\left(\int_a^b H_u dB_u \right)^2 \right] = \mathbb{E} \left[\int_a^b H_u^2 du \right].$$

Integral of L^2 integrable process.

Theorem 8.2.5. If $M \in \mathcal{M}_0^{2,c}$ and H is a progressively measurable process such that

$$\mathbb{E} \left[\int_0^T H_s^2 d\langle M \rangle_s \right] < \infty, \quad (8.1)$$

for all $T \geq 0$, then there exists a sequence of predictable simple processes $H^{(n)}$ such that

$$\sup_{T > 0} \lim_{n \rightarrow \infty} \mathbb{E} \left[\int_0^T |H_s^{(n)} - H_s|^2 d\langle M \rangle_s \right] = 0.$$

Remark 8.2.6. Note that because $\langle M \rangle$ is increasing, it can use Riemannian-Stieltjes to define

$$\int_0^T H_s^2 d\langle M \rangle_s.$$

Remark 8.2.7. For $M \in \mathcal{M}_0^{2,c}$ and any H satisfies above condition and a corresponding sequence of predictable simple processes $H^{(n)}$, because

$$\mathbb{E} \left[\left(\int_0^T H_s^{(n)} dM_s - \int_0^T H_s^{(m)} dM_s \right)^2 \right] = \mathbb{E} \left[\int_0^T |H_s^{(n)} - H_s^{(m)}|^2 d\langle M \rangle_s \right] \rightarrow 0,$$

due to $\int_0^T H_s^{(n)} d\langle M \rangle_s$ converges in L^2 . So $\int_0^T H_s^{(n)} dM_s$ is Cauchy in L^2 and it is convergent in L^2 .

Notation: For any $0 \leq T < \infty$, let $\mathcal{L}_T^*(M)$ be the set of all bounded progressively measurable process satisfying condition (8.1) and $\mathcal{L}^*(M) = \bigcap_{T \geq 0} \mathcal{L}_T^*$

Definition 8.2.8 (Stochastic Integral). For $H \in \mathcal{L}_T^*$, the stochastic integral w.s.t. $M \in \mathcal{M}^{2,c}$ is defined by

$$\int_0^T H_s dM_s = \lim_{n \rightarrow \infty} \int_0^T H_s^{(n)} dM_s,$$

where $H^{(n)}$ is a sequence satisfying (8.1) and the convergence is in L^2 .

Remark 8.2.9. (i) Note that the convergence is also in L^1 .

- (ii) This definition is well-defined, i.e., independent of the choice of $H^{(n)}$. If there is another $K^{(n)}$, then we can construct $Z^{(n)}$ such that $Z^{(2n)} = H^{(n)}$ and $Z^{(2n+1)} = K^{(n)}$. So

$$\sup_{T>0} \lim_{n \rightarrow \infty} \mathbb{E} \left[\int_0^T |Z_s^{(n)} - H_s|^2 d\langle M \rangle_s \right] = 0,$$

which implies that $\int_0^T Z_s^{(n)} dM_s$ is also Cauchy in L^2 and thus

$$\lim_{n \rightarrow \infty} \int_0^T Z_s^{(n)} dM_s = \lim_{n \rightarrow \infty} \int_0^T H_s^{(n)} dM_s = \lim_{n \rightarrow \infty} \int_0^T K_s^{(n)} dM_s.$$

- (iii) If $t \mapsto \langle M \rangle_t$ is absolutely continuous a.e., then $\int_0^T H_s dM_s$ is well-defined if H is bounded, measurable and $\mathbb{F}$ -adapted.

Proposition 8.2.10. *Let $M \in \mathcal{M}^{2,c}$ and $H, K \in \mathcal{L}_T^*(M)$ and $\alpha, \beta \in \mathbb{R}$.*

- (1) *Since H is $\mathbb{F}$ -adapted, $\left(\int_0^t H_u dM_u \right)_{0 \leq t \leq T} \in \mathcal{M}_0^{2,c}$*

- (2) *Linearity:*

$$\int_0^T \alpha H_u + \beta K_u dM_u = \alpha \int_0^T H_u dM_u + \beta \int_0^T K_u dM_u.$$

- (3) *Isometry:*

$$\mathbb{E} \left[\left| \int_0^T H_u dM_u \right|^2 \right] = \mathbb{E} \left[\int_0^T H_u^2 d\langle M \rangle_u \right].$$

- (4) *Moreover,*

$$\mathbb{E} \left[\left| \int_s^t H_u dM_u \right|^2 \mid \mathcal{F}_s \right] = \mathbb{E} \left[\int_s^t H_u^2 d\langle M \rangle_u \mid \mathcal{F}_s \right].$$

- (5)

$$\left\langle \int_0^\cdot H_u dM_u \right\rangle_t = \int_0^t H_u^2 d\langle M \rangle_u.$$

Proof. (1) Choose a simple process $H^{(n)}$ to approximate H . Because

$$\int_0^T H_s^{(n)} dM_s = \sum_{i=0}^{n-1} h^i (M_{T \wedge t_{i+1}} - M_{T \wedge t_i}) \in \mathcal{F}_T,$$

and

$$\int_0^T H_s dM_s = \lim_{n \rightarrow \infty} \int_0^T H_s^{(n)} dM_s,$$

it is $\mathcal{F}_T$ -measurable. Moreover, because above convergence is in L^2 (L^2 implies L^1), $\int_0^T H_s dM_s \in L^1$. For any $0 \leq s < t$ and any $A \in \mathcal{F}_s$, it suffice to prove

$$\mathbb{E} \left[\int_0^t H_u dM_u \mathbb{I}_A \right] = \mathbb{E} \left[\int_0^s H_u dM_u \mathbb{I}_A \right] \Leftrightarrow \mathbb{E} \left[\int_s^t H_u dM_u \mathbb{I}_A \right] = 0$$

Note that by L^1 -convergence,

$$\mathbb{E} \left[\int_s^t H_u dM_u \mathbb{I}_A \right] = \lim_{n \rightarrow \infty} \mathbb{E} \left[\int_s^t H_u^{(n)} dM_u \mathbb{I}_A \right].$$

But because $(\int_0^t H_u^{(n)} dM_u)$ is a martingale, $\mathbb{E} \left[\int_s^t H_u^{(n)} dM_u \mathbb{I}_A \right] = 0$. So

$$\mathbb{E} \left[\int_s^t H_u dM_u \mathbb{I}_A \right] = 0.$$

(2) It is directly obtained by the linearity of $\int_0^t H_u^{(n)} dM_u$ and L^1 -convergence.

(3) It is directly obtained by the same property of $\int_0^t H_u^{(n)} dM_u$ and also the L^1 -convergence.

(4) For $A \in \mathcal{F}_s$, because $\mathbb{I}_A^2 = \mathbb{I}_A$, by (3),

$$\mathbb{E} \left[\left| \int_s^t H_u dM_u \right|^2 \mathbb{I}_A \right] = \mathbb{E} \left[\left| \int_s^t H_u \mathbb{I}_A dM_u \right|^2 \right] = \mathbb{E} \left[\int_s^t H_u^2 d\langle M \rangle_u \mathbb{I}_A \right].$$

(5) For $0 \leq s < t$, by (1), $(\int_0^t H_u dM_u)_{t \geq 0}$ is a martingale. So by (4),

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^t H_u dM_u \right)^2 - \left(\int_0^s H_u dM_u \right)^2 \mid \mathcal{F}_s \right] &= \mathbb{E} \left[\left(\int_0^t H_u dM_u - \int_0^s H_u dM_u \right)^2 \mid \mathcal{F}_s \right] \\ &= \mathbb{E} \left[\int_s^t H_u^2 d\langle M \rangle_u \mid \mathcal{F}_s \right]. \end{aligned}$$

It follows that

$$\mathbb{E} \left[\left(\int_0^t H_u dM_u \right)^2 - \int_0^t H_u^2 d\langle M \rangle_u \mid \mathcal{F}_s \right] = \left(\int_0^s H_u dM_u \right)^2 - \int_0^s H_u^2 d\langle M \rangle_u.$$

Because $\int_0^t H_u^2 d\langle M \rangle_u$ is an increasing process and $\left(\int_0^t H_u dM_u \right)^2 - \int_0^t H_u^2 d\langle M \rangle_u$ is a martingale, by the uniqueness of Doob-Meyer decomposition,

$$\left\langle \int_0^\cdot H_u dM_u \right\rangle_t = \int_0^t H_u^2 d\langle M \rangle_u. \quad \square$$

Remark 8.2.11. On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, if $\mathcal{G} \subset \mathcal{F}$, then by Jensen's inequality,

$$|\mathbb{E}[X \mid \mathcal{G}]|^p \leq \mathbb{E}[|X|^p \mid \mathcal{G}] \Rightarrow \|\mathbb{E}[X \mid \mathcal{G}]\|_p \leq \|X\|_p$$

for $1 \leq p < \infty$, which is also true for $p = \infty$. So for $\mathcal{G}$ -measurable $X_n \rightarrow X$ in $L^p(\mathcal{F})$, by

$$\|\mathbb{E}[X \mid \mathcal{G}] - X\|_p \leq \|\mathbb{E}[X - X_n \mid \mathcal{G}]\|_p + \|X_n - X\|_p \leq 2\|X_n - X\|_p \rightarrow 0,$$

$X = \mathbb{E}[X \mid \mathcal{G}]$ is $\mathcal{G}$ -measurable.

Corollary 8.2.12. Consider a Brownian motion B and $H \in \mathcal{L}^*(B)$,

$$\begin{aligned} \mathbb{E} \left[\int_s^t H_u dM_u \mid \mathcal{F}_s \right] &= 0, \\ \mathbb{E} \left[\left(\int_s^t H_u dM_u \right)^2 \mid \mathcal{F}_s \right] &= \mathbb{E} \left[\int_s^t H_u^2 du \mid \mathcal{F}_s \right] = \int_s^t \mathbb{E} [H_u^2] du \end{aligned}$$

Theorem 8.2.13. Let $M, N \in \mathcal{M}^{2,c}$ and $H \in \mathcal{L}^*(M)$ and $K \in \mathcal{L}^*(N)$.

(1) For stopping times $S \leq T$,

$$\mathbb{E} \left[\int_0^{t \wedge T} H_u dM_u \mid \mathcal{F}_S \right] = \int_0^{t \wedge S} H_u dM_u.$$

(2) For a stopping time T ,

$$\int_0^{t \wedge T} H_u dM_u = \int_0^t H_u \mathbb{I}_{[0, T]} dM_u = \int_0^t H_u dM_{u \wedge T}.$$

(3) For stopping times $S \leq T$,

$$\mathbb{E} \left[\left(\int_{t \wedge S}^{t \wedge T} H_u dM_u \right) \left(\int_{t \wedge S}^{t \wedge T} K_u dN_u \right) \mid \mathcal{F}_S \right] = \mathbb{E} \left[\left(\int_{t \wedge S}^{t \wedge T} H_u K_u d\langle M, N \rangle_u \right) \mid \mathcal{F}_S \right].$$

In particular, if S, T are constant

$$\mathbb{E} \left[\left(\int_s^t H_u dM_u \right) \left(\int_s^t K_u dN_u \right) \mid \mathcal{F}_s \right] = \mathbb{E} \left[\left(\int_s^t H_u K_u d\langle M, N \rangle_u \right) \mid \mathcal{F}_s \right].$$

Moreover, it follows that

$$\left\langle \int_0^\cdot H_u dM_u, \int_0^\cdot K_u dN_u \right\rangle_t = \int_0^t H_u K_u d\langle M, N \rangle_u$$

In particular,

$$\left\langle \int_0^\cdot H_u dM_u, N \right\rangle_t = \int_0^t H_u d\langle M, N \rangle_u$$

(4) If $G \in \mathcal{L}^*(\int_0^\cdot H_u dM_u)$, then $GH \in \mathcal{L}^*(M)$ and

$$\int_0^t G_s d\left(\int_0^s H_u dM_u\right) = \int_0^t G_u H_u dM_u.$$

Proposition 8.2.14 (Kunita-Watanabe). Let $M, N \in \mathcal{M}_0^{2,c}$ and $H \in \mathcal{L}^*(M)$ and $K \in \mathcal{L}^*(N)$. Then

$$\int_0^t |H_u K_u| d\langle M, N \rangle_u \leq \left(\int_0^t H_u^2 d\langle M \rangle_u \right)^{\frac{1}{2}} \left(\int_0^t K_u^2 d\langle N \rangle_u \right)^{\frac{1}{2}}.$$

Remark 8.2.15. Condition (8.1) can be weaker as

$$\mathbb{P} \left(\int_0^T H_u^2 d\langle M \rangle_u < \infty \right) = 1,$$

but the convergence

$$\int_0^T H_u dM_u = \lim_{n \rightarrow \infty} \int_0^T H_u^{(n)} dM_u$$

is weaker to in probability. In such case, $\left(\int_0^t H_u dM_u \right)_{t \geq 0}$ is not a martingale, but a local martingale.

8.3 Integral w.s.t. Local (Semi) Martingale

Local martingale.

Definition 8.3.1. For $M \in \mathcal{M}^{c,loc}$ and $X \in \mathcal{L}^*(M)$, i.e.

$$\mathbb{E} \left[\int_0^T X_u^2 d\langle M \rangle_u \right] < \infty, \quad \forall T,$$

the stochastic integral of X w.s.t. M is defined by

$$\int_0^t X_s dM_s := \int_0^t X_s \mathbb{I}_{\{T_n \geq s\}} dM_{s \wedge T_n}$$

on $\{0 \leq t \leq T_n\}$, where $T_n \uparrow \infty$ is the sequence such that $(M_{t \wedge T_n})_{t \geq 0}$ is a UI martingale.

Remark 8.3.2. Here we do not need the condition of L^2 -integrability for martingale because such T_n can be chosen such that $(M_{t \wedge T_n})_{t \geq 0}$ is L^2 -integrable.

Theorem 8.3.3. Let $M \in \mathcal{M}^{c,loc}$ and $X, Y \in \mathcal{L}^*(M)$.

(1) $\left(\int_0^t X_u dM_u \right)$ is a continuous local martingale, i.e., in $\mathcal{M}_0^{c,loc}$.

(2) *Linearity:*

$$\int_0^t (\alpha X_s + \beta Y_s) dM_s = \alpha \int_0^t X_s dM_s + \beta \int_0^t Y_s dM_s.$$

(3) *Quadratic variation:*

$$\left\langle \int_0^t X_s dM_s \right\rangle_t = \int_0^t X_s^2 d\langle M \rangle_s.$$

(4) *For stopping time T ,*

$$\int_0^{t \wedge T} X_s dM_s = \int_0^t X_s I_{\{s \leq T\}} dM_s.$$

Remark 8.3.4. Note that for the properties related to expectation cannot be extended to local martingale, like

$$\mathbb{E} \left[\left(\int_0^t X_u dM_u \right)^2 \right] \neq \mathbb{E} \left[\int_0^t X_u^2 d\langle M \rangle_u \right], \quad \mathbb{E} \left[\left(\int_s^t X_u dM_u \right)^2 \mid \mathcal{F}_s \right] \neq \mathbb{E} \left[\int_s^t X_u^2 d\langle M \rangle_u \mid \mathcal{F}_s \right],$$

in general.

Semimartingale. Recall $X = (X_t)_{t \geq 0}$ is a semimartingale if

$$X_t = X_0 + M_t^X + A_t^X,$$

where $M^X = (M_t^X)_{t \geq 0}$ is a local martingale with $M_0^X = 0$ and $A^X = (A_t^X)_{t \geq 0}$ is a càdlàg, adapted process of bounded variation. Note that this decomposition is not unique unless X is continuous. In the following, we consider continuous X .

Definition 8.3.5. Let X be a continuous semimartingale. For $H \in \mathcal{L}^*(M^X)$, define

$$\int_0^t H_s dX_s = \int_0^t H_s dM_s^X + \int_0^t H_s dA_s^X,$$

where the second integral is the Riemannian-Stieltjes integral.

Next, we need to define the quadratic variation for general case.

Definition 8.3.6. Let X, Y be semimartingales.

(1) The quadratic variation of X is defined as

$$[X, X]_t := \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi_n, t_i \leq t} (X_{t_{i+1} \wedge t} - X_{t_i \wedge t})^2.$$

(2) The cross variation of X, Y is

$$[X, Y]_t = \frac{1}{4} ([X + Y, X + Y]_t - [X - Y, X - Y]_t).$$

Remark 8.3.7. If X, Y are two continuous local martingale, then $[X, X]_t = \langle X \rangle_t$ and $[X, Y]_t = \langle X, Y \rangle_t$.

Theorem 8.3.8. If X, Y are semimartingales and let $M^{X,c}, M^{Y,c}$ be their continuous local martingale parts, then

$$[X, Y]_t = \langle M^{X,c}, M^{Y,c} \rangle_t + \sum_{s \leq t} \Delta X_s \Delta Y_s,$$

where $\Delta X_s = X_s - X_{s-}$.

Remark 8.3.9. In particular, if X, Y are continuous semimartingales, then

$$[X, Y]_t = \langle M^X, M^Y \rangle_t.$$

Corollary 8.3.10. If X, Y are continuous semimartingale and $Y \in \mathcal{L}^*(M^X)$, then

$$\left[\int_0^\cdot H_s dX_s, Y \right]_t = \int_0^t H_s d[X, Y]_s.$$

Theorem 8.3.11 (DCT). Let X be a continuous semimartingale with the decomposition $X_t = X_0 + M_t + A_t$. Let $H^{(n)}$ and H be locally bounded progressive processes, and let K be a nonnegative progressive process. If

(i) $H_s^{(n)} \rightarrow H_s$ a.e. for any $s \in [0, t]$,

(ii) $|H_s^{(n)}| \leq K_s$ a.e. for any n and $s \in [0, t]$,

(iii) $K_s \in \mathcal{L}^*$ and $\int_0^t |K_s| |dA_s| < \infty$,

then

$$\int_0^t H_s^{(n)} dX_s \rightarrow \int_0^t H_s dX_s.$$

8.4 Itô Formula

Theorem 8.4.1 (1-dim, Continuous Form). Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a C^2 -function and $X = (X_t)_{t \geq 0}$ be a continuous semimartingale with the decomposition $X_t = X_0 + M_t + A_t$. Then

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t f'(X_u) dX_u + \frac{1}{2} \int_0^t f''(X_u) d[X, X]_u \\ &= f(X_0) + \int_0^t f'(X_u) dM_u + \int_0^t f'(X_u) dA_u + \frac{1}{2} \int_0^t f''(X_u) d\langle M \rangle_u. \end{aligned}$$

Sketch of Proof. By Taylor formula,

$$f(X_{t_{i+1} \wedge t}) - f(X_{t_i}) = f'(X_{t_i})\Delta_i X + \frac{1}{2}f''(X_{t_i})(\Delta_i X)^2 + R_i.$$

By taking the summation,

$$f(X_t) - f(X_0) = \sum_i f'(X_{t_i})\Delta_i X + \frac{1}{2} \sum_i f''(X_{t_i})(\Delta_i X)^2 + \sum_i R_i.$$

By the definition of integral,

$$\sum_i f'(X_{t_i})\Delta_i X \rightarrow \int_0^t f'(X_u) dX_u.$$

By the definition of quadratic variation,

$$\sum_i f''(X_{t_i})(\Delta_i X)^2 \rightarrow \int_0^t f''(X_u) d[X, X]_u.$$

For R_i , because $f \in C^2$

$$\left| \sum_i R_i \right| \leq \frac{1}{2} \sum_i |f''(\xi_i) - f''(X_{t_i})| |\Delta_i X| \leq \frac{1}{2} \sum_i |\Delta_i X|^2 \rightarrow 0. \quad \square$$

Remark 8.4.2. (i) Note that $\int_0^t f'(X_0) dM_u$ is a continuous local martingale, and $\int_0^t f'(X_0) dA_u + \int_0^t f''(X_u) d\langle M \rangle_u$ is of bounded variation (because integral of bounded variation is still of bounded variation). So $f(X_t)$ is also a continuous semi-martingale.

(ii) It has the differential form

$$\begin{aligned} df(X_t) &= f'(X_t) dX_t + \frac{1}{2} f''(X_t) d[X, X]_t \\ &= f'(X_t) dM_t + f'(X_t) dA_t + \frac{1}{2} f''(X_t) d\langle X \rangle_t, \end{aligned}$$

which is just a formula.

(iii) If X is a continuous semimartingale, we denote $(dX_t)^2 = d[X, X]_t$. Therefore, the different form becomes

$$df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) (dX_t)^2.$$

In particular, for $X = B$ a Brownian motion, $(dB_t)^2 = dt$. For example, if $dX_t = f dB_t + g dt$, then

$$(dX_t)^2 = f^2 (dB_t)^2 + 2fg dB_t dt + g^2 (dt)^2 = f^2 (dB_t)^2 = f^2 dt.$$

Example 8.4.3. (1) Let $f(x) = x^2$ and $X = B$ a Brownian motion. Then

$$B_t^2 = B_0^2 + 2 \int_0^t B_s dB_s + \int_0^t ds \Rightarrow \int_0^t B_s dB_s = \frac{1}{2} (B_t^2 - t).$$

(2) Let W be a Brownian motion and $X \in \mathcal{L}^*(W)$. Consider the process

$$Z_t = \exp \left(\int_0^t X_u dW_u - \frac{1}{2} \int_0^t X_u^2 du \right).$$

Let

$$Y_t = \int_0^t X_u dW_u - \frac{1}{2} \int_0^t X_u^2 du,$$

which is a semimartingale, or informally, $dY_t = X_t dW_t - \frac{1}{2} X_t^2 dt$. Let $f(x) = e^x$. Then

$$\begin{aligned} dZ_t &= f'(Y_t) dY_t + \frac{1}{2} f''(Y_t) (dY_t)^2 \\ &= Z_t \left(X_t dW_t - \frac{1}{2} X_t^2 dt \right) + \frac{1}{2} Z_t X_t^2 dt \\ &= Z_t X_t dW_t. \end{aligned}$$

Therefore,

$$Z_t = Z_0 + \int_0^t Z_u X_u dW_u = 1 + \int_0^t Z_u X_u dW_u$$

Moreover, $Z_t = \exp \left(\int_0^t X_u dW_u - \frac{1}{2} \int_0^t X_u^2 du \right)$ is a solution of SDE

$$dZ_t = Z_t X_t dW_t.$$

In particular, if $X_t \equiv \sigma$, then

$$Z_t = \exp \left(\sigma W_t - \frac{1}{2} \sigma^2 t \right)$$

is a solution of $dZ_t = \sigma Z_t dW_t$.

Remark 8.4.4. Note that Z is a continuous local martingale. In fact, for X with $\mathbb{P}(\int_0^T X_u^2 du < \infty) = 1$, it is also a local martingale.

Theorem 8.4.5 (Multi-dim, Local Martingale, Continuous Form). *Let $\mathbf{X} = (X^1, \dots, X^n)$ be a vector of continuous local martingales. Let $f: [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}$ be $C^{1,2}$.*

$$\begin{aligned} f(\mathbf{X}_t) &= f(0, \mathbf{X}_0) + \int_0^t \frac{\partial}{\partial t} f(s, \mathbf{X}_s) dt + \sum_{i=1}^n \int_0^t \frac{\partial}{\partial x_i} f(s, \mathbf{X}_s) dX_s^i \\ &\quad + \frac{1}{2} \sum_{i,j=1}^n \int_0^t \frac{\partial^2}{\partial x_i \partial x_j} f(s, \mathbf{X}_s) d\langle X^i, X^j \rangle_s. \end{aligned}$$

or in differential form

$$df(X_t) = \frac{\partial}{\partial t} f(t, \mathbf{X}_t) dt + \sum_{i=1}^n \frac{\partial}{\partial x_i} f(t, \mathbf{X}_t) dX_t^i + \sum_{i,j=1}^n \frac{\partial^2}{\partial x_i \partial x_j} f(t, \mathbf{X}_t) d\langle X^i, X^j \rangle_t.$$

Example 8.4.6. Let $\mathbf{W} = (W^1, \dots, W^n)$ be a n -dimensional standard Brownian motion with $n \geq 2$. Let

$$R_t = \|\mathbf{W}\| = \sqrt{\sum_{i=1}^n (W^i)^2 + \dots + (W^n)^2},$$

called the Bessel process. Let $f(\mathbf{x}) = \|\mathbf{x}\|$. Then $\frac{\partial}{\partial x_i} f(\mathbf{x}) = x_i/f(\mathbf{x})$ and

$$\frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{x}) = \begin{cases} -\frac{x_i x_j}{f(\mathbf{x})^3}, & i \neq j \\ \frac{f(\mathbf{x})^2 - x_i^2}{f(\mathbf{x})^3}, & i = j. \end{cases}$$

Then we have

$$\begin{aligned} dR_t &= \sum_{i=1}^n \frac{\partial}{\partial x_i} f(\mathbf{W}_t) dW_t^i = \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{W}_t) d\langle B^i, B^j \rangle_t \\ &= \sum_{i=1}^n \frac{W_t^i}{R_t} dW_t^i + \frac{1}{2} \sum_{i=1}^n \frac{R_t^2 - (W_t^i)^2}{R_t^3} dt \\ &= \sum_{i=1}^n \frac{W_t^i}{R_t} dW_t^i + \frac{1}{2} \frac{n-1}{R_t} dt. \end{aligned}$$

Therefore,

$$R_t dR_t = \sum_{i=1}^n W_t^i dW_t^i + \frac{n-1}{2} dt.$$

Theorem 8.4.7 (Itô Formula). *Let $\mathbf{X} = (X^1, \dots, X^n)$ be a n -dim semimartingale with decomposition*

$$X_t^i = X_0^i + M_t^i + A_t^i, \quad i = 1, \dots, n$$

(Note that because the decomposition is not unique, it should be given.) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 . Then $f(\mathbf{X})$ is a semimartingale and

$$\begin{aligned} f(\mathbf{X}_t) &= f(\mathbf{X}_0) + \sum_{i=1}^n \int_0^t \frac{\partial}{\partial x_i} f(\mathbf{X}_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^n \int_0^t \frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{X}_s) d\langle M^{i,c}, M^{j,c} \rangle_s \\ &\quad + \sum_{s \leq t} \left(f(\mathbf{X}_s) - f(\mathbf{X}_{s-}) - \sum_{i=1}^n \frac{\partial}{\partial x_i} f(\mathbf{X}_{s-}) \Delta X_{s-}^i \right). \end{aligned}$$

Application: Integration by parts.

Example 8.4.8. Given a standard Brownian motion W . Consider $\int_0^t s dW_s$. Let $f(t, x) = tx$. Then

$$\frac{\partial}{\partial t} f(t, x) = x, \quad \frac{\partial}{\partial x} f(t, x) = t, \quad \frac{\partial^2}{\partial t \partial x} f(t, x) = 0.$$

So

$$tW_t = \int_0^t W_s ds + \int_0^t s dW_s \Rightarrow \int_0^t s dW_s = tW_t - \int_0^t W_s ds.$$

Theorem 8.4.9. *Let $f(s, \omega)$ is continuous of bounded variation w.s.t. for a.e. ω . Then*

$$\int_0^t f(s) dW_s = f(t)W_t - \int_0^t W_s df(s).$$

Theorem 8.4.10 (Integration by parts). *Suppose X, Y are continuous semimartingale, then*

$$\int_0^t X_s dY_s = X_t Y_t - X_0 Y_0 + \int_0^t Y_s dX_s - [X, Y]_t.$$

on informally,

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t.$$

Proof. It can be obtained by Itô formula on $f(x, y) = xy$. □

Remark 8.4.11. In general, it can be written as

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + (dX_t)(dY_t),$$

where X_t, Y_t can be continuous semimartingales ($M_t = 0$, i.e., as above theorem, or $M_t \neq 0$) or deterministic and for calculating $(dX_t)(dY_t)$, we can use

$$(dB_t)^2 = 0, \quad dt dB_t = (dt)^2 = 0.$$

Remark 8.4.12. If X, Y are semimartingale,

$$X_t Y_t = X_0 Y_0 + \int_0^t X_{u-} dY_u + \int_0^t Y_{u-} dX_u + [X, Y]_t.$$

In particular,

$$X_t^2 = X_0^2 + 2 \int_0^t X_{u-} dX_u + [X, X]_t.$$

8.5 Martingale Representation Theorem

The problem is whether a martingale can be represented as

$$M_t = M_0 + \int_0^t H_s dB_s.$$

Example 8.5.1. Let W^1, W^2 be two independent Brownian motions and $\mathcal{F}_t = \sigma(W_s^1, W_s^2: s \leq t)$. Then W^1, W^2 are two martingales w.s.t. $\mathbb{F} = (\mathcal{F}_t)$. If

$$W_t^2 = \int_0^t H_s dW_s^1$$

for some $H \in \mathcal{L}^*(W^1)$, then

$$t = \langle W^2, W^2 \rangle_t = \left\langle \int_0^t H_s dW_s^1, W^2 \right\rangle_t = \int_0^t H_s d\langle W^1, W^2 \rangle_s = 0,$$

which induces a contradiction.

Theorem 8.5.2 (Martingale Representation Theorem). *Let $\mathbf{B}$ be a n -dimensional Brownian motion w.s.t. its natural filtration $\mathbb{F}^{\mathbf{B}}$. Let M be a martingale w.s.t. $\mathbb{F}$ and $\mathbb{F}^{\mathbf{B}}$ that is in $\mathcal{M}^2$ and càdlàg. Then there exist $H^i \in \mathcal{L}^*$ for all i such that*

$$M_t = M_0 + \sum_{i=1}^n \int_0^t H_s^i dB_s^i.$$

Remark 8.5.3. Note that $\mathcal{F}_t^{\mathbf{B}} = \sigma(B_s^1, \dots, B_s^n: s \leq t)$.

8.6 Girsanov Theorem

Fix $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ satisfying the usual condition. Let $W = (W_t)_{t \geq 0}$ be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$. Denote $\mathbb{E} = \mathbb{E}_{\mathbb{P}}$.

Example 8.6.1. Let $Z_1, \dots, Z_n \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$ on $(\Omega, \mathcal{F}, \mathbb{P})$. Let

$$d\tilde{\mathbb{P}}(\omega) = \exp \left(\sum_{i=1}^n \mu_i Z_i - \frac{1}{2} \sum_{i=1}^n \mu_i^2 \right) d\mathbb{P}(\omega).$$

Denote $\tilde{\mathbb{E}} = \mathbb{E}_{\tilde{\mathbb{P}}}$. Consider the characteristic equation,

$$\begin{aligned} \tilde{\mathbb{E}} [\exp(it_1 Z_1 + \dots + it_n Z_n)] &= \int_{\Omega} \exp(it_1 Z_1 + \dots + it_n Z_n) d\tilde{\mathbb{P}} \\ &= \int_{\Omega} \exp(it_1 Z_1 + \dots + it_n Z_n) \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} d\mathbb{P} \\ &= \int_{\Omega} \exp \left(\sum_{j=1}^n ((it_j + \mu_j) Z_j - \frac{1}{2} \mu_j^2) \right) d\mathbb{P} \\ &= \mathbb{E} \left[\prod_{j=1}^n \exp \left((it_j + \mu_j) Z_j - \frac{1}{2} \mu_j^2 \right) \right] \\ &= \prod_{j=1}^n \mathbb{E} \left[\exp \left((it_j + \mu_j) Z_j - \frac{1}{2} \mu_j^2 \right) \right] \\ &= \prod_{j=1}^n \exp \left(-\frac{t_j^2}{2} + it_j \mu_j \right). \end{aligned}$$

It follows that $Z_1, \dots, Z_n \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu_i, 1)$ on $\tilde{\mathbb{P}}$.

Let X be a measurable, $\mathbb{F}$ -adapted stochastic process such that

$$\mathbb{P} \left(\int_0^T X_u^2 du < \infty \right) = 1, \quad \forall T \geq 0.$$

Define

$$Z_t := \exp \left(\int_0^t X_u dW_u - \frac{1}{2} \int_0^t X_u^2 du \right).$$

Then by above

$$Z_t = 1 + \int_0^t Z_u X_u dW_u,$$

$Z = (Z_t)_{t \geq 0}$ is a continuous local martingale.

Proposition 8.6.2 (Novikov Condition). *If X be a measurable, $\mathbb{F}$ -adapted stochastic process such that*

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T X_u^2 du \right) \right] < \infty, \quad \forall 0 \leq T < \infty,$$

then Z is a martingale.

Remark 8.6.3. Note that by Jensen's inequality, if X satisfies Novikov condition, it is in $\mathcal{L}^*(W)$. In fact, Novikov Condition is equivalent to $\mathbb{E}[Z_t] = 1$ for all t .

Define $\tilde{\mathbb{P}}_t$ on $\mathcal{F}_t$ by,

$$\tilde{\mathbb{P}}_t(A) := \int_A Z_t d\mathbb{P}, \quad \forall A \in \mathcal{F}_t, \Rightarrow Z_t = \frac{d\tilde{\mathbb{P}}_t}{d\mathbb{P}}.$$

Remark 8.6.4. If $Z = (Z_t)_{t \geq 0}$ is a martingale, then

(i) $\tilde{\mathbb{P}}_t$ is a probability measure because $\mathbb{E}[Z_t] = \mathbb{E}[Z_0] = 1$.

(ii) for any $s \leq t$ and any $A \in \mathcal{F}_s$,

$$\tilde{\mathbb{P}}_s(A) = \tilde{\mathbb{P}}_t(A)$$

because of the martingale property of Z .

Theorem 8.6.5 (Girsanov Theorem). *Assume that $Z = (Z_t)_{t \geq 0}$ defined as above is a martingale. Define a process $\tilde{W}$ as*

$$\tilde{W}_t = W_t - \int_0^t X_u du.$$

Then for each fixed $T \in [0, \infty)$, $(\tilde{W}_t)_{t \in [0, T]}$ is a Brownian motion on $(\Omega, \mathcal{F}_T, \tilde{P}_T)$.

Corollary 8.6.6. *Under the same assumption of above theorem, suppose f is a measurable function such that $f(W_t) \in L^1$. Then*

$$\mathbb{E}_{\mathbb{Q}}[f(\tilde{W}_t)] = \mathbb{E}[f(W_t)].$$

Example 8.6.7. (1) Suppose $X_t = h(t)$, a deterministic function, that is in $L^2([0, T])$. Since

$$\mathbb{E} \left[\frac{1}{2} \exp \left(\int_0^T h^2(u) du \right) \right] = \frac{1}{2} \exp \left(\int_0^T h^2(u) du \right) < \infty,$$

by above theorem

$$Z_t = \exp \left(\int_0^t h(u) dW_u - \frac{1}{2} \int_0^t h^2(u) du \right)$$

is a martingale. So

$$\tilde{W}_t = W_t - \int_0^t h(u) du$$

is a Brownian motion w.s.t. $\tilde{\mathbb{P}}$ defined as,

$$d\tilde{\mathbb{P}} = \exp \left(\int_0^T h(u) dW_u - \frac{1}{2} \int_0^T h^2(u) du \right) d\mathbb{P}.$$

(2) Consider $X_t = \text{sign}(W_t)$ (i.e. $X_t = 1$ for $W_t \geq 0$ and $X_t = -1$ for $W_t < 0$).

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T X_u^2 du \right) \right] = \exp \left(\frac{T}{2} \right) < \infty.$$

So

$$\tilde{W}_t = W_t - \int_0^t \text{sign}(W_u) du$$

is a Brownian motion w.s.t. $\mathbb{Q}$ given by

$$d\mathbb{Q} = \exp \left(\int_0^T \text{sign}(W_u) du - \frac{T}{2} \right) d\mathbb{P}.$$

Remark 8.6.8. Suppose $Y = (Y_t)_{t \geq 0}$ is a martingale w.s.t. to $d\mathbb{Q} = Z_T d\mathbb{P}$, i.e.,

$$Y_s = \mathbb{E}_{\mathbb{Q}}[Y_t | \mathcal{F}_s] = \frac{\mathbb{E}[Y_t Z_T | \mathcal{F}_s]}{\mathbb{E}[Z_T | \mathcal{F}_s]} = \frac{\mathbb{E}[Y_t Z_T | \mathcal{F}_s]}{Z_s}.$$

In particular,

$$\mathbb{E}[Y_T Z_T | \mathcal{F}_s] = Y_s Z_s.$$

When considering continuous process $X = (X_t)_{t \in [0, T]}$ on some probability space equipping with $\mathbb{P}$, it is equivalent to consider its distribution $\mathbb{P}_X = X_{\#}\mathbb{P}$ on canonical path space $C([0, T], \mathbb{R}^n)$. The Girsanov theorem can be applied on $C([0, T], \mathbb{R}^n)$; see Liptser et al. [2013](#). But it only considers fixed $X_0 = 0$. For general X_0 , it can be extended.

Proposition 8.6.9. *Let X, Y be two continuous processes on probability space equipping with $\mathbb{P}$. Let $\mathbb{P}_X = X_{\#}\mathbb{P}$ and $\mathbb{P}_Y = Y_{\#}\mathbb{P}$ on $C([0, T], \mathbb{R}^n)$ be their distributions and assume $\mathbb{P}_X \ll \mathbb{P}_Y$. Moreover, let $\mathbb{P}_X^0, \mathbb{P}_Y^0$ be the corresponding distributions with fixing $X_0 = 0$. Let $\mathbb{P}_0 = (X_0)_{\#}\mathbb{P}, \mathbb{Q}_0 = (Y_0)_{\#}\mathbb{P}$ on $\mathbb{R}^n$. Then*

$$\frac{d\mathbb{P}_X}{d\mathbb{P}_Y}(\omega) = \frac{d\mathbb{P}_0}{d\mathbb{Q}_0}(\omega(0)) \cdot \frac{d\mathbb{P}_X^0}{d\mathbb{P}_Y^0}(\omega).$$

Proof. Let $\mathbb{P}_X^x = X_{\#}^x \mathbb{P}$ and $\mathbb{P}_Y^x = Y_{\#}^x \mathbb{P}$. Let

$$\frac{d\mathbb{P}_X^0}{d\mathbb{P}_Y^0}(\omega) = K(\omega), \quad L(x) \frac{d\mathbb{P}_0}{d\mathbb{Q}_0}(x).$$

Then for

$$R(\omega) = \frac{d\mathbb{P}_0}{d\mathbb{Q}_0}(\omega(0)) \cdot \frac{d\mathbb{P}_X^0}{d\mathbb{P}_Y^0}(\omega),$$

we have

$$\begin{aligned} \int_A R(\omega) \mathbb{P}_Y(d\omega) &= \int_{\mathbb{R}^n} \int_A L(\omega(0)) K(\omega) \mathbb{P}_Y^x(d\omega) \mathbb{Q}_0(dx) \\ &= \int_{\mathbb{R}^n} L(x) \int_A K(\omega) \mathbb{P}_Y^x(d\omega) \mathbb{Q}_0(dx) \quad (\text{since } \omega(0) = x \mathbb{P}_Y^x\text{-a.s.}) \\ &= \int_{\mathbb{R}^n} L(x) \mathbb{P}_X^x(A) \mathbb{Q}_0(dx) \quad (\text{because } K = d\mathbb{P}_X^x / d\mathbb{P}_Y^x) \\ &= \int_{\mathbb{R}^n} \mathbb{P}_X^x(A) \mathbb{P}_0(dx) \quad (\text{since } L = d\mathbb{P}_0 / d\mathbb{Q}_0) \\ &= \mathbb{P}_X(A). \end{aligned}$$

Therefore,

$$\frac{d\mathbb{P}_X}{d\mathbb{P}_Y}(\omega) = \frac{d\mathbb{P}_0}{d\mathbb{Q}_0}(\omega(0)) \cdot \frac{d\mathbb{P}_X^0}{d\mathbb{P}_Y^0}(\omega). \quad \square$$

8.7 Local Times

Let $W = (W_t)$ be a standard Brownian motion. Define the level set

$$\{0 \leq t < \infty : W_t(\omega) = x\}.$$

Obviously, its Lebesgue measure is 0.

Definition 8.7.1. For any Borel set $B \in \mathcal{B}$, define the occupation time of B as

$$\Gamma_t(B) = \int_0^t \mathbb{I}_{W_s \in B} ds = m(\{s \in [0, t] : W_s \in B\}).$$

Note that it is a random variable.

Remark 8.7.2. The stochastic process $\Gamma(B) = (\Gamma_t(B))_{t \geq 0}$ is adapted and continuous.

Definition 8.7.3 (Local Time). For a given Brownian motion W , the local time is defined as

$$L_t(x) = L_t(x, \omega) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \Gamma_t([x - \varepsilon, x + \varepsilon]).$$

Note it is a random variable.

Remark 8.7.4. Local time serves as a density function w.s.t. the Lebesgue measure for the occupation time, i.e.,

$$\Gamma_t(B, \omega) = \int_B L_t(x, \omega) dx.$$

Theorem 8.7.5. *The local times of a Brownian motion exist.*

Consider $g(x) = |x|$. For $\varepsilon > 0$, let

$$g_\varepsilon(x) = \begin{cases} |x|, & |x| > \varepsilon \\ \frac{1}{2} \left(\varepsilon + \frac{x^2}{\varepsilon} \right), & |x| \leq \varepsilon. \end{cases}$$

Then $g_\varepsilon \in C^1$ and

$$g'_\varepsilon(x) = \begin{cases} 1, & x \geq \varepsilon \\ \frac{x}{\varepsilon}, & |x| < \varepsilon \\ -1, & x < -\varepsilon. \end{cases}$$

Moreover, for $|x| \neq \varepsilon$,

$$g''_\varepsilon(x) = \begin{cases} 0, & |x| > \varepsilon \\ \frac{1}{\varepsilon}, & |x| < \varepsilon. \end{cases}$$

Therefore,

$$\begin{aligned} g_\varepsilon(W_t) &= g_\varepsilon(W_0) + \int_0^t g'_\varepsilon(W_s) dW_s + \frac{1}{2} \int_0^t g''_\varepsilon(W_s) ds \\ &= g_\varepsilon(0) + \int_0^t g'_\varepsilon(W_s) dW_s + \frac{1}{2} \int_0^t \frac{1}{\varepsilon} \mathbb{I}_{\{|W_s| < \varepsilon\}} ds \\ &= \frac{\varepsilon}{2} + \int_0^t g'_\varepsilon(W_s) dW_s + \frac{1}{2\varepsilon} \Gamma_t([-\varepsilon, \varepsilon]). \end{aligned}$$

For the second term,

$$\begin{aligned} \int_0^t g'_\varepsilon(W_s) dW_s &= \int_0^t g'_\varepsilon(W_s) \mathbb{I}_{\{|W_s| \leq \varepsilon\}} dW_s + \int_0^t g'_\varepsilon(W_s) \mathbb{I}_{\{|W_s| > \varepsilon\}} dW_s \\ &= \int_0^t g'_\varepsilon(W_s) \mathbb{I}_{\{|W_s| \leq \varepsilon\}} dW_s + \int_0^t \text{sign}(W_s) \mathbb{I}_{\{|W_s| > \varepsilon\}} dW_s. \end{aligned}$$

Note that as $\varepsilon \rightarrow 0$, by the DCT for stochastic integral,

$$\int_0^t \text{sign}(W_s) \mathbb{I}_{\{|W_s| > \varepsilon\}} dW_s \rightarrow \int_0^t \text{sign}(W_s) dW_s.$$

For the other one,

$$I = \int_0^t g'_\varepsilon(W_s) \mathbb{I}_{\{|W_s| \leq \varepsilon\}} dW_s = \int_0^t \frac{W_s}{\varepsilon} \mathbb{I}_{\{|W_s| \leq \varepsilon\}} dW_s$$

Note that because I is a martingale, $\mathbb{E}[I] = 0$ and

$$\begin{aligned} \mathbb{E}[I^2] &= \mathbb{E} \left[\int_0^t \frac{W_s^2}{\varepsilon^2} \mathbb{I}_{\{|W_s| \leq \varepsilon\}} ds \right] \\ &= \int_0^t \frac{1}{\varepsilon^2} \mathbb{E} [W_s^2 \mathbb{I}_{\{|W_s| \leq \varepsilon\}}] ds \\ &\leq \int_0^t \mathbb{E} [\mathbb{I}_{\{|W_s| \leq \varepsilon\}}] ds \\ &= \int_0^t \mathbb{P}(|W_s| \leq \varepsilon) ds \\ &= \int_0^t \mathbb{P}(|W_1| \leq \varepsilon/\sqrt{s}) ds \\ &= \int_0^t \frac{1}{\sqrt{2\pi}} \int_{-\varepsilon/\sqrt{s}}^{\varepsilon/\sqrt{s}} e^{-y^2} dy ds \rightarrow 0. \end{aligned}$$

Therefore, as $\varepsilon \rightarrow 0$, we get the Tanaka formula,

$$|W_t| = \int_0^t \text{sign}(W_s) dW_s + L_t.$$

Corollary 8.7.6. Fix $a \in \mathbb{R}$,

$$\begin{aligned} |W_t - a| &= |a| + \int_0^t \text{sign}(W_s - a) dW_s + L_t(a) \\ (W_t - a)^+ &= (-a)^+ + \int_0^t \mathbb{I}_{(a, \infty)}(W_s) dW_s + \frac{1}{2} L_t(a) \\ (W_t - a)^- &= (-a)^- - \int_0^t \mathbb{I}_{(-\infty, a]}(W_s) dW_s + \frac{1}{2} L_t(a). \end{aligned}$$

Remark 8.7.7. For every Borel measurable function $f: \mathbb{R} \rightarrow [0, \infty)$,

$$\int_0^t f(W_s) ds = \int_{-\infty}^{\infty} f(x) L_t(x) dx.$$

Remark 8.7.8. For any semimartingale $X = X_0 + M + A$, we have the similar definition of local time Λ_t , which satisfies

$$\int_0^t f(X_s) d\langle M \rangle_s = \int_{-\infty}^{\infty} f(x) \Lambda_t(x) dx, \quad 0 \leq t < \infty,$$

and the Tanaka-Meyer formula

$$|X_t - a| = |X_0 - a| + \int_0^t \text{sign}(X_s - a) dX_s + \Lambda_t(a).$$

Chapter 9

Stochastic Differential Equation

Consider a stochastic process satisfying:

$$dX_t = b(t, X_t, W_t)dt + \sigma(t, X_t, W_t)dW_t. \quad (*)$$

for two problems:

- (1) existence and uniqueness and properties of solutions,
- (2) how to solve for particular cases.

9.1 Examples

Example 9.1.1 (Geometric Brownian Motion). Solving

$$dX_t = \alpha X_t dt + \sigma X_t dW_t.$$

where the initial value X_0 is given, and α, σ are constant.

Solution: Dividing X_t on the both side and integrating,

$$\int_0^t \frac{dX_u}{X_u} = \alpha t + \sigma W_t.$$

Let $f(x) = \log x$. Then by Itô formula,

$$\begin{aligned} d \log X_t &= \frac{1}{X_t} dX_t - \frac{1}{2} \frac{1}{X_t^2} (dX_t)^2 \\ &= \frac{1}{X_t} dX_t - \frac{1}{2} \frac{1}{X_t^2} \sigma^2 X_t^2 dt \\ &= \frac{1}{X_t} dX_t - \frac{1}{2} \sigma^2 dt. \end{aligned}$$

So

$$\log X_t - \log X_0 = \int_0^t \frac{dX_u}{X_u} - \frac{1}{2} \sigma^2 t.$$

It follows that

$$X_t = X_0 \exp \left(\sigma W_t + \left(\alpha - \frac{1}{2} \sigma^2 \right) t \right).$$

Remark 9.1.2. (1) If $(W_t)_{t \geq 0}$ is independent of X_0 , then

$$\begin{aligned}\mathbb{E}[X_t] &= \mathbb{E}[X_0] \mathbb{E} \left[\exp \left(\sigma W_t + \left(\alpha - \frac{1}{2} \sigma^2 \right) t \right) \right] \\ &= e^{\alpha t} \mathbb{E}[X_0] \mathbb{E} \left[\exp \left(\sigma W_t - \frac{1}{2} \sigma^2 t \right) \right] \\ &= e^{\alpha t} \mathbb{E}[X_0],\end{aligned}$$

because $\exp(\sigma W_t - \frac{1}{2} \sigma^2 t)$ is a martingale.

(2) If $\alpha > \frac{1}{2} \sigma^2$, then $X_t \rightarrow \infty$ as $t \rightarrow \infty$. If $\alpha < \frac{1}{2} \sigma^2$, $X_t \rightarrow 0$ as $t \rightarrow \infty$. If $\alpha = \frac{1}{2} \sigma^2$, X_t will fluctuate between arbitrary large and arbitrary small values.

Definition 9.1.3. A stochastic process $(X_t)_{t \geq 0}$ of the form

$$X_t = X_0 \exp(\sigma W_t + \mu t)$$

is called the geometric Brownian motion.

Example 9.1.4 (Hull-White Interest Rate Model). Consider SDE

$$dR_t = (a_t - b_t R_t) + \sigma_t dW_t, \quad R_0 = r,$$

where a_t, b_t, σ_t are deterministic.

Solution:

$$dR_t + b_t dR_t = a_t dt + \sigma_t dW_t.$$

Multiplying $e^{\int_0^t b_u du}$,

$$e^{\int_0^t b_u du} dR_t + e^{\int_0^t b_u du} b_t dR_t = e^{\int_0^t b_u du} a_t dt + \sigma_t e^{\int_0^t b_u du} dW_t.$$

Because $e^{\int_0^t b_u du}$ is of bounded variation on any interval, by Itô formula

$$d \left(e^{\int_0^t b_u du} \right) dR_t = e^{\int_0^t b_u du} a_t dt + \sigma_t e^{\int_0^t b_u du} dW_t.$$

So

$$R_t = r e^{-\int_0^t b_u du} + r \int_0^t e^{\int_t^s b_u du} a_s ds + \int_0^t \sigma_s e^{\int_t^s b_u du} dW_s.$$

Example 9.1.5. Consider SDE

$$dX_t = r X_t (K - X_t) dt + \beta X_t dW_t, \quad X_0 = x > 0.$$

Solution:

$$\frac{1}{X_t} dX_t + r X_t dt = r K dt + \beta dW_t.$$

Therefore,

$$\int_0^t \frac{1}{X_t} dX_t + \int_0^t r X_t dt = r K t + \beta W_t.$$

For the left hand side, first

$$\int_0^t \frac{1}{X_t} dX_t = \log X_t - \log X_0 + \frac{1}{2} \int_0^t \frac{1}{X_s^2} (dX_s)^2$$

$$\begin{aligned}
&= \log X_t - \log X_0 + \frac{1}{2} \int_0^t \frac{1}{X_s^2} \beta^2 X_s^2 ds \\
&= \log \frac{X_t}{x} + \frac{1}{2} \beta^2 t.
\end{aligned}$$

Therefore,

$$X_t \exp \left(r \int_0^t X_s ds \right) = x \exp \left(\beta W_t + \left(rK - \frac{1}{2} \beta^2 \right) t \right).$$

Integrating w.s.t. t ,

$$\begin{aligned}
\int_0^t \exp \left(r \int_0^s X_s ds \right) d \left(\int_0^s X_s ds \right) &= \frac{1}{r} \left(\exp \left(r \int_0^t X_s ds \right) - 1 \right) \\
&= x \int_0^t \exp \left(\beta W_s + \left(rK - \frac{1}{2} \beta^2 \right) s \right) ds.
\end{aligned}$$

It follows that

$$\exp \left(r \int_0^t X_s ds \right) = 1 + rx \int_0^t \exp \left(\beta W_s + \left(rK - \frac{1}{2} \beta^2 \right) s \right) ds,$$

and so

$$\int_0^t X_s ds = \frac{1}{r} \log \left(1 + rx \int_0^t \exp \left(\beta W_s + \left(rK - \frac{1}{2} \beta^2 \right) s \right) ds \right).$$

Differentiating w.s.t. t ,

$$X_t = \frac{x \exp \left(\beta W_t + \left(rK - \frac{1}{2} \beta^2 \right) t \right)}{1 + rx \int_0^t \exp \left(\beta W_s + \left(rK - \frac{1}{2} \beta^2 \right) s \right) ds}.$$

Example 9.1.6. Consider SDE

$$dX_t = \alpha_t dt + b_t X_t dW_t.$$

Solution: Trying to find an integrator ρ_t for

$$\rho_t dX_t - b_t \rho_t X_t dW_t = \alpha_t \rho_t dt.$$

By Itô formula,

$$d(\rho_t X_t) = \rho_t dX_t + X_t d\rho_t + (dX_t)(d\rho_t)$$

First, try to find ρ_t such that

$$X_t d\rho_t = -b_t \rho_t X_t dW_t \Rightarrow \frac{d\rho_t}{\rho_t} = -b_t dW_t.$$

Then by above

$$\log \frac{\rho_t}{\rho_0} + \frac{1}{2} \int_0^t \frac{1}{\rho_u^2} (d\rho_u)^2 = \log \frac{\rho_t}{\rho_0} + \frac{1}{2} \int_0^t b_u^2 du = - \int_0^t b_u dW_u,$$

which implies that

$$\rho_t = \exp \left(- \int_0^t b_u dW_u - \frac{1}{2} \int_0^t b_u^2 du \right)$$

by setting $\rho_0 = 1$. Then

$$d(\rho_t X_t) = \rho_t dX_t + X_t d\rho_t + (dX_t)(d\rho_t)$$

$$\begin{aligned}
&= \rho_t dX_t - b_t \rho_t X_t dW_t - b_t^2 \rho_t X_t dt \\
&= \alpha_t \rho_t dt - b_t^2 \rho_t X_t dt.
\end{aligned}$$

So

$$d(\rho_t X_t) + b_t^2 \rho_t X_t dt = \alpha_t \rho_t dt.$$

Multiplying $e^{\int_0^t b_u^2 du}$, then we have

$$d\left(e^{\int_0^t b_u^2 du} \rho_t X_t\right) = e^{\int_0^t b_u^2 du} \alpha_t \rho_t dt.$$

It follows that

$$\rho_t X_t - X_0 = \int_0^t e^{\int_s^t b_u^2 du} \alpha_s \rho_s ds,$$

i.e.,

$$X_t = X_0 \exp\left(\int_0^t b_s dW_s - \frac{1}{2} \int_0^t b_s^2 ds\right) + \int_0^t \alpha_s \exp\left(\int_s^t b_u dW_u - \frac{1}{2} \int_s^t b_u^2 du\right) ds.$$

Example 9.1.7. Consider SDE

$$LX_t'' + RX_t' + \frac{1}{2}X_t = G_t + \alpha \tilde{W}_t,$$

where $\tilde{W}_t$ is white noise, i.e., $\tilde{W}_t dt = dW_t$ (in distribution meaning).

Solution: Introduce

$$\mathbf{X}_t = \begin{pmatrix} X_t \\ X_t' \end{pmatrix}$$

and it follows that

$$d\mathbf{X}_t = A\mathbf{X}_t dt + \mathbf{H}_t dt + \mathbf{K} dW_t,$$

where

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{1}{CL} & -\frac{R}{L} \end{pmatrix}, \quad \mathbf{H}_t = \begin{pmatrix} 0 \\ \frac{G_t}{L} \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 0 \\ \frac{\alpha}{L} \end{pmatrix}.$$

So

$$\mathbf{X}_t = \exp(At)\mathbf{X}_0 + \exp(At) \int_0^t \exp(-As) (\mathbf{H}_s ds + \mathbf{K} dW_s).$$

9.2 Existence and Uniqueness of Solution

Theorem 9.2.1 (Existence and Uniqueness). *Fix $T > 0$ (T could be ∞). Let $b: [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma: [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ be measurable such that*

$$\|b(t, x)\| + \|\sigma(t, x)\|_F \leq K(1 + \|x\|), \quad \forall x \in \mathbb{R}^n, t \in [0, T] \quad (9.1)$$

and

$$\|b(t, x) - b(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\|_F \leq K \|x - y\|, \quad \forall x, y \in \mathbb{R}^n, t \in [0, T] \quad (9.2)$$

for some constant K . Let $\mathbf{Z}$ be a random variable independent of $\mathcal{F}_\infty^W$ and $\mathbf{Z} \in L^2$. Then

$$d\mathbf{X}_t = b(t, \mathbf{X}_t)dt + \sigma(t, \mathbf{X}_t)d\mathbf{W}_t, \quad 0 \leq t \leq T, \quad \mathbf{X}_0 = \mathbf{Z},$$

has a unique (strong) solution $\mathbf{X} = (\mathbf{X}_t)_{t \in [0, T]}$ that is continuous w.s.t. t with properties

(i) $\mathbf{X}_t$ is measurable w.s.t. $\mathcal{F}_t^W \vee \sigma(\mathbf{Z})$ for all t ,

(ii) $\mathbb{E} [\sup_{0 \leq s \leq T} \|\mathbf{X}_s\|^2] < \infty$.

Remark 9.2.2. Note that condition (9.2) implies condition (9.1). For the uniqueness, it means that indistinguishable.

Proof. First, let $\mathcal{E}_T$ be the set of all continuous and $\mathbb{F}^W$ -adapted process $\mathbf{X}$ such that

$$\mathbb{E} \left[\sup_{0 \leq s \leq T} \|\mathbf{X}_s\|^2 \right] < \infty.$$

Then define a norm on $\mathcal{E}_T$ as

$$\|\mathbf{X}\|_E^2 := \mathbb{E} \left[\sup_{0 \leq s \leq T} \|\mathbf{X}_s\|^2 \right].$$

It is not hard to see $(\mathcal{E}_T, \|\cdot\|_E)$ is a Banach space. Note $\mathbf{X} = \mathbf{Y}$ in $\mathcal{E}_T$ means that they are equal a.e. in the sense of

$$\mathbf{X}, \mathbf{Y}: \Omega \rightarrow C([0, T], \mathbb{R}^n).$$

Therefore, $\mathbf{X} = \mathbf{Y}$ in $\mathcal{E}_T$ is equivalent to

$$\mathbb{P}(\mathbf{X}_t(\omega) = \mathbf{Y}_t(\omega), t \in [0, T]) = 1,$$

in other words, process $\mathbf{X}$ and $\mathbf{Y}$ are indistinguishable.

- **Check:** If $\mathbf{X}$ is a solution, then for any $T > 0$, $\mathbf{X} \in \mathcal{E}_T$.

Fix $T > 0$. Define a stopping time

$$\tau_n = \inf \{t \geq 0: \|\mathbf{X}_t\| > n\}.$$

and consider $\mathbf{X}_{t \wedge \tau_n}$ for $t \leq T$, i.e.,

$$\mathbf{X}_{t \wedge \tau_n} = \mathbf{X}_0 + \int_0^{t \wedge \tau_n} b(s, \mathbf{X}_s) ds + \int_0^{t \wedge \tau_n} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s.$$

Because $\|\mathbf{a} + \mathbf{b} + \mathbf{c}\|^2 \leq 3(\|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 + \|\mathbf{c}\|^2)$,

$$\|\mathbf{X}_{t \wedge \tau_n}\|^2 \leq 3 \left(\|\mathbf{X}_0\|^2 + \left\| \int_0^{t \wedge \tau_n} b(s, \mathbf{X}_s) ds \right\|^2 + \left\| \int_0^{t \wedge \tau_n} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s \right\|^2 \right).$$

It follows that

$$\begin{aligned} \mathbb{E} \left[\sup_{0 \leq u \leq t \wedge \tau_n} \|\mathbf{X}_u\|^2 \right] &\leq 3 \left(\mathbb{E} [\|\mathbf{X}_0\|^2] + \mathbb{E} \left[\sup_{0 \leq u \leq t \wedge \tau_n} \left\| \int_0^u b(s, \mathbf{X}_s) ds \right\|^2 \right] \right. \\ &\quad \left. + \mathbb{E} \left[\sup_{0 \leq u \leq t \wedge \tau_n} \left\| \int_0^u \sigma(s, \mathbf{X}_s) d\mathbf{W}_s \right\|^2 \right] \right). \end{aligned}$$

First, by condition (9.1),

$$\mathbb{E} \left[\sup_{0 \leq u \leq t \wedge \tau_n} \left\| \int_0^u b(s, \mathbf{X}_s) ds \right\|^2 \right] \leq K^2 \mathbb{E} \left[\left(\int_0^{t \wedge \tau_n} (1 + \|\mathbf{X}_s\|) ds \right)^2 \right]$$

$$\begin{aligned}
&\leq K^2(t \wedge \tau_n) \mathbb{E} \left[\int_0^t (1 + \|\mathbf{X}_{s \wedge \tau_n}\|)^2 ds \right] \\
&\leq 2K^2T \int_0^t (1 + \mathbb{E} [\|\mathbf{X}_{s \wedge \tau_n}\|^2]) ds \\
&\leq 2K^2T \int_0^t \left(1 + \mathbb{E} \left[\sup_{0 \leq u \leq s \wedge \tau_n} \|\mathbf{X}_u\|^2 \right] \right) ds,
\end{aligned}$$

where the second inequality is by

$$\int_0^{t \wedge \tau_n} g(X_s) ds \leq \int_0^t g(X_{s \wedge \tau_n}) ds$$

for any positive g . Because

$$\int_0^{t \wedge \tau_n} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s$$

is a martingale, by Doob's maximal inequality,

$$\begin{aligned}
\mathbb{E} \left[\sup_{0 \leq u \leq t} \left\| \int_0^{u \wedge \tau_n} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s \right\|^2 \right] &\leq 4\mathbb{E} \left[\left\| \int_0^{t \wedge \tau_n} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s \right\|^2 \right] \\
&= 4\mathbb{E} \left[\int_0^{t \wedge \tau_n} \|\sigma(s, \mathbf{X}_s)\|_F^2 ds \right] \\
&\leq 4K^2 \mathbb{E} \left[\int_0^{t \wedge \tau_n} (1 + \|\mathbf{X}_s\|)^2 ds \right] \\
&\leq 8K^2 \int_0^t \left(1 + \mathbb{E} \left[\sup_{0 \leq u \leq s \wedge \tau_n} \|\mathbf{X}_u\|^2 \right] \right) ds.
\end{aligned}$$

Therefore, we have

$$\mathbb{E} \left[\sup_{0 \leq u \leq t \wedge \tau_n} \|\mathbf{X}_u\|^2 \right] \leq 3 \left(\mathbb{E} [\|\mathbf{X}_0\|^2] + 2(K^2T + 4K^2) \int_0^t \left(1 + \mathbb{E} \left[\sup_{0 \leq u \leq s \wedge \tau_n} \|\mathbf{X}_u\|^2 \right] \right) ds \right).$$

Note that $\mathbf{X}_0 = \mathbf{Z} \in L^2$, so $\mathbb{E} [\|\mathbf{X}_0\|^2] < \infty$. By applying Grönwall's lemma to

$$t \mapsto \mathbb{E} \left[\sup_{0 \leq u \leq t \wedge \tau_n} \|\mathbf{X}_u\|^2 \right],$$

we have

$$\mathbb{E} \left[\sup_{0 \leq u \leq T \wedge \tau_n} \|\mathbf{X}_u\|^2 \right] < C,$$

for some C . As $n \rightarrow \infty$, by Fatou's lemma,

$$\mathbb{E} \left[\sup_{0 \leq u \leq T} \|\mathbf{X}_u\|^2 \right] < C,$$

and so $\mathbf{X} \in \mathcal{E}_t$.

- Fix $T = \delta$, for a small δ . Let $\mathbf{X} \in \mathcal{E}_T$. Define

$$\Phi(\mathbf{X})_t = \mathbf{Z} + \int_0^t b(s, \mathbf{X}_s) ds + \int_0^t \sigma(s, \mathbf{X}_s) d\mathbf{W}_s.$$

Note that it is obviously $\Phi(\mathbf{X})_t$ is $\mathcal{F}_t^W \vee \sigma(\mathbf{Z})$ -measurable. In particular, by above we have

$$\Phi(\mathbf{X}) \leq 3 \left(\mathbb{E}[\mathbf{Z}^2] + 2 (K^2 T^2 + 4K^2 T) (1 + \|\mathbf{X}\|_E^2) \right) < \infty,$$

i.e., $\Phi: \mathcal{E}_T \rightarrow \mathcal{E}_T$. Moreover, by applying condition (9.2) and similarly reasoning as above, we have

$$\|\Phi(\mathbf{X}) - \Phi(\mathbf{Y})\|^2 \leq 2 (K^2 T^2 + 4K^2 T) \|\mathbf{X} - \mathbf{Y}\|^2.$$

When $T = \delta$ small enough, Φ is a contraction. So by Banach fixed point theorem, there exists a unique fixed point

$$\mathbf{X}_t = \mathbf{Z} + \int_0^t b(s, \mathbf{X}_s) ds + \int_0^t \sigma(s, \mathbf{X}_s) d\mathbf{W}_s,$$

which provides a desired solution locally.

- In order to get a solution on $[0, \infty)$, we can apply previous step to get a solution on $[T_n, T_{n+1}]$, for $T_{n+1} - T_n$ sufficiently small. Then as $T_n \rightarrow \infty$, we get the solution. Moreover, the uniqueness is because of the uniqueness of each interval $[T_n, T_{n+1}]$. $\square$

Remark 9.2.3. In the proof, the truncation by τ_n is important, because in order to apply Doob's maximal inequality, we need the L^2 -integrability.

Remark 9.2.4. In fact, if $\mathbf{Z} \in L^p$, then we can get

$$\mathbb{E} \left[\sup_{0 \leq s \leq T} \|\mathbf{X}_s\|^p \right] < \infty,$$

for all $p \geq 1$, by using by replacing Doob's inequality with Burkholder–Davis–Gundy (BDG) inequality on $\mathbf{M}_{t \wedge \tau_n} = \int_0^{t \wedge \tau_n} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s$, which gets that

$$\mathbb{E} \left[\sup_{0 \leq u \leq t \wedge \tau_n} \left\| \int_0^u \sigma(s, \mathbf{X}_s) d\mathbf{W}_s \right\|^p \right] \leq C_p \mathbb{E} \left[\left(\int_0^{t \wedge \tau_n} \|\sigma(s, \mathbf{X}_s)\|_F^2 ds \right)^{p/2} \right].$$

Theorem 9.2.5 (Burkholder–Davis–Gundy). *Let M be a continuous local martingale with $M_0 = 0$ and τ is a stopping time. Then for any $1 \leq p < \infty$, there exists c_p and C_p such that*

$$c_p \mathbb{E} \left[\langle M \rangle_\tau^{p/2} \right] \leq \mathbb{E} [(M_\tau^*)^p] \leq C_p \mathbb{E} \left[\langle M \rangle_\tau^{p/2} \right],$$

where $M_t^* = \sup_{0 \leq s \leq t} |M|_s$.

If we only have the local Lipschitz continuity, then the solution also uniquely exists up to a stopping time.

Definition 9.2.6. Consider a SDE

$$d\mathbf{X}_t = b(t, \mathbf{X}_t) dt + \sigma(t, \mathbf{X}_t) d\mathbf{W}_t, \quad 0 \leq t \leq T, \quad \mathbf{X}_0 = \mathbf{Z}.$$

Let τ be a $\mathbb{F}^W$ -stopping time. X is a solution up to τ if

- $\mathbf{X}_t$ defined on $\{t \leq \tau\}$ is continuous and $\mathcal{F}_t^W \vee \sigma(\mathbf{Z})$ -measurable.
- $\mathbb{P} \left(\int_0^T |b_i(s, \mathbf{X}_s)| + |\sigma_{ij}(s, \mathbf{X}_s)|^2 ds < \infty \right) = 1$.
- $\mathbf{X}_t = \mathbf{Z} + \int_0^t b(s, \mathbf{X}_s) ds + \int_0^t \sigma(s, \mathbf{X}_s) d\mathbf{W}_s$ on $\{t < \tau\}$ a.e..

Remark 9.2.7. Simply, we denote $\mathbf{X} = (\mathbf{X}_t)_{t < \tau}$. Also, we have another expression

$$\mathbf{X}_{t \wedge \tau} = \mathbf{Z} + \int_0^{t \wedge \tau} b(s, \mathbf{X}_s) ds + \int_0^{t \wedge \tau} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s.$$

Note that these two are actually equivalent. It seems the second one is stronger, because first for $\omega \in \{t < \omega\}$, it has the above expression

$$\mathbf{X}_t(\omega) = \mathbf{Z}(\omega) + \int_0^t b(s, \mathbf{X}_s(\omega)) ds + \int_0^t \sigma(s, \mathbf{X}_s(\omega)) d\mathbf{W}_s$$

but for $\omega \in \{t \geq \tau\}$, it further requires

$$\mathbf{X}_{\tau(\omega)}(\omega) = \mathbf{Z}(\omega) + \int_0^{\tau(\omega)} b(s, \mathbf{X}_s(\omega)) ds + \int_0^{\tau(\omega)} \sigma(s, \mathbf{X}_s(\omega)) d\mathbf{W}_s.$$

In fact, the first expression also has this formula by continuity.

Theorem 9.2.8. Fix $T > 0$ (T could be ∞). Let $b: [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma: [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ be locally Lipschitz, i.e., for every $N \geq 0$, there exists $C_N > 0$ such that

$$\|b(t, x) - b(t, y)\| + \|\sigma(t, x) - \sigma(t, y)\|_F \leq C_N \|x - y\|, \quad \forall B(\mathbf{0}, N), \quad t \in [0, T].$$

Then for any $\mathbf{Z} \in L^2$, there exists a stopping time $\tau > 0$ a.e., and a solution $\mathbf{X} = (\mathbf{X}_t)_{t < \tau}$ up to τ for SDE

$$\mathbf{X}_t = \mathbf{Z} + \int_0^t b(s, \mathbf{X}_s) ds + \int_0^t \sigma(s, \mathbf{X}_s) d\mathbf{W}_s, \quad t < \tau$$

This solution unique in the sense that if there exist another stopping time τ' and a continuous process $\mathbf{X}'$ such that

$$\mathbf{X}'_t = \mathbf{Z} + \int_0^t b(s, \mathbf{X}'_s) ds + \int_0^t \sigma(s, \mathbf{X}'_s) d\mathbf{W}_s, \quad t < \tau',$$

then $\tau' \leq \tau$ and

$$\mathbf{X}'_t \mathbb{I}_{\{t < \tau'\}} = \mathbf{X}_t \mathbb{I}_{\{t < \tau'\}}.$$

Then $\tau = \tau(\mathbf{Z})$ is called the explosion time.

Proof. • Fix $N > 0$. Define

$$\pi_N(\mathbf{x}) := \begin{cases} \mathbf{x}, & \|\mathbf{x}\| \leq N \\ \frac{N}{\|\mathbf{x}\|} \mathbf{x}, & \|\mathbf{x}\| > N \end{cases}$$

and let

$$b_N(t, \mathbf{x}) := b(t, \pi_N(\mathbf{x})), \quad \sigma_N(t, \mathbf{x}) := \sigma(t, \pi_N(\mathbf{x})).$$

Then we have

$$\|b_N(t, \mathbf{x}) - b_N(t, \mathbf{y})\| = \|b(t, \pi_N(\mathbf{x})) - b(t, \pi_N(\mathbf{y}))\| \leq C_N \|\pi_N(\mathbf{x}) - \pi_N(\mathbf{y})\| \leq C_N \|\mathbf{x} - \mathbf{y}\|$$

for any $t \in [0, T]$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Similarly, $\|\sigma_N(t, \mathbf{x}) - \sigma_N(t, \mathbf{y})\| \leq C_N \|\mathbf{x} - \mathbf{y}\|$. Then by above theorem, there exists a unique solution $\mathbf{X}^N$ such that

$$\mathbf{X}_t^N = \mathbf{Z} + \int_0^t b_N(s, \mathbf{X}_s^N) ds + \int_0^t \sigma_N(s, \mathbf{X}_s^N) d\mathbf{W}_s, \quad t \in [0, T].$$

Now define

$$\tau_N := \inf \{t \geq 0 : \|\mathbf{X}_t^N\| \geq N\} \wedge T.$$

Then on $\{t < \tau_N\}$, because $\|\mathbf{X}_t^N\| < N$,

$$b_N(t, \mathbf{X}_t^N) = b(t, \mathbf{X}_t^N), \quad \sigma_N(t, \mathbf{X}_t^N) = \sigma(t, \mathbf{X}_t^N).$$

which implies that $\mathbf{X}^N$ is a solution of original SDE on $\{t < \tau_N\}$.

- For $M > N$, we can have τ_M and $\mathbf{X}^M$. First, for all $\|\mathbf{x}\| \leq N < M$, $\pi_N(\mathbf{x}) = \pi_M(\mathbf{x})$, hence

$$b_N(t, x) = b_M(t, x) = b(t, x), \quad \sigma_N(t, x) = \sigma_M(t, x) = \sigma(t, x).$$

Let

$$\tau_N^M := \inf \{t \geq 0 : \|\mathbf{X}_t^M\| \geq N\} \wedge T$$

On $t < \tau_M^N \wedge \tau_N$, because $\mathbf{X}^M, \mathbf{X}^N \in B(\mathbf{0}, N)$, they solve the same SDE, So

$$\mathbf{X}_t^N = \mathbf{X}_t^M, \quad \forall 0 \leq t \leq \tau_N \wedge \tau_N^M,$$

which implies that $\tau_N = \tau_N^M$. Moreover, it obvious that $\tau_N^M \leq \tau_M$. So $\{\tau_N\}_N$ is an increasing sequence of stopping times. Let

$$\tau = \lim_{N \rightarrow \infty} \tau_N \in [0, \infty].$$

$\tau > 0$ because we can choose $N > \|\mathbf{Z}\| + 1$.

- On $\omega \in \{t < \tau\}$, choose N such that $t < \tau_N(\omega)$, define

$$\mathbf{X}_t(\omega) := \mathbf{X}_t^N(\omega),$$

which is well-defined because for $t < \tau_N \leq \tau_M$, $\mathbf{X}_t^N = \mathbf{X}_t^M$. Clearly, $\mathbf{X}_0 = \mathbf{Z}$ and on $\{t < \tau_N\}$,

$$\mathbf{X}_t = \mathbf{Z} + \int_0^t b(s, \mathbf{X}_s) ds + \int_0^t \sigma(s, \mathbf{X}_s) d\mathbf{W}_s,$$

i.e.,

$$\mathbf{X}_{t \wedge \tau_N} = \mathbf{Z} + \int_0^{t \wedge \tau_N} b(s, \mathbf{X}_s) ds + \int_0^{t \wedge \tau_N} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s.$$

As $N \rightarrow \infty$,

$$\mathbf{X}_{t \wedge \tau} = \mathbf{Z} + \int_0^{t \wedge \tau} b(s, \mathbf{X}_s) ds + \int_0^{t \wedge \tau} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s.$$

- Uniqueness: If we have $\mathbf{X}'$ and τ' such that

$$\mathbf{X}'_{t \wedge \tau'} = \mathbf{Z} + \int_0^{t \wedge \tau'} b(s, \mathbf{X}'_s) ds + \int_0^{t \wedge \tau'} \sigma(s, \mathbf{X}'_s) d\mathbf{W}_s$$

Let

$$\rho_N := \inf \{t \geq 0 : \|\mathbf{X}_t\| \geq N\}, \quad \rho'_N := \inf \{t \geq 0 : \|\mathbf{X}'_t\| \geq N\}.$$

Then on $\{t < \rho_N \wedge \rho'_N \wedge \tau \wedge \tau'\}$, both satisfies the original SDE. Define

$$\mathbf{Y}_t := \mathbf{X}_{t \wedge \rho_N \wedge \rho'_N \wedge \tau \wedge \tau'} - \mathbf{X}'_{t \wedge \rho_N \wedge \rho'_N \wedge \tau \wedge \tau'},$$

and so

$$\mathbf{Y}_t = \int_0^{t \wedge \rho_N \wedge \rho'_N \wedge \tau \wedge \tau'} [b(s, \mathbf{X}_s) - b(s, \mathbf{X}'_s)] ds + \int_0^{t \wedge \rho_N \wedge \rho'_N \wedge \tau \wedge \tau'} [\sigma(s, \mathbf{X}_s) - \sigma(s, \mathbf{X}'_s)] d\mathbf{W}_s$$

By similar reasoning, for $f(t) := \mathbb{E}(\sup_{0 \leq u \leq t} \|\mathbf{Y}_u\|^2)$,

$$f(t) \leq K \int_0^t f(s) ds.$$

Because $f(t) \geq 0$ and $f(0) = 0$, $f(t) \equiv 0$, i.e.,

$$\mathbb{E} \left(\sup_{0 \leq u \leq t} \|\mathbf{Y}_u\|^2 \right) = 0 \quad \mathbf{Y}_t = 0,$$

which implies that

$$\mathbf{X}_{t \wedge \rho_N \wedge \rho'_N \wedge \tau \wedge \tau'} = \mathbf{X}'_{t \wedge \rho_N \wedge \rho'_N \wedge \tau \wedge \tau'}$$

Then as $N \rightarrow \infty$,

$$\mathbf{X}_t = \mathbf{X}'_t \quad \forall t < \tau \wedge \tau'.$$

Finally, we can show $\tau' \leq \tau$. By continuity on $[0, \tau'(\omega)]$ a.e.,

$$\sup_{0 \leq s \leq \tau'(\omega)} \|\mathbf{X}'_s(\omega)\| < N.$$

Then by the uniqueness of solution,

$$\mathbf{X}'_t(\omega) = \mathbf{X}_t^N(\omega) \quad \forall t \leq \tau'(\omega).$$

Therefore,

$$\sup_{0 \leq s \leq \tau'(\omega)} \|\mathbf{X}_s^N(\omega)\| < N,$$

which implies that $\tau(\omega) \geq \tau_N(\omega) > \tau'(\omega)$ a.e.. □

Remark 9.2.9. Note that on our construction, we only care about the definition of $\mathbf{X}$ on $\{t < \tau\}$. For $t \geq \tau$, we can extend it by adding a cemetery state $\Delta = \infty$,

$$\tilde{\mathbf{X}}_t = \begin{cases} \mathbf{X}_t, & t < \tau \\ \Delta, & t \geq \tau. \end{cases}$$

Therefore, we consider solution valued on $\mathbb{R}^n \cup \infty$, moreover, it is continuous by definition of τ .

Moreover, a solution of SDE is a Markov process

Theorem 9.2.10 (Markov Property). *Assume b, σ are locally Lipschitz continuous. And $\mathbf{X}$ is a solution up to τ . Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a bounded Borel function. Then for any $t, s \geq 0$, we have*

$$\mathbb{E}_x [f(\mathbf{X}_{t+s}) \mid \mathcal{F}_t^B] = \mathbb{E}_{\mathbf{X}_t} [f(\mathbf{X}_s)],$$

on $\{t < \tau\}$.

Theorem 9.2.11 (Strong Markov Property). *Assume b, σ are locally Lipschitz continuous. And $\mathbf{X}$ is a solution up to τ . Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a bounded Borel function and $T < \infty$ be a stopping time w.s.t. $\mathbb{F}^B$. Then*

$$\mathbb{E}_x [f(\mathbf{X}_{T+s}) \mid \mathcal{F}_T^B] = \mathbb{E}_{\mathbf{X}_T} [f(\mathbf{X}_s)],$$

on $\{T < \tau\}$.

9.3 Weak Solution

Definition 9.3.1. Given SDE,

$$d\mathbf{X}_t = b(t, \mathbf{X}_t)dt + \sigma(t, \mathbf{X}_t)d\mathbf{W}_t. \quad (9.3)$$

- (1) A strong solution of (9.3) on a give probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and w.s.t. the fixed Brownian motion $\mathbf{W}$ and initial value $\mathbf{Z}$, is a stochastic process $\mathbf{X}$ with continuous paths and with the following properties:

- (i) $\mathbf{X}$ is adapted to $\mathbb{F}^W = (\mathcal{F}_t^W)_{t \geq 0}$,
- (ii) $\mathbb{P}(\mathbf{X}_0 = \mathbf{Z}) = 1$,
- (iii) $\mathbb{P}\left(\int_0^T |b_i(s, \mathbf{X}_s)| + |\sigma_{ij}(s, \mathbf{X}_s)|^2 ds < \infty\right) = 1$, for all i, j ,
- (iv) $\mathbf{X}$ satisfies the integral version

$$\mathbf{X}_t = \mathbf{X}_0 + \int_0^t b(s, \mathbf{X}_s)ds + \int_0^t \sigma(s, \mathbf{X}_s)d\mathbf{W}_s.$$

- (2) A weak solution of (9.3) is a triple $((\mathbf{X}, \mathbf{W}), (\Omega, \mathcal{F}, \mathbb{P}), \mathbb{F} = (\mathcal{F}_t))$ such that

- (i) $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space,
- (ii) $\mathbb{F}$ is a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$ with the usual condition,
- (iii) $\mathbf{X}$ is a continuous $\mathbb{F}$ -adapted process,
- (iv) $\mathbf{W} = (\mathbf{W}_t, \mathcal{F}_t)$ is a Brownian motion,
- (v) $\mathbb{P}\left(\int_0^T |b_i(s, \mathbf{X}_s)| + |\sigma_{ij}(s, \mathbf{X}_s)|^2 ds < \infty\right) = 1$, for all i, j ,
- (vi) $\mathbf{X}$ satisfies the integral version

$$\mathbf{X}_t = \mathbf{X}_0 + \int_0^t b(s, \mathbf{X}_s)ds + \int_0^t \sigma(s, \mathbf{X}_s)d\mathbf{W}_s.$$

Remark 9.3.2. It is obviously that the existence of strong solution implies that the solution is also a weak solution. But the existence of weak solution does not implies the existence of strong solution.

Example 9.3.3 (Tanaka Equation). Consider the SDE

$$dX_t = \text{sign}(X_t)dW_t.$$

Note that $\sigma(t, x) = \text{sign}(x)$ does not satisfy the Lipschitz condition.

Claim: (9.3.3) has no strong solution.

Suppose (9.3.3) has a strong solution X . Then

$$X_t = \int_0^t \text{sign}(X_s)dW_s \Rightarrow \langle X \rangle_t = t.$$

By Lévy theorem, X is a Brownian motion. On the other hand,

$$dW_t = \text{sign}(X_t)dX_t \Rightarrow W_t = \int_0^t \text{sign } X_s ds,$$

which means W is a Brownian motion w.s.t. $\mathbb{F}^X$, i.e., $\mathbb{F}^W \subset \mathbb{F}^X$. By the Tanaka equation,

$$W_t = |X_t| - L_t^X,$$

which implies that $\mathcal{F}_t^W \neq \mathcal{F}_t^X$, contradicting to X adapted to $\mathbb{F}^W$.

(9.3.3) has a weak solution. Choose $B = (B_t)_{t \geq 0}$ be a Brownian motion. Define

$$\tilde{W}_t = \int_0^t \text{sign}(B_u) dB_u,$$

which is a Brownian motion. Then let $X = B$,

$$d\tilde{W}_t = \text{sign}(X_t) dX_t \Rightarrow dX_t = \text{sign}(X_t) d\tilde{W}_t.$$

Definition 9.3.4. For a given SDE,

$$d\mathbf{X}_t = b(t, \mathbf{X}_t)dt + \sigma(t, \mathbf{X}_t)d\mathbf{W}_t, \quad \mathbf{X}_0 \sim \mu,$$

- Weak uniqueness means that any two weak solutions has the same law.
- Pathwise uniqueness means that when fix $(\Omega, \mathcal{F}, \mathbb{P}), \mathbb{F} = (\mathcal{F}_t)$, Brownian motion $\mathbf{W} = (\mathbf{W}_t, \mathcal{F}_t)$, and $\mathbf{X}_0$, two solutions are indistinguishable, i.e. strong uniqueness.

Theorem 9.3.5 (Yamada–Watanabe). *If both weak existence and pathwise uniqueness hold, then weak uniqueness also holds.*

Corollary 9.3.6. *If b, σ are locally Lipschitz continuous, then the solution is weakly unique.*

9.4 Feynman-Kac Formula

Theorem 9.4.1 (Feynman-Kac Formula). *Consider SDE*

$$dX_t = \beta(t, X_t)dt + \sigma(t, X_t)dW_t.$$

Let f be a Borel-measurable function. Fix $T > 0$ and $t \in [0, T]$. Define

$$g(t, x) = \mathbb{E}[f(X_T) \mid X_t = x] = \mathbb{E}^{t,x}[f(X_T)].$$

Assume $g(t, x) < \infty$. Then $g(t, x)$ satisfies PDE

$$\frac{\partial}{\partial t}g + \beta \frac{\partial}{\partial x}g + \frac{1}{2}\sigma^2 \frac{\partial^2}{\partial x^2}g = 0$$

with terminal $g(T, x) = f(x)$.

Remark 9.4.2. Note that $(g(t, X_t))_{0 \leq t \leq T}$ is a martingale, because by Itô formula

$$\begin{aligned} dg(t, X_t) &= g_t(t, X_t)dt + g_x(t, X_t)dX_t + \frac{1}{2}g_{xx}(t, X_t)(dX_t)^2 \\ &= g_t dt + g_x(\beta dt + \sigma dW_t) + \frac{1}{2}g_{xx}\sigma^2 dt \\ &= g_x \sigma dW_t + \left(g_t + \beta g_x + \frac{1}{2}\sigma^2 g_{xx} \right) dt \\ &= g_x \sigma dW_t. \end{aligned}$$

Theorem 9.4.3 (Discounted Feynman-Kac Formula). *Consider SDE*

$$dX_t = \beta(t, X_t)dt + \sigma(t, X_t)dW_t.$$

Let f be a Borel-measurable function and r be a constant. Fix $T > 0$ and $t \in [0, T]$. Define

$$h(t, x) = \mathbb{E}^{t,x} [e^{-r(T-t)} f(X_T)].$$

Then h satisfies

$$h_t(t, x) + \beta(t, x)h_x(t, x) + \frac{1}{2}\sigma(t, x)h_{xx}(t, x) = rh(t, x)$$

with terminal $h(T, x) = f(x)$.

9.5 Discretization

Euler–Maruyama Algorithm. For a given SDE,

$$d\mathbf{X}_t = b(t, \mathbf{X}_t)dt + \sigma(t, \mathbf{X}_t)d\mathbf{W}_t, \quad t \in [0, T], \quad \mathbf{X}_0 = \mathbf{Z},$$

with globally Lipschitz continuous b, σ , and we always assume $\mathbf{W}$ is independent of $\mathbf{Z}$. Let $h = T/N$ for a large $N \in \mathbb{N}$ and $t_k = kh$ for $k = 0, 1, \dots, N$. Construction $\{\mathbf{X}_k^h\}_{k=0}^N$ by setting $\mathbf{X}_0^h = \mathbf{Z} \in L^2$ and

$$\mathbf{X}_{k+1}^h = \mathbf{X}_k^h + b(t_k, \mathbf{X}_k^h)h + \sigma(t_k, \mathbf{X}_k^h)\Delta\mathbf{W}_k.$$

Note that $\Delta\mathbf{W}_k$ is replaced by $\sqrt{h}\boldsymbol{\xi}$ for $\boldsymbol{\xi} \sim \mathcal{N}(0, I)$ in practice.

Theorem 9.5.1 (Strong Convergence of EM). *Using notations as above, there exists a constant $C > 0$ such that*

$$\left(\mathbb{E} \left[\max_{0 \leq k \leq N} \|\mathbf{X}_{t_k} - \mathbf{X}_k^h\|^2 \right] \right)^{1/2} \leq Ch^{1/2}.$$

Proof. First, from the proof of the existence and uniqueness of solution, we have

$$\sup_{0 \leq t \leq T} \mathbb{E} [\|\mathbf{X}_t\|^2] < \infty, \quad \max_{0 \leq k \leq N} \mathbb{E} [\|\mathbf{X}_k^h\|^2] < \infty.$$

More, by the similar reasoning, there exists C_1 such for $s \in [t_k, t_{k+1}]$,

$$\mathbb{E} [\|\mathbf{X}_s - \mathbf{X}_{t_k}\|^2] \leq C_1 h,$$

when h is small.

Let $\mathbf{e}_k := \mathbf{X}_{t_k} - \mathbf{X}_k^h$. Note that

$$\mathbf{X}_{t_{k+1}} = \mathbf{X}_{t_k} + \int_{t_k}^{t_{k+1}} b(s, \mathbf{X}_s) ds + \int_{t_k}^{t_{k+1}} \sigma(s, \mathbf{X}_s) d\mathbf{W}_s$$

and $\mathbf{X}_{k+1}^h = \mathbf{X}_k^h + b(t_k, \mathbf{X}_k^h)h + \sigma(t_k, \mathbf{X}_k^h)\Delta\mathbf{W}_k$. Therefore, $\mathbf{e}_{k+1} = \mathbf{e}_k + \mathbf{A}_k + \mathbf{M}_k$, where

$$\begin{aligned} \mathbf{A}_k &:= \int_{t_k}^{t_{k+1}} (b(s, \mathbf{X}_s) - b(t_k, \mathbf{X}_k^h)) ds \\ \mathbf{M}_k &:= \int_{t_k}^{t_{k+1}} (\sigma(s, \mathbf{X}_s) - \sigma(t_k, \mathbf{X}_k^h)) d\mathbf{W}_s. \end{aligned}$$

Therefore, by K -Lipschitz continuity,

$$\begin{aligned}
\mathbb{E} [\|\mathbf{A}_k\|^2] &= \mathbb{E} \left[\left\| \int_{t_k}^{t_{k+1}} (b(s, \mathbf{X}_s) - b(t_k, \mathbf{X}_k^h)) ds \right\|^2 \right] \\
&\leq K^2 \mathbb{E} \left[\left(\int_{t_k}^{t_{k+1}} \|\mathbf{X}_s - \mathbf{X}_k^h\| ds \right)^2 \right] \\
&\leq K^2 h \int_{t_k}^{t_{k+1}} \mathbb{E} [\|\mathbf{X}_s - \mathbf{X}_k^h\|^2] ds \\
&\leq K^2 h \int_{t_k}^{t_{k+1}} \mathbb{E} [\|\mathbf{X}_s - \mathbf{X}_{t_k}\|^2] + \mathbb{E} [\|\mathbf{e}_k\|^2] ds \\
&\leq K^2 h^2 (C_1 h + \mathbb{E} [\|\mathbf{e}_k\|^2]),
\end{aligned}$$

and similarly,

$$\mathbb{E} [\|\mathbf{M}_k\|^2] = \int_{t_k}^{t_{k+1}} \mathbb{E} [\|\sigma(s, \mathbf{X}_s) - \sigma(t_k, \mathbf{X}_k^h)\|^2] ds \leq K^2 h (C_1 h + \mathbb{E} [\|\mathbf{e}_k\|^2]),$$

Writing $\mathbf{e}_{k+1} = \mathbf{e}_k + \mathbf{A}_k + \mathbf{M}_k$, so

$$\begin{aligned}
\mathbb{E} [\|\mathbf{e}_{k+1}\|^2] &= \mathbb{E} [\|\mathbf{e}_k\|^2] + \mathbb{E} [\|\mathbf{A}_k\|^2] + \mathbb{E} [\|\mathbf{M}_k\|^2] \\
&\quad + 2(\mathbb{E} [\langle \mathbf{e}_k, \mathbf{A}_k \rangle] + \mathbb{E} [\langle \mathbf{e}_k, \mathbf{M}_k \rangle] + \mathbb{E} [\langle \mathbf{M}_k, \mathbf{A}_k \rangle])
\end{aligned}$$

- For $\mathbb{E} [\langle \mathbf{e}_k, \mathbf{M}_k \rangle]$, let $\mathbb{F} = (\mathcal{F}_t)$ be $\mathcal{F}_t = \sigma(\mathbf{X}_0, \mathbf{W}_s : s \leq t)$. Then we know X_k^h, X_{t_k} are $\mathcal{F}_{t_k}$ -measurable and so is $\mathbf{e}_k$. Moreover, because $\mathbf{W}$ is independent of $\mathbf{X}_0$, $\mathbf{W}$ is a Brownian motion w.s.t. $\mathbb{F}$, which implies that

$$\mathbb{E} [\mathbf{M}_k \mid \mathcal{F}_{t_k}] = \mathbb{E} \left[\int_{t_k}^{t_{k+1}} (\sigma(s, \mathbf{X}_s) - \sigma(t_k, \mathbf{X}_k^h)) d\mathbf{W}_s \mid \mathcal{F}_{t_k} \right] = 0.$$

So

$$\begin{aligned}
\mathbb{E} [\langle \mathbf{e}_k, \mathbf{M}_k \rangle] &= \mathbb{E} [\mathbb{E} [\langle \mathbf{e}_k, \mathbf{M}_k \rangle \mid \mathcal{F}_{t_k}]] \\
&= \mathbb{E} [\langle \mathbf{e}_k, \mathbb{E} [\mathbf{M}_k \mid \mathcal{F}_{t_k}] \rangle] = 0
\end{aligned}$$

- For $\mathbb{E} [\langle \mathbf{M}_k, \mathbf{A}_k \rangle]$, by $2ab \leq a^2 + b^2$,

$$2 |\mathbb{E} [\langle \mathbf{M}_k, \mathbf{A}_k \rangle]| \leq 2 (\mathbb{E} [\|\mathbf{A}_k\|^2])^{1/2} (\mathbb{E} [\|\mathbf{M}_k\|^2])^{1/2} \leq \mathbb{E} [\|\mathbf{A}_k\|^2] + \mathbb{E} [\|\mathbf{M}_k\|^2].$$

- For $\mathbb{E} [\langle \mathbf{e}_k, \mathbf{A}_k \rangle]$,

$$2 |\mathbb{E} [\langle \mathbf{e}_k, \mathbf{A}_k \rangle]| \leq 2 (\mathbb{E} [\|\mathbf{e}_k\|^2])^{1/2} (\mathbb{E} [\|\mathbf{A}_k\|^2])^{1/2} \leq \varepsilon \mathbb{E} [\|\mathbf{e}_k\|^2] + \frac{1}{\varepsilon} \mathbb{E} [\|\mathbf{A}_k\|^2],$$

for any $\varepsilon > 0$. In particular, choose $\varepsilon = h$,

$$2 |\mathbb{E} [\langle \mathbf{e}_k, \mathbf{A}_k \rangle]| \leq h \mathbb{E} [\|\mathbf{e}_k\|^2] + \frac{1}{h} \mathbb{E} [\|\mathbf{A}_k\|^2].$$

Finally, combining this with $\mathbb{E} [\|\mathbf{e}_0\|^2] = 0$, we have

$$\mathbb{E} [\|\mathbf{e}_{k+1}\|^2] \leq (1 + h) \mathbb{E} [\|\mathbf{e}_k\|^2] + \left(2 + \frac{1}{h}\right) \mathbb{E} [\|\mathbf{A}_k\|^2] + 2 \mathbb{E} [\|\mathbf{M}_k\|^2]$$

$$\begin{aligned}
&\leq (1+h) \mathbb{E} [\|\mathbf{e}_k\|^2] + (2K^2h^2 + K^2h) (C_1h + \mathbb{E} [\|\mathbf{e}_k\|^2]) + 2K^2h(C_1h + \mathbb{E} [\|\mathbf{e}_k\|^2]) \\
&\leq (1+C_2h) \mathbb{E} [\|\mathbf{e}_k\|^2] + C_3h^2
\end{aligned}$$

By discrete Grönwall's inequality,

$$\mathbb{E} [\|\mathbf{e}_k\|^2] \leq C_3h^2 \sum_{j=0}^{k-1} (1+C_2h)^{k-1-j} \leq C_3h^2 \frac{(1+C_2h)^k - 1}{(1+C_2h) - 1} \leq Ch. \quad \square$$

Remark 9.5.2. For the discrete Grönwall's inequality, if

$$u_{k+1} \leq (a+bh)u_k + ch^2, \quad u_0 = 0,$$

where $a, b, c > 0$, then we have

$$u_k \leq \alpha_h^k u_0 + ch^2 \sum_{j=0}^{k-1} \alpha_h^j = ch^2 \sum_{j=0}^{k-1} \alpha_h^j, \quad \alpha_h := a + bh.$$

- For $a < 1$, for small enough h , we can have $\alpha_h = a + bh < 1$. So

$$\sum_{j=0}^{k-1} \alpha_h^j \leq \frac{1}{1 - \alpha_h} \leq c',$$

as $h \rightarrow 0$. It follows that

$$u_k \leq cc'h^2 = \mathcal{O}(h^2),$$

as $h \rightarrow 0$.

- For $a = 1$, as above

$$u_k \leq ch \left((1+bh)^k - 1 \right).$$

Because $kh = t_k$ and $bh > 0$,

$$(1+bh)^k \leq e^{bhk} = e^{bt_k} \leq e^{bT}.$$

So

$$u_k \leq (e^{bT} - 1)ch = \mathcal{O}(h).$$

- For $a > 1$, then we cannot get the convergence.

Weak Convergence. For a given SDE,

$$d\mathbf{X}_t = b(t, \mathbf{X}_t)dt + \sigma(t, \mathbf{X}_t)d\mathbf{W}_t,$$

we construct a discrete $(\mathbf{X}_k^h)_{k=0}^N$ with $h = T/N < 1$ and $t_k = kh$, which is said a weak approximation if

$$\mathbb{E}[\varphi(\mathbf{X}_N^h)] \rightarrow \mathbb{E}[\varphi(\mathbf{X}_T)].$$

Consider the following example. Let $b(t, x) = \mu(x)$ and $\sigma(t, x) = \sigma(x)$. Assume there exists a family of probability measures $\{\nu_x\}_{x \in \mathbb{R}^d}$ such that

$$\mu(x) = \mathbb{E}_{\mathbf{Y} \sim \nu_x}[\mathbf{Y}], \quad \sigma(x)\sigma(x)^\top = \text{Cov}_{\mathbf{Y} \sim \nu_x}[\mathbf{Y}].$$

Let $(P_t)_{t \in [0, T]}$ be the corresponding semigroup. Note that the corresponding generator is given by

$$Af(x) = \sum_i \mu_i(x) \frac{\partial f}{\partial x^i}(x) + \frac{1}{2} \sum_{i,j} a_{ij}(x) \frac{\partial^2 f}{\partial x^i \partial x^j}(x) = \langle \mu, \nabla f \rangle + \frac{1}{2} a : D^2 \varphi,$$

where $a(x) = \sigma(x)\sigma(x)^\top$; see in the next chapter.

We construct $\mathbf{X}_k^h$ by the following rule: first, drawing $\mathbf{V}_k$ such that

$$\mathbf{V}_k \mid \mathbf{X}_k^h = x \sim \nu_x,$$

and then updating

$$\mathbf{X}_{k+1}^h = \mathbf{X}_k^h + \mu(\mathbf{X}_k^h)h + \sqrt{h}(\mathbf{V}_k - \mu(\mathbf{X}_k^h)).$$

Let $(R_k^h)_{k=0}^N$ be the corresponding semigroup.

Theorem 9.5.3. *Using notations as above, assume $\mu_i, \sigma_{ij} \in C_b^3(\mathbb{R}^n)$ and $a(x)$ is uniformly elliptic and regular enough. Further assume that $\sup_x \mathbb{E}_{\mathbf{Y} \sim \nu_x} [\|\mathbf{Y}\|^3] < \infty$. Then there exists a $C_T > 0$ such that*

$$\|R_N^h f - P_T f\|_\infty \leq C_T h^{1/2}, \quad \forall f \in C_b^3(\mathbb{R}^n)$$

Lemma 9.5.4. *Let $S_h = R_1^h$. For any $\varphi \in C_b^3$,*

$$\sup_x |(S_h \varphi)(x) - \varphi(x) - h(A\varphi)(x)| \leq Ch^{3/2}.$$

Proof. Fix $x \in \mathbb{R}^n$. Let $\eta_x = V - \mu(x)$ for $V \sim \nu_x$. Then by definition,

$$\mathbb{E}[\eta_x] = 0, \quad \mathbb{E}[\eta_x \eta_x^\top] = a(x), \quad \mathbb{E}[|\eta_x|^3] \leq C.$$

Let

$$\Delta := \Delta(x, h) := \mu(x)h + \sqrt{h}\eta_x.$$

We have

$$(S_h \varphi)(x) = \mathbb{E}[\varphi(x + \Delta)].$$

(because $\mathbb{E}[\varphi(X, Y) \mid X = x] = \mathbb{E}[\varphi(x, Z)]$ for $Z \sim Y \mid X = x$). By Taylor expansion,

$$\varphi(x + \Delta) = \varphi(x) + \nabla \varphi(x) \cdot \Delta + \frac{1}{2} \Delta^\top D^2 \varphi(x) \Delta + R(x, \Delta),$$

where

$$|R(x, \Delta)| \leq C_\varphi |\Delta|^3,$$

by $\sup_x |D^3 \varphi(x)| < C_\varphi$. Therefore,

$$(S_h \varphi)(x) - \varphi(x) = \nabla \varphi(x) \cdot \mathbb{E}[\Delta] + \frac{1}{2} \mathbb{E}[\Delta^\top D^2 \varphi(x) \Delta] + \mathbb{E}[R(x, \Delta)].$$

For the three terms on the RHS,

- $\mathbb{E}[\Delta] = \mu(x)h + \sqrt{h}\mathbb{E}[\eta_x] = \mu(x)h$.
- because $\mathbb{E}[\Delta \Delta^\top] = ha(x) + h^2 \mu(x) \mu(x)^\top$,

$$\frac{1}{2} \mathbb{E}[\Delta^\top D^2 \varphi(x) \Delta] = \frac{1}{2} (ha(x) + h^2 \mu \mu^\top) : D^2 \varphi(x) = \frac{1}{2} ha(x) : D^2 \varphi(x) + O(h^2)$$

because $D^2 \varphi$ is uniformly bounded.

- by $|a + b|^3 \leq C_1(|a|^3 + |b|^3)$,

$$|\Delta|^3 \leq C(h^3 |\mu(x)|^3 + h^{3/2} |\eta_x|^3).$$

Since μ and $\mathbb{E}[|\eta_x|^3]$ are uniformly bounded,

$$\mathbb{E}|\Delta|^3 \leq C_2(h^3 + h^{3/2}) \leq C_3 h^{3/2}.$$

for $h < 1$.

Then we have

$$(S_h \varphi)(x) - \varphi(x) = h(A\varphi)(x) + O(h^2) + O(h^{3/2}). \quad \square$$

Lemma 9.5.5. Fix $f \in C_b^3(\mathbb{R}^n)$. Let $u(t, x) = \mathbb{E}[f(X_{T-t}) : X_0 = x] = P_{T-t}f(x)$ and $u_k(x) = u(t_k, x)$. There exists a C' such that

$$\sup_x |u_k(x) - u_{k+1}(x) - hAu_{k+1}(x)| \leq C'h^2.$$

Proof. Note that by the backward Kolmogorov equation

$$\partial_t u(t, x) + Au(t, x) = 0, \quad u(T, x) = f(x).$$

Therefore,

$$u_k(x) - u_{k+1}(x) = \int_{t_k}^{t_{k+1}} Au(s, x) ds,$$

which implies that

$$u_k(x) - u_{k+1}(x) - hAu_{k+1}(x) = \int_{t_k}^{t_{k+1}} (Au(s, x) - Au(t_{k+1}, x)) ds.$$

Moreover, our “regular enough” for a means that $u \in C_b^{1,3}([0, T] \times \mathbb{R}^n)$, so

$$|Au(s, x) - Au(t_{k+1}, x)| \leq C_4 |s - t_{k+1}|$$

by the boundedness of μ, σ . Therefore,

$$|u_k(x) - u_{k+1}(x) - hAu_{k+1}(x)| \leq \int_{t_k}^{t_{k+1}} C_4 |s - t_{k+1}| ds \leq C'h^2. \quad \square$$

Proof of Theorem 9.5.3. Fix $f \in C_b^3(\mathbb{R}^n)$. Let $u(t, x) = \mathbb{E}[f(X_{T-t}) : X_0 = x] = P_{T-t}f(x)$ and $u_k = u(t_k, \cdot)$. We need to measure $\|R_N^h f - u(0, \cdot)\|$.

- First, let

$$\delta_k(x) := (S_h u_{k+1})(x) - u_k(x).$$

Then

$$\delta_k(x) = (S_h u_{k+1}(x) - u_{k+1}(x) - hAu_{k+1}(x)) + (u_{k+1}(x) + hAu_{k+1}(x) - u_k(x)).$$

For the first term, by Lemma 9.5.4,

$$|S_h u_{k+1}(x) - u_{k+1}(x) - hAu_{k+1}(x)| \leq Ch^{3/2}.$$

By Lemma 9.5.5,

$$|u_{k+1}(x) + hAu_{k+1}(x) - u_k(x)| \leq C'h^2.$$

Therefore,

$$|\delta_k(x)| \leq C_5 h^{3/2} \implies \|\delta_k\|_\infty \leq C_5 h^{3/2}.$$

- Note that $u(T, x) = f(x)$ and $R_N^h f(x) = \mathbb{E}[f(\mathbf{X}_N^h) | X_0 = x] = \mathbb{E}_x[u(T, \mathbf{X}_N^h)]$. So

$$R_N^h f(x) - P_T f(x) = \mathbb{E}_x[u(T, \mathbf{X}_N^h) - u(0, x)] = \sum_{k=0}^{N-1} \mathbb{E}_x[u(t_{k+1}, \mathbf{X}_{k+1}^h) - u(t_k, \mathbf{X}_k^h)].$$

For each k , we have

$$\mathbb{E}_x [u(t_{k+1}, \mathbf{X}_{k+1}^h) - u(t_k, \mathbf{X}_k^h)] = \mathbb{E}_x [\mathbb{E} [u_{k+1}(\mathbf{X}_{k+1}^h) - u_k(\mathbf{X}_k^h) \mid \mathbf{X}_k^h]].$$

Because

$$\mathbb{E} [u_{k+1}(\mathbf{X}_{k+1}^h) \mid \mathbf{X}_k^h = y] = (S_h u_{k+1})(y),$$

we get

$$\mathbb{E}_x [u(t_{k+1}, \mathbf{X}_{k+1}^h) - u(t_k, \mathbf{X}_k^h)] = \mathbb{E}_x [\delta_k(\mathbf{X}_k^h)],$$

which implies that

$$\|R_N^h f - P_T f\|_\infty = \sup_x \left| \sum_{k=0}^{N-1} \mathbb{E}_x [\delta_k(\mathbf{X}_k^h)] \right| \leq \sum_{k=0}^{N-1} \sup_x |\mathbb{E}_x [\delta_k(\mathbf{X}_k^h)]|.$$

Moreover, since $|\mathbb{E}_x [\delta_k(\mathbf{X}_k^h)]| \leq \|\delta_k\|_\infty$,

$$\|R_N^h f - P_T f\|_\infty \leq \sum_{k=0}^{N-1} \|\delta_k\|_\infty \leq \sum_{k=0}^{N-1} C_5 h^{3/2} = N C_5 h^{3/2} = \frac{T}{h} C_5 h^{3/2} = C_T h^{1/2}. \quad \square$$

9.6 Diffusion Process

In the following, we do not distinguish vector-valued X and scale-valued X .

We mainly consider the time-homogeneous case, i.e.,

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t,$$

where $X_t \in \mathbb{R}^n$, $b: \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\sigma: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ are time-independent, and $B = (B_t)_{t \geq 0}$ be a m -dimensional Brownian motion. Any process $X = (X_t)_t$ satisfies (9.6) is called a (Itô) diffusion. We further assume b, σ are locally Lipschitz continuous, which implies that a solution X to (9.6) is a Markov process satisfying the strong Markov property.

Semigroup. So it has the corresponding transition semigroup $(Q_t)_{t \geq 0}$, i.e.,

$$Q_t g(x) = \mathbb{E}_x [g(X_t) \mathbb{I}_{\{t < \tau(x)\}}] = \mathbb{E}[g(X_t) \mathbb{I}_{\{t < \tau(x)\}} \mid X_0 = x], \quad \forall g \in \mathcal{B}_b(\mathbb{R}^n)$$

Note that if for the globally Lipschitz continuous case, $(Q_t)_{t \geq 0}$ is a true Markov semigroup. But for the locally Lipschitz continuous case, $(Q_t)_{t \geq 0}$ is a sub-Markov semigroup, i.e, $Q_t: \mathcal{B}_b(\mathbb{R}^n) \rightarrow \mathcal{B}_b(\mathbb{R}^n)$ is a linear contraction, positive preserving operator with $Q_t Q_s = Q_{t+s}$, and

$$0 < Q_t \mathbb{1}(x) = \mathbb{P}_x(t < \tau(x)) \leq 1$$

instead of $= 1$. But if we consider $\mathbb{R}^n \cup \{\infty\}$ and extend the solution $\tilde{X}_t$ valued on $\mathbb{R}^n \cup \{\infty\}$, then define

$$\tilde{Q}_t \tilde{g}(x) = \mathbb{E}_x [\tilde{g}(\tilde{X}_t)], \quad \tilde{g} \in \mathcal{B}_b(\mathbb{R}^n \cup \{\infty\}),$$

and so such $(\tilde{Q}_t)_{t \geq 0}$ is a true Markov semigroup. Moreover, for $g \in \mathcal{B}_b(\mathbb{R}^n)$, we can extend g to $\tilde{g} \in \mathcal{B}_b(\mathbb{R}^n \cup \{\infty\})$ by define $\tilde{g}(\infty) = 0$. Then

$$Q_t g = \tilde{Q}_t \tilde{g}.$$

Furthermore, for any x , define

$$p_t(x, A) = \mathbb{P}_x(X_t^x \in A, t < \tau(x)), \quad \forall A \in \mathcal{B}(\mathbb{R}^n),$$

which is a transition kernel because $x \mapsto p_t(x, A)$ is Borel measurable. Then we have for all $g \in \mathcal{B}_b(\mathbb{R}^n)$

$$Q_t g(x) = \mathbb{E}_x [g(X_t) \mathbb{I}_{\{t < \tau(x)\}}] = \int g(y) p_t(x, dy).$$

Remark 9.6.1. Similarly, by introducing ∞ and $\tilde{X}_t$, the definition is as usual,

$$\tilde{p}_t(x, A) := \mathbb{P}_x \left(\tilde{X}_t \in A \right),$$

a proper Markov kernel defined on $\mathbb{R}^n \cup \{\infty\}$.

Note that, for locally Lipschitz continuous case, Q_t does not satisfy the Jensen's inequality, unlike the global Lipschitz case, but for $1 \leq p < \infty$, we still have

$$|Q_t f|^p \leq Q_t |f|^p.$$

Proof. Let

$$m_t(x) = p_t(x, \mathbb{R}^n) = \mathbb{P}_x(X_t^x \in \mathbb{R}^n, t < \tau(x)) < 1, \quad \text{and} \quad \pi_x(dy) = \frac{p_t(x, dy)}{m_t(x)}.$$

So π_x is a probability measure, which implies that

$$\left| \int f \pi_x \right|^p \leq \int |f|^p \pi_x$$

by Jensen's inequality. Note that $P_t f(x) = \int f(y) p_t(x, dy) = m_t(x) \int f \pi_x(dy)$. Therefore,

$$\begin{aligned} |Q_t f(x)|^p &= m_t(x)^p \left| \int f \pi_x \right|^p \\ &\leq m_t(x)^p \int |f|^p \pi_x = m_t(x)^{p-1} \int |f|^p p_t(x, dy) \\ &\leq \int |f|^p p_t(x, dy). \end{aligned}$$

□

Generator. We can also define the generator

$$Lg(x) = \lim_{t \downarrow 0} \frac{Q_t g(x) - g(x)}{t},$$

for

$$g \in \mathcal{D}(L) := \left\{ g \in \mathcal{B}_b(\mathbb{R}^n) : \exists f \in \mathcal{B}_b(\mathbb{R}^n), \lim_{t \downarrow 0} \left\| \frac{Q_t g(x) - g(x)}{t} - f \right\|_\infty = 0 \right\}.$$

Such L can be extended by solving the martingale problem, for

$$f \in D(L) \subset \mathcal{B}_b(\mathbb{R}^n) \Leftrightarrow f(X_t^x) - f(x) - \int_0^t g(X_s^x) ds, \quad \text{a local martingale}$$

for some measurable g with

$$\int_0^t |g(X_s^x)| ds < \infty, \quad \text{a.e. on } \{t < \tau(x)\}, \quad \forall x, t$$

First, g is unique up to in the sense that if g_1, g_2 satisfy above, then

$$g_1(X_t^x) = g_2(X_t^x), \quad \forall \text{ Lebesgue a.e. } -t \geq 0,$$

by differentiating $\int_0^t (g_1(X_s^x) - g_2(X_s^x)) ds = 0$. And it is independent of the choice of X_0 . If in $C(\mathbb{R}^n)$, g is unique up to pointwise, i.e., $g_1(x) = g_2(x)$, because the differentiating exists at each t and we can choose $X_0^x = x$.

Theorem 9.6.2. *If b and σ are globally Lipschitz in x , then $(Q_t)_{t \geq 0}$ is a Feller-Dynkin semigroup.*

Proof. For

$$X_t^x = x + \int_0^t \sigma(X_s^x) dB_s + \int_0^t b(X_s^x) ds,$$

by the global Lipschitz of b, σ and similar reasoning as in the proof of the existence and uniqueness of the solution, we have

$$\mathbb{E} [\|X_t^x - x\|^2] \leq C(t + t^2),$$

for some C . By Markov's inequality,

$$\sup_{x \in \mathbb{R}^d} P(\|X_t^x - x\| > R) \xrightarrow{R \rightarrow \infty} 0.$$

Moreover, writting

$$|Q_t f(x)| = |\mathbb{E}[f(X_t^x)]| \leq |\mathbb{E}[f(X_t^x) \mathbf{1}_{\{\|X_t^x - x\| \leq A\}}]| + \|f\| P(\|X_t^x - x\| > R),$$

Since $f \in C_0$, $f(x) \rightarrow 0$ as $x \rightarrow \infty$. So

$$\limsup_{x \rightarrow \infty} |Q_t f(x)| \leq \|f\| \sup_{x \in \mathbb{R}^d} P(\|X_t^x - x\| > R).$$

As $R \rightarrow \infty$, $\limsup_{x \rightarrow \infty} |Q_t f(x)| = 0$, and so $Q_t f \in C_0$.

Similarly, for any $\varepsilon > 0$,

$$\sup_{x \in \mathbb{R}^d} |E[f(X_t^x)] - f(x)| \leq \sup_{x, y, \|x-y\| \leq \varepsilon} |f(x) - f(y)| + 2\|f\| \sup_x P(\|X_t^x - x\| > \varepsilon).$$

As above, by Markov's inequality,

$$\sup_x P(\|X_t^x - x\| > \varepsilon) \xrightarrow{t \rightarrow 0} 0.$$

Therefore, by continuity for f ,

$$\begin{aligned} \limsup_{t \rightarrow 0} \|Q_t f - f\| &= \limsup_{t \rightarrow 0} \left(\sup_x |E[f(X_t^x)] - f(x)| \right) \\ &\leq \sup_{x, y, \|x-y\| \leq \varepsilon} |f(x) - f(y)| \rightarrow 0 \end{aligned}$$

as $\varepsilon \rightarrow 0$. □

Remark 9.6.3. For global Lipschitz case, because Q is Feller-Dynkin, the definition of generator coincides with the usual definition on C_0 .

Remark 9.6.4. If b and σ are only locally Lipschitz, because of the explosion time, we cannot guarantee $Q_t f$ in C_0 for $f \in C_0$.

Note that by Itô formula, for any $f \in C_c^2(\mathbb{R}^n)$

$$\begin{aligned} f(X_t) &= f(X_0) + \sum_{i=1}^n \int_0^t \frac{\partial f}{\partial x_i} dX_s^i + \frac{1}{2} \sum_{i,j=1}^n \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j} d[X^i, X^j]_s \\ &= f(X_0) + \sum_{i=1}^n \int_0^t b^i \frac{\partial f}{\partial x_i} ds + \sum_{i,j=1}^n \int_0^t \sigma_{ij} \frac{\partial f}{\partial x_i} dB_s^j \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \sum_{i,j=1}^n \int_0^t (\sigma \sigma^\top)_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} dt \\
& = f(X_0) + \int_0^t \left(\sum_{i=1}^n b^i \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^n (\sigma \sigma^\top)_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} \right) ds + \sum_{i,j=1}^n \int_0^t \sigma_{ij} \frac{\partial f}{\partial x_i} dB_s^j,
\end{aligned}$$

because

$$d[X^i, X^j]_s = \left(\sum_k \sigma_{ik} dB^k \right) \left(\sum_\ell \sigma_{j\ell} dB^\ell \right) = (\sigma \sigma^\top)_{ij} dt.$$

This implies the following theorem.

Theorem 9.6.5. *If $f \in C_c^2(\mathbb{R}^n)$, then $f \in \mathcal{D}(L)$ and*

$$Lf(x) = \sum_i b_i(x) \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i,j} (\sigma \sigma^\top)_{i,j}(x) \frac{\partial^2 f}{\partial x_i \partial x_j},$$

where L is the generator of Markov process X .

For $f \in C_c^2(\mathbb{R}^n)$, we have

$$f(X_t) = f(X_0) + \int_0^t Lf(X_s) ds + \int_0^t \nabla f(X_s)^\top \sigma(X_s) dB_s.$$

So

$$M_t = f(X_t) - \int_0^t Lf(X_s) ds$$

is a martingale, which is a particular case of Theorem 7.3.11. Note that here X_0 is not fixed to a point. Moreover, if $f \in C^2$, then we just know $(M_t)_{t \geq 0}$ is a local martingale.

Theorem 9.6.6 (Dynkin's formula). *If $f \in C_c^2(\mathbb{R}^n)$ and τ is a stopping time with $\mathbb{E}_x[\tau] < \infty$, then*

$$\mathbb{E}_x[f(X_\tau)] = f(x) + \mathbb{E}_x \left[\int_0^\tau Lf(X_s) ds \right].$$

Example 9.6.7 (Bessel Process). Let B be a m -dimensional standard Brownian motion. Consider

$$R_t = \|B_t\| = \sqrt{(B_t^1)^2 + \cdots + (B_t^m)^2},$$

the Bessel process. Then we know

$$dR_t = \frac{n-1}{2R_t} dt + \sum_{i=1}^m \frac{B_t^i}{R_t} dB_t^i.$$

Let

$$\tilde{B}_t = \sum_{i=1}^m \int_0^t \frac{B_s^i}{\|B_s\|} dB_s^i.$$

Then by Lévy's theorem, $\tilde{B} = (\tilde{B}_t)_{t \geq 0}$ is a 1-dimensional Brownian motion. Therefore,

$$dR_t = \frac{n-1}{2R_t} dt + d\tilde{B}_t.$$

So by the uniqueness of weak solution, $R = (R_t)_{t \geq 0}$ is also an Itô diffusion with generator

$$Lf(x) = \frac{1}{2} f''(x) + \frac{n-1}{2x} f'(x).$$

Example 9.6.8. Let $U \in C^1(\mathbb{R}^n)$ and

$$L = \Delta \cdot + \langle \nabla U, \nabla \cdot \rangle$$

on $C_c^\infty(\mathbb{R}^n)$. Then by the divergence theorem,

$$\mu(dx) = e^{U(x)} dx$$

is symmetric for L . Moreover, L is essentially self-adjoint on $L^2(\mathbb{R}^n, \mu)$.

Invariant Measure. For the semigroup P_t induced by a solution of SDE with locally Lipschitz continuous coefficient, a measure μ is invariant if

$$\int Q_t f d\mu = \int f d\mu,$$

for all bounded measurable function f , which is equivalent to $\int Q_t f d\mu = \int f d\mu$ for all $f \in C_c^\infty$, since on a locally compact Hausdorff space, Radon measures are uniquely determined by their integrals against C_c . Because $C_c^\infty \subset D(L)$, it is equivalent to

$$\int Lf d\mu = 0, \quad \forall f \in C_c^\infty(\mathbb{R}^n).$$

Moreover, μ is called sub-invariant if

$$\int Q_t f d\mu \leq \int f d\mu,$$

for all bounded measurable function f . Note that, if μ is sub-invariant, then by above $|Q_t f|^p \leq Q_t |f|^p$, for any $f \in \mathcal{B}_b \cap L^p(\mu)$,

$$\int |Q_t f|^p d\mu \leq \|f\|_p^p < \infty,$$

i.e., $Q_t f \in \mathcal{B}_b \cap L^p(\mu)$, because $\mathcal{B}_b \cap L^p(\mu) \subset L^p(\mu)$ dense, so

$$Q_t: L^p(\mu) \rightarrow L^p(\mu)$$

a contraction, for all $1 \leq p \leq \infty$.

A measure μ is called symmetric to Q_t if for any $f, g \in \mathcal{B}_b(\mathbb{R}^n)$,

$$\int f Q_t g d\mu = \int g Q_t f d\mu.$$

In kernel representation,

$$\mu(dx) p_t(x, dy) = \mu(dy) p_t(dx, y).$$

It is obvious that μ is sub-invariant, so $Q_t: L^2(\mu) \rightarrow L^2(\mu)$ is self-adjoint. On the other hand, because $C_c^\infty \subset L^2(\mu)$ is dense, μ is symmetric if and only if

$$\int f Q_t g d\mu = \int g Q_t f d\mu, \quad \forall f, g \in C_c^\infty,$$

so it is equivalent to

$$\int f L g d\mu = \int g L f d\mu, \quad \forall f, g \in C_c^\infty.$$

Note that because $P_t: L^2(\mu) \rightarrow L^2(\mu)$ is a contractive semigroup that satisfies

$$\lim_{t \rightarrow 0} \|Q_t f - f\|_2 = 0, \quad \forall f \in L^2(\mu),$$

by the continuity of X_t and DCT, the Hille–Yosida theory provides a generator A with $D(A)$ that is self-adjoint on $L^2(\mu)$ and

$$Q_t = e^{tA}.$$

By above, it is obvious that $L \subset A$ when setting considering $D(L) = C_c^\infty$. So A is a self-adjoint extension of L . For L , we can consider its Friedrichs extension A_F , $A = A_F$ by the construction. Moreover, if L is essentially self-adjoint on $C_c^\infty(M)$ w.s.t. μ , it has only one self-adjoint extension that is $A = A_F$. On the other hand, we can consider the martingale extension $\mathcal{L}$ on $L^2(\mu)$ similar as above, that is, $f \in D(\mathcal{L})$ if $f \in L^2(\mu)$ and there exists a $g \in L^2(\mu)$ such that for any x

$$f(X_t^x) - f(x) - \int_0^t g(X_s^x) ds$$

is a local martingale, then $g := \mathcal{L}f$. Note that $A = \mathcal{L}$, because $A \subset \mathcal{L}$ is directly by Dynkin's formula, and $\mathcal{L} \subset A$ is by taking expectation, which implies that

$$Q_t f(x) - f(x) = \int_0^t Q_s g(x) ds \Rightarrow \lim_{t \rightarrow 0} \frac{Q_t f(x) - f(x)}{t} = g(x).$$

Weak uniqueness. On $\mathbb{R}^n$ (a Riemannian manifold M), when given a smooth second order elliptic, but not necessarily nondegenerate, differential operator L , an F_* -adapted stochastic process $X: \Omega \rightarrow W(\mathbb{R}^n \cup \{\infty\})$ defined on a filtered probability space $(\Omega, F_*, \mathbb{P})$ is called a diffusion process generated by L if X is an $\mathbb{R}^n \cup \{\infty\}$ -valued F_* semimartingale up to stopping time $e(X)$ and

$$M^f(X)_t = f(X_t) - f(X_0) - \int_0^t Lf(X_s) ds, \quad 0 \leq t < e(X),$$

and if $X_0 \sim \mu$, then $\mathbb{P}_\mu = X_\# \mathbb{P}$ is called a L -diffusion measure on $W(\mathbb{R}^n \cup \{\infty\})$. For such L and given μ , $\mathbb{P}_\mu$ uniquely exists (Hsu 2002).

For a given diffusion process

$$dX_t = \mu(X_t)dt + \sigma(X_t)dB_t,$$

we know its solution is an L -diffusion process with

$$Lf = \sum_i \mu_i(x) \frac{\partial f}{\partial x^i}(x) + \frac{1}{2} \sum_{i,j} a_{ij}(x) \frac{\partial^2 f}{\partial x^i \partial x^j}(x).$$

Conversely, for a given L ,

$$L = \sum_i \mu_i(x) \frac{\partial}{\partial x^i} + \frac{1}{2} \sum_{i,j} a_{ij}(x) \frac{\partial^2}{\partial x^i \partial x^j},$$

and X is a semimartingale on a probability space $(\Omega, \mathbb{P}, F_*)$ satisfying that

$$M^f(X)_t = f(X_t) - f(X_0) - \int_0^t Lf(X_s) ds, \quad 0 \leq t < e(X),$$

is a local martingale for any $f \in C^\infty(M)$, then X is a solution of

$$dX_t = \mu(X_t)dt + \sigma(X_t)dB_t,$$

for $\sigma = a^{1/2}$ and some Brownian motion B .

Chapter 10

Symmetric Markov Operator

In this chapter, let E be a Polish space that is a separable complete metric space and let $\mathcal{F}$ be equipped with the Borel σ -field. Then the measure decomposition theorem implies that for any probability measure μ on the product σ -field $\mathcal{F} \otimes \mathcal{F}$ on $E \times E$ with $\mu_1 = \pi_1^\# \mu$, the first projection, then

$$\mu(dx, dy) = k(x, dy)\mu_1(dx)$$

for some probability transition kernel $k: E \times \mathcal{F} \rightarrow [0, 1]$. Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space.

Remark 10.0.1. Moreover, because of the existence of kernels, by Ionescu–Tulcea theorem, for any probability measure μ on E^n , there are kernels k_i from E^{i-1} to E such that

$$\mu(dx_1, dx_2, \dots, dx_n) = \mu_1(dx_1)k_2(x_1, dx_2)k_3(x_1, x_2, dx_3) \cdots k_n(x_1, \dots, x_{n-1}, dx_n).$$

For now on any measure μ is assumed to be σ -finite.

10.1 Markov Operator

Definition 10.1.1. A Markov operator P on $(E, \mathcal{F})$ is a linear operator $P: \mathcal{B}_b(E) \rightarrow \mathcal{B}_b(E)$ such that

- (1) (*Mass conservation*) $P\mathbb{1} = \mathbb{1}$ for constant function $\mathbb{1}(x) \equiv 1$,
- (2) (*Positivity preserving*) for $f \geq 0$, $Pf \geq 0$.

Remark 10.1.2. For $0 \leq f \leq 1$,

$$P(\mathbb{1} - f) \geq 0 \Rightarrow 0 \leq Pf \leq P\mathbb{1} \leq \mathbb{1}.$$

Therefore, $\|Pf\|_\infty \leq \|f\|_\infty$ for all $f \in \mathcal{B}_b(E)$.

Proposition 10.1.3 (Jensen's inequality). *For any convex $\phi: \mathbb{R} \rightarrow \mathbb{R}$ and any $f \in \mathcal{B}_b(E)$, if P is a Markov operator, then*

$$P(\phi(f)) \geq \phi(Pf)$$

Proof. Because ϕ is convex, for any $b \in \mathbb{R}$, there is $a = a(b)$ such that

$$\phi(c) \geq \phi(b) + a(b)(c - b), \quad \forall c \in \mathbb{R}$$

For any $x \in E$, let $c = f(x)$. We have

$$\phi(f(x)) \geq \phi(b) + a(b)(f(x) - b) \Rightarrow \phi(f) \geq \phi(b) + a(b)(f - b)$$

By the positivity and mass properties of P , we have

$$P(\phi(f)) \geq \phi(b) + a(b)(Pf - b)$$

So for any $x \in E$,

$$P(\phi(f))(x) \geq \phi(b) + a(b)(Pf(x) - b)$$

Then let $b = Pf(x)$, we get

$$P(\phi(f))(x) \geq \phi(Pf(x))$$

which is true for any $x \in E$. □

Definition 10.1.4 (Invariant Measure). A measure μ on $(E, \mathcal{F})$ is called invariant for a Markov operator P if

$$\int_E Pf d\mu = \int_E f d\mu,$$

for all $f \in \mathcal{B}_b(E)$.

Remark 10.1.5. When $f \in \mathcal{B}_b(E)$ is 0 μ -a.e., $Pf = 0$ μ -a.e.. Therefore, P can be extended on $L^\infty(\mu)$. Moreover, μ is invariant for P if

$$\int_E Pf d\mu = \int_E f d\mu, \quad \forall f \in L^1(\mu) \cap L^\infty(\mu).$$

Note that for $1 \leq p < \infty$, $L^1(\mu) \cap L^\infty(\mu) \subset L^p(\mu)$ that is because

$$\|f\|_p^p = \int |f|^p d\mu \leq \|f\|_\infty^{p-1} \int |f| d\mu = \|f\|_\infty^{p-1} \|f\|_1.$$

So by Jensen's inequality for $\phi(x) = |x|^p$ ($1 \leq p < \infty$),

$$\int |Pf|^p d\mu \leq \int P(|f|^p) d\mu = \int |f|^p d\mu, \quad \forall f \in L^1(\mu) \cap L^\infty(\mu).$$

Lemma 10.1.6. For any $1 \leq p < \infty$,

$$L^1(\mu) \cap L^\infty(\mu) \subset L^p(\mu)$$

is dense.

Proof. Let $f \in L^p$ and $\varepsilon > 0$.

- Step 1: For any $n \in \mathbb{N}$, let

$$g_n(x) := \max(-n, \min(f(x), n)) \in [-n, n].$$

Therefore, $g_n \in L^\infty$ and

$$\|f - g_n\|_p^p = \int_{|f|>n} ||f| - n|^p d\mu \leq \int_{|f|>n} |f|^p d\mu \xrightarrow{n \rightarrow \infty} 0.$$

So let n be sufficiently large such that $\|f - g_n\|_p \leq \varepsilon/2$.

- By σ -finiteness, choose $E_k \uparrow E$ with $\mu(E_k) < \infty$ and put $h_k = g_n \mathbb{I}_{E_k}$. So $h_k \in L^1 \cap L^\infty$. Because $\mathbb{I}_{E_k^c} \rightarrow 0$ as $k \rightarrow \infty$, by DCT,

$$\|g_n - h_k\|_p^p = \int_{E_k^c} |g_n|^p d\mu \leq \int_{E_k^c} |f|^p d\mu \xrightarrow{k \rightarrow \infty} 0.$$

Therefore,

$$\|f - h_k\|_p \leq \|f - g_n\|_p + \|g_n - h_k\|_p < \varepsilon. \quad \square$$

Then because

$$\|Pf\|_p \leq \|f\|_p, \quad \forall f \in L^1(\mu) \cap L^\infty(\mu),$$

and the density,

$$P: L^p(\mu) \rightarrow L^p(\mu)$$

for all $1 \leq p \leq \infty$. Note that this definition should fix an invariant μ .

Definition 10.1.7 (Reversible Measure). A measure μ is called reversible for a Markov operator P if

$$\int fPg d\mu = \int gPfd\mu, \quad \forall f, g \in L^2(\mu).$$

Remark 10.1.8. It is obviously that if μ is reversible, then it is invariant, because it can choose $g_n \in L^2(\mu)$ such that $g_n \uparrow \mathbf{1}$ by the σ -finiteness of μ .

Definition 10.1.9. A symmetric Markov semigroup on $(E, \mathcal{F}, \mu)$ is a family of $(P_t)_{t \geq 0}$ of Markov operators such that

- (i) (*Initial Condition*) $P_0 f = f$ for all $f \in L^\infty$;
- (ii) (*Semigroup*) for every $t, s \geq 0$, $P_t P_s = P_{t+s}$;
- (iii) (*Symmetry*) for every $t \geq 0$, μ is reversible for P_t ;
- (iv) (*Strong Continuity*) for all $f \in L^2(\mu)$, $P_t f \rightarrow f$ in $L^2(\mu)$ as $t \rightarrow 0$.

Remark 10.1.10. Note that strong continuity implies that $P_t \rightarrow P_{t_0}$ in the strong operator topology on $L^2(\mu)$ as $t \rightarrow t_0$ with the help of the initial condition and the semigroup property.

Theorem 10.1.11 (Kernel Representation). *Let P be a Markov operator on $(E, \mathcal{F})$ that is continuous on $L^1(\nu)$. Then there exists a probability kernel p on $(E, \mathcal{F})$ such that for every $f \in L^\infty(\nu)$ and ν -a.e. $x \in E$,*

$$Pf(x) = \int_E f(y)p(x, dy).$$

10.2 Generator

For a given symmetric Markov semigroup $(P_t)_{t \geq 0}$ on $(E, \mathcal{F}, \mu)$, we can similarly define the generator but the domain is different,

$$\mathcal{D}(L) := \left\{ f \in L^2(\mu) : \lim_{t \rightarrow 0} \frac{P_t f - f}{t} \text{ exists in } L^2(\mu) \right\}.$$

And we also define $\mathcal{D}_p(L)$ for considering the convergence in $L^p(\mu)$. Except for the domain, some properties are as same as the generator of a Feller semigroup, like, $\mathcal{D}(L) \subset L^2(\mu)$ dense, and $L(P_t f) = P_t(Lf)$. So

$$P_t f - f = \int_0^t P_s(Lf) ds = \int_0^t L(P_s f) ds.$$

Moreover, by the symmetry of $(P_t)_{t \geq 0}$, for any $f, g \in \mathcal{D}(L)$

$$\int_E f Lg d\mu = \int_E g Lf d\mu,$$

and for every $f \in \mathcal{D}_1(L)$,

$$\int Lf d\mu = 0.$$

Assume there exists an algebra $\mathcal{A} \subset \mathcal{D}(L)$, for example, $\mathcal{A} = C_c^\infty(\mathbb{R}^n)$.

Definition 10.2.1 (Carré du Champ). The carré du champ associated to L is the bilinear form Γ on $\mathcal{A} \times \mathcal{A}$ defined by

$$\Gamma(f, g) = \frac{1}{2}(L(fg) - fLg - gLf).$$

$$\Gamma(f) = \Gamma(f, f).$$

Note that

$$\frac{d}{dt} (P_t f)^2 = 2f \frac{d}{dt} P_t f = 2f L P_t f,$$

and by Jensen's inequality,

$$L(f^2) = \lim_{t \rightarrow 0} \frac{P_t(f^2) - f^2}{t} \geq \lim_{t \rightarrow 0} \frac{(P_t f)^2 - f^2}{t} = \left. \frac{d(P_t f)^2}{dt} \right|_{t=0} \leq 2f L f,$$

which implies that $\Gamma(f) \geq 0$. Then by the Cauchy-Schwartz inequality,

$$\Gamma(f, g)^2 \leq \Gamma(f) \Gamma(g).$$

Proposition 10.2.2. *Let $(P_t)_{t \geq 0}$ on $(E, \mathcal{F}, \mu)$ be a given symmetric Markov semigroup and L be its generator. Then L is a self-adjoint operator on L^2 and so it is closed.*

Moreover, because

$$0 \leq \int_E \Gamma(f) d\mu = - \int_E f L f d\mu,$$

L is non-positive definite.

Construct semigroup from generator L . Let's assume

$$L = \sum_{i,j=1}^n \sigma_{ij}(x) \frac{\partial^2}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i(x) \frac{\partial}{\partial x_i},$$

where b_i and σ_{ij} are continuous functions and $\sigma = (\sigma_{ij}(x)) \in \mathbb{R}^{n \times n}$ is symmetric and nonnegative. $\mathcal{D}(L) = C_c^\infty(\mathbb{R}^n)$. Moreover, if σ is invertible, L is called an elliptic diffusion operator. A Borel measure μ is called symmetric for L if for any $f, g \in C_c^\infty(\mathbb{R}^n)$,

$$\int_{\mathbb{R}^n} g L f d\mu = \int_{\mathbb{R}^n} f L g d\mu.$$

In the following, let's fix a measure μ symmetric for L .

Note that because $\mathcal{D}(L) = C_c^\infty(\mathbb{R}^n) \subset L^2(\mathbb{R}^n, \mu)$ is dense, L is a non-positive symmetric operator that is densely defined on $L^2(\mathbb{R}^n, \mu)$. But it is not self-adjoint. However, it can be extended to a self-adjoint operator.

Theorem 10.2.3 (Friedrichs Extension). *On the Hilbert space $L^2(\mathbb{R}^n, \mu)$, for L defined above, there exists a densely defined non-positive self-adjoint extension of L .*

In fact, if L is essentially self-adjoint, then the Friedrichs extension is the closed operator $\bar{L}$. In such case,

$$\ker(-L^* + \lambda I) = \{0\}, \quad \lambda > 0.$$

It means

$$-L f + \lambda f = 0 \Rightarrow f = 0,$$

where $L f$, the differential in the sense of distribution.

Therefore, in the following, we assume L is essentially self-adjoint and replace $\bar{L}$ by L . Then L is self-adjoint on $L^2(\mathbb{R}^n, \mu)$. So we can define

$$P_t = e^{tL} = \int_{\mathbb{R}} e^{t\lambda} dE_L(\lambda) = \int_0^\infty e^{-t\lambda} dE_L(\lambda), \forall t \geq 0,$$

where E_L is the spectral measure associated with L . The $P_t: L^2(\mathbb{R}^n, \mu) \rightarrow L^2(\mathbb{R}^n, \mu)$ is a bounded operator. Note that

(i) $P_t P_s = P_{t+s}$ for all $t, s \geq 0$.

(ii) for all $f \in L^2$,

$$\|P_t f\|_2 \leq \|f\|_2.$$

(iii) for all $f \in L^2$, $t \mapsto P_t f$ is continuous in $L^2(\mu, \mathbb{R}^n)$.

(iv) for all $f, g \in L^2$,

$$\int_{\mathbb{R}^n} f P_t g d\mu = \int_{\mathbb{R}^n} g P_t f d\mu,$$

i.e., μ is reversible for P_t .

(v) for all $f \in L^2$,

$$\lim_{t \rightarrow 0} \|P_t f - f\|_2 = 0$$

(vi) for all any $f \in \mathcal{D}(L)$,

$$\lim_{t \rightarrow 0} \left\| \frac{P_t f - f}{t} - Lf \right\|_2 = 0.$$

(vii) if $\mathbb{1} \in \mathcal{D}(L)$ and $L\mathbb{1} = 0$, then $P_t \mathbb{1} = \mathbb{1}$.

10.3 Compact Markov Operators

Definition 10.3.1 (Diffusion Carré du Champ). Let $\mathcal{A} \subset \mathbb{R}^E$ be an algebra such that for any $k \in \mathbb{N}$, any $f_1, \dots, f_k \in \mathcal{A}$, and any $\Psi \in C^\infty(\mathbb{R}^k)$, $\Psi(f_1, \dots, f_k) \in \mathcal{A}$. We say a bilinear form $\Gamma: \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ is called a diffusion carré du champ if for any Ψ and f_i as above,

$$\Gamma(\Psi(f_1, \dots, f_k), g) = \sum_{i=1}^k \partial_i \Psi(f_1, \dots, f_k) \Gamma(f_i, g).$$

Consider a symmetric Markov semigroup with generator L and the corresponding carré du champ

$$\Gamma(f, g) = \frac{1}{2}(L(fg) - fLg - gLf).$$

If Γ is a diffusion carré du champ, then

$$L\Psi(f_1, \dots, f_k) = \sum_{i=1}^k \partial_i \Psi(f_1, \dots, f_k) Lf_i + \sum_{i,j=1}^k \partial_i \partial_j \Psi(f_1, \dots, f_k) \Gamma(f_i, f_j). \quad (10.1)$$

In particular, for $k = 1$,

$$\begin{aligned} \Gamma(\psi(f), g) &= \psi'(f) \Gamma(f, g) \\ L\psi(f) &= \psi'(f) Lf + \psi''(f) \Gamma(f, f). \end{aligned}$$

Definition 10.3.2 (Diffusion Semigroup). An operator L satisfying (10.1) is called a diffusion generator. A symmetric Markov semigroup whose generator is a diffusion generator is called a diffusion semigroup.

Definition 10.3.3 (Dirichlet Form). A bilinear form $\mathcal{E}: \mathcal{D}(\mathcal{E}) \times \mathcal{D}(\mathcal{E}) \rightarrow \mathbb{R}$ is called a Dirichlet form if

- (i) $\mathcal{D}(\mathcal{E}) \subset L^2(\mu)$ dense for some μ ,
- (ii) $\mathcal{E}(f, g) = \mathcal{E}(g, f)$ for $f, g \in \mathcal{D}(\mathcal{E})$,
- (iii) $\mathcal{E}(f) = \mathcal{E}(f, f) \geq 0$ for $f \in \mathcal{D}(\mathcal{E})$,
- (iv) $\mathcal{D}(\mathcal{E})$ is complete w.s.t.

$$\langle f, g \rangle_{\mathcal{E}} := \int_E fg d\mu + \mathcal{E}(f, g),$$

- (v) for any $f \in \mathcal{D}(\mathcal{E})$, $0 \vee f \wedge 1 \in \mathcal{D}(\mathcal{E})$ and

$$\mathcal{E}(0 \vee f \wedge 1) \leq \mathcal{E}(f).$$

Remark 10.3.4. Note that for any symmetric, non-negative bilinear form $\mathcal{E}$ defined on some dense $D \subset L^2(\mu)$, if $\mathcal{E}$ satisfies that for any $f_n \rightarrow 0$ in $L^2(\mu)$ and f_n Cauchy w.s.t. $\langle \cdot, \cdot \rangle_{\mathcal{E}}$, $\mathcal{E}(f_n) \rightarrow 0$, then D can be extended to the closure of D w.s.t. $\|\cdot\|_2 + \mathcal{E}(\cdot)$, and such $\mathcal{E}$ is called closable.

If Γ is a diffusion carré du champ on an algebra $\mathcal{A} \subset L^2(\mu)$ dense and $\Gamma(f, f) \geq 0$ for all $f \in \mathcal{A}$, then let

$$\mathcal{E}(f, g) := \int_E \Gamma(f, g) d\mu$$

and taking $\mathcal{D}(\mathcal{E})$ be the closure of $\mathcal{A}$ w.s.t. $\langle \cdot, \cdot \rangle_{\mathcal{E}}$. It can prove that such $\mathcal{E}$ is a Dirichlet form. Moreover, by the symmetric and positivity of $\mathcal{E}$, Riesz representation theorem implies that we can define a non-positive, symmetric operator L by

$$\int g L f d\mu = -\mathcal{E}(f, g)$$

on the domain

$$\mathcal{D}(L) = \{f \in \mathcal{D}(\mathcal{E}) : \exists C \text{ such that } \mathcal{E}(f, g) \leq C \|g\|_2 \text{ for all } g \in \mathcal{D}(\mathcal{E})\}.$$

Moreover, it can be extended to a self-adjoint operator L by using Friedrichs extension.

Definition 10.3.5 (Compact Markov Diffusion Triple). Let $(E, \mathcal{F}, \mu)$ be a polished measure space and μ be a probability measure. For $\mathcal{A} \subset L^2(\mu)$, let

$$\Gamma: \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$$

be a symmetric bilinear form. We say (E, μ, Γ) a compact Markov diffusion triple the followings are satisfied.

- (a) $\mathcal{A}$ is dense in $L^2(\mu)$,
- (b) $\mathcal{A}$ is an algebra closed under composition with smooth functions,
- (c) $\Gamma(f) = \Gamma(f, f) \geq 0$ for all $f \in \mathcal{A}$,

(d) Γ is a diffusion carré du champ,

(e) $\Gamma(f) = 0$ implies that f is a constant,

and let $\mathcal{E}(f, g) = \int_E \Gamma(f, g) d\mu$ for all $f, g \in \mathcal{A}$, which satisfies

(f) for every $f \in \mathcal{A}$, there exist a $C > 0$ such that $\mathcal{E}(f, g) \leq C \|g\|_2$ for all $g \in \mathcal{A}$.

It follows that $\mathcal{E}$ can be extended to a Dirichlet form. Let L be the self-adjoint operator defined on $\mathcal{D}(L)$ such that

$$\int g L f d\mu = -\mathcal{E}(f, g).$$

Note that $\mathcal{A} \subset \mathcal{D}(L)$. Let $P_t = e^{tL}$ called the semigroup be assumed that

(g) $L\mathcal{A} \subset \mathcal{A}$,

(h) $P_t\mathcal{A} \subset \mathcal{A}$.

Proposition 10.3.6. *Let $(E, \mathcal{F}, \mu)$ be a compact Markov diffusion triple and P_t be its semigroup.*

(1) P_t is a symmetric Markov semigroup for μ .

(2) For any $f \in L^2(\mu)$,

$$\lim_{t \rightarrow \infty} P_t f = \int_E f d\mu$$

in L^2 , which is called the ergodic property.

Curvature.

Definition 10.3.7. Given a compact Markov diffusion triple $(E, \mathcal{F}, \mu)$. For any $f, g \in \mathcal{A}$,

$$\Gamma_2(f, g) = \frac{1}{2} (L\Gamma(f, g) - \Gamma(f, Lg) - \Gamma(g, Lf)),$$

and $\Gamma_2(f) = \Gamma_2(f, f)$.

Definition 10.3.8 (Curvature Dimension). For $\rho \in \mathbb{R}$ and $n \in [1, \infty]$, a compact Markov diffusion triple $(E, \mathcal{F}, \mu)$ is said to satisfy the curvature-dimension condition $\text{CD}(\rho, n)$ if

$$\Gamma_2(f) \geq \rho \Gamma(f) + \frac{1}{n} (Lf)^2.$$

10.4 Poincaré Inequality

Proposition 10.4.1. *Let P_t be the semigroup of a compact Markov triple. TFAE.*

(1) $\text{CD}(\rho, \infty)$ holds for some $\rho \in \mathbb{R}$.

(2) For every $f \in \mathcal{A}$ and $t \geq 0$,

$$\Gamma(P_t f) \leq e^{-2\rho t} P_t \Gamma(f).$$

(3) For every $f \in \mathcal{A}$ and $t \geq 0$,

$$P_t(f^2) - (P_t f)^2 \leq \frac{1 - e^{-2\rho t}}{\rho} P_t \Gamma(f).$$

(4) For every $f \in \mathcal{A}$ and $t \geq 0$,

$$P_t(f^2) - (P_t f)^2 \geq \frac{e^{2\rho t} - 1}{\rho} \Gamma(P_t f).$$

For the last two conditions, if $\rho = 0$, then the coefficients in RHS can be taken as $2t$.

Proof. (1) $\Rightarrow$ (2): for $f \in \mathcal{A}$, let

$$\Lambda(s) = e^{-2\rho s} P_s \Gamma(P_{t-s} f).$$

Then by chain rule,

$$\Lambda'(s) = 2e^{-2\rho s} P_s (\Gamma_2(P_{t-s} f) - \rho \Gamma(P_{t-s} f)) \geq 0,$$

because of $\text{CD}(\rho, \infty)$. Therefore, $\Lambda(t) \geq \Lambda(0)$.

(2) $\Rightarrow$ (3): Let

$$\Lambda(s) = P_s (P_{t-s} f)^2.$$

So $\Lambda'(s) = 2P_s \Gamma(P_{t-s} f)$ and

$$\begin{aligned} \Lambda(t) - \Lambda(0) &= 2 \int_0^t P_s \Gamma(P_{t-s} f) ds \\ &\leq 2 \int_0^t e^{-2\rho(t-s)} P_t \Gamma(f) ds \\ &= \frac{1 - e^{-2\rho t}}{\rho} P_t \Gamma(f). \end{aligned}$$

(2) $\Rightarrow$ (4): Similarly, as above, by using $P_s \Gamma(P_{t-s} f) \geq e^{2\rho s} \Gamma(P_t f)$,

$$\Lambda(t) - \Lambda(0) \geq 2 \int_0^t e^{2\rho s} \Gamma(P_t f) ds = \frac{e^{2\rho t} - 1}{\rho} \Gamma(P_t f).$$

(3) $\Rightarrow$ (1): Note that for any $h \in \mathcal{A}$,

$$P_t h = h + tLh + \frac{t^2}{2} L^2 h + o(t^2), \quad t \rightarrow 0.$$

Therefore choosing $h = f$ and $h = f^2$, we have

$$\begin{aligned} P_t(f^2) - (P_t f)^2 &= tL(f^2) + \frac{t^2}{2} L^2(f^2) - 2t f L f - t^2 (L f)^2 - t^2 f L^2 f + o(t^2) \\ &= 2t \Gamma(f) + \frac{t^2}{2} L^2(f^2) - t^2 (L f)^2 - t^2 f L^2 f + o(t^2). \end{aligned}$$

On the other hand,

$$\frac{1 - e^{-2\rho t}}{\rho} P_t \Gamma(f) = 2t \Gamma(f) - 2\rho t^2 \Gamma(f) + 2t^2 L \Gamma(f) + o(t^2).$$

Therefore, by (3),

$$\frac{1}{2} L^2(f^2) - (L f)^2 - f L^2 f + o(1) \leq -2\rho \Gamma(f) + 2L \Gamma(f) + o(1).$$

As $t \rightarrow 0$, we have

$$\frac{1}{2} L^2(f^2) - (L f)^2 - f L^2 f \leq -2\rho \Gamma(f) + 2L \Gamma(f).$$

Then by arranging,

$$L \Gamma(f) - 2\Gamma(f, L f) \geq 2\rho \Gamma(f).$$

(4) $\Rightarrow$ (1): It is similarly as above. □

For (3) in above proposition, which is called local Poincaré inequality, if $\rho > 0$, by the ergodic property, as $t \rightarrow \infty$,

$$\int f^2 d\mu - \left(\int f d\mu \right)^2 \leq \frac{1}{\rho} \int \Gamma(f) d\mu,$$

which is called a Poincaré inequality.

Definition 10.4.2 (Poincaré inequality). Let μ be a probability measure and $\mathcal{E}$ be a Dirichlet form on $L^2(\mu)$. We say that μ and $\mathcal{E}$ satisfy a Poincaré inequality with constant C (PI(C)) if

$$\int_E f^2 d\mu - \left(\int_E f d\mu \right)^2 \leq C \mathcal{E}(f),$$

for any $f \in \mathcal{D}(\mathcal{E})$. The Poincaré constant of μ and $\mathcal{E}$ is the smallest C such that above inequality holds for all $f \in \mathcal{D}(\mathcal{E})$.

Remark 10.4.3. When considering a compact Markov triple, because $\mathcal{A} \subset \mathcal{D}(\mathcal{E})$ is dense, it suffices to check PI on $\mathcal{A}$. Moreover, if a compact Markov triple satisfies $\text{CD}(\rho, \infty)$, it satisfies $\text{PI}(1/\rho)$.

Corollary 10.4.4. *The compact Markov triple (E, μ, Γ) satisfies $\text{CD}(\rho, \infty)$ if and only if for any $t \geq 0$ and $x \in E$ μ -a.e., the measure $p_t(x, \cdot)$ satisfies PI with constant $(1 - e^{-2\rho t})/\rho$.*

Proposition 10.4.5 (Spectral Gap). *If the compact Markov triple (E, μ, Γ) satisfies PI(C) for some constant C , then the spectrum of L*

$$\sigma(L) \subset (-\infty, -\frac{1}{C}] \cup \{0\}.$$

Proof. Let $\lambda \in \sigma(L)$ such that $\lambda \neq 0$. Because L is self-adjoint, i.e., $\sigma(L) = \sigma_{ap}(L)$, there exists $f_n \in \mathcal{D}(L)$ such that $\|f_n\|_2 = 1$ and

$$\|Lf_n - \lambda f_n\|_2 \rightarrow 0, \quad n \rightarrow \infty.$$

Note that because μ is a probability measure, $\|Lf_n - \lambda f_n\|_1 \rightarrow 0$. It follows that $\int f_n d\mu \rightarrow 0$ by $\int Lf_n d\mu = 0$ for all n . Then PI implies that

$$\int_E f_n^2 d\mu - \left(\int_E f_n d\mu \right)^2 \leq C \int_E \Gamma(f_n) d\mu = -C \int_E f_n Lf_n d\mu.$$

As $n \rightarrow \infty$,

$$\lambda = \int_E f_n Lf_n d\mu \leq -\frac{1}{C}. \quad \square$$

PI under $\text{CD}(\rho, n)$.

Lemma 10.4.6. *Suppose the compact Markov triple (E, μ, Γ) satisfies $\text{CD}(\rho, \infty)$ with some $\rho > 0$. It satisfies PI(C) for some $C > 0$ if and only if*

$$\int_E \Gamma(f) d\mu \leq C \int_E (Lf)^2 d\mu, \quad \forall f \in \mathcal{D}(L).$$

Proof. $\Rightarrow$: Let

$$\Lambda(t) = \int_E (P_t f)^2 d\mu.$$

Then

$$\Lambda'(t) = -2 \int_E \Gamma(P_t f) d\mu, \quad \Lambda''(t) = 4 \int_E (LP_t f)^2 d\mu.$$

Because it satisfies $\text{CD}(\rho, \infty)$ with some $\rho > 0$, by

$$\Gamma(P_t f) \leq e^{-2\rho t} P_t \Gamma(f) \leq P_t \Gamma(f) \leq \Gamma(f)$$

Then by DCT, $\lim_{t \rightarrow \infty} \Lambda'(t)$ exists. And by ergodicity,

$$\lim_{t \rightarrow \infty} \Lambda(t) = \int_E f d\mu,$$

$\lim_{t \rightarrow \infty} \Lambda'(t) = 0$. By assumption,

$$\Lambda''(t) \geq -\frac{2}{C} \Lambda'(t).$$

Therefore,

$$\begin{aligned} \int f^2 d\mu - \left(\int f d\mu \right)^2 &= - \int_0^\infty \Lambda'(t) dt \\ &\leq \frac{C}{2} \int_0^\infty \Lambda''(t) dt \\ &= -\frac{C}{2} \Lambda'(0) \\ &= C \int_E \Gamma(f) d\mu. \end{aligned}$$

$\Leftarrow$: Choosing $f \in \mathcal{D}(L)$ with mean 0 (otherwise, let $f - \int f d\mu$ and note that $\Gamma(c, g) = 0$ for any constant c and function g). By Cauchy-Schwartz inequality,

$$\begin{aligned} \int_E \Gamma(f) d\mu &= \int_E f(-Lf) d\mu \\ &\leq \sqrt{\int_E f^2 d\mu \int_E (Lf)^2 d\mu} \\ &\leq \sqrt{C \int_E \Gamma(f) d\mu \int_E (Lf)^2 d\mu}. \end{aligned} \quad \square$$

Theorem 10.4.7. *Let (E, μ, Γ) be a compact Markov triple. If it satisfies $\text{CD}(\rho, n)$ for some $\rho > 0$ and $n > 1$, then μ satisfies $\text{PI}(C)$ with $C = \frac{n-1}{\rho n}$.*

Proof. Because of $\text{CD}(\rho, n)$,

$$\int_E \Gamma_2(f) d\mu \geq \rho \int_E \Gamma(f) d\mu + \frac{1}{n} \int_E (Lf)^2 d\mu.$$

Because $\int Lh d\mu = 0$,

$$\begin{aligned} \int_E \Gamma_2(f) &= \frac{1}{2} \left(\int_E L\Gamma(f) d\mu - \int_E \Gamma(f, Lf) d\mu \right) \\ &= \frac{1}{2} \int_E L\Gamma(f) d\mu - \frac{1}{2} \int_E L(fLf) d\mu + \frac{1}{2} \int_E (Lf)^2 + fL^2 f d\mu \\ &= \int_E (Lf)^2 d\mu. \end{aligned}$$

Therefore,

$$\frac{n-1}{\rho n} \int_E (Lf)^2 d\mu \geq \int_E \Gamma(f) d\mu.$$

Then by above lemma, it has the result. $\square$

10.5 Applications with PI

Decay of Variance. For a probability measure μ and $f \in L^2(\mu)$, let

$$\mathrm{Var}_\mu(f) = \int_E f^2 d\mu - \left(\int_E f d\mu \right)^2.$$

Proposition 10.5.1. *The compact Markov triple (E, μ, Γ) satisfies PI(C) if and only if*

$$\mathrm{Var}_\mu(P_t f) \leq e^{-\frac{2t}{C}} \mathrm{Var}_\mu(f), \quad f \in L^2(\mu).$$

Proof. $\Rightarrow$: For $f \in \mathcal{A}$,

$$\frac{d}{dt} \int_E (P_t f)^2 d\mu = 2 \int_E P_t f L P_t f d\mu = -2\mathcal{E}(P_t f).$$

Define

$$\Lambda(t) = e^{2t/C} \mathrm{Var}_\mu(P_t f),$$

so

$$\Lambda'(t) = \frac{2}{C} \mathrm{Var}_\mu(P_t f) - 2\mathcal{E}(P_t f) \leq 0$$

by PI(C). It follows that $\Lambda(t) \leq \Lambda(0)$. For general $f \in L^2(\mu)$, it can get by density.

$\Leftarrow$: It suffices to prove that for $f \in \mathcal{A}$ with $\int f d\mu = 0$. Note that

$$P_t f = f + tLf + o(t),$$

and so

$$\mathrm{Var}(P_t f) = \int_E f^2 d\mu + 2t \int_E f L f d\mu + o(t).$$

On the other hand,

$$e^{-2t/C} \mathrm{Var}_\mu(f) = \left(1 - \frac{2t}{C} + o(t) \right) \mathrm{Var}_\mu(f).$$

Therefore,

$$2t \int_E f L f d\mu + o(t) \leq \left(-\frac{2t}{C} + o(t) \right) \mathrm{Var}_\mu(f).$$

Then dividing t on the both sides and taking $t \rightarrow 0$,

$$2 \int_E f L f d\mu \leq -\frac{2}{C} \mathrm{Var}_\mu(f). \quad \square$$

Log-concave measures.

Definition 10.5.2. The probability measure μ on $\mathbb{R}^n$ defined by

$$d\mu(x) = e^{-W(x)} dx$$

is called log-concave if $W: \mathbb{R}^n \rightarrow \mathbb{R}$ is convex. For $\rho > 0$, μ is called ρ -strongly log-concave if $W(x) - \rho|x|^2$ is convex.

Assume $W \in C^\infty(\mathbb{R}^n)$. And on $\mathbb{R}^n$,

$$\Gamma(f, g) = \langle \nabla f, \nabla g \rangle$$

is a carré du champ. Then by divergence theorem,

$$-\int_{\mathbb{R}^n} \Gamma(f, g) d\mu = -\int_{\mathbb{R}^n} \langle e^{-W} \nabla f, \nabla g \rangle dx = \int_{\mathbb{R}^n} f(\Delta g - \langle \nabla W, \nabla g \rangle) d\mu.$$

Therefore,

$$Lg = \Delta g - \langle \nabla W, \nabla g \rangle.$$

If all derivatives of $W(x)$ grow at most polynomially fast as $|x| \rightarrow \infty$, then $(\mathbb{R}^n, \mu, \Gamma)$ is a compact Markov triple with $\mathcal{A}$ being the class of smooth, bounded functions whose derivatives all vanish super-polynomially fast.

Moreover,

$$\Gamma_2(f, g) = \langle \nabla^2 f, \nabla^2 g \rangle + (\nabla f)^\top (\nabla^2 W) \nabla g,$$

By the strongly convexity of W ,

$$\Gamma_2(f, f) \geq \rho \|f\|_2 = \rho \Gamma(f),$$

i.e., $(\mathbb{R}^n, \mu, \Gamma)$ is $\text{CD}(\rho, \infty)$.

Corollary 10.5.3. *Every ρ -strongly log-concave probability measure satisfies $\text{PI}(1/\rho)$, i.e.,*

$$\text{Var}_\mu(f) \leq \frac{1}{\rho} \mathbb{E}_\mu[\|\nabla f\|^2].$$

10.6 Log-Sobolev Inequality

Theorem 10.6.1 (Strong Gradient Bound). *Let (E, μ, Γ) be a compact Markov triple that satisfies $\text{CD}(\rho, \infty)$. Then for every $f \in \mathcal{A}$ and $t \geq 0$,*

$$\sqrt{\Gamma(P_t f)} \leq e^{-\rho t} P_t \sqrt{\Gamma(f)}.$$

Proof. First, assume $f > 0$. Fix t and define

$$\Lambda(s) = P_s \sqrt{\Gamma(P_{t-s} f)}.$$

by the chain rule for L ,

$$\begin{aligned} \Lambda'(s) &= P_s L \sqrt{\Gamma(P_{t-s} f)} + P_s \frac{\frac{d}{ds} \Gamma(P_{t-s} f)}{2 \sqrt{\Gamma(P_{t-s} f)}} \\ &= P_s \left[\frac{L \Gamma(P_{t-s} f)}{2 \sqrt{\Gamma(P_{t-s} f)}} - \frac{\Gamma(\Gamma(P_{t-s} f))}{4 \Gamma(P_{t-s} f)^{3/2}} - \frac{\Gamma(P_{t-s} f, L P_{t-s} f)}{\sqrt{\Gamma(P_{t-s} f)}} \right] \\ &= P_s \left[\frac{\Gamma_2(P_{t-s} f)}{\sqrt{\Gamma(P_{t-s} f)}} - \frac{\Gamma(\Gamma(P_{t-s} f))}{4 \Gamma(P_{t-s} f)^{3/2}} \right]. \end{aligned}$$

and so

$$\frac{d}{ds} (e^{-\rho s} \Lambda(s)) = e^{-\rho s} (\Lambda'(s) - \rho \Lambda(s))$$

$$= e^{-\rho s} P_s \left[\frac{\Gamma_2(P_{t-s}f) \Gamma(P_{t-s}f) - \rho \Gamma(P_{t-s}f)^2 - \frac{1}{4} \Gamma(\Gamma(P_{t-s}f))}{4 \Gamma(P_{t-s}f)^{3/2}} \right].$$

Let $g = P_{t-s}f$.

$$\frac{d}{ds} (e^{-\rho s} \Lambda(s)) \geq 0 \Leftrightarrow \Gamma(g) (\Gamma_2(g) - \rho \Gamma(g)) \geq \frac{1}{4} \Gamma(\Gamma(g)).$$

Therefore, it suffices to prove above inequality, which is followed by the diffusion property of Γ and $\text{CD}(\rho, \infty)$.

For general f , let $\psi(x) = \sqrt{x + \varepsilon}$ and replace $\Lambda(s)$ by

$$\Lambda(s) = P_s \psi(e^{-2\rho s} \Gamma(P_{t-s}f)).$$

□

Let μ be a probability measure on E and $f: E \rightarrow [0, \infty)$ measurable. Define the entropy by

$$\text{Ent}_\mu(f) := \int_E f \log f d\mu - \int_E f d\mu \log \left(\int_E f d\mu \right),$$

where we adopt the convention that $0 \log 0 = 0$. By Jensen's inequality for $\psi(x) = x \log x$, $\text{Ent}_\mu f \geq 0$. Because ψ is strictly convex, $\text{Ent}_\mu f = 0$ if and only if f is constant μ -a.e. Also,

$$\text{Ent}_\mu(cf) = c \text{Ent}_\mu(f), \quad \forall c > 0.$$

Remark 10.6.2. Note that if $\nu \ll \mu$ is another probability measure and let $f = \frac{d\nu}{d\mu}$,

$$\text{Ent}_\mu(f) = \int_E f \log f d\mu = \text{KL}(\nu \parallel \mu).$$

Definition 10.6.3 (Log-Sobolev Inequality). If μ is a probability measure and $\mathcal{E}$ is a Dirichlet form, we say they satisfy a log-Sobolev inequality with constant C ($\text{LSI}(C)$) if for all $f \in \mathcal{D}(\mathcal{E})$,

$$\text{Ent}_\mu(f^2) \leq 2C \mathcal{E}(f).$$

The smallest C for which μ and $\mathcal{E}$ satisfy a $\text{LSI}(C)$ is called the log-Sobolev constant of $\mu, \mathcal{E}$.

Assume $f > \varepsilon > 0$, i.e., f is bounded below. Then

$$\mathcal{E}(\sqrt{f}) = \int_E \Gamma(\sqrt{f}) d\mu = \frac{1}{4} \int_E \frac{\Gamma(f)}{f} d\mu.$$

Then $\text{LSI}(C)$ is equivalent to

$$\text{Ent}_\mu(f) \leq 2C \mathcal{E}(\sqrt{f}) = \frac{C}{2} \int_E \frac{\Gamma(f)}{f} d\mu.$$

Definition 10.6.4. Let $\nu \ll \mu$ be another probability measure and $f = \frac{d\nu}{d\mu}$. The Fisher information of ν w.s.t. μ is defined as

$$I(\nu \mid \mu) = I_\mu(f) = \int_E \frac{\Gamma(f)}{f} d\mu$$

and the entropy of ν w.s.t. μ (i.e., KL divergence) is defined as

$$H(\nu \mid \mu) = \text{Ent}_\mu(f).$$

By $\text{Ent}_\mu(cf) = c \text{Ent}_\mu(f)$, μ, Γ satisfy $\text{LSI}(C)$ if and only if

$$H(\nu \mid \mu) \leq \frac{C}{2} I(\nu \mid \mu)$$

for every probability measure $\nu \ll \mu$, where we allow infinity on the both sides. Moreover, if $\frac{d\nu}{d\mu} \notin \mathcal{D}(\mathcal{E})$, the RHS is ∞ .

Proposition 10.6.5. *If $\mu, \mathcal{E}$ satisfy $\text{LSI}(C)$, then they satisfy $\text{PI}(C)$.*

Proof. Given $f \in \mathcal{A}$ with mean 0 and $\varepsilon > 0$. Because $\log(1+x) = x - \frac{1}{2}x^2 + o(x^2)$,

$$\begin{aligned} \text{Ent}_\mu((1+\varepsilon f)^2) &= \int_E (1+\varepsilon f)^2 \left(\varepsilon f - \frac{\varepsilon^2}{2} f^2 \right) d\mu - \int_E (1+\varepsilon f)^2 d\mu \log \int_E (1+\varepsilon f)^2 d\mu + o(\varepsilon^2) \\ &= 2\varepsilon^2 \int_E f^2 d\mu + o(\varepsilon^2) \end{aligned}$$

Moreover,

$$\mathcal{E}(1+\varepsilon f) = \varepsilon^2 \mathcal{E}(f)$$

Apply LSI to $1+\varepsilon f$.

$$\mathcal{E}(1+\varepsilon f) \leq 2C\varepsilon^2 \mathcal{E}(f).$$

Dividing ε and taking $\varepsilon \rightarrow 0$,

$$\int_E f^2 d\mu \leq C\mathcal{E}(f). \quad \square$$

LSI under $\text{CD}(\rho, \infty)$.

Proposition 10.6.6. *For a compact Markov triple (E, μ, Γ) , TFAE.*

(1) *It satisfies $\text{CD}(\rho, \infty)$ for some $\rho \in \mathbb{R}$.*

(2) *For all $f \in \mathcal{A}$,*

$$\Gamma(f) (\Gamma_2(f) - \rho \Gamma(f)) \geq \frac{1}{4} \Gamma(\Gamma(f)).$$

(3) *For every $f \in \mathcal{A}$ and $t \geq 0$,*

$$\sqrt{\Gamma(P_t f)} \leq e^{-\rho t} P_t \sqrt{\Gamma(f)}.$$

(4) *For every positive $f \in \mathcal{A}$ and $t \geq 0$,*

$$P_t(f \log f) - P_t f \log P_t f \leq \frac{1 - e^{-2\rho t}}{2\rho} P_t \frac{\Gamma(f)}{f}.$$

(5) *For every positive $f \in \mathcal{A}$ and $t \geq 0$,*

$$P_t(f \log f) - P_t f \log P_t f \geq \frac{e^{2\rho t} - 1}{2\rho} \frac{\Gamma(P_t f)}{P_t f}.$$

Note that for (4), it is called a local LSI. As $t \rightarrow \infty$, by ergodicity, it obtains $\text{LSI}(1/\rho)$.

Corollary 10.6.7. *If (E, μ, Γ) is a compact Markov triple satisfies $\text{CD}(\rho, \infty)$ for some $\rho > 0$, then μ, Γ satisfy a $\text{LSI}(1/\rho)$.*

Proof of Proposition 10.6.6. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) is by the strong gradient bound. (3) together with Jensen's inequality implies that

$$\Gamma(P_t f) \leq e^{-2\rho t} P_t \Gamma(f),$$

which follows that (1).

(3) $\Rightarrow$ (4): Fix $t > 0$ and define

$$\Lambda(s) = P_s [P_{t-s} f \log P_{t-s} f] = P_s \psi(P_{t-s} f),$$

for $\psi(x) = x \log x$. So by (3),

$$\begin{aligned} \Lambda'(s) &= P_s [\psi''(P_{t-s} f) \Gamma(P_{t-s} f)] \\ &= P_s \frac{\Gamma(P_{t-s} f)}{P_{t-s} f} \\ &\leq e^{-2\rho(t-s)} P_s \frac{\left(P_{t-s} \sqrt{\Gamma(f)}\right)^2}{P_{t-s} f}, \end{aligned}$$

Note that by Cauchy-Schwartz inequality,

$$\mathbb{E}X = \mathbb{E} \left[\sqrt{Y} \frac{X}{\sqrt{Y}} \right] \leq \sqrt{\mathbb{E}Y \mathbb{E} \frac{X^2}{Y}} \Rightarrow \frac{(\mathbb{E}X)^2}{\mathbb{E}Y} \leq \mathbb{E} \frac{X^2}{Y}.$$

By setting $X = \sqrt{\Gamma(f)}$, $Y = f$ and taking expectation w.s.t. $p_{t-s}(x, \cdot)$,

$$\Lambda'(s) \leq e^{-2\rho(t-s)} P_s P_{t-s} \frac{\Gamma(f)}{f} = e^{-2\rho(t-s)} P_t \frac{\Gamma(f)}{f}.$$

So

$$\Lambda(t) - \Lambda(0) \leq P_t \frac{\Gamma(f)}{f} \int_0^t e^{-2\rho(t-s)} ds = \frac{1 - e^{-2\rho t}}{2\rho} P_t \frac{\Gamma(f)}{f}.$$

(3) $\Rightarrow$ (5): It is similar as (3) $\Rightarrow$ (4), except for taking $X = \sqrt{\Gamma(P_{t-s} f)}$, $Y = P_{t-s} f$ and the expectation w.s.t. $p_s(x, \cdot)$, which implies that

$$\Lambda'(s) \geq \frac{\left(P_s \sqrt{\Gamma(P_{t-s} f)}\right)^2}{P_t f} \geq e^{2\rho s} \frac{\Gamma(P_t f)}{P_t f}.$$

(4) $\Rightarrow$ (1): (4) is local $\text{LSI}(\rho)$, which implies local PI as similar as above proposition. Then it implies $\text{CD}(\rho, \infty)$. (5) $\Rightarrow$ (1) is as similar as above. $\square$

LSI under $\text{CD}(\rho, n)$.

Theorem 10.6.8. *If a compact Markov triple (E, μ, Γ) satisfies $\text{CD}(\rho, n)$ for some $\rho > 0$, then μ, Γ satisfy a $\text{LSI}(C)$.*

Lemma 10.6.9. *Suppose that*

$$\int_E f \Gamma(\log f) d\mu \leq C \int_E f \Gamma_2(\log f) d\mu$$

for some $C > 0$ and all positive $f \in \mathcal{A}$. Then μ, Γ satisfy a $\text{LSI}(C)$.

Proof. For fix some positive $f \in \mathcal{A}$. Let

$$\Lambda(t) = \int_E P_t f \log P_t f d\mu.$$

Then

$$\Lambda'(t) = - \int_E \frac{\Gamma(P_t f)}{P_t f} d\mu = - \int_E P_t f \Gamma(\log P_t f) d\mu,$$

and

$$\Lambda''(t) = \int_E \frac{\Gamma(P_t f) L P_t f}{(P_t f)^2} - \frac{2\Gamma(P_t f, L P_t f)}{P_t f} d\mu.$$

Taking $g = P_t f$, by the diffusion property of L ,

$$L \frac{\Gamma(g)}{g} = - \frac{\Gamma(g) L g}{g^2} + \frac{L \Gamma(g)}{g} - \frac{2\Gamma(g, \Gamma(g))}{g^2} + \frac{2\Gamma(g)^2}{g^3}.$$

Since $\int L h d\mu = 0$,

$$\int_E \frac{\Gamma(g) L g}{g^2} = \int_E \frac{L \Gamma(g)}{g} - 2 \frac{\Gamma(g, \Gamma(g))}{g^2} + 2 \frac{\Gamma(g)^2}{g^3} d\mu.$$

Note that $L \Gamma(g) - 2\Gamma(g, L g) = 2\Gamma_2(g)$. So

$$\Lambda''(t) = 2 \int_E \frac{\Gamma_2(g)}{g} - \frac{\Gamma(g, \Gamma(g))}{g^2} + \frac{\Gamma(g)^2}{g^3} d\mu = \int_E g \Gamma_2(\log g).$$

Therefore,

$$\begin{aligned} \Lambda'(t) &= - \int_E P_t f \Gamma(\log P_t f) d\mu \\ \Lambda''(t) &= 2 \int_E P_t f \Gamma_2(\log P_t f) d\mu. \end{aligned}$$

By assumption of $-\Lambda'(t) \leq \frac{C}{2} \Lambda''(t)$, $\Lambda'(t) \geq \Lambda'(0) \exp(-\frac{2t}{C})$, i.e.,

$$\int_E \frac{\Gamma(P_t f)}{P_t f} d\mu \leq e^{-2t/C} \int_E \frac{\Gamma(f)}{f} d\mu.$$

Therefore,

$$\Lambda(0) - \Lambda(t) = - \int_0^t \Lambda'(s) ds \leq -\Lambda'(0) \int_0^t e^{-2s/C} ds = \frac{C(1 - e^{-2t/C})}{2} \mathbb{I}_\mu(f),$$

then it can prove that by taking $t \rightarrow \infty$. □

Proof of Theorem 10.6.8. It suffices to check the condition of above lemma. By the diffusion property of Γ ,

$$\begin{aligned} \Gamma_2(e^{ag}) &= a^2 e^{2ag} [\Gamma_2(g) + a\Gamma(g, \Gamma(g)) + a^2 \Gamma(g)^2] \\ \Gamma(e^{ag}) &= a^2 e^{2ag} \Gamma(g) \\ L e^{ag} &= a e^{ag} [Lg + a\Gamma(g)]. \end{aligned}$$

Therefore, $\text{CD}(\rho, n)$ implies that

$$\Gamma_2(e^{ag}) = a^2 e^{2ag} [\Gamma_2(g) + a\Gamma(g, \Gamma(g)) + a^2 \Gamma(g)^2]$$

$$\begin{aligned}\Gamma(e^{ag}) &= a^2 e^{2ag} \Gamma(g) \\ Le^{ag} &= ae^{ag}[Lg + a\Gamma(g)].\end{aligned}$$

So

$$\int_E e^g \left[\Gamma_2(g) + a\Gamma(g, \Gamma(g)) + a^2\Gamma(g)^2 - \rho\Gamma(g) - \frac{1}{n}[Lg + a\Gamma(g)]^2 \right] d\mu \geq 0.$$

Note that

$$4(L e^{g/2})^2 = e^g \left[Lg + \frac{1}{2}\Gamma(g) \right]^2, \quad \text{and} \quad \int_E (Lf)^2 d\mu = \int_E \Gamma_2(f) d\mu.$$

It implies that

$$\begin{aligned}& \int_E [Lg + a\Gamma(g)]^2 d\mu \\&= \int_E 4(L e^{g/2})^2 + e^g \left[(2a-1)\Gamma(g)Lg + \frac{4a^2-1}{4}\Gamma(g)^2 \right] d\mu \\&= \int_E 4\Gamma_2(e^{g/2}) + e^g \left[(2a-1)\Gamma(g)Lg + \frac{4a^2-1}{4}\Gamma(g)^2 \right] d\mu \\&= \int_E 4\Gamma_2(e^{g/2}) + e^g \left[(2a-1)\Gamma(g)Lg + \frac{4a^2-1}{4}\Gamma(g)^2 \right] d\mu \\&= \int_E e^g \left[\Gamma_2(g) + \frac{1}{2}\Gamma(g, \Gamma(g)) + (2a-1)\Gamma(g)Lg + a^2\Gamma(g)^2 \right] d\mu,\end{aligned}$$

and so

$$\int_E e^g \Gamma(g)Lg d\mu = - \int_E \Gamma(g, e^g \Gamma(g)) d\mu = - \int_E e^g \Gamma(g, \Gamma(g)) + e^g \Gamma(g)^2 d\mu.$$

Then the inequality implies that

$$\int_E e^g \left[\frac{n-1}{n}\Gamma_2(g) + b_n\Gamma(g, \Gamma(g)) + c_n\Gamma(g)^2 - \rho\Gamma(g) \right] d\mu \geq 0$$

for

$$b_n = \frac{2an + 4a - 3}{2n}, \quad c_n = \frac{na^2 - (a-1)^2}{n}.$$

By choosing $a = \frac{3}{2n+4}$, it has

$$\int_E e^g \Gamma_2(g) d\mu \geq \frac{n\rho}{n-1} \int_E e^g \Gamma(g) d\mu. \quad \square$$

10.7 Applications with LSI

Decay of Entropy.

Proposition 10.7.1. *The compact Markov triple (E, μ, Γ) satisfies a $\text{LSI}(C)$ if and only if*

$$\text{Ent}_\mu(P_t f) \leq e^{-2t/C} \text{Ent}_\mu(f)$$

for every $t \geq 0$ and every $f \in L^1(\mu)$ with finite entropy.

Proof. It suffices to consider $f \in \mathcal{A}$ with finite entropy. Define

$$\Lambda(t) = \text{Ent}_\mu(P_t f) = \int_E P_t f \log P_t f d\mu - \int_E f d\mu \log \int_E f d\mu.$$

$\Rightarrow$: Note that

$$\Lambda'(t) = - \int_E \frac{\Gamma(P_t f)}{P_t f} d\mu = -I_\mu(P_t f),$$

so by LSI, $\Lambda'(t) \leq -\frac{2}{C}\Lambda(t)$ that implies that $\Lambda(t) \leq e^{-2t/C}\Lambda(0)$.

$\Leftarrow$: By Taylor expansion,

$$\Lambda(t) = \Lambda(0) + t\Lambda'(0) + o(t) = \Lambda(0) - tI_\mu(f) + o(t).$$

Because

$$\Lambda(t) \leq e^{-2t/C}\Lambda(0) = \left(1 - \frac{2t}{C} + o(t)\right)\Lambda(0),$$

as $t \rightarrow 0$, we have

$$I_\mu(f) \geq \frac{2}{C}\Lambda(0). \quad \square$$

If $f = \frac{d\nu_0}{d\mu}$, then for $\nu_t = P_t^* \mu_0$,

$$d\nu_t = P_t f d\mu.$$

Suppose μ, Γ satisfies LSI(C). Then we have

$$H(\nu_t | \mu) \leq e^{-\frac{2t}{C}} H(\nu_0 | \mu).$$

Moreover, by the following Pinsker-Csizsár-Kullback inequality,

$$d_{\text{TV}}(\mu, \nu_t)^2 \leq \frac{1}{2} e^{-\frac{2t}{C}} H(\nu_0 | \mu),$$

where

$$d_{\text{TV}}(\mu, \nu) = \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)| = \frac{1}{2} \int_E \left| 1 - \frac{d\nu}{d\mu} \right| d\mu.$$

Moreover, PI can also be applied to consider the convergence, because

$$d_{\text{TV}}(\mu, \nu)^2 \leq \frac{1}{4} \text{Var}_\mu \left(\frac{d\nu}{d\mu} \right),$$

but it needs $\frac{d\nu}{d\mu} \in L^2(\mu)$.

Proposition 10.7.2 (Pinsker-Csizsár-Kullback). *For any probability measure μ, ν on the same space,*

$$d_{\text{TV}}(\mu, \nu)^2 \leq \frac{1}{2} H(\nu | \mu),$$

where $H(\nu | \mu) = \infty$ if ν is not absolutely continuous to μ .

Proof. WTLG, assume $f = \frac{d\nu}{d\mu} \in L^1$. Therefore, it suffices to show

$$\left(\int_E |1 - f| d\mu \right)^2 \leq 2 \text{Ent}_\mu(f).$$

Define $f_s = 1 + s(f - 1)$ for $s \in [0, 1]$ and

$$\Lambda(s) = 2 \text{Ent}_\mu(f_s) - \left(\int_E |1 - f_s| d\mu \right)^2 = 2 \text{Ent}_\mu(f_s) - s^2 \left(\int_E |1 - f| d\mu \right)^2.$$

Since $\int f_s d\mu = 1$ for all s , it follows that

$$\frac{d}{ds} \text{Ent}_\mu(f_s) = \int_E (f - 1) (1 + \log f_s) d\mu, \quad \frac{d^2}{ds^2} \text{Ent}_\mu(f_s) = \int_E \frac{(f - 1)^2}{f_s} d\mu.$$

In particular, $\Lambda(0) = \Lambda'(0) = 0$ and

$$\Lambda''(s) = 2 \int_E \frac{(f - 1)^2}{f_s} d\mu - 2 \left(\int_E |1 - f| d\mu \right)^2 \geq 0,$$

by Cauchy-Schwartz inequality. Hence, $\Lambda(s) \geq 0$ for all $s \in [0, 1]$. $\square$

Hypercontractivity. We already shown that if μ is an invariant measure, then $P_t: L^p(\mu) \rightarrow L^p(\mu)$ is contractive.

Theorem 10.7.3. *For a compact Markov triple with semigroup P_t , TFAE.*

(1) *It satisfies LSI(C).*

(2) *(Hypercontractivity) For some (or every) $1 < p < \infty$, every $t \geq 0$ and $f \in L^p(\mu)$,*

$$\|P_t f\|_{q(t)} \leq \|f\|_p,$$

where $q(t)$ satisfies $\frac{q(t)-1}{p-1} = e^{2t/C}$.

(3) *(Reverse hypercontractivity) For some (or every) $-\infty < p < 1$, every $t \geq 0$ and every positive, bounded $f \in \mathcal{A}$,*

$$\|P_t f\|_{q(t)} \geq \|f\|_p,$$

where $q(t)$ is as above.

Remark 10.7.4. All eigenvectors of $L: L^2 \rightarrow L^2$ belong to L^q for all $2 < q < \infty$. If $Lf = -\lambda f$ for $\lambda > 0$ then $P_t f = e^{-\lambda t} f$. By above theorem with $p = 2$ and $t = \frac{C}{2} \log(q-1)$,

$$e^{-\lambda t} \|f\|_q = \|P_t f\|_q \leq \|f\|_2 \Rightarrow \|f\|_q \leq (q-1)^{C\lambda/2} \|f\|_2.$$

10.8 Riemannian Markov Operator

Let (M, g) be a compact Riemannian manifold and $W \in C^\infty(M)$. WTLG, assume $\int_M e^{-W} dV = 1$, where dV is the canonical volume form on M . Let M be equipped with Borel σ -algebra and for any Borel set A , define

$$\mu(A) := \int_M \mathbb{I}_A e^{-W} dV,$$

which is a probability measure. Let $\mathcal{A} = C^\infty(M)$ and define $\Gamma: \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ by

$$\Gamma(f, g) = \langle \nabla f, \nabla g \rangle.$$

Locally, $\Gamma(f, g) = g^{ij} \partial_i f \partial_j g$, which obviously a diffusion carré du champ. We can obtain

$$Lg = \Delta g - \langle \nabla W, \nabla g \rangle, \quad \forall g \in C^\infty(M),$$

because by the convergence theorem,

$$\int_M f Lg d\mu = - \int_M \Gamma(f, g) d\mu.$$

Moreover,

$$|\mathcal{E}(f, g)| = \left| \int_M \Gamma(f, g) d\mu \right| \leq \|Lf\|_{L^2(\mu)} \|g\|_{L^2(\mu)},$$

which implies that $\mathcal{E}$ is closable. So it is a Dirichlet form.

Remark 10.8.1. For L , we can similarly consider the L^* acting on probability measure on M . If $d\nu = p dV$, then by definition of $L^* \nu$,

$$d(L^* \nu) = (L^* p) dV,$$

where on the right hand side L^*p coincide with the adjoint of L on $L^2(dV)$. Assume $p \in C^\infty$. Then for any $g \in C^\infty$,

$$\begin{aligned} \int_M g d(L^* \nu) &= \int_M g(L^* p) dV = \int_M (Lg) p dV \\ &= \int_M p \Delta g dV - \int_M \langle \nabla W, p \nabla g \rangle dV \\ &= \int_M g \Delta p dV + \int_M p g \Delta W + \int_M \langle \nabla W, \nabla p \rangle g dV, \\ &= \int_M (\Delta p + \operatorname{div}(p \nabla W)) g dV. \end{aligned}$$

because $\operatorname{div}(f \nabla W) = \langle \nabla W, \nabla f \rangle + f \Delta W$. Therefore,

$$L^* p = \Delta p + \operatorname{div}(p \nabla W).$$

Theorem 10.8.2 (Parabolic regularity theorem). *Suppose that $u: M \times \mathbb{R} \rightarrow \mathbb{R}$ is a bounded solution to*

$$\begin{cases} \frac{\partial u(x, t)}{\partial t} = \Delta u - \langle \nabla W, \nabla u \rangle \\ u(x, 0) = f(x), \end{cases}$$

where $f, W \in C^\infty(M)$. Then $u(t, \cdot) \in C^\infty(M)$.

Because $u(x, t) = (P_t f)(x)$ satisfies above PDE, $P_t f \in C^\infty(M)$ when $f \in C^\infty(M)$. Therefore, we have $P_t \mathcal{A} \subset \mathcal{A}$. It follows that (M, μ, Γ) is a compact Markov triple.

Lemma 10.8.3. *Let (M, g) be a Riemannian manifold with the Levi-Civita connection ∇ and $f \in C^\infty(M)$.*

(1) *Let E_i be an orthonormal frame in TM . Then*

$$\|\nabla^2 f\|^2 = \sum_i \|\nabla_{E_i}(\nabla f)\|^2,$$

where note that $\|\nabla^2 f\|^2 = g^{kh} g^{lj} f_{;hl} f_{;kj}$.

(2) *For any $p \in M$, choosing a normal coordinate centered p ,*

$$(\Delta f)(p) = \sum_{i=1}^n (\partial_i^2 f)(p).$$

Proof. (1) Locally, let

$$E_i = E_i^\ell \frac{\partial}{\partial x^\ell}.$$

Let

$$\nabla_{\frac{\partial}{\partial x^\ell}} dx^k = -\Gamma_{\ell h}^k dx^h,$$

where $\Gamma_{\ell h}^k$ is the Christoffel symbol. Then

$$\begin{aligned} \nabla_{E_i}(\nabla f) &= \nabla_{E_i} \left(\frac{\partial f}{\partial x^k} dx^k \right) \\ &= E_i^\ell \frac{\partial^2 f}{\partial x^k \partial x^\ell} dx^k - E_i^\ell \frac{\partial f}{\partial x^k} \Gamma_{\ell h}^k dx^h \end{aligned}$$

$$\begin{aligned}
&= E_i^\ell \left(\frac{\partial^2 f}{\partial x^h \partial x^\ell} - \frac{\partial f}{\partial x^k} \Gamma_{\ell h}^k \right) dx^h \\
&= E_i^\ell f_{;h\ell} dx^h.
\end{aligned}$$

So we have

$$\|\nabla_{E_i}(\nabla f)\|^2 = g^{kh} (E_i^\ell f_{;h\ell}) (E_i^j f_{;kj}),$$

and

$$\sum_i \|\nabla_{E_i}(\nabla f)\|^2 = g^{kh} \left(\sum_i E_i^\ell E_i^j \right) f_{;h\ell} f_{;kj}.$$

Because E_i is orthonormal,

$$\begin{aligned}
\langle E_i, E_m \rangle &= \left\langle E_i^\ell \frac{\partial}{\partial x^\ell}, E_m^j \frac{\partial}{\partial x^j} \right\rangle \\
&= E_i^\ell g_{\ell j} E_m^j = \delta_{im},
\end{aligned}$$

which means that $E^\top g E = I$ for matrix $E = (E_j^m)_{m \times j}$. It follows that

$$g^{-1} = E E^\top \Rightarrow g^{\ell j} = \sum_i E_i^\ell E_i^j.$$

Therefore,

$$\sum_i \|\nabla_{E_i}(\nabla f)\|^2 = g^{kh} g^{\ell j} f_{;h\ell} f_{;kj} = \|\nabla^2 f\|^2.$$

(2) Choose a normal coordinate centered at p , i.e. $g^{ij}(p) = \delta_{ij}$ and $\Gamma_{ij}^h(p) = 0$. Then

$$\begin{aligned}
(\Delta f)(p) &= g^{ij}(p) f_{;ij}(p) \\
&= g^{ij}(p) \left(\frac{\partial^2 f}{\partial x^i \partial x^j}(p) - \frac{\partial f}{\partial x^h}(p) \Gamma_{ij}^h(p) \right) \\
&= \frac{\partial^2 f}{(\partial x^i)^2}(p).
\end{aligned}$$

□

Remark 10.8.4. In general, for $\omega_1, \omega_2 \in \Gamma(T^*M)$,

$$\langle \omega_1, \omega_2 \rangle := \langle \sharp \omega_1, \sharp \omega_2 \rangle = g(\sharp \omega_1, \sharp \omega_2).$$

Note that for $\omega_1 = f_i^1 dx^i$ and $\omega_2 = f_j^2 dx^j$,

$$\begin{aligned}
\langle \omega_1, \omega_2 \rangle &= \left\langle g^{\ell i} f_i^1 \frac{\partial}{\partial x^\ell}, g^{kj} f_j^2 \frac{\partial}{\partial x^k} \right\rangle \\
&= f_i^1 f_j^2 g^{\ell i} g^{kj} g_{\ell k} \\
&= f_i^1 f_j^2 g^{ji},
\end{aligned}$$

which means that $\langle \cdot, \cdot \rangle$ on T^*M with the matrix expression g^{-1} . In this notation,

$$\langle \omega, X \rangle := \langle \sharp \omega, X \rangle = \langle \omega, \flat X \rangle = \omega(X).$$

Note that $\langle X, \text{grad } f \rangle = X(f)$ and by above notation

$$\langle \omega, \nabla f \rangle = \langle \omega, \text{grad } f \rangle = \omega(\text{grad } f).$$

Moreover, for any $\omega \in T^*M$ and $X \in \Gamma(TM)$, because ∇ is Levi-Civita,

$$\nabla_X(\sharp\omega) = \sharp(\nabla_X\omega).$$

It is because for any $Y \in \Gamma(TM)$,

$$\begin{aligned} Xg(\sharp\omega, Y) &= g(\nabla_X\sharp\omega, Y) + g(\sharp\omega, \nabla_XY) \\ &= g(\nabla_X\sharp\omega, Y) + \omega(\nabla_XY). \end{aligned}$$

On the other hand,

$$\begin{aligned} Xg(\sharp\omega, Y) &= X(\omega(Y)) = (\nabla_X\omega)(Y) + \omega(\nabla_XY) \\ &= g(\sharp(\nabla_X\omega), Y) + \omega(\nabla_XY). \end{aligned}$$

Therefore,

$$g(\sharp(\nabla_X\omega), Y) = g(\nabla_X\sharp\omega, Y) \Rightarrow \nabla_X(\sharp\omega) = \sharp(\nabla_X\omega).$$

It follows that

$$\langle \nabla_X(\sharp\omega_1), \sharp\omega_2 \rangle = \langle \nabla_X\omega_1, \omega_2 \rangle,$$

which implies that

$$\nabla_X \langle \omega_1, \omega_2 \rangle = \langle \nabla_X\omega_1, \omega_2 \rangle + \langle \omega_1, \nabla_X\omega_2 \rangle.$$

Lemma 10.8.5. Choose a orthonormal basis $\{E_i\}$ and let

$$H_{ij} = \nabla^2 f(E_i, E_j).$$

Then

$$\|\nabla^2 f\|^2 = \sum_{i,j} H_{ij}^2, \quad \Delta f = \sum_j H_{ii},$$

i.e., $\|\nabla^2 f\|^2 = \|H\|_F^2$ and $\Delta f = \text{tr } H$, which induces

$$(\Delta f)^2 \leq n \|\nabla^2 f\|^2$$

if $\dim M = n$.

Proof. First, for any $\omega \in \Gamma(T^*M)$,

$$\begin{aligned} \sum_j \omega(E_j)^2 &= \sum_j \left(\sum_i \omega_i E_j^i \right) \left(\sum_\ell \omega_\ell E_j^\ell \right) \\ &= \sum_{i,\ell} \omega_i \omega_\ell \left(\sum_j E_j^i E_j^\ell \right) \\ &= \sum_{i,\ell} \omega_i \omega_\ell g^{i\ell} = \|\omega\|^2. \end{aligned}$$

By above lemma,

$$\|\nabla^2 f\|^2 = \sum_{i,j} (\nabla^2 f(E_i, E_j))^2.$$

And $\Delta f = \text{tr } H$ is just by definition. □

Definition 10.8.6 (Weighted Ricci Curvature). Given $W \in C^\infty(M)$, the weighted Ricci curvature is a $(2, 0)$ -tensor Ric_W defined by

$$\text{Ric}_W(X, Y) := \text{Ric}(X, Y) + (\nabla^2 W)(X, Y).$$

Proposition 10.8.7. *Let (M, g) be a compact Riemannian manifold and $W \in C^\infty(M)$. Define*

$$\begin{aligned} Lf &= \Delta f - \langle \nabla W, \nabla f \rangle \\ \Gamma(f, g) &= \langle \nabla f, \nabla g \rangle \\ \Gamma_2(f) &= \frac{1}{2}L\Gamma(f) - \Gamma(f, Lf). \end{aligned}$$

Then

$$\Gamma_2(f) = \text{Ric}_W(\text{grad } f, \text{grad } f) + \|\nabla^2 f\|^2.$$

Proof. By Bochner's formula, we have

$$\frac{1}{2}\Delta \|\nabla f\|^2 = \text{Ric}(\text{grad } f, \text{grad } f) + \langle \nabla f, \nabla(\Delta f) \rangle + \|\nabla^2 f\|^2.$$

By definition,

$$\begin{aligned} \Gamma_2(f) &= \frac{1}{2}L(\|\nabla f\|^2) - \langle \nabla f, \nabla Lf \rangle \\ &= \frac{1}{2}\Delta(\|\nabla f\|^2) - \frac{1}{2}\langle \nabla W, \nabla(\|\nabla f\|^2) \rangle - \langle \nabla f, \nabla(\Delta f) \rangle + \langle \nabla f, \nabla \langle \nabla W, \nabla f \rangle \rangle \\ &= \text{Ric}(\text{grad } f, \text{grad } f) + \|\nabla^2 f\|^2 - \frac{1}{2}\langle \nabla W, \nabla(\|\nabla f\|^2) \rangle + \langle \nabla f, \nabla \langle \nabla W, \nabla f \rangle \rangle. \end{aligned}$$

Moreover,

$$\begin{aligned} \frac{1}{2}\langle \nabla W, \nabla(\|\nabla f\|^2) \rangle &= \frac{1}{2}\nabla_{\text{grad } W} \langle \nabla f, \nabla f \rangle \\ &= \langle \nabla_{\text{grad } W} \nabla f, \nabla f \rangle. \end{aligned}$$

On the other hand,

$$\begin{aligned} \langle \nabla f, \nabla \langle \nabla W, \nabla f \rangle \rangle &= \nabla_{\text{grad } f} \langle \nabla W, \nabla f \rangle \\ &= \langle \nabla_{\text{grad } f} \nabla W, \nabla f \rangle + \langle \nabla W, \nabla_{\text{grad } f} \nabla f \rangle \\ &= (\nabla^2 W)(\text{grad } f, \text{grad } f) + \langle \nabla W, \nabla_{\text{grad } f} \nabla f \rangle, \end{aligned}$$

because

$$(\nabla^2 W)(\text{grad } f, \text{grad } f) = (\nabla_{\text{grad } f} \nabla W)(\text{grad } f) = \langle \nabla_{\text{grad } f} \nabla W, \nabla f \rangle$$

Moreover,

$$\langle \nabla_{\text{grad } W} \nabla f, \nabla f \rangle = \nabla^2 f(\text{grad } f, \text{grad } W)$$

and

$$\langle \nabla W, \nabla_{\text{grad } f} \nabla f \rangle = \nabla^2 f(\text{grad } W, \text{grad } f),$$

they are same by the symmetry of $\nabla^2 f$. Therefore,

$$\begin{aligned} \Gamma_2(f) &= \text{Ric}(\text{grad } f, \text{grad } f) + \|\nabla^2 f\|^2 + (\nabla^2 W)(\text{grad } f, \text{grad } f) \\ &= \text{Ric}_W(\text{grad } f, \text{grad } f) + \|\nabla^2 f\|^2. \end{aligned}$$

□

Corollary 10.8.8. *Consider a compact Markov triple (M, μ, Γ) defined as above, where M has dimension n .*

(1) *If $W \equiv c$ and $\text{Ric} \geq \rho g$, then (M, μ, Γ) satisfies $\text{CD}(\rho, n)$.*

(2) *If $\text{Ric}_W \geq \rho g$, then (M, μ, Γ) satisfies $\text{CD}(\rho, \infty)$.*

(3) If $\text{Ric}_W \geq \rho g$ and $m > n$ such that

$$\text{Ric}_W \geq \rho g + \frac{1}{m-n} \nabla W \otimes \nabla W,$$

then (M, μ, Γ) satisfies $\text{CD}(\rho, m)$.

Proof. (1) It is directly obtained by

$$(\Delta f)^2 \leq n \|\nabla f\|^2.$$

(2) It is obvious.

(3) Similarly,

$$\begin{aligned} \Gamma_2(f) &\geq \rho \Gamma(f) + \frac{1}{m-n} \langle \nabla W, \nabla f \rangle^2 + \frac{1}{n} (\Delta f)^2 \\ &\geq \rho \Gamma(f) + \frac{1}{m} (\Delta f - \langle \nabla W, \nabla f \rangle), \end{aligned}$$

where the last inequality is because $\frac{a^2}{m} + \frac{b^2}{m-n} \geq \frac{1}{m}(a+b)^2$. □

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